

Keumoe Fozeu

Solutions to Tao's Analysis I

Chapters 1 – 7

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Disclaimer
While every effort has been made to ensure the mathematical accuracy of these solutions, they have not been peer-reviewed. Readers are encouraged to verify the arguments independently.

To all my mathematics teachers

Preface

Analysis I by Terence Tao was my first introduction to real analysis. After my first year at university, I wanted a preview of the subject before taking the formal course the following semester. I chose this book for two reasons. First, I sought a different perspective from the one I would eventually see in class. Second, I was drawn to the author, Professor Terence Tao, a leading figure in contemporary mathematics. Since his primary work is at a level of mathematics that I do not yet fully grasp, I was excited to learn that he had written an introductory undergraduate textbook, allowing me to engage with a work of his that I could understand.

Upon reading the book, I was won over by Professor Tao's method of rigorously building everything from the foundations. It fosters a deeper understanding of the subject, a more rigorous mind, and a particularly greater capacity for abstract reasoning. Although I eventually had to pivot to my university's prescribed text when the semester began, I returned to this book later as part of a personal endeavor. I then decided to compile the solutions to the chapters I had completed into a structured, clean, and professional document.

I am principally writing this for undergraduate students seeking a different perspective on these problems. It is generally worthwhile to consult alternative methods of reaching the truth. I would recommend the reader to look at these solutions only after reaching their own solution or after sustained effort trying to come up with one. In the latter case, it would be best to read only the initial steps before attempting to complete the argument independently. This ensures the reader derives the maximum learning benefit from this manual.

In an attempt to be as helpful as possible, emphasis has been placed on clarity and completeness. All arguments are written in full detail, with explicit references to definitions, theorems, propositions, corollaries and lemmas where they are applied

(except in cases where frequently used results have become so familiar that they can be invoked without explicit mention). Intermediate steps are generally not omitted, and results are applied only after their introduction in the text, with the exception of Exercise 3.4.6. While this level of detail may seem redundant and slow-paced to some, the objective is to provide a very accessible learning support material. Also, the notations and conventions mirror the original textbook, allowing for seamless use alongside it. Please, note that this work contains only the solutions. The exercise prompts as well as the definition and propositions are only available in Professor Tao's textbook.

While one does not need concepts above high school mathematics to begin this book, familiarity with elementary logic would be critically helpful not only to understand the text but also to understand the solutions here. It is especially the case in Chapter 3. The reader is thus encouraged to go through Appendix A (The Basics of Mathematical Logic) of the textbook as early as possible.

At present, this manual covers up to Chapter 7. Chapters 8, 9 and 10 have already been solved and will be added soon. Chapter 11 will be added afterwards as well. Furthermore, this manual may be extended in the future to include *Analysis II*. To get the latest version and updates, please visit the manual's GitHub: https://www.github.com/KeumoeFozeu/solutions-tao-analysis.

In a work of this magnitude, mistakes may remain despite all my proofreading and reviewing efforts. I would be grateful for any correction, comment, or question, which may be sent to: keumoe.fozeu.r@gmail.com.

I am deeply thankful to Professor Terence Tao for this masterful introduction to mathematical analysis. I am also very grateful to all my mathematics teachers, in particular Mr. Ulrich Moyo, Mr. Michel Fotso, Mr. Raoul Koung, and Mr. Coleens Nyengka, who opened my mind to the elegance of mathematics and encouraged me to grow and explore beyond the standard curriculum.

Yaoundé, Cameroon
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About the Author

Born in Cameroon, **Keumoe Fozeu** went to South Korea to study mathematical sciences at the Korea Advanced Institute of Science and Technology (KAIST). He won the National Award at the 2021 International Youth Math Challenge. Alongside his mathematical studies, he has pursued independent projects on African thought and civilization, including the curation of *African Wisdom: 888 Quotes from the Cradle of Humanity* in both French and English.

Chapter 2
Starting at the Beginning: The Natural Numbers

2.2 Addition

Exercise 2.2.1

Let us prove that the addition of natural numbers is associative, i.e., let us prove that for any natural numbers a, b, c, we have: $(a+b)+c = a+(b+c)$. We can proceed by induction on a (keeping b and c fixed).

The base case is $a = 0$. By the definition of addition (Definition 2.2.1), $0+b = b$ and $0+(b+c) = b+c$, thus:

$$(0+b)+c = b+c = 0 = 0+(b+c).$$

Hence, the associativity holds for $a = 0$.

Now, suppose inductively that the associativity holds for a. Then,

$$\begin{aligned}
((a{+}{+})+b)+c &= ((a+b){+}{+})+c && \text{by Definition 2.2.1}\\
&= ((a+b)+c){+}{+} && \text{by Definition 2.2.1}\\
&= ((a+(b+c)){+}{+} && \text{by induction hypothesis}\\
&= (a{+}{+})+(b+c) && \text{by Definition 2.2.1.}
\end{aligned}$$

Thus, the associativity property is also true for $a{+}{+}$.

We conclude that addition is associative over the natural numbers.

Exercise 2.2.2

Let us prove that any positive natural number is the successor of exactly one natural number.

To do this, we can just prove that every natural number is either 0 or the successor of exactly one natural number. We proceed by induction. The base case is readily validated since we start at 0.

Now, suppose that a is a natural number that is either 0 or the successor of exactly one natural number. In the case where a is 0, then $a{+}{+} = 0{+}{+}$, i.e., $a{+}{+}$ is the successor of 0. By Axiom 2.4, there is no natural number distinct from 0 whose successor is $a{+}{+}$. Hence, $a{+}{+}$ is the successor of exactly one natural number.

In the case where a is the successor of exactly one natural number, let b be that natural number. We have: $a = b{+}{+}$. This implies that: $a{+}{+} = (b{+}{+}){+}{+}$. Given that $b{+}{+}$ is a natural number (by Axiom 2.2), $a{+}{+}$ is the successor of a natural number. Plus, Axiom 2.4 guarantees that $b{+}{+}$ is the only natural number whose successor is $a{+}{+}$. Thus, $a{+}{+}$ is the successor of exactly one natural number. This closes the induction.

We conclude that all nonzero natural numbers, that is all positive natural numbers, are the successor of exactly one natural number.

Exercise 2.2.3

Let a, b, c be natural numbers. Let us show that:

(a) Order is reflexive.

By Lemma 2.2.2, $a = a + 0$. Since 0 is a natural number, we have $a \geq a$, according to Definition 2.2.11.

(b) Order is transitive.

Suppose that $a \geq b$ and $b \geq c$. By Definition 2.2.11,

$$a \geq b \Longrightarrow a = b + x \text{ for some natural number } x$$

and

$$b \geq c \Longrightarrow b = c + y \text{ for some natural number } y.$$

So, $a = (c+y)+x$, which implies that $a = c+(y+x)$ by the associativity of addition (Proposition 2.2.5). Since $y+x$ is a natural number (as the addition of two natural numbers), we see that $a \geq c$ by Definition 2.2.11.

(c) Order is antisymmetric.

Suppose that $a \geq b$ and $b \geq c$. By Definition 2.2.11,

$$a \geq b \Longrightarrow a = b + x \text{ for some natural number } x$$

and

$$b \geq a \Longrightarrow b = a + y \text{ for some natural number } y.$$

So, $b = (b+x)+y = b+(x+y)$ (by the associativity of addition—Proposition 2.2.5), and thus $b+0 = b+(x+y)$ (see Lemma 2.2.2). Now, the cancellation law (Proposition 2.2.6) yields $0 = x+y$. Thus, $x = 0$ and $y = 0$ by Corollary 2.2.9. But this implies that $a = b+x = b+0 = b$. Therefore, $a = b$.

(d) Addition preserves order.

On the one hand, suppose that $a \geq b$. According to Definition 2.2.11, this means that there exists a natural number x such that $a = b+x$. Using the commutativity and the associativity of addition (Propositions 2.2.4 and 2.2.5), we get

$$a+c = (b+x)+c = b+(x+c) = b+(c+x) = (b+c)+x.$$

Thus, $a+c \geq b+c$ by Definition 2.2.11.

On the other hand, suppose that $a+c \geq b+c$. Then, it follows from Definition 2.2.11 that there exists a natural number y such that $a+c = (b+c)+y$. Using the commutativity and associativity of addition (Propositions 2.2.4 and 2.2.5), we see that

$$a+c = (b+c)+y \iff c+a = (c+b)+y \iff c+a = c+(b+y).$$

Next, the cancellation law (Proposition 2.2.6) yields $a = b+y$. This implies that $a \geq b$ by Definition 2.2.11.

In conclusion, we see that $a \geq b$ if and only if $a+c \geq b+c$.

(e) We have $a < b$ if and only if $a{+}{+} \leq b$.

We will prove the two implications between the statements to establish their equivalence.

Suppose first that $a < b$. By Definition 2.2.11, this means that there exists a natural number x such that $b = a + x$ and $b \neq a$. If $x = 0$, then $b = a + 0 = a$, which is impossible. Hence, $x \neq 0$ and there is a natural number y such that $y{+}{+} = x$ (Lemma 2.2.10). Thus, by Lemma 2.2.3 and Definition 2.2.1,

$$b = a + (y{+}{+}) = (a + y){+}{+} = (a{+}{+}) + y.$$

So, $a{+}{+} \leq b$ following Definition 2.2.11.

Second, suppose that $a{+}{+} \leq b$. Then, there exists a natural number k for which $b = (a{+}{+}) + k$, by Definition 2.2.11. By Definition 2.2.1 and Lemma 2.2.3, we have $b = a + (k{+}{+})$. As $k{+}{+}$ is a natural number (Axiom 2.2) and $b \neq a$ (for $b = a$ implies by the cancellation law that $k{+}{+} = 0$, a clear contradiction given Axiom 2.3), we conclude that $a < b$ by Definition 2.2.11.

Therefore, $a < b$ if and only if $a{+}{+} \leq b$.

(f) We have $a < b$ if and only if $b = a + d$ for some positive number d.

On the one side, suppose that $a < b$. It means that $a \leq b$ and $a \neq b$, that is $b = a + d$ for some natural number d and $a \neq b$ (by Definition 2.2.11). If $d = 0$, then $b = a + d$ yields $b = a$, a contradiction. Hence, d is a positive natural number.

On the other side, suppose that $b = a + d$ for some positive natural number d. Then by Definition 2.2.11, $a \leq b$. Furthermore, $a \neq b$ for otherwise we have a contradiction:

$$\begin{aligned} a = b &\Longrightarrow a = a + d \\ &\Longrightarrow a + 0 = a + d && \text{by Lemma 2.2.2} \\ &\Longrightarrow 0 = d && \text{by Proposition 2.2.6.} \end{aligned}$$

Thus, $a < b$.

We conclude that $a < b$ if and only if $b = a + d$ for some positive number d.[1]

[1] Here is another, simpler way to prove this equivalence. Proposition 2.2.12(e) says that $a < b$ if and only if $a{+}{+} \leq b$. According to Definition 2.2.11, $a{+}{+} \leq b$ if and only if $b = (a{+}{+}) + x$, for some natural number x. But $b = (a{+}{+}) + x$ if and only if $b = a + (x{+}{+})$ (by Definition 2.2.1 and Lemma 2.2.3). Therefore, $a < b$ if and only if $b = a + (x{+}{+})$, where x is a natural number. Given that $x{+}{+}$ is non-zero (Axiom 2.3) and thus positive (Definition 2.2.7), we see that $a < b$ if and only if $b = a + d$, for some positive natural number d.

Exercise 2.2.4

1. We are to prove that any natural number b is greater than or equal to 0.

For any natural number b, we have $b = 0 + b$ by Definition 2.2.1, and hence, $0 \leq b$ by Definition 2.2.11.

2. Let us show that for any natural numbers a and b, if $a > b$, then $a{+}{+} > b$.

Let a and b be two natural numbers such that $a > b$. Then, by Proposition 2.2.12(f), $a = b + d$ for some positive natural number d. So, $a{+}{+} = (b+d){+}{+} = b + (d{+}{+})$, by Lemma 2.2.3. Since $d{+}{+}$ is a positive natural number (Axiom 2.3), this implies that $a{+}{+} > b$ by Proposition 2.2.12(f).

3. Let us prove that for any natural numbers a and b, if $a = b$, then $a{+}{+} > b$.

Let a and b be two natural numbers such that $a = b$. We know that $b{+}{+} = b + 1$ (see text after Lemma 2.2.3).

As 1 is a positive natural number (because it is different from zero which is the only natural number that is not positive), $b{+}{+} > b$ by Proposition 2.2.12(f).

In addition, $a = b$ implies that $a{+}{+} = b{+}{+}$. To see this, assume toward contradiction that $a{+}{+} > b{+}{+}$. Then, there exists a positive natural number d such that $a{+}{+} = (b{+}{+}) + d$ by Proposition 2.2.12(f). We see that $a{+}{+} = (b+d){+}{+}$ by Definition 2.2.1. In other words, $a + 1 = (b+d) + 1$ (see text after Lemma 2.2.3). It follows from Proposition 2.2.4 that $1 + a = 1 + (b+d)$. Hence, $a = b + d$ by Proposition 2.2.6, and so $a > b$ by Proposition 2.2.12(f). But this implies that $a \neq b$ by Definition 2.2.11, a contradiction. It is easy to see that, similarly, $a{+}{+} < b{+}{+}$ yields a contradiction. So we have $b{+}{+} > b$ and $a{+}{+} = b{+}{+}$. Therefore, $a{+}{+} > b$.

Exercise 2.2.5

Let m_0 be a natural number and let $P(m)$ be a property pertaining to an arbitrary natural number m. Let $Q(n)$ be the property "$P(m)$ is true for all $m_0 \leq m < n$". Suppose that for each $m \geq m_0$, we have the following implication: if $Q(m)$ is true, then $P(m)$ is also true. To prove Proposition 2.2.14, we just have to show that $P(m)$ is true for all natural numbers $m \geq m_0$ under these conditions. We start by proving that $Q(m)$ is true for all natural numbers $m \geq m_0$ using induction (Axiom 2.5).

To start with, the statement $Q(0)$: "$P(m')$ is true for all $m_0 \leq m' < 0$" is vacuously

true[2] because all natural numbers are greater than or equal to 0[3] . This validates the base case.

Now, note that there is no number m' for which $m_0 \leq m' < d$, where d is a natural number less than or equal to m_0, because of the transitivity, the asymmetry, and the trichotomy of order in $\mathbf{N}$. Indeed, $m_0 \leq m' < d \leq m_0$ implies that $m_0 \leq m' \leq m_0$, which implies that $m' = m_0$; thus $m' < d$ means that $m_0 < d$, a contradiction. Therefore, $Q(d)$ is vacuously true for all natural numbers $0 \leq d \leq m_0$. Hence, we can safely restrict the inductive step of our proof to numbers greater than or equal to m_0.

Accordingly, let k be a natural number greater than or equal to m_0 and suppose that $Q(k)$ is true. Given that $Q(m) \Longrightarrow P(m)$ for each $m \geq m_0$, we see that $P(k)$ is true as well. Since $Q(k)$ and $P(k)$ are true, $P(k)$ is true for all natural numbers m such that $m_0 \leq m \leq k$, i.e., $m_0 \leq m < k{+}{+}$ (see Proposition 2.2.12(e)[4]). But this just means that $Q(k{+}{+})$ is true. This closes the induction.

We have just shown that $Q(m)$ is true for all natural numbers m. In particular, $Q(m)$ is true for all natural numbers $m \geq m_0$. Given the implication $Q(m) \Longrightarrow P(m)$ for each $m \geq m_0$, we conclude that $P(m)$ is also true for all $m \geq m_0$.

Exercise 2.2.6

Let $P(m)$ be a property pertaining to the natural numbers such that whenever $P(m{+}{+})$ is true, $P(m)$ is true. We will prove the following statement by induction on n: "If $P(n)$ is true, then $P(m)$ is true for all natural numbers $m \leq n$."

For the base case, it is obvious that if $P(0)$ is true, then $P(m)$ is true for all natural numbers $m \leq 0$.

Now, suppose inductively that the statement is true for a natural number k. This means that if $P(k)$ is true, then $P(m)$ is true for all natural numbers $m \leq k$. Suppose that $P(k{+}{+})$ is true. Then $P(k)$ is also true. Therefore, $P(m)$ is true for all $m \leq k$ (by

[2] A statement is said to be *vacuously* true if there are no counterexamples to contradict it. For example, the statement "blue apples do not grow in Yaounde" is vacuously true because there exists no real blue apple to contradict it. In our particular case, the statement "$P(m')$ is true for all $m_0 \leq m' < 0$" is vacuously true because there is no natural number m' greater than or equal to m_0 and less than 0 to contradict it. See Appendix A.2 in the textbook to learn more about vacuous statements.

[3] We do not start the induction at $m = m_0$ because we do not yet know that it is possible; so far, we only know how to do induction beginning at 0 (Axiom 2.5). We will have to wait for Exercise 2.2.7.

[4] Note that for any natural numbers a and b, the statement "$a < b \iff a{+}{+} \leq b$" is logically equivalent to the statement "$a \geq b \iff a{+}{+} > b$".

induction hypothesis). In other words, $P(m)$ is true for all $m < k{+}{+}$ (see Proposition 2.2.12(e)[4]). And since $P(k{+}{+})$ is also true, $P(m)$ is true for all $m \leq k{+}{+}$. So, the statement is true for $k{+}{+}$ as well. This closes the induction.

Exercise 2.2.7

Let $P(m)$ be a property pertaining to the natural numbers such that whenever $P(m)$ is true, $P(m{+}{+})$ is true. Let n be a natural number. We want to prove that if $P(n)$ is true, then $P(k)$ is true for all $k \geq n$. To do this, we consider the statement $Q(m)$: "$P(m)$ is false", pertaining to natural numbers m. It is enough to show that if $Q(k)$ is true for some $k \geq n$, then $Q(n)$ is also true[5] .

Note that for any natural number m, whenever $P(m)$ is true, $P(m{+}{+})$ is also true; thus, whenever $P(m{+}{+})$ is false, $P(m)$ is false (contrapositive); that is, whenever $Q(m{+}{+})$ is true, $Q(m)$ is also true.

Now, suppose that $Q(k)$ is true for some $k \geq n$. By the principle of backwards induction (see Exercise 2.2.6), $Q(m)$ is true for all natural numbers $m \leq k$. Since $n \leq k$, $Q(n)$ is true as well.

In summary, we have shown that "if $Q(k)$ is true for some $k \geq n$, then $Q(n)$ is also true" or, in other words, "if $P(k)$ is false for some $k \geq n$, then $P(n)$ is also false." But this is just equivalent to "if $P(n)$ is true, then $P(k)$ is true for all $k \geq n$."

[5] Note that for any mathematical statements p and q, the implication p true implies q true is logically equivalent to the implication q false implies p false (its contrapositive). So, we can prove any of these two implications by proving the other one. In our particular case, the implication we want to prove is $\big(P(n)\big)$ true implies $\big(P(k)$ is true for all $k \geq n\big)$ true. Its contrapositive is $\big(P(k)$ is false for some $k \geq n\big)$ true implies $\big(P(n)\big)$ false. In other words, $\big(Q(k)$ is true for some $k \geq n\big)$ true implies $\big(Q(n)\big)$ true. See Appendix A.2 in the textbook to learn more about contrapositives.

2.3 Multiplication

Exercise 2.3.1

We are to prove that the statement $P(n)$: "$n \times m = m \times n$, for all natural numbers m" is true for all natural numbers n. We will use induction.

The base case is $n = 0$. First, $0 \times m := 0$ for all natural numbers m by Definition 2.3.1. Second, we claim that $m \times 0 = 0$ for all natural numbers m and show this by induction as follows. For the base case ($m = 0$), we have $0 \times 0 := 0$. For the inductive step, if $m \times 0 = 0$ for some natural number m, then $(m{+}{+}) \times 0 = (m \times 0) + 0 = 0 + 0 = 0$ by Definition 2.3.1. This establishes the claim. Hence,

$$0 \times m = 0 = m \times 0$$

for all natural numbers m. Therefore, $P(0)$ is true and we are done with the base case.

Now, suppose inductively that $P(n)$ is true for some natural number n[6]. By Definition 2.3.1, $(n{+}{+}) \times m = (n \times m) + m$, for all natural numbers m. By induction hypothesis, $(n{+}{+}) \times m = (n \times m) + m = (m \times n) + m$.

We claim that $m \times (n{+}{+}) = m \times n + m$, for all natural numbers m. We show this by induction. For the base case ($m = 0$), we have $0 \times (n{+}{+}) := 0 = 0 \times n + 0$ by Definition 2.3.1. For the inductive step, suppose that $m \times (n{+}{+}) = m \times n + m$, for some natural number m. We have:

$$\begin{aligned}
(m{+}{+}) \times (n{+}{+}) &= m \times (n{+}{+}) + (n{+}{+}) && \text{by Definition 2.3.1} \\
&= m \times n + m + (n{+}{+}) && \text{by the induction hypothesis} \\
&= m \times n + m + n + 1 \\
&= (m \times n + n) + (m + 1) \\
&= (m{+}{+}) \times n + (m{+}{+}) && \text{by Definition 2.3.1.}
\end{aligned}$$

This proves our claim by induction.

Therefore,

$$(n{+}{+}) \times m = n \times m + m = m \times n + m = m \times (n{+}{+}),$$

[6] Note that, so far, we refrain from using the set **N** as well as set-theoritic notations and terms like $\in$, "belongs to", $n \in \mathbf{N}$. This is because we have not talked about sets yet. Even in Assumption 2.6, **N** was presented as a "number system", and not a "set." What is a set (even informally)? We will have to wait till Chapter 3 to inquire about and manipulate sets.

for all natural numbers m. This closes the induction and proves that $P(n)$ is true for all natural numbers n.

Exercise 2.3.2

Let n and m be natural numbers.

Suppose that n and m are both positive and consider their product nm. Since n is positive, there exists a natural number a such that $a{+}{+} = n$ (Lemma 2.2.10). Hence,

$$nm = (a{+}{+}) \times m := a \times m + m.$$

Since m is positive, $a \times m + m$ is also positive by Proposition 2.2.8. Thus, nm is a positive natural number.

Now, if $n = 0$, then $n \times m = 0 \times m := 0$ (Definition 2.3.1). If instead, $m = 0$, we have $n \times m = m \times n = 0 \times n := 0$, according to Definition 2.3.1 and Lemma 2.3.2. So,

$$n = 0 \text{ or } m = 0 \Longrightarrow nm = 0.$$

Besides, we proved earlier that $(n > 0 \text{ and } m > 0) \Longrightarrow nm > 0$. But this is just equivalent to:

$$nm = 0 \Longrightarrow n = 0 \text{ or } m = 0.$$

Therefore,

$$nm = 0 \Longleftrightarrow n = 0 \text{ or } m = 0.$$

Exercise 2.3.3

We will prove by induction on a that $(a \times b) \times c = a \times (b \times c)$ for all natural numbers a, b, and c.

In the base case, $a = 0$. On the one hand, we have $(0 \times b) \times c = 0 \times c = 0$, by using the definition of multiplication by 0 twice. On the other hand, $0 \times (b \times c) := 0$, by using the definition once. So, $(0 \times b) \times c = 0 \times (b \times c)$, and the base case is done.

Now suppose inductively that $(a \times b) \times c = a \times (b \times c)$. Then,

$$\begin{aligned}((a{+}{+}) \times b) \times c &= ((a \times b) + b) \times c && \text{by Definition 2.3.1}\\ &= (a \times b) \times c + b \times c && \text{by Proposition 2.3.4}\\ &= a \times (b \times c) + b \times c && \text{by induction hypothesis}\\ &= (a{+}{+}) \times (b \times c) && \text{by Definition 2.3.1.}\end{aligned}$$

This closes the induction.

Exercise 2.3.4

Let a and b be natural numbers. With a triple application of Definition 2.3.11, we have

$$(a+b)^2 = (a+b)^1(a+b) = (a+b)^0(a+b)(a+b) = 1 \times (a+b) \times (a+b).$$

By Definition 2.3.1,

$$1 \times (a+b) \times (a+b) = (0 \times (a+b) + (a+b)) \times (a+b) = (0 + (a+b)) \times (a+b).$$

Hence,

$$(a+b)^2 = (a+b) \times (a+b).$$

Next, we use Proposition 2.3.4 twice to have

$$(a+b)^2 = (a+b) \times a + (a+b) \times b = a \times a + b \times a + a \times b + b \times b.$$

By the same reasoning we used to show that $(a+b)^2 = (a+b) \times (a+b)$, we see that

$$a^2 = a \times a \quad \text{and} \quad b^2 = b \times b.$$

In addition,

$$b \times a + a \times b = ba + ab = ab + ab$$

by Lemma 2.3.2. Since $1ab = (0{+}{+})ab := 0 \times ab + ab = 0 + ab = ab$ by Definition 2.3.1, we have

$$ab + ab = 1ab + 1ab.$$

By Proposition 2.3.4,

$$1ab + 1ab = (1+1)ab = 2ab.$$

Hence,

$$b \times a + a \times b = 2ab.$$

Therefore,

$$(a+b)^2 = a^2 + 2ab + b^2.$$

Exercise 2.3.5

We will prove that, for any natural number n and any positive natural number q, there exist natural numbers m, r such that $0 \leq r < q$ and $n = mq + r$, by induction on n (keeping q fixed).

For the base case ($n = 0$), we can just take $m = 0$ and $r = 0$. Zero is less than any positive natural number and Definition 2.3.1 guarantees that $0 \times q + 0 = 0$.

For the inductive step, suppose that there are natural numbers m, r such that $0 \leq r < q$ and $n = mq + r$ for some natural number n. By Lemma 2.2.3, we have:

$$n{+}{+} = (mq + r){+}{+} = mq + (r{+}{+}).$$

Because $r < q$, $r{+}{+} \leq q$ (Proposition 2.2.12(e)). If $r{+}{+} < q$, then we are done. But if $r{+}{+} = q$, then $mq + (r{+}{+}) = mq + q = (m{+}{+}) \times q$ by Definition 2.3.1. So, $n{+}{+} = (m{+}{+})q + 0$ and we are done. The proposition is thus true for $n{+}{+}$ as well. This closes the induction.

Chapter 3
Set Theory

3.1 Fundamentals

Exercise 3.1.1

Let a, b, c, d be objects such that $\{a, b\} = \{c, d\}$. We will check if any of the statements "$a = c$ and $b = d$" and "$a = d$ and $b = c$" is true.

To start with, assume that $a = b$. Then, by Axiom 3.2 and Axiom 3.4, the pair set $\{a, b\}$ is equal to the singleton set $\{a\}$, since both contain only the object a. In this case, $\{a, b\} = \{c, d\}$ implies that $\{a\} = \{c, d\}$. As $c \in \{c, d\}$ and $d \in \{c, d\}$ (Axiom 3.4), we have $c \in \{a\}$ and $d \in \{a\}$ by the substitution axiom. But this means that $c = a$ and $d = a$. Hence, $c = d = a = b$. Therefore, both statements are true. Following this same line of reasoning, it is easy to see that when $c = d$, then $a = b$ and the two statements are true.

Now, assume that $a \neq b$. Then $c \neq d$ (for if $c = d$, then $a = b$ as seen above). By Axiom 3.4, $a \in \{a, b\}$ and $b \in \{a, b\}$. Since $\{a, b\} = \{c, d\}$, using the substitution axiom yields $a \in \{c, d\}$ and $b \in \{c, d\}$. Hence a is equal to c or d, and b is equal to c or d (Axiom 3.4). If $a = c$, then $a \neq b$ implies that $c \neq b$, and so $b = d$. In this case, the first of the two statements to check is true. If $a = d$, then $a \neq b$ implies that $d \neq b$, and so $b = c$. Thus, the second of these two statements is true. Had we assumed at first that $c \neq d$, then we would also have $a \neq b$ and, using the same course of reasoning, we would also see that one of the two statements is true.

We conclude that at least one of the two statements holds in any case.

Exercise 3.1.2

By Axiom 3.3, the empty set $\emptyset$ exists and contains no element. By Axiom 3.1, the empty set, being a set, is an object. Therefore, there exists a set containing only the empty set, namely the singleton set $\{\emptyset\}$, by Axiom 3.4. Given that $\emptyset \notin \emptyset$ (Axiom 3.3) while $\emptyset \in \{\emptyset\}$, it follows from Axiom 3.2 that $\emptyset \neq \{\emptyset\}$.

Being a set, $\{\emptyset\}$ is an object according to Axiom 3.1. Therefore, the set $\{\{\emptyset\}\}$ that contains only one element, namely $\{\emptyset\}$, exists by Axiom 3.4. Since $\{\emptyset\} \notin \emptyset$ (by Axiom 3.3) and $\{\emptyset\} \in \{\{\emptyset\}\}$, we have $\emptyset \neq \{\{\emptyset\}\}$ by Axiom 3.2.

Besides, we saw earlier that $\emptyset \neq \{\emptyset\}$. Given that the only element in $\{\emptyset\}$ is $\emptyset$ and that the only element in $\{\{\emptyset\}\}$ is $\{\emptyset\}$, the sets $\{\emptyset\}$ and $\{\{\emptyset\}\}$ do not have any element in common. So, $\{\emptyset\} \neq \{\{\emptyset\}\}$ by Axiom 3.2.

Now, as $\emptyset$ and $\{\emptyset\}$ are objects, there exists a pair set $\{\emptyset, \{\emptyset\}\}$ whose only elements are $\emptyset$ and $\{\emptyset\}$ by Axiom 3.4. On the one hand, $\emptyset \notin \emptyset$ by Axiom 3.3, and on the other hand, $\emptyset \in \{\emptyset, \{\emptyset\}\}$. Thus, $\emptyset \neq \{\emptyset, \{\emptyset\}\}$ by Axiom 3.2.

Next, note that $\{\emptyset\} \neq \emptyset$ implies that $\{\emptyset\} \notin \{\emptyset\}$ because $\emptyset$ is the only element of $\{\emptyset\}$. Since $\{\emptyset\} \notin \{\emptyset\}$ and $\{\emptyset\} \in \{\emptyset, \{\emptyset\}\}$, we have $\{\emptyset\} \neq \{\emptyset, \{\emptyset\}\}$ by Axiom 3.2.

Finally, we note that $\emptyset \notin \{\{\emptyset\}\}$ (because $\emptyset \neq \{\emptyset\}$ and only $\{\emptyset\}$ belongs to $\{\{\emptyset\}\}$) and $\emptyset \in \{\emptyset, \{\emptyset\}\}$, and use Axiom 3.2 to deduce that $\{\{\emptyset\}\} \neq \{\emptyset, \{\emptyset\}\}$.

In summary, the sets $\emptyset$, $\{\emptyset\}$, $\{\{\emptyset\}\}$, and $\{\emptyset, \{\emptyset\}\}$ are all distinct.

Exercise 3.1.3

We have three claims to prove.

1. Let a and b be objects. Let us prove that $\{a, b\} = \{a\} \cup \{b\}$.

By Axiom 3.5, the elements of $\{a\} \cup \{b\}$ are the elements which belong to $\{a\}$ or $\{b\}$ or both. By Axiom 3.4, the only element in $\{a\}$ is a and the only element in $\{b\}$ is b. Hence, the elements in $\{a\} \cup \{b\}$ are just a and b. But these are precisely all the elements of $\{a, b\}$ according to Axiom 3.4. Therefore, $\{a, b\} = \{a\} \cup \{b\}$ by Axiom 3.2.

2. Let A, B be sets. We will show that $A \cup B = B \cup A$.

Let x be a mathematical object. According to Axiom 3.5, we have:

$$x \in A \cup B \iff x \in A \text{ or } x \in B$$

and

$$x \in B \cup A \iff x \in B \text{ or } x \in A.$$

But the statements "$x \in A$ or $x \in B$" and "$x \in B$ or $x \in A$" are logically equivalent. Therefore,

$$x \in A \cup B \iff x \in B \cup A.$$

This means that

$$A \cup B = B \cup A$$

by Axiom 3.2.

3. Let A be a set. Let us prove that $A \cup A = A \cup \emptyset = \emptyset \cup A = A$.

Consider an object x. To start with,

$$x \in A \cup A \iff x \in A \text{ or } x \in A \iff x \in A^{1},$$

in light of Axiom 3.5. Hence, $A \cup A = A$, according to Axiom 3.2. Now, note that $x \notin \emptyset$ (Axiom 3.3), hence the statement "$x \in \emptyset$" is false. Thus, by applying Axiom 3.5, we have

$$x \in A \cup \emptyset \iff x \in A \text{ or } x \in \emptyset \iff x \in A^{2}.$$

So, $A \cup \emptyset = A$ by Axiom 3.2. Since the pairwise union of sets is commutative (as shown above), $A \cup \emptyset = \emptyset \cup A$. In conclusion,

$$A \cup A = A \cup \emptyset = \emptyset \cup A = A.$$

Exercise 3.1.4

Let A, B, C be sets.

Suppose that $A \subseteq B$ and $B \subseteq A$. Then, by Definition 3.1.14 (subsets), all the elements of A are elements of B and all the elements of B are elements of A. It follows from Axiom 3.2 that $A = B$.

[1] For any mathematical statement p, the statement "p or p" is logically equivalent to the statement p.
[2] If p is a true statement and q is a false statement, then the statement "p or q" is logically equivalent to the statement p. You are highly encouraged to read Appendix A (The Basics of Mathematical Logic) if you are not yet familiar with these notions.

Now, suppose that $A \subsetneq B$ and $B \subsetneq C$. By Definition 3.1.14, this means that $A \subseteq B$ and $A \neq B$, and $B \subseteq C$ and $B \neq C$. If $A = C$, then $A \subseteq B$ is equivalent to $C \subseteq B$ (axiom of substitution). But based on the claim we proved above, $C \subseteq B$ and $B \subseteq C$ yield $B = C$, a contradiction. Hence, $A \neq C$. Moreover, $A \subseteq B$ and $B \subseteq C$ imply that $A \subseteq C$ according to the first claim of Proposition 3.1.17[3]. Given that $A \subseteq C$ and $A \neq C$, we have $A \subsetneq C$ by Definition 3.1.14.

Exercise 3.1.5

Let A and B be sets and consider the statements:

$$(E_1)\text{: “}A \subseteq B\text{”}, \quad (E_2)\text{: “}A \cup B = B\text{”}, \text{ and } (E_3)\text{: “}A \cap B = A\text{”}.$$

We will show that (E_1), (E_2) and (E_3) are equivalent statements.

Let us first deal with the case where A is the empty set. We have $x \notin A$ for any object x by Axiom 3.3. So for any object x, the implication “$x \in \emptyset \implies x \in B$” is vacuously true. Hence, $A \subseteq B$ by Definition 3.1.14. Also, for any object x, $x \in A \cap B$ implies that $x \in A$, a contradiction; hence, $x \notin A \cap B$. Thus $A \cap B = \emptyset = A$ by Axiom 3.2. In addition, for any object x, we have

$$x \in A \cup B \iff x \in A \text{ or } x \in B;$$

by Axiom 3.5; but as $x \notin A$, the statement “$x \in A$ or $x \in B$” is equivalent to “$x \in B$”, and so

$$x \in A \cup B \iff x \in B.$$

This means that $A \cup B = B$ according to Axiom 3.2. Therefore, when $A = \emptyset$, (E_1), (E_2), and (E_3) are equivalent as they are equally true statements.

Now, we assume that $A \neq \emptyset$. First of all, suppose that (E_2) is true, i.e., $A \cup B = B$. Then, by Axiom 3.5,

$$x \in A \text{ or } x \in B \iff x \in B$$

for any object x. As “$x \in A$” implies “$x \in A$ or $x \in B$”, we have

$$x \in A \implies x \in B.$$

This means that $A \subseteq B$ by Definition 3.1.14. Therefore, (E_2) implies (E_1).

[3] That claim has been proved in the textbook.

Second, suppose that (E_1) is true, i.e., $A \subseteq B$. This means that any element of A belongs to B (Definition 3.1.14). Thus, any element of A belongs to both A and B. So, any element of A belongs to $A \cap B$ by Definition 3.1.22. The same definition also tells us that all the elements of $A \cap B$ belong in A. Therefore, by Axiom 3.2, $A = A \cap B$. This shows that (E_1) implies (E_3).

Third, suppose that (E_3) is true, i.e., $A \cap B = A$. For any object x, the statements "$x \in A$ and $x \in B$" and "$x \in A$" are thus equivalent. Let z be an element of A. Then, $z \in A$ and $z \in B$ (as per the latter equivalence), and so $z \in B$. Hence, "$z \in A$ or $z \in B$" implies that $z \in B$. In other words, $z \in A \cup B$ implies that $z \in B$ (Axiom 3.5). This means that $A \cup B \subseteq B$ according to Definition 3.1.14. Besides, it is clear that for any object x, $x \in B$ implies that $x \in A$ or $x \in B$. That is to say, $B \subseteq A \cup B$ by Definition 3.1.14. So, $A \cup B = B$ by Proposition 3.1.17. Therefore, (E_3) implies (E_2).

These three implications establish the equivalence of the statements (E_1), (E_2) and (E_3), when $A \neq \emptyset$.

Exercise 3.1.6

Let A, B, C be sets, and let X be a set containing A, B, C as subsets. We will prove the following eight statements.

(a) $A \cup \emptyset = A$ and $A \cap \emptyset = \emptyset$.

Since $\emptyset \subseteq A$, we know from Exercise 3.1.5 that $A \cup \emptyset = A$ and $A \cap \emptyset = \emptyset$.

(b) $A \cup X = X$ and $A \cap X = A$

Given that $A \subseteq X$, we have $A \cup X = X$ and $A \cap X = A$, as shown in Exercise 3.1.5.

(c) $A \cap A = A$ and $A \cup A = A$

It is obvious that $A \subseteq A$ (any element of A belongs to A). Using the result of Exercise 3.1.5 once again, we have: $A \cap A = A$ and $A \cup A = A$. (You can also easily prove this using the definitions of intersection and pairwise union directly.)

(d) $A \cup B = B \cup A$ and $A \cap B = B \cap A$

By the commutativity of the union operation (Lemma 3.1.12), we know that $A \cup B = B \cup A$. Applying Definition 3.1.22, for any object x, we have

$$x \in A \cap B \iff x \in A \text{ and } x \in B \iff x \in B \text{ and } x \in A \iff x \in B \cap A.$$

Thus, $A \cap B = B \cap A$.

(e) $(A \cup B) \cup C = A \cup (B \cup C)$ and $(A \cap B) \cap C = A \cap (B \cap C)$

By Lemma 3.1.12, we have $(A \cup B) \cup C = A \cup (B \cup C)$. Now using Definition 3.1.22, we see that for any object x,

$$\begin{aligned} x \in (A \cap B) \cap C &\iff x \in A \cap B \text{ and } x \in C \\ &\iff (x \in A \text{ and } x \in B) \text{ and } x \in C \\ &\iff x \in A \text{ and } x \in B \text{ and } x \in C \\ &\iff x \in A \text{ and } (x \in B \text{ and } x \in C) \\ &\iff x \in A \text{ and } x \in B \cap C \\ &\iff x \in A \cap (B \cap C) \end{aligned}$$

Therefore, $(A \cap B) \cap C = A \cap (B \cap C)$.

(f) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ and $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

Let x be an object. On the one hand,

$$\begin{aligned} x \in A \cap (B \cup C) &\iff x \in A \text{ and } x \in B \cup C \\ &\iff x \in A \text{ and } (x \in B \text{ or } x \in C) \\ &\iff (x \in A \text{ and } x \in B) \text{ or } (x \in A \text{ and } x \in C) \\ &\iff x \in A \cap B \text{ or } x \in A \cap C \\ &\iff x \in (A \cap B) \cup (A \cap C). \end{aligned}$$

Thus, $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$. On the other hand,

$$\begin{aligned} x \in A \cup (B \cap C) &\iff x \in A \text{ or } x \in B \cap C \\ &\iff x \in A \text{ or } (x \in B \text{ and } x \in C) \\ &\iff (x \in A \text{ or } x \in B) \text{ and } (x \in A \text{ or } x \in C) \\ &\iff x \in A \cup B \text{ and } x \in A \cup C \\ &\iff x \in (A \cup B) \cap (A \cup C). \end{aligned}$$

Thus, $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

(g) $A \cup (X \setminus A) = X$ and $A \cap (X \setminus A) = \emptyset$

Since $A \subseteq X$ and $X \setminus A \subseteq X$ (from Definition 3.1.26), we have $A \cup (X \setminus A) \subseteq X$ as any element of A or $X \setminus A$ or both is also an element of X. For any x in X, if $x \notin A$, then $x \in X \setminus A$ (Definition 3.1.26), otherwise $x \in A$. So, any object in X also belongs to $(X \setminus A) \cup A$. Hence, $X \subseteq (X \setminus A) \cup A$. Therefore, $(X \setminus A) \cup A = X$.

By Definition 3.1.26, any element of $X \setminus A$ does not belong to A. So, for any object x, the statement "$x \in A$ and $x \in X \setminus A$" is false. But that statement is logically equivalent to "$x \in A \cap B$" (Definition 3.1.22). Therefore, $A \cap (X \setminus A) = \emptyset$.

(h) $X \setminus (A \cup B) = (X \setminus A) \cap (X \setminus B)$ and $X \setminus (A \cap B) = (X \setminus A) \cup (X \setminus B)$

Let x be an object. On the one hand, we have:

$$\begin{aligned} x \in X \setminus (A \cup B) &\iff x \in X \text{ and } x \notin (A \cup B) && \text{by Definition 3.1.26} \\ &\iff x \in X \text{ and } (x \notin A \text{ and } x \notin B) && \text{by Axiom 3.5.} \end{aligned}$$

On the other hand,

$$\begin{aligned} x \in (X \setminus A) \cap (X \setminus B) &\iff x \in X \setminus A \text{ and } x \in X \setminus B && \text{by Definition 3.1.22} \\ &\iff (x \in X \text{ and } x \notin A) \text{ and } (x \in X \text{ and } x \notin B) \\ & && \text{by Definition 3.1.26} \\ &\iff x \in X \text{ and } (x \notin A \text{ and } x \notin B). \end{aligned}$$

So,

$$x \in X \setminus (A \cup B) \iff x \in (X \setminus A) \cap (X \setminus B).$$

Therefore, $X \setminus (A \cup B) = (X \setminus A) \cap (X \setminus B)$ by Axiom 3.2.

Note that

$$\begin{aligned} x \in (X \setminus A) \cup (X \setminus B) &\iff x \in X \setminus A \text{ or } x \in X \setminus B && \text{by Axiom 3.5} \\ &\iff (x \in X \text{ and } x \notin A) \text{ or } (x \in X \text{ and } x \notin B) \\ & && \text{by Definition 3.1.26} \\ &\iff x \in X \text{ and } (x \notin A \text{ or } x \notin B) \\ &\iff x \in X \text{ and } x \notin A \cap B && \text{by Definition 3.1.22} \\ &\iff x \in X \setminus (A \cap B) && \text{by Definition 3.1.26.} \end{aligned}$$

Therefore, $X \setminus (A \cap B) = (X \setminus A) \cup (X \setminus B)$ by Axiom 3.2.

Exercise 3.1.7

Let A, B, C be sets. We will prove the following four claims. We will use Definition 3.1.14 throughout without explicitly mentioning it, but it should be very clear.

1. $A \cap B \subseteq A$ and $A \cap B \subseteq B$

By Definition 3.1.22, any element of $A \cap B$ is an element of A and an element of B. Hence, $A \cap B \subseteq A$ and $A \cap B \subseteq B$.

2. $C \subseteq A$ and $C \subseteq B$ if and only if $C \subseteq A \cap B$

Since $A \cap B \subseteq A$ and $A \cap B \subseteq B$, we have:

$$C \subseteq A \cap B \implies C \subseteq A \text{ and } C \subseteq B,$$

by the transitivity of $\subseteq$ (Proposition 3.1.17).

If $C \subseteq A$ and $C \subseteq B$, then every element of C is also an element of both A and B. That is, every element of C belongs to $A \cap B$ (Definition 3.1.22). Thus, $C \subseteq A \cap B$.

Therefore, $C \subseteq A$ and $C \subseteq B$ if and only if $C \subseteq A \cap B$.

3. $A \subseteq A \cup B$ and $B \subseteq A \cup B$

Every element of A and every element of B belong to $A \cup B$ according to Axiom 3.5. So, $A \subseteq A \cup B$ and $B \subseteq A \cup B$.

4. $A \subseteq C$ and $B \subseteq C$ if and only if $A \cup B \subseteq C$

On the one hand, given that $A \subseteq A \cup B$ and $B \subseteq A \cup B$, Proposition 3.1.17 ensures that

$$A \cup B \subseteq C \implies A \subseteq C \text{ and } B \subseteq C.$$

On the other hand, "$A \subseteq C$ and $B \subseteq C$" implies that every element of A and every element in B belong to C. Thus, every element of $A \cup B$ belongs to C. In other words, $A \cup B \subseteq C$.

Therefore, $A \subseteq C$ and $B \subseteq C$ if and only if $A \cup B \subseteq C$.

Exercise 3.1.8

Let A, B be sets. To start with, note that $A \subseteq A$ and $A \subseteq A \cup B$ (by the third claim proved in Exercise 3.1.7). So, $A \subseteq A \cap (A \cup B)$, according to the second claim proved in Exercise 3.1.7. In addition, $A \cap (A \cup B) \subseteq A$ by the first claim proved in that same exercise. So, $A \cap (A \cup B) = A$.

Now, the third claim proved in Exercise 3.1.7 shows that $A \subseteq A \cup (A \cap B)$. Plus, we have $A \subseteq A$ and $A \cap B \subseteq A$ (Definition 3.1.22). So, $A \cup (A \cap B) \subseteq A$ by the fourth claim proved in Exercise 3.1.7. Therefore, $A \cup (A \cap B) = A$.

Exercise 3.1.9

Let A, B, X be sets such that $A \cup B = X$ and $A \cap B = \emptyset$. Since $A \cup B = B \cup A$ and $A \cap B = B \cap A$ (by Proposition 3.1.27(d)), we can just show that $A = X \setminus B$ to prove not only that $A = X \setminus B$ but also that $B = X \setminus A$, without loss of generality.

By Definition 3.1.26,

$$X \setminus B := \{x \in X : x \notin B\} = \{x \in A \cup B : x \notin B\}.$$

Let x be an object. We have:

$$\begin{aligned} x \in X \setminus B &\iff x \in A \cup B \text{ and } x \notin B \\ &\iff (x \in A \text{ or } x \in B) \text{ and } x \notin B \\ &\iff (x \in A \text{ and } x \notin B) \text{ or } (x \in B \text{ and } x \notin B). \end{aligned}$$

Since "$x \in B$ and $x \notin B$" is false, the statement "$(x \in A$ and $x \notin B)$ or $(x \in B$ and $x \notin B)$" is equivalent to the statement "$x \in A$ and $x \notin B$". Thus,

$$x \in X \setminus B \iff x \in A \text{ and } x \notin B \implies x \in A.$$

Now, we have $A \cap B = \emptyset$. So, no element of A belongs to B and vice versa. Hence, if $x \in A$, then $x \notin B$. As $x \in A$ (trivially) implies that $x \in A$, we see that $x \in A$ implies "$x \in A$ and $x \notin B$". So,

$$x \in X \setminus B \iff x \in A \text{ and } x \notin B \iff x \in A.$$

We conclude that $A = X \setminus B$ and, by symmetry, $B = X \setminus A$.

Exercise 3.1.10

Let A and B be sets. Note that for any object x,

$$\begin{aligned} x \in A \setminus B &\iff x \in A \text{ and } x \notin B, \\ x \in A \cap B &\iff x \in A \text{ and } x \in B, \\ x \in B \setminus A &\iff x \notin A \text{ and } x \in B. \end{aligned}$$

Since the statement "$x \in A$ and $x \notin A$" and the statement "$x \in B$ and $x \notin B$" are both false, there is no object that belongs to any two of the sets $A \setminus B$, $A \cap B$, and $B \setminus A$. In other words, these sets are disjoint.

From the third claim proved in Exercise 3.1.7, we know that $A \subseteq A \cup B$ and $B \subseteq A \cup B$. Plus, it is clear from Definition 3.1.26 that $A \setminus B \subseteq A$ and $B \setminus A \subseteq B$ because for any object x,

$$x \in A \setminus B \implies x \in A \text{ and } x \notin B \implies x \in A$$

and

$$x \in B \setminus A \implies x \in B \text{ and } x \notin A \implies x \in B.$$

It follows from Proposition 3.1.17 that

$$A \setminus B \subseteq A \cup B \text{ and } B \setminus A \subseteq A \cup B.$$

Hence,

$$(A \setminus B) \cup (B \setminus A) \subseteq A \cup B$$

by the fourth claim proved in Exercise 3.1.7. In addition, given that $A \cap B \subseteq A$ (by the first claim proved in Exercised 3.1.7) and $A \subseteq A \cup B$, we have

$$A \cap B \subseteq A \cup B$$

by Proposition 3.1.17. Therefore, applying the fourth claim proved in Exercise 3.1.7 again, we have

$$((A \setminus B) \cup (B \setminus A)) \cup (A \cap B) \subseteq A \cup B,$$

which leads to

$$(A \setminus B) \cup (A \cap B) \cup (B \setminus A) \subseteq A \cup B.$$

Now, for any object x, $x \in A \cup B$ if and only if $x \in A$ or $x \in B$. In the case where $x \in A$, if $x \in B$, then $x \in A \cap B$, otherwise $x \notin B$, and so $x \in A \setminus B$. Similarly, when $x \in B$, either $x \in A \cap B$ or $x \in B \setminus A$. Thus, any element of $A \cup B$ is in $A \setminus B$, $A \cap B$, or $B \setminus A$. That is to say,

$$A \cup B \subseteq (A \setminus B) \cup (A \cap B) \cup (B \setminus A).$$

In conclusion:

$$(A \setminus B) \cup (A \cap B) \cup (B \setminus A) = A \cup B.$$

Exercise 3.1.11

Let A be a set. For any object $x \in A$, and any object y, consider the statement

$$Q(x, y) : \text{“}P(x) \text{ is true and } y = x\text{”}$$

where $P(x)$ is a property pertaining to $x \in A$. Then, $Q(x, y)$ pertains to x and y, such that for each $x \in A$, there is at most one y for which $Q(x, y)$ is true.

By the axiom of replacement (Axiom 3.7), there exists a set

$$M := \{y : Q(x, y) \text{ is true for some } x \in A\}.$$

That is,

$$M = \{y : P(x) \text{ is true and } y = x, \text{ for some } x \in A\}.$$

Using the axiom of substitution, we see that if an object y belongs to M, then $y \in A$ (as $y = x$ for some $x \in A$) and $P(y)$ is true (as for some $x \in A$, $y = x$ and $P(x)$ is true). Also, for any $x \in A$ such that $P(x)$ is true, it is clear that $x \in M$ since $Q(x, x)$ is true. Therefore, M consists of all the elements x in A for which $P(x)$ is true. But this is just the set we would have obtained from the axiom of specification (Axiom 3.6) by specifying A using $P(x)$.

Since we can derive any set obtained from the axiom of specification by using the axiom of replacement, we conclude that the axiom of replacement implies the axiom of specification.

Exercise 3.1.12

Let A, B, A', and B' be sets such that $A' \subseteq A$ and $B' \subseteq B$.

(i) We will prove that $A' \cup B' \subseteq A \cup B$ and $A' \cap B' \subseteq A \cap B$.

Here, we will freely use Proposition 3.1.17 and the results of Exercise 3.1.7, without precisely pointing out where they are used. However, it should be quite clear at this point. Note that

$$A' \subseteq A \text{ and } A \subseteq A \cup B \implies A' \subseteq A \cup B$$

and

$$B' \subseteq B \text{ and } B \subseteq A \cup B \implies B' \subseteq A \cup B.$$

Next,

$$A' \subseteq A \cup B \text{ and } B' \subseteq A \cup B \implies A' \cup B' \subseteq A \cup B.$$

Similarly, we have

$$A' \cap B' \subseteq A' \text{ and } A' \subseteq A \implies A' \cap B' \subseteq A$$

and

$$A' \cap B' \subseteq B' \text{ and } B' \subseteq B \implies A' \cap B' \subseteq B;$$

and then,

$$A' \cap B' \subseteq A \text{ and } A' \cap B' \subseteq B \implies A' \cap B' \subseteq A \cap B.$$

(ii) The statement $A' \setminus B' \subseteq A \setminus B$ is false. For a counterexample, suppose that $A = \{1,2,3\}$, $A' = \{3\}$, $B = \{2,3,4\}$, and $B' = \{4\}$. Then, we have $A' \setminus B' = \{3\}$ while $A \setminus B = \{1\}$, and so $A' \setminus B' \nsubseteq A \setminus B$.

A modification of this statement which is true is: "$A' \setminus B \subseteq A \setminus B'$." It is easy to see that this statement holds by noting that for any object x:

$$x \in A' \setminus B \implies (x \in A' \text{ and } x \notin B) \implies (x \in A \text{ and } x \notin B') \implies x \in A \setminus B',$$

where the second implication holds because $A' \subseteq A$ and $B' \subseteq B$ respectively.

Exercise 3.1.13

Let A be a non-empty set.

First, suppose that A does not have any non-empty proper subset. By the single choice lemma (Lemma 3.1.5), there exists an object a in A. As $a \in A$, the singleton set $\{a\}$ is a subset of A. However, $\{a\}$ is not a proper subset of A since $\{a\} \neq \emptyset$. Thus, $\{a\} = A$.

Second, suppose that A is of the form $A = \{x\}$ for some object x. Then, for any object y, y belongs to A if and only if $y = x$. Let B be a non-empty subset of A. Every element of B belongs to A, and so every element of B is equal to x. Hence, B consists of only one element, namely x. This shows that $B = \{x\} = A$. So, the only non-empty subset of A is A itself. Therefore, A does not have any non-empty proper subsets.

In conclusion, A does not have any non-empty proper subsets if and only if A is of the form $A = \{x\}$ for some object x.

3.2 Russell's Paradox (Optional)

Exercise 3.2.1

Assume that Axiom 3.9, the axiom of universal specification, is true.

First, consider the property $P_3(x)$: "x is not an object" that pertains to every object x. By Axiom 3.9, the set $S_3 := \{x : P_3(x) \text{ is true}\}$ exists. The set S_3 contains no element as $P_3(x)$ is false for every object x. We see that S_3 is the empty set of Axiom 3.3.

Second, let $P_4(x)$: "x is a" and $Q_4(x)$: "x is a or x is b" be two properties, where a and b are objects. According to Axiom 3.9, the sets $S_4 := \{x : P_4(x) \text{ is true}\}$ and $T_4 := \{x : Q_4(x) \text{ is true}\}$ exist. Since $P_4(x)$ is true only when $x = a$ and $Q_4(x)$ is true only when $x = a$ or $x = b$, we see that S_4 is a singleton set and T_4 is a pair set, following the definitions given in Axiom 3.4.

Third, let A and B be sets and consider the property $P_5(x)$: "$x \in A$ or $x \in B$". By Axiom 3.9, the set $S_5 := \{x : P_5(x) \text{ is true}\}$ exist. The S_5 consists of the objects that belong in A or B or both. Thus, S_5 is the union of A and B as defined in Axiom 3.5.

Fourth, let F be a set. Let $P_6(y)$: "$Q(x,y)$ is true for some $x \in F$" be a property, where $Q(x,y)$ is a property pertaining to x and y, such that for each $x \in F$ there is at most one y for which $Q(x,y)$ is true. By Axiom 3.9, the set $S_7 := \{y : P_6(y) \text{ is true}\}$ exists. Given that $y \in S_7$ if and only if $Q(x,y)$ is true for some $x \in F$, S_7 is the set we would have obtained if we used Axiom 3.7 on F with the statement $Q(x,y)$.

Finally, we proved in Exercise 3.1.11 that Axiom 3.7 implies Axiom 3.6. Since Axiom 3.9 implies Axiom 3.7, it is also true that Axiom 3.9 implies Axiom 3.6.

Exercise 3.2.2

Let A be a set.

First, assume that A is empty. Then, $A \notin A$ since no object belongs in the empty set by Axiom 3.3.

Now, assume that A is a non-empty set. Suppose that $A \in A$. Since A is an object, there exists a singleton set $C := \{A\}$ according to Axiom 3.4. The set C has one element which is A. But, A is a set and $C \cap A = \{A\} \neq \emptyset$. This contradicts the axiom

of regularity (Axiom 3.10). Thus, $A \notin A$.

Let E and F be two sets. If any one of them, say E, is the empty set, then we have $F \notin E$ because no object belongs in the empty set. Now, let's assume that they are both non-empty. Suppose that $E \in F$ and $F \in E$ (note that $E \notin E$ and $F \notin F$, as proven above.) By Axiom 3.4, the pair set $G := \{E, F\}$ exists. The elements of G are the sets E and F and we have $G \cap E = \{F\} \neq \emptyset$ and $G \cap F = \{E\} \neq \emptyset$. But this is a contradiction according to the axiom of regularity (Axiom 3.10). Therefore, the statement "$E \in F$ and $F \in E$" is false, or equivalently, the statement "$E \notin F$ or $F \notin E$"[4] is true.

Exercise 3.2.3

On the one hand, assume that Axiom 3.9 is true. Let $P(x)$ be the statement "x is an object". By Axiom 3.9, there exists a set Ω defined by $\Omega := \{x : P(x) \text{ is true}\}$. We see that Ω is the universal set as it contains all objects.

On the other hand, suppose that there exists a universal set Ω. By the axiom of specification (Axiom 3.6), for any property $P(x)$ pertaining to an object x, the set $L := \{x \in \Omega : P(x) \text{ is true}\}$ exists; and given that for any object y,

$$y \in L \iff P(y) \text{ is true},$$

we see that L is the set we would have obtained from Axiom 3.9 by using $P(x)$. Therefore, the existence of the universal set implies Axiom 3.9.

In conclusion, Axiom 3.9 is equivalent to an axiom postulating the existence of a universal set consisting of all objects.

[4] Recall that in mathematics, "or" is understood to be inclusive by default.

3.3 Functions

Exercise 3.3.1

We are going to show that the definition of the equality of functions in Definition 3.3.8 is reflexive, symmetric, and transitive, in that order.

But before that, we have to prove that the equality of sets, as defined in Axiom 3.2, is reflexive, symmetric, and transitive. Take three sets A, B, and C.

For reflexivity, note that every element x of A is an element of A as there is no element x of A that is also not an element of A. Otherwise, there is some object x_0 such that $x_0 \in A$ and $x_0 \notin A$, a contradiction. So, $A = A$.

For symmetry, suppose that $A = B$. Then, the definition tells us that every element x of A belongs in B and every element y of B belongs to A. So clearly, we also have $B = A$.

For transitivity, suppose that $A = B$ and $B = C$. Then, every element of A belongs in B and every element of B belongs in C. This implies that every element of A is an element of C. Similarly, every element of C belongs in B and every element of B belongs in C. Thus, every element of C is an element of A. It follows that $A = C$ according to Axiom 3.2.

Now, let us show that the equality of functions as defined in Definition 3.3.8 is an equivalence relation.

For reflexivity, let $f : X \to Y$ be a function defined by a property P. If $f \neq f$, then either $X \neq X$ or $Y \neq Y$ or $P(x, y) \not\iff P(x, y)$ for some $x \in X$ and some $y \in Y$ (i.e., $f(x) \neq f(x)$ for some x in X), by Definition 3.3.1. But, since the equality of sets is reflexive, $X = X$ and $Y = Y$. Furthermore, any statement is logically equivalent to itself, and so $P(x, y) \iff P(x, y)$ for all $x \in X$ and all $y \in Y$, so that $f(x) = f(x)$ for all $x \in X$. Therefore, f must be equal to f.

To show symmetry, let $f : X \to Y$ and $g : X' \to Y'$ be functions such that $f = g$. Let P a property defining f. Since $f = g$, we have $X = X'$ and $Y = Y'$, by Definition 3.3.1. Because of the symmetry of the equality of sets, we also have $X' = X$ (E_1) and $Y' = Y$ (E_2). In addition, we have $f(x) = g(x)$ for all $x \in X$ by Definition 3.3.1, so that $P(x, f(x)) \iff P(x, g(x))$ for all $x \in X$ (E_3), in light of the axiom of substitution. If $x_0 \in X'$ such that $g(x_0) \neq f(x_0)$, then $x_0 \in X$ (as $X' = X$) and $g(x_0) \in Y$ (as $g(x_0) \in Y'$ and $Y' = Y$), moreover, $P(x_0, g(x_0))$ must be false because for any $y \in Y$, $P(x_0, y)$ is true if and only if $y = f(x_0)$ (see Definition 3.3.1); hence, $P(x_0, f(x_0)) \not\iff P(x_0, g(x_0))$. But this contradicts (E_3). Therefore, $g(x) = f(x)$ for all $x \in X'$ (E_4). From (E_1), (E_2),

and (E_4), we deduce that $g = f$ in light of Definition 3.3.1.

For transitivity, let $f : X \to Y$, $g : X' \to Y'$ and $h : X'' \to Y''$ be functions such that $f = g$ and $g = h$. According to Definition 3.3.8, $f = g$ implies that $X = X'$ and $Y = Y'$, while $g = h$ implies that $X' = X''$ and $Y' = Y''$. By the transitivity of the equality of sets, we have $X = X''$ and $Y = Y''$. Let P be a property defining f. For every $x \in X$, $f(x) = g(x)$, and so $P(x, f(x)) \iff P(x, g(x))$ by the axiom of substitution. Plus, $g(x) = h(x)$, thus $P(x, g(x)) \iff P(x, h(x))$ by the axiom of substitution, and so $P(x, f(x)) \iff P(x, h(x))$. Therefore, $f(x) = h(x)$ for every $x \in X$ (for, if $f(x) \neq h(x)$, then $P(x, h(x))$ would be false while $P(x, f(x))$ holds true, and that would contradict the equivalence). We deduce that $f = h$.

Our second task is to verify the substitution property of the composition of functions. Let $f, \tilde{f} : X \to Y$ and $g, \tilde{g} : Y \to Z$ be functions such that $f = \tilde{f}$ and $g = \tilde{g}$. First, we know from Definition 3.1.13 that the domain and codomain of $g \circ f$ are X and Z respectively, and the domain and codomain of $\tilde{g} \circ \tilde{f}$ are also X and Z respectively. Second, for any $x \in X$,

$$\begin{aligned}
(g \circ f)(x) &= g(f(x)) && \text{by Definition 3.3.13} \\
&= g(\tilde{f}(x)) && \text{since } f = \tilde{f} \\
&= \tilde{g}(\tilde{f}(x)) && \text{since } g = \tilde{g} \\
&= (\tilde{g} \circ \tilde{f})(x) && \text{by Definition 3.3.13.}
\end{aligned}$$

Therefore,

$$g \circ f = \tilde{g} \circ \tilde{f}.$$

Exercise 3.3.2

Let $f : X \to Y$ and $g : Y \to Z$ be functions.

Suppose that f and g are both injective. There are two cases to consider here.

In the first case, $X = \emptyset$. Then, $g \circ f$ is the empty function : $\emptyset \to Z$. As $\emptyset$ has no element, it is vacuously true that different elements map to different elements under $g \circ f$. That is to say, $g \circ f$ is injective.

In the second case, $X \neq \emptyset$. Here, we also have $Y \neq \emptyset$ and $Z \neq \emptyset$ for if $Y = \emptyset$, then X must be the empty set for f to be a well-defined function, and if $Z = \emptyset$, then Y must be the empty set for g to be well-defined (otherwise, elements of the domain would have no corresponding element in the codomain). Now, suppose that x, x' are elements of X such that $g \circ f(x) = g \circ f(x')$, i.e., $g(f(x)) = g(f(x'))$. Then, $f(x) = f(x')$ since g is

injective. This in turn implies that $x = x'$ since f is injective. So, $g \circ f$ is an injective function as well.

Suppose that f and g are both surjective.

If $Z = \emptyset$, then $Y = \emptyset$ and $X = \emptyset$ (otherwise f and g are not well-defined). In this case, $g \circ f$ is surjective because $g \circ f = f = g$ (they are all the unique empty function from $\emptyset$ to $\emptyset$, which is subjective because it is vacuously true that every element of the codomain has a corresponding element in the domain).

If $Z \neq \emptyset$, let $z \in Z$. Since g is surjective, there exists a y in Y such that $g(y) = z$. Since f is surjective, there exists an x in X such that $f(x) = y$. Hence, $g(f(x)) = z$, i.e., $g \circ f(x) = z$ by the axiom of substitution. We see that $g \circ f$ is a surjective function.

Exercise 3.3.3

Let X be a set and consider the empty function $f : \emptyset \to X$.

As we noted in Exercise 3.3.2, the empty function is always injective because there are no two distinct elements in $\emptyset$ that have the same image in X by f. This is vacuously true since $\emptyset$ does not have two elements.

The empty function is surjective only when its codomain is the empty set $\emptyset$. To see this clearly, suppose that the codomain X is non-empty. For any x in X, there is no element t in the domain $\emptyset$ such that $f(t) = x$, since the domain contains no element. Hence, the empty function is not surjective when X is non-empty. Suppose now that X is empty. The statement "if $y \in \emptyset$, then there exists $x \in \emptyset$ such that $f(x) = y$" is vacuously true because there is no object y that lies in $\emptyset$. So, f is surjective in this case.

Given that the empty function is always injective but surjective only when its codomain X is the empty set, it is clear that the empty function is bijective only when its codomain is the empty set.

Exercise 3.3.4

Let $f : X \to Y$, $\tilde{f} : X \to Y$, $g : Y \to Z$, and $\tilde{g} : Y \to Z$ be functions. In the following proofs, we will skip the cases where a domain or codomain is empty as we believe that they are quite straightforward given the results of Exercise 3.3.3 and the uniqueness of the empty function from $\emptyset$ to a given set (see Example 3.3.11).

Suppose that $g \circ f = g \circ \tilde{f}$ and g is injective. Let $x \in X$. Given that $g \circ f = g \circ \tilde{f}$,

we have $g \circ f(x) = g \circ \tilde{f}(x)$, i.e., $g(f(x)) = g(\tilde{f}(x))$. The injectivity of g implies that $f(x) = \tilde{f}(x)$. Besides, f and $\tilde{f}$ have the same domain and codomain. Hence, $f = \tilde{f}$.

Without the assumption that g is injective, this result is not true. To see this, consider the following counter example. Let $f : \{0,1\} \to \{0,1\}$, $\tilde{f} : \{0,1\} \to \{0,1\}$, and $g : \{0,1\} \to \{2\}$ be three functions such that $f(0) = 1$, $f(1) = 0$, $\tilde{f}(0) = 0$ and $\tilde{f}(1) = 1$. Even though $g \circ f = g \circ \tilde{f}$, we have $f \neq \tilde{f}$.

Suppose that $g \circ f = \tilde{g} \circ f$ and f is surjective. Let $y \in Y$. By the surjectivity of f, there exists an $x \in X$ such that $y = f(x)$. We have:

$$g(y) = g((f(x)) = g \circ f(x) = \tilde{g} \circ f(x) = \tilde{g}(f(x)) = \tilde{g}(y),$$

where the third equality follows because $g \circ f = \tilde{g} \circ f$. Given that g and $\tilde{g}$ also have the same domain and codomain, we conclude that $g = \tilde{g}$.

In the case where f is not surjective, this result does not hold. Here is a simple counter example. Define $g : \{0,1\} \to \{0,1\}$, $\tilde{g} : \{0,1\} \to \{0,1\}$, and $f : \{0\} \to \{0,1\}$ such that $g(0) = \tilde{g}(0) = 0$, $g(1) = 0$, $\tilde{g}(1) = 1$, and $f(0) = 0$. It is clear that $g \circ f = \tilde{g} \circ f$ while $g \neq \tilde{g}$.

Exercise 3.3.5

Let $f : X \to Y$ and $g : Y \to Z$ be functions. In this is exercise, we will skip the cases where a domain or a codomain is the empty set for the same reasons mentioned in Exercise 3.3.4.

Suppose that $g \circ f$ is injective. Then, for any x_1, x_2 in X,

$$x_1 \neq x_2 \implies g(f(x_1)) \neq g(f(x_2)).$$

Let x_1, x_2 be two distinct elements of X. If $f(x_1) = f(x_2)$, then $g(f(x_1)) = g(f(x_2))$ by the axiom of substitution. But this is contradictory. So, we must have $f(x_1) \neq f(x_2)$. Therefore, f is injective.

Besides, it is not true that g must also be injective under these conditions. To see this, define $f : \{0\} \to \{2,3\}$ and $g : \{2,3\} \to \{4\}$ with $f(0) := 3$, $g(2) := 4$, and $g(3) := 4$. Then, $g \circ f : \{0\} \to \{4\}$ is injective while g is not.

Suppose that $g \circ f$ is surjective. Let $z \in Z$. By the surjectivity of $g \circ f$, there is some x in X such that $g \circ f(x) = z$, i.e., $g(f(x)) = z$. We just found an element y of Y, namely $f(x)$, such that $g(y) = z$. This shows that g is surjective.

Moreover, the function f does not need to be surjective here. Consider this counter example. Suppose that the functions are $f : \{5\} \to \{6,7\}$ and $g : \{6,7\} \to \{8\}$ with $f(5) := 7$, $g(6) := 8$, and $g(7) := 8$. Even though $g \circ f$ and g are surjective, the function f is not.

Exercise 3.3.6

Let $f : X \to Y$ be a bijective function.

By Definition 3.3.1, for every $x \in X$, there exists a unique $y \in Y$ such that $f(x) = y$, and, as f is bijective, x is the only element in X such that $y = f(x)$, hence $x = f^{-1}(y)$ (see text after Remark 3.3.27). Therefore, for each $x \in X$, there exists a unique $y \in Y$ such that $y = f(x)$ and we have $x = f^{-1}(y)$; so, $f^{-1}(f(x)) = x$ by the axiom of substitution. The first cancellation law is thus proven.

Because f is bijective, for each $y \in Y$, there exists a unique $x \in X$ such that $y = f(x)$, and that element x is denoted $f^{-1}(y)$ (see text after Remark 3.3.27). So, for every $y \in Y$ there exists a unique $x \in X$ such that $y = f(x)$ and we have $x = f^{-1}(y)$; therefore we can apply the axiom of substitution to see that $y = f(f^{-1}(y))$. This proves the second cancellation law.

On the one hand, $f^{-1} \circ f$ is the function defined from X to X that associates any element of X to itself, given of the first cancellation law. It is clear that $f^{-1} \circ f$ is invertible because for every element x in the codomain X, there is exactly one element in the domain X, namely x itself, which is mapped to x under this function. In particular, $f^{-1} \circ f$ is a surjective function as guaranteed by Definition 3.3.23. By the results of Exercise 3.3.5, this implies that f^{-1} is a surjective function. On the other hand, since $f \circ f^{-1}$ is invertible (the proof of this is analogous to that of the invertibility of $f^{-1} \circ f$), it is also an injective function by Definition 3.3.23. Given the claims we proved in Exercise 3.3.5, it follows that f^{-1} is injective. Therefore, f^{-1} is a bijective function.

Since f^{-1} is a well-defined function, for every $y \in Y$, there exists exactly one $x \in X$ such that $f^{-1}(y) = x$. But given that f^{-1} is bijective, $f^{-1}(y) = x$ implies that $y = (f^{-1})^{-1}(x)$ (see text after Remark 3.3.27) and that $y = f(x)$ (by the second cancellation law). So, for every $y \in Y$, there exists exactly one $x \in X$ such that $f^{-1}(y) = x$ and we have $y = (f^{-1})^{-1}(x) = f(x)$. We see that the function f defined from X to Y maps every x in X to the unique y in Y such that $f^{-1}(y) = x$. This means that $(f^{-1})^{-1} = f$.

Exercise 3.3.7

Let $f : X \to Y$ and $g : Y \to Z$ be bijective functions.

By the results of Exercise 3.3.2, the injectivity of both f and g implies the injectivity of $g \circ f$, and the surjectivity of both f and g implies the surjectivity of $g \circ f$. Therefore, $g \circ f$ is bijective.

Now, observe that the domain of $g \circ f$ is the codomain of $f^{-1} \circ g^{-1}$ and vice versa. Also, since f and g are bijective, so are f^{-1} and g^{-1} (see Exercise 3.3.6); and thus $f^{-1} \circ g^{-1}$ is bijective as well (see the paragraph above). Furthermore, for all $z \in Z$, we have:

$$(g \circ f)(f^{-1} \circ g^{-1}(z)) = (g \circ f)(f^{-1}(g^{-1}(z))) = g\big(f(f^{-1}(g^{-1}(z)))\big) = g(g^{-1}(z)) = z,$$

where the last two equalities follow from the second cancellation law proven in Exercise 3.3.6. As $g \circ f$ is bijective, this shows that for every $z \in Z$, the unique element x in X such that $g \circ f(x) = z$ is $x = f^{-1} \circ g^{-1}(z)$. Therefore,

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}.$$

Exercise 3.3.8

(a) Let X, Y, and Z be sets such that $X \subseteq Y \subseteq Z$.

To start with, the function $\iota_{Y \to Z} \circ \iota_{X \to Y}$ has the same domain and codomain as the function $\iota_{X \to Z}$. Next, for all x in the domain X,

$$\iota_{Y \to Z} \circ \iota_{X \to Y}(x) = \iota_{Y \to Z}(x) = x$$

and

$$\iota_{X \to Z}(x) = x.$$

So,

$$\iota_{Y \to Z} \circ \iota_{X \to Y} = \iota_{X \to Z}.$$

(b) Consider a function $f : A \to B$.

For every $x \in A$, we have

$$f \circ \iota_{A \to A}(x) = f(\iota_{A \to A}(x)) = f(x),$$

and, as $f(x) \in B$, we also have

$$\iota_{B\to B} \circ f(x) = \iota_{B\to B}(f(x)) = f(x).$$

In addition, the functions $f \circ \iota_{A\to A}$, $\iota_{B\to B} \circ f$, and f all have the same domain and codomain. Therefore,

$$f = f \circ \iota_{A\to A} = \iota_{B\to B} \circ f.$$

(c) Let $f : A \to B$ be a bijective function.

First, $f \circ f^{-1}$ has the same domain and codomain as $\iota_{B\to B}$. Second, for any y in the domain B, $f \circ f^{-1}(y) = y = \iota_{B\to B}(y)$ because of the second cancellation law proven in Exercise 3.3.6. Therefore, $f \circ f^{-1} = \iota_{B\to B}$. An analogous argument shows that $f^{-1} \circ f = \iota_{A\to A}$.

(d) Let X, Y, Z be sets such that X and Y are disjoint, and let $f : X \to Z$ and $g : Y \to Z$ be functions.

Consider the function $h : X \cup Y \to Z$ defined by

$$h(t) := \begin{cases} f(t), & t \in X \\ g(t), & t \in Y \end{cases}.$$

This is a well-defined function because for each t in $X \cup Y$, there is exactly one z in Z such that $h(t) = z$ since t is either in X or in Y but not both (as $X \cap Y = \emptyset$), and f and g are well-defined functions. Thus, $h \circ \iota_{X\to X\cup Y}$ and $h \circ \iota_{Y\to X\cup Y}$, being compositions of well-defined functions, are well-defined functions as well. The functions $h \circ \iota_{X\to X\cup Y}$ and f have the same domain and codomain. Plus, $h \circ \iota_{X\to X\cup Y}(x) = h(x) = f(x)$ for every $x \in X$. Hence,

$$h \circ \iota_{X\to X\cup Y} = f.$$

An analogous argument yields

$$h \circ \iota_{Y\to X\cup Y} = g.$$

Moreover, this function h is unique. To see this, suppose that there is another function $h' : X \cup Y \to Z$ such that $h' \circ \iota_{X\to X\cup Y} = f$ and $h' \circ \iota_{Y\to X\cup Y} = g$. Then, for any $t \in X$,

$$h'(t) = h'(\iota_{X\to X\cup Y}(t)) = h' \circ \iota_{X\to X\cup Y}(t) = f(t) = h(t)$$

because $\iota_{X\to X\cup Y}(t) = t$; and for any $t \in Y$,

$$h'(t) = h'(\iota_{Y\to X\cup Y}(t)) = h' \circ \iota_{Y\to X\cup Y}(t) = g(t) = h(t)$$

because $\iota_{Y\to X\cup Y}(t) = t$. We see that for any $t \in X \cup Y$, $h'(t) = h(t)$. Thus, $h' = h$.

(e) Let X, Y, Z be sets, and let $f : X \to Z$ and $g : Y \to Z$ be functions such that $f(x) = g(x)$ for all $x \in X \cap Y$.

Consider the function $h : X \cup Y \to Z$ defined by

$$h(t) = \begin{cases} f(t), & t \in X \\ g(t), & t \in Y \end{cases}.$$

Note that $X \cup Y = (X \setminus Y) \cup (X \cap Y) \cup (Y \setminus X)$ and the three sets on the right-hand side of this equality are pairwise disjoint (see Exercise 3.1.10). The function h is a well-defined function because for each t in $X \cup Y$, there is exactly one z in Z such that $h(t) = z$ since $X \setminus Y$, $X \cap Y$, and $Y \setminus X$ are disjoint, and f and g are well-defined functions that coincide in $X \cap Y$. Thus, $h \circ \iota_{X \to X \cup Y}$ and $h \circ \iota_{Y \to X \cup Y}$ are well-defined functions as well. From this point, we can just follow the same line of reasoning as in (d) and obtain the same results since we do not make use of any property pertaining to $X \cap Y$ anymore.

3.4 Images and Inverse Images

Exercise 3.4.1

Let $f : X \to Y$ be a bijective function, and let $f^{-1} : Y \to X$ be its inverse. Let V be any subset of Y. Let W_1 be the forward image of V under f^{-1} and W_2 be the inverse image of V under f.

By Definition 3.4.1, W_1 is the subset of X defined by

$$W_1 := \{f^{-1}(y) : y \in V\}.$$

Note that for any $x \in X$, $x \in W_1$ if and only if $x = f^{-1}(y)$ for some $y \in V$. So, we can write

$$W_1 = \{x \in X : x = f^{-1}(y) \text{ for some } y \in V\}.$$

Similarly, Definition 3.4.5 tells us that W_2 is the subset of X defined by

$$W_2 := \{x \in X : f(x) \in V\}.$$

Given that $f(x) \in V$ if and only if $f(x) = y$ for some $y \in V$, we can write

$$W_2 = \{x \in X : f(x) = y \text{ for some } y \in V\}.$$

Since f is bijective, for any $x \in X$ and $y \in Y$, we have:

$$x = f^{-1}(y) \iff f(x) = f(f^{-1}(y)) \iff f(x) = y.$$

So for any $x \in X$, the statement "$x = f^{-1}(y)$ for some $y \in V$" is logically equivalent to the statement "$f(x) = y$ for some $y \in V$", and hence

$$x \in W_1 \iff x \in W_2.$$

In other words,

$$W_1 = W_2.$$

Exercise 3.4.2

Let $f : X \to Y$ be a function from one set X to another set Y, let S be a subset of X, and let U be a subset of Y.

(i) By Definition 3.4.5,

$$f^{-1}(f(S)) = \{x \in X : f(x) \in f(S)\}.$$

For any $x \in S$, we have $x \in X$ (as $S \subseteq X$) and $f(x) \in f(S)$ (by Definition 3.4.1), and so $x \in f^{-1}(f(S))$. Hence,

$$S \subseteq f^{-1}(f(S)).$$

It is not necessarily true that $f^{-1}(f(S)) \subseteq S$.[5] A straightforward counter example is $f : \{1, 2\} \to \{0\}$ with $f(1) = f(2) = 0$ and $S = \{1\}$, where $f^{-1}(f(S)) = f^{-1}(f(\{1\})) = \{1, 2\} \not\subseteq S$.

(ii) By Definition 3.4.1,

$$f(f^{-1}(U)) = \{f(x) : x \in f^{-1}(U)\}.$$

However, $x \in f^{-1}(U)$ implies that $f(x) \in U$ according to Definition 3.4.5. Thus, $y \in f(f^{-1}(U))$ implies that $y \in U$. Therefore,

$$f(f^{-1}(U)) \subseteq U.$$

It is not generally true that $U \subseteq f(f^{-1}(U))$.[6] To illustrate, observe that if f was defined from $\{10\}$ to $\{20, 30\}$ by $f(10) := 20$ and that U was the set $\{20, 30\}$, then we would have $f(f^{-1}(U)) = \{20\}$.

(iii) From part (i), we have:

$$f^{-1}(U) \subseteq f^{-1}(f(f^{-1}(U))). \qquad (\star)$$

Next, by Definition 3.4.5, $f^{-1}(f(f^{-1}(U))) := \{x \in X : f(x) \in f(f^{-1}(U))\}$. But by part (ii), $f(f^{-1}(U)) \subseteq U$. So, $f(x) \in f(f^{-1}(U))$ implies that $f(x) \in U$. Hence, for any object z

$$z \in \{x : f(x) \in f(f^{-1}(U))\} \implies z \in \{x : f(x) \in U\}.$$

Thus,

$$\{x : f(x) \in f(f^{-1}(U))\} \subseteq \{x : f(x) \in U\},$$

that is

$$f^{-1}(f(f^{-1}(U))) \subseteq f^{-1}(U). \qquad (\star\star)$$

It follows from $(\star)$ and $(\star\star)$ that

$$f^{-1}(f(f^{-1}(U))) = f^{-1}(U).$$

[5] We will see in Exercise 3.4.5 when $f^{-1}(f(S)) \subseteq S$.
[6] We will see in Exercise 3.4.5 when $U \subseteq f(f^{-1}(U))$.

Exercise 3.4.3

Let A and B be subsets of a set X, and let $f : X \to Y$ be a function.

First, we will show that $f(A \cap B) \subseteq f(A) \cap f(B)$.

By Definition 3.4.1, $f(A \cap B) := \{f(x) : x \in A \cap B\}$. In other words, for any object y, $y \in f(A \cap B)$ if and only if $y = f(x)$ for some $x \in A \cap B$. Observe that if $y = f(x)$ for some $x \in A \cap B$, then it is true that $y = f(a)$ for some $a \in A$ (as $A \cap B \subseteq A$) and so it is true that $y \in f(A)$ by Definition 3.4.1. Similarly, if $y = f(x)$ for some $x \in A \cap B$, then it is true that $y = f(b)$ for some $b \in B$ (as $A \cap B \subseteq B$) and so it is true that $y \in f(B)$ by Definition 3.4.1. We see that if $y \in f(A \cap B)$, then $y \in f(A)$ and $y \in f(B)$, so that $y \in f(A) \cap f(B)$. This shows that

$$f(A \cap B) \subseteq f(A) \cap f(B).$$

It is not true that the $\subseteq$ relation can be improved to $=$ here. To see this, suppose that A and B are non-empty disjoint sets, and $Y = \{0\}$. Then, $A \cap B = \emptyset$ and we have $f(A \cap B) = f(\emptyset) = \{f(x) : x \in \emptyset\} = \emptyset$ by Definition 3.4.1. In addition, $f(A)$ and $f(B)$ as defined by Definition 3.4.1 are non-empty subsets of Y because A and B are non-empty. Hence, $f(A) = f(B) = \{0\}$, which leads to $f(A) \cap f(B) = \{0\}$. We see that $f(A \cap B) \neq f(A) \cap f(B)$.

Second, we will show that $f(A) \setminus f(B) \subseteq f(A \setminus B)$.

Consider an object y. Suppose that $y \in f(A) \setminus f(B)$. Then $y \in f(A)$ and $y \notin f(B)$. Note that $y \notin f(B)$ implies that $y \neq f(x)$ for all $x \in B$ according to Definition 3.4.1. In particular,

$$y \neq f(x) \text{ for all } x \in A \cap B \qquad (\star)$$

as $A \cap B \subseteq B$. Now, $y \in f(A)$ implies that there exists some x_0 in A for which $y = f(x_0)$, by Definition 3.4.1. Since $A = (A \cap B) \cup (A \setminus B)$[7], $x_0 \in A \cap B$ or $x_0 \in A \setminus B$. If $x_0 \in A \cap B$, we have a contradiction by $(\star)$, thus x_0 must belong to $A \setminus B$. Hence, $y \in f(A \setminus B)$ by Definition 3.4.1. Therefore,

$$f(A) \setminus f(B) \subseteq f(A \setminus B).$$

This statement is not true when $\subseteq$ is replaced with $=$. We can illustrate this by reconsidering the counterexample used in the previous paragraph. There, we have $f(A) \setminus f(B) = \{0\} \setminus \{0\} = \emptyset$ while $f(A \setminus B) = f(A) = \{0\}$.

Third, we will show that $f(A \cup B) = f(A) \cup f(B)$. Let y be an object.

[7] Note that, by Proposition 3.1.27(h), $A \setminus (A \cap B) = (A \setminus A) \cup (A \setminus B) = \emptyset \cup (A \setminus B) = A \setminus B$, and see Proposition 3.1.27(g).

If $y \in f(A \cup B)$, we have that $y = f(x)$ for some $x \in A \cup B$ by Definition 3.4.1. Either $x \in A$ or $x \in B$. By Definition 3.4.1, we have $y \in f(A)$ in the former case, and $y \in f(B)$ in the latter case. So in any case, $y \in f(A) \cup f(B)$. Hence $f(A \cup B) \subseteq f(A) \cup f(B)$.

If $y \in f(A) \cup f(B)$, then either $y \in f(A)$ or $y \in f(B)$. By Definition 3.4.1, we have $y = f(x)$ for some $x \in A$ in the first case, and $y = f(x)$ for some $x \in B$ in the second case. In either case, $x \in A \cup B$ since both A and B are subsets of $A \cup B$. Thus, $y \in f(A \cup B)$ by Definition 3.4.1. Therefore, $f(A) \cup f(B) \subseteq f(A \cup B)$.

We deduce from these two inclusions that

$$f(A \cup B) = f(A) \cup f(B).$$

Exercise 3.4.4

Let $f : X \to Y$ be a function from a set X to a set Y, and let U, V be subsets of Y.

First, let us show that $f^{-1}(U \cup V) = f^{-1}(U) \cup f^{-1}(V)$. For any object x,

$$\begin{aligned}
x \in f^{-1}(U) \cup f^{-1}(V) &\iff x \in f^{-1}(U) \text{ or } x \in f^{-1}(V) \\
&\iff (x \in X \text{ and } f(x) \in U) \text{ or } (x \in X \text{ and } f(x) \in V) \\
&\qquad\qquad \text{by Definition 3.4.5} \\
&\iff x \in X \text{ and } (f(x) \in U \text{ or } f(x) \in V) \\
&\iff x \in X \text{ and } f(x) \in U \cup V \\
&\iff x \in f^{-1}(U \cup V) \qquad \text{by Definition 3.4.5.}
\end{aligned}$$

Therefore,

$$f^{-1}(U \cup V) = f^{-1}(U) \cup f^{-1}(V).$$

Second, we will prove that $f^{-1}(U \cap V) = f^{-1}(U) \cap f^{-1}(V)$. For any object x,

$$\begin{aligned}
x \in f^{-1}(U \cap V) &\iff x \in X \text{ and } f(x) \in U \cap V \qquad \text{by Definition 3.4.5} \\
&\iff x \in X \text{ and } (f(x) \in U \text{ and } f(x) \in V) \\
&\iff (x \in X \text{ and } f(x) \in U) \text{ and } (x \in X \text{ and } f(x) \in V) \\
&\iff x \in f^{-1}(U) \text{ and } x \in f^{-1}(V) \qquad \text{by Definition 3.4.5} \\
&\iff x \in f^{-1}(U) \cap f^{-1}(V).
\end{aligned}$$

So,

$$f^{-1}(U \cap V) = f^{-1}(U) \cap f^{-1}(V).$$

Lastly, we will show that $f^{-1}(U \setminus V) = f^{-1}(U) \setminus f^{-1}(V)$. For any object x, if $x \in X$, we have:

$$\begin{aligned} x \in f^{-1}(U \setminus V) &\iff f(x) \in U \setminus V && \text{by Definition 3.4.5} \\ &\iff f(x) \in U \text{ and } f(x) \notin V \\ &\iff x \in f^{-1}(U) \text{ and } x \notin f^{-1}(V) && \text{by Definition 3.4.5} \\ &\iff x \in f^{-1}(U) \setminus f^{-1}(V); \end{aligned}$$

and if $x \notin X$, then in particular, $x \notin f^{-1}(U \setminus V)$ and $x \notin f^{-1}(U) \setminus f^{-1}(V)$ by Definition 3.4.5, and so the statements "$x \in f^{-1}(U \setminus V)$" and "$x \in f^{-1}(U) \setminus f^{-1}(V)$" are equivalent as they are equally false. We see that in any case,

$$x \in f^{-1}(U \setminus V) \iff x \in f^{-1}(U) \setminus f^{-1}(V).$$

Hence,

$$f^{-1}(U \setminus V) = f^{-1}(U) \setminus f^{-1}(V).$$

Exercise 3.4.5

Let $f : X \to Y$ be a function. We have two statements to prove.

First, let us show that $f(f^{-1}(S)) = S$ for every $S \subseteq Y$ if and only if f is surjective. In the case where Y is the empty set, X is also the empty set and f is surjective (see Exercise 3.3.3). Plus, since $f(\emptyset) = f^{-1}(\emptyset) = \emptyset$ and $\emptyset$ is the only subset of the $\emptyset$, it is clear that $f(f^{-1}(S)) = S$ for every $S \subseteq Y$. The equivalence is thus established. Now assume that $Y \neq \emptyset$.

On the one side, suppose that $f(f^{-1}(S)) = S$ for every subset $S \subseteq Y$. For any $y \in Y$, the singleton set $\{y\}$ exists and is a subset of Y, so $f(f^{-1}(\{y\})) = \{y\} \neq \emptyset$; hence $f^{-1}(\{y\}) \neq \emptyset$ since $f(\emptyset) = \emptyset$. For any $y \in Y$, given that $f^{-1}(\{y\}) \neq \emptyset$, Definition 3.4.5 guaranties the existence of some $x \in X$ such that $f(x) \in \{y\}$, i.e, $f(x) = y$. Therefore, f is surjective.

On the other side, suppose that f is surjective. This means that for every $y \in Y$, there is some $x \in X$ such that $f(x) = y$. Consider a subset S of Y. Assume that S is non-empty (if $S = \emptyset$, we have $f(f^{-1}(S)) = f(f^{-1}(\emptyset)) = f(\emptyset) = \emptyset = S$). For every $y \in S$, the surjectivity of f ensures that there exists some $x \in X$ such that $y = f(x)$, and by Definition 3.4.5, this x belongs in $f^{-1}(S)$. So, for every $y \in S$, there exists some $x \in f^{-1}(S)$ such that $y = f(x)$; but by Definition 3.4.1, this implies that $y \in f(f^{-1}(S))$. Thus, $S \subseteq f(f^{-1}(S))$. Besides, it follows from Exercise 3.4.2 part (ii) that $f(f^{-1}(S)) \subseteq S$. Hence, $f(f^{-1}(S)) = S$.

In summary, $f(f^{-1}(S)) = S$ for every $S \subseteq Y$ if and only if f is surjective.

Second, we will prove that $f^{-1}(f(S)) = S$ for every $S \subseteq X$ if and only if f is injective. If X is the empty set, then f is injective (see Exercise 3.3.3) and it is true that $f^{-1}(f(S)) = S$ for every $S \subseteq X$ (since $f(\emptyset) = f^{-1}(\emptyset) = \emptyset$). As these statements are equally true, they are equivalent to each other. Now, we may assume that $X \neq \emptyset$.

On the one side, suppose that $f^{-1}(f(S)) = S$ for every $S \subseteq X$. For any $x_0 \in X$, the singleton set $\{x_0\}$ exists and is a subset of X; moreover:

$$\begin{aligned} f^{-1}(f(\{x_0\})) = \{x_0\} &\iff \{x \in X : f(x) \in f(\{x_0\})\} = \{x_0\} && \text{by Definition 3.4.5} \\ &\iff \{x \in X : f(x) \in \{f(x_0)\}\} = \{x_0\} && \text{by Definition 3.4.1} \\ &\iff \{x \in X : f(x) = f(x_0)\} = \{x_0\} \\ &\iff x_0 \text{ is the only element in } X \text{ that is mapped to } f(x_0) \text{ by } f. \end{aligned}$$

Given that f does not map different elements of X to the same element of Y, it is an injective function.

On the other side, suppose that f is injective. Let us take an arbitrary subset S of X. We can assume that S is non-empty given that when $S = \emptyset$, it is straightforward to see that $f^{-1}(f(S)) = S$. We already know from Exercise 3.4.2 part (i) that $S \subseteq f^{-1}(f(S))$. Now, let $x_0 \in f^{-1}(f(S))$ (we can select an element in $f^{-1}(f(S))$ because this set is non-empty as it includes the non-empty set S). Given that $f(x_0) \in f(S)$ by Definition 3.4.5, there is some $x_1 \in S$ such that $f(x_0) = f(x_1)$. Since f is injective, $f(x_0) = f(x_1)$ implies that $x_0 = x_1$. As $x_1 \in S$, we have $x_0 \in S$. Hence, $f^{-1}(f(S)) \subseteq S$. Therefore, $f^{-1}(f(S)) = S$.

We conclude that $f^{-1}(f(S)) = S$ for every $S \subseteq X$ if and only if f is injective.

Exercise 3.4.6

(i) Let us prove Lemma 3.4.10.

Let X be a set. Consider the set $A := \{0,1\}^X$ that is indeed a well-defined set by the power set axiom (Axiom 3.11).

For each function f in A, there exists exactly one set S which consists of the elements in the domain X that are mapped to 1; that set is the inverse image of $\{1\}$, i.e. $f^{-1}(\{1\})$. So, by the axiom of replacement (Axiom 3.7),

$$B := \{S : S = f^{-1}(\{1\}) \text{ for some } f \in A\}$$

is a well-defined set. Every element of B, being an inverse image of a subset of $\{0,1\}$ under a function defined from X to $\{0,1\}$, is a subset of X.

Now, let W be a subset of X. Define the function $f_W : X \to \{0,1\}$ by $f_W(x) = 1$ if $x \in W$ and $f_W(x) = 0$ otherwise. Then, $f_W \in A$ and $f_W^{-1}(\{1\}) = W$. Hence, $W \in B$.

In sum, every element of B is a subset of X and every subset of X is an element of B. This means that for all objects Y,

$$Y \in B \iff Y \subseteq X.$$

We see that B is the set presented in Lemma 3.4.10 and conclude the proof.

(ii) Let us show that Axiom 3.11 can be deduced from the preceding axioms of set theory and Lemma 3.1.10.

Actually, we do not think that it is possible to prove Axiom 3.11 using only Lemma 3.1.10 and Axioms 3.1 – 3.10 (excluding Axiom 3.9). This is because there are cases where the set containing all the functions from X to Y is larger than any set we can build using only these lemma and axioms.[8] For example, the set of all functions from $\mathbf{N}$ to $\mathbf{N}$ is larger than any set we can construct using only Lemma 3.1.10 and Axioms 3.1 – 3.10 (excluding Axiom 3.9). Note that Axiom 3.3 only yield a set of 0 element. Using Axiom 3.4, we can only build sets of 1 or 2 elements. With Axioms 3.6 and 3.7, we can only construct sets equally large or smaller than a given set (in terms of cardinality[9]). Axiom 3.8 gives $\mathbf{N}$. With Lemma 3.4.10, we cannot build a set larger that $\{0,1\}^X$ given a set X (see part (i)). Finally, using Axiom 3.5, we can only build finite unions of sets made from the previous axioms and lemma. We see that there is no way to build the set $(2^{\mathbf{N}})^{\mathbf{N}}$ using only Axioms 3.1 - 3.10 (excluding Axiom 3.9) and Lemma 3.4.10. To solve this problem, we think that we need at least Axiom 3.12 (Union). So, the following proof will make use of Axiom 3.12.[10]

Let X, Y be sets. Given any function $f : X \to Y$ defined from X to Y, the object $f(x)$ exists and is unique for each object x in X, and so the set $\{\{x\}, \{x, f(x)\}\}$ not only exists (by applying Axiom 3.4 thrice) but is also unique for each x in X. Therefore,

[8] Note that none of these axioms allow us to create the set $\{(x,y) : x \in X \text{ and } y \in Y\}$ or the set $\{\{\{x\}, \{x,y\}\} : x \in X \text{ and } y \in Y\}$, where X and Y are uncountable sets such as the set of real numbers $\mathbf{R}$.

[9] See Sections 3.6, 8.1, and 8.3.

[10] If you believe that you found a way to deduce Axiom 3.11 only from Lemma 3.1.10 and Axioms 3.1 – 3.10 (excluding Axiom 3.9), please let me know at keumoe.fozeu.r@gmail.com or on the manual's GitHub.

for any function $f : X \to Y$, we can define the set

$$M_f := \big\{\{\{x\}, \{x, f(x)\}\} : x \in X\big\}$$

with the axiom of replacement (Axiom 3.7). Note that for every function $f : X \to Y$, the set M_f is unique (as any set constructed by the axiom of replacement is unique).

Using the axiom of replacement (Axiom 3.7) and the axiom of singleton sets and pair sets (Axiom 3.4), we can define the set

$$K_x := \big\{\{\{x\}, \{x, y\}\} : y \in Y\big\}$$

for each $x \in X$. By the axiom of union (Axiom 3.12), the object M defined by

$$M := \bigcup_{x \in X} K_x$$

is a well-defined set. For all objects m, we have

$$m \in M \iff m = \{\{x\}, \{x, y\}\} \text{ for some } x \in X \text{ and } y \in Y.$$

By Lemma 3.4.10, the object $T := \{S : S \subseteq M\}$ exists. By the axiom of specification (Axiom 3.6), there exists a set U defined by

$$U := \{S \in T : \text{ for every } x \in X, \text{ there exists a unique } s \in S \text{ such that } \{x\} \in s\}.$$

Note that for any function function f, M_f belongs in U. Besides, for any $S \in U$, we can define the function $g : X \to Y$ that maps each x in X to the only object y in Y for which $\{\{x\}, \{x, y\}\} \in S$, and we have $S = M_g$. We see that

$$S \in U \iff S = M_f \text{ for some function } f \text{ defined from } X \text{ to } Y.$$

In other words, if f is any function from X to Y, then M_f belongs in U and if $S \in U$ then $S = M_g$ for some function g defined from X to Y. So, U consists of all the sets M_f, where f is some function defined from X to Y. Given that M_f is unique for every f, we can use the axiom of replacement (Axiom 3.7) to construct the set

$$F := \{f : M_f \in U\}.$$

The set F consists of all the functions defined from X to Y. This proves the power set axiom (Axiom 3.11).

Exercise 3.4.7

Let X, Y be sets. According to Lemma 3.4.10, there exist sets A and B defined by:

$$A := \{X' : X' \subseteq X\} \quad \text{and} \quad B := \{Y' : Y' \subseteq Y\}.$$

For each $X' \in A$ and each $Y' \in B$, there exists a set $Y'^{X'}$ which consists of all the functions from X' to Y' by Axiom 3.11. By the replacement axiom (Axiom 3.7), $\{Y'^{X'} : X' \in A\}$ is a set for a given $Y' \in B$. Now, by the union axiom (Axiom 3.12), the object C defined by

$$C := \bigcup_{Y' \in B} \{Y'^{X'} : X' \in A\}$$

is a set. But this set consists of all functions from a subset X' of X to a subset Y' of Y. Therefore, the collection of these functions is indeed a set.

Exercise 3.4.8

Let A, B be sets. By Axiom 3.1, A and B are objects. So, we can use the axiom of singleton sets and pair sets (Axiom 3.4) to build the set $T := \{A, B\}$, which contains two elements (if A and B are distinct) or one element (if A is equal to B). Since all the elements of T are sets, by Axiom 3.12 there exists a set $\bigcup T$ whose elements are precisely those objects which are elements of the elements of T. Thus, for any object x,

$$x \in \bigcup T \iff x \in S \text{ for some } S \in T,$$

in other words

$$x \in \bigcup T \iff x \in A \text{ or } x \in B.$$

We see that $\bigcup T$ is the pairwise union of the sets A, B given in Axiom 3.5. Therefore, Axiom 3.5 can be deduced from Axiom 3.1, Axiom 3.4, and Axiom 3.12.

Exercise 3.4.9

Let I be a set containing at least two elements, and to each $\alpha \in I$ assign a set A_α. Let β and β' be two elements of I. Define the sets B and B' by

$$B := \{x \in A_\beta : x \in A_\alpha \text{ for all } \alpha \in I\} \text{ and } B' := \{x \in A_{\beta'} : x \in A_\alpha \text{ for all } \alpha \in I\}.$$

To start with, it is clear that for any object x,

$$x \in B \implies x \in A_\alpha \text{ for all } \alpha \in I \qquad (\star)$$

and

$$x \in B' \implies x \in A_\alpha \text{ for all } \alpha \in I. \qquad (\star\star)$$

Also, if x is any object that belongs to A_α for all $\alpha \in I$, then we also have $x \in A_\beta$ since $\beta \in I$, and it follows that $x \in B$. Similarly, any object that belongs to A_α for all $\alpha \in I$ also belongs to B'. Thus, for any object x,

$$x \in A_\alpha \text{ for all } \alpha \in I \implies x \in B \text{ and } x \in B'. \qquad (\star\star\star)$$

From $(\star)$, $(\star\star)$ and $(\star\star\star)$, we deduce that for any object x,

$$x \in B \iff x \in A_\alpha \text{ for all } \alpha \in I$$

and

$$x \in B' \iff x \in A_\alpha \text{ for all } \alpha \in I,$$

and so

$$x \in B \iff x \in B'.$$

This shows that $B = B'$. In other words,

$$\{x \in A_\beta : x \in A_\alpha \text{ for all } \alpha \in I\} = \{x \in A_{\beta'} : x \in A_\alpha \text{ for all } \alpha \in I\}.$$

Now, observe that $\bigcap_{\alpha\in I} A_\alpha = B$ according to (3.3). Hence,

$$y \in \bigcap_{\alpha\in I} A_\alpha \iff y \in A_\alpha \text{ for all } \alpha \in I$$

for any object y. This explains why (3.4) is true.

Exercise 3.4.10

Let I and J be two sets. For all $\alpha \in I \cup J$, let A_α be a set. For any object x,

$$\begin{aligned}
x \in \left(\bigcup_{\alpha\in I} A_\alpha\right) \cup \left(\bigcup_{\alpha\in J} A_\alpha\right) &\iff x \in \bigcup_{\alpha\in I} A_\alpha \text{ or } x \in \bigcup_{\alpha\in J} A_\alpha \\
&\iff (x \in A_\alpha \text{ for some } \alpha \in I) \\
&\qquad \text{or } (x \in A_\alpha \text{ for some } \alpha \in J) \\
&\iff x \in A_\alpha \text{ for some } \alpha \text{ such that } (\alpha \in I \text{ or } \alpha \in J) \\
&\iff x \in A_\alpha \text{ for some } \alpha \in I \cup J \\
&\iff x \in \bigcup_{\alpha\in I\cup J} A_\alpha.
\end{aligned}$$

So,

$$\left(\bigcup_{\alpha\in I} A_\alpha\right) \cup \left(\bigcup_{\alpha\in J} A_\alpha\right) = \bigcup_{\alpha\in I\cup J} A_\alpha.$$

Now, suppose that I and J are non-empty. For any object x,

$$\begin{aligned} x \in \left(\bigcap_{\alpha\in I} A_\alpha\right) \cap \left(\bigcap_{\alpha\in J} A_\alpha\right) &\iff x \in \bigcap_{\alpha\in I} A_\alpha \text{ and } x \in \bigcap_{\alpha\in J} A_\alpha \\ &\iff (x \in A_\alpha \text{ for all } \alpha \in I) \\ &\qquad \text{and } (x \in A_\alpha \text{ for all } \alpha \in J) \\ &\iff x \in A_\alpha \text{ for all } \alpha \text{ such that } (\alpha \in I \text{ or } \alpha \in J) \\ &\iff x \in A_\alpha \text{ for all } \alpha \in I \cup J \\ &\iff x \in \bigcap_{\alpha\in I\cup J} A_\alpha. \end{aligned}$$

So,

$$\left(\bigcap_{\alpha\in I} A_\alpha\right) \cap \left(\bigcap_{\alpha\in J} A_\alpha\right) = \bigcap_{\alpha\in I\cup J} A_\alpha.$$

Exercise 3.4.11

Let X be a set, let I be a non-empty set, and for all $\alpha \in I$ let A_α be a subset of X.

Consider an object x. We have

$$x \in X \setminus \bigcup_{\alpha\in I} A_\alpha \iff x \in X \text{ and } x \notin \bigcup_{\alpha\in I} A_\alpha.$$

Since for any object y,

$$y \in \bigcup_{\alpha\in I} A_\alpha \iff y \in A_\alpha \text{ for some } \alpha \in I,$$

we have

$$y \notin \bigcup_{\alpha\in I} A_\alpha \iff y \notin A_\alpha \text{ for all } \alpha \in I.$$

Hence,

$$\begin{aligned} x \in X \setminus \bigcup_{\alpha\in I} A_\alpha &\iff x \in X \text{ and } (x \notin A_\alpha \text{ for all } \alpha \in I) \\ &\iff (x \in X \text{ and } x \notin A_\alpha) \text{ for all } \alpha \in I \\ &\iff x \in X \setminus A_\alpha \text{ for all } \alpha \in I \\ &\iff x \in \bigcap_{\alpha\in I} (X \setminus A_\alpha), \end{aligned}$$

where the second equivalence holds because the statement "$x \in X$" is independent from the index α. Therefore,

$$X \setminus \bigcup_{\alpha \in I} A_\alpha = \bigcap_{\alpha \in I} (X \setminus A_\alpha).$$

We also have

$$\begin{aligned} x \in \bigcup_{\alpha \in I} (X \setminus A_\alpha) &\iff x \in X \setminus A_\alpha \text{ for some } \alpha \in I \\ &\iff (x \in X \text{ and } x \notin A_\alpha) \text{ for some } \alpha \in I \\ &\iff x \in X \text{ and } (x \notin A_\alpha \text{ for some } \alpha \in I) \end{aligned}$$

For any object y, given that

$$y \in \bigcap_{\alpha \in I} A_\alpha \iff y \in A_\alpha \text{ for all } \alpha \in I,$$

we have

$$y \notin \bigcap_{\alpha \in I} A_\alpha \iff y \notin A_\alpha \text{ for some } \alpha \in I.$$

Hence,

$$\begin{aligned} x \in \bigcup_{\alpha \in I} (X \setminus A_\alpha) &\iff x \in X \text{ and } x \notin \bigcap_{\alpha \in I} A_\alpha \\ &\iff x \in X \setminus \bigcap_{\alpha \in I} A_\alpha. \end{aligned}$$

Therefore,

$$X \setminus \bigcap_{\alpha \in I} A_\alpha = \bigcup_{\alpha \in I} (X \setminus A_\alpha).$$

Now, we will make a comparison of the two statements we just proved with the De Morgan's law seen in Proposition 3.1.27.

Suppose that $I = \{1, 2\}$. Then,

$$\bigcup_{\alpha \in I} A_\alpha = A_1 \cup A_2 \quad \text{and} \quad \bigcap_{\alpha \in I} A_\alpha = A_1 \cap A_2.$$

We have:

$$X \setminus \bigcup_{\alpha \in I} A_\alpha = X \setminus (A_1 \cup A_2),$$

$$\bigcap_{\alpha \in I} (X \setminus A_\alpha) = \bigcap \{X \setminus A_1, X \setminus A_2\} = (X \setminus A_1) \cap (X \setminus A_2),$$

$$X \setminus \bigcap_{\alpha \in I} A_\alpha = X \setminus (A_1 \cap A_2),$$

and

$$\bigcup_{\alpha \in I} (X \setminus A_\alpha) = \bigcup \{X \setminus A_1, X \setminus A_2\} = (X \setminus A_1) \cup (X \setminus A_2).$$

Using the first statement we proved in this exercise, we get

$$X \setminus (A_1 \cup A_2) = (X \setminus A_1) \cap (X \setminus A_2)$$

while the second statement yields

$$X \setminus (A_1 \cap A_2) = (X \setminus A_1) \cup (X \setminus A_2).$$

These are the De Morgan laws. As we can see, the De Morgan laws are just special cases of the first two statements we proved in this exercise.

3.5 Cartesian Products

Exercise 3.5.1

(i) Let us show that the Kuratowski definition of an ordered pair satisfies property (3.5).

Let x, x', y, y' be objects and consider the ordered pairs (x, y) and (x', y') constructed from the Kuratowski definition.

First, assume that $x = y$. Then, $(x, y) = (x, x) = \{\{x\}\}$. Furthermore,

$$\begin{aligned}
(x, y) = (x', y') &\iff \{\{x\}\} = \{\{x'\}, \{x', y'\}\} \\
&\iff \{x\} = \{x'\} \text{ and } \{x\} = \{x', y'\} \\
&\iff x = x' \text{ and } (x = x' \text{ and } x = y') \\
&\iff x = x' \text{ and } x = y' \\
&\iff x = x' \text{ and } y = y' \qquad \text{since } y = x.
\end{aligned}$$

Now, assume that x and y are not equal objects. Then, the set $\{x, y\}$ is made of two elements, while the sets $\{x\}$ and $\{x'\}$ are each made of only one element. Thus,

$$\{x, y\} \neq \{x\} \quad \text{and} \quad \{x, y\} \neq \{x'\}. \tag{$\star$}$$

So,

$$\begin{aligned}
(x, y) = (x', y') &\iff \{\{x\}, \{x, y\}\} = \{\{x'\}, \{x', y'\}\} \\
&\iff \{x\} = \{x'\} \text{ and } \{x, y\} = \{x', y'\} \qquad \text{given } (\star) \\
&\iff x = x' \text{ and } \{x, y\} = \{x', y'\} \\
&\iff x = x' \text{ and } ((x = x' \text{ and } y = y') \text{ or } (x = y' \text{ and } y = x')) \\
&\iff x = x' \text{ and } (x = x' \text{ and } y = y') \\
&\iff x = x' \text{ and } y = y',
\end{aligned}$$

where the fourth equivalence stems from the result of Exercise 3.1.1 and the fifth equivalence follows because the statement "$x = y'$ and $y = x'$" is false (as it would imply that $x = y$, a contradiction).

We see that in any case, the Kuratowski definition obeys property (3.5).

(ii) Let us show that the short definition of an ordered pair also satisfies property (3.5).

Let x, x', y, y' be objects and consider the ordered pairs (x, y) and (x', y') constructed from the short definition. First of all, we deduced from the axiom of regularity (Axiom 3.10) that no set contains itself (see Exercise 3.2.2). So, $x \neq \{x, y\}$ since $x \in \{x, y\}$, and $x' \neq \{x', y'\}$ since $x' \in \{x', y'\}$. Hence, the sets $\{x, \{x, y\}\}$ and $\{x', \{x', y'\}\}$ are each made of two elements. Also, it is clear with the axiom of substitution that if $x = x'$ and $y = y'$, then $\{x, \{x, y\}\} = \{x', \{x', y'\}\}$, i.e.,

$$x = x' \text{ and } y = y' \implies (x, y) = (x', y'). \tag{$*$}$$

Now,

$$\begin{aligned}(x, y) = (x', y') &\iff \{x, \{x, y\}\} = \{x', \{x', y'\}\} \\ &\iff \{x, \{x, y\}\} \subseteq \{x', \{x', y'\}\} \text{ and } \{x', \{x', y'\}\} \subseteq \{x, \{x, y\}\}.\end{aligned}$$

Note that if $x = \{x', y'\}$ and $x' = \{x, y\}$, then x and x' are sets such that $x \in x'$ and $x' \in x$, which is contradictory as shown in Exercise 3.2.2. Hence, "$x = \{x', y'\}$ and $x' = \{x, y\}$" is a false statement. Consequently, $\{x, \{x, y\}\} \subseteq \{x', \{x', y'\}\}$ if and only if $x = x'$ and $\{x, y\} = \{x', y'\}$; in addition, $\{x', \{x', y'\}\} \subseteq \{x, \{x, y\}\}$ if and only if $x' = x$ and $\{x', y'\} = \{x, y\}$. Therefore,

$$\begin{aligned}(x, y) = (x', y') &\iff x = x' \text{ and } \{x, y\} = \{x', y'\} \\ &\implies x = x' \text{ and } \{x, y\} = \{x, y'\} \\ &\implies x = x' \text{ and } ((x = x \text{ and } y = y') \text{ or } (x = y' \text{ and } y = x)) \\ &\implies x = x' \text{ and } y = y'. \end{aligned} \tag{$**$}$$

We conclude from $(*)$ and $(**)$ that $(x, y) = (x', y')$ if and only if $x = x'$ and $y = y'$, as stated by property (3.5).

(iii) Let X, Y be sets. We will show that their Cartesian product $X \times Y$ is indeed a set using the axioms of set theory.

For any $x \in X$ and any $y \in Y$, the object (x, y) exists and is unique regardless of the definition (among the three we saw). Hence, by the axiom of replacement (Axiom 3.7), the object $\{(x, y) : y \in Y\}$ is a well-defined set for each $x \in X$. Now, by the union axiom (Axiom 3.12), we can define the set $\bigcup_{x \in X} \{(x, y) : y \in Y\}$. But this set is precisely the collection of all ordered pairs (x, y), where $x \in X$ and $y \in Y$. That is,

$$\bigcup_{x \in X} \{(x, y) : y \in Y\} = X \times Y.$$

Thus, $X \times Y$ is indeed a set.

Exercise 3.5.2

Let X, Y be sets, and n a natural number. Let $x : \{i \in \mathbf{N} : 1 \leq i \leq n\} \to X$ and $y : \{i \in \mathbf{N} : 1 \leq i \leq n\} \to Y$ be two surjective functions. As x and y are functions,

$$x = y \iff X = Y \text{ and } (x(i) = y(i) \text{ for all } 1 \leq i \leq n).$$

Note that since x and y are surjective,

$$x(\{i \in \mathbf{N} : 1 \leq i \leq n\}) = X \quad \text{and} \quad y(\{i \in \mathbf{N} : 1 \leq i \leq n\}) = Y.$$

So,

$$\begin{aligned} x(i) = y(i) \text{ for all } 1 \leq i \leq n &\implies \{x(i) : 1 \leq i \leq n\} = \{y(i) : 1 \leq i \leq n\} \\ &\implies x(\{i \in \mathbf{N} : 1 \leq i \leq n\}) = y(\{i \in \mathbf{N} : 1 \leq i \leq n\}) \\ &\implies X = Y; \end{aligned}$$

thus,

$$X = Y \text{ and } (x(i) = y(i) \text{ for all } 1 \leq i \leq n) \iff x(i) = y(i) \text{ for all } 1 \leq i \leq n.^{11}$$

Therefore,

$$x = y \iff x(i) = y(i) \text{ for all } 1 \leq i \leq n.$$

In another notation, we can write this as:

$$(x_i)_{1 \leq i \leq n} = (y_i)_{1 \leq i \leq n} \iff x_i = y_i \text{ for all } 1 \leq i \leq n.$$

Let $X_1, \ldots, X_n$ be n sets. Let A and B be sets defined by

$$A := \{i \in \mathbf{N} : 1 \leq i \leq n\} \quad \text{and} \quad B := \bigcup_{i \in A} X_i,$$

using Axiom 3.6 and Axiom 3.12. As shown in Exercise 3.4.7, the collection of all partial functions from A to B is a set, which we denote by P. By the axiom of specification (Axiom 3.6), the object Q defined by

$$Q := \{f : A' \to B' \in P \mid A' = A, f(i) \in X_i \text{ for all } 1 \leq i \leq n, f \text{ is surjective}\}$$

is a set.

[11] Given two mathematical statements p and q such that $q \implies p$, we have (p and q) $\iff q$.

Now, take a function $f : A' \to B'$ in Q. This function is an ordered n-tuple according to the new definition, since $A' = A$ and f is surjective. Also, the ith component f_i of f belongs to X_i for all $1 \leq i \leq n$.

Next, let $z : A \to Z$ be an ordered n-tuple such that $z_i \in X_i$ for each $1 \leq i \leq n$. Then, by the definition of an ordered n-tuple, z is surjective. In addition, the surjectivity of z implies that $X = z(A) = \{z_i : 1 \leq i \leq n\}$, and so $Z \subseteq B$. Hence, $z \in Q$.

We deduce that the set Q consists precisely of the ordered n-tuples whose ith component belongs to X_i for each $1 \leq i \leq n$. But this is the definition of $\prod_{1 \leq i \leq n} X_i$ according to Definition 3.5.6. So, $\prod_{1 \leq i \leq n} X_i$ is a set.

Exercise 3.5.3

An ordered pair is an ordered 2-tuple. So, we can just work with ordered n-tuples. Let $x = (x_i)_{1 \leq i \leq n}$, $y = (y_i)_{1 \leq i \leq n}$, and $z = (z_i)_{1 \leq i \leq n}$ be ordered n-tuples. We assume that the reflexivity, symmetry, and transitivity axioms of equality all hold for every individual component of x, y, and z.

First, given that $x_i = x_i$ for every i between 1 and n, Definition 3.5.6 tells us that $(x_i)_{1 \leq i \leq n} = (x_i)_{1 \leq i \leq n}$, i.e., $x = x$. Hence, the definition of equality for ordered n-tuples is reflexive.

Second, suppose that $x = y$. Then, by Definition 3.5.6, $x_i = y_i$ for all $1 \leq i \leq n$. As the equality for individual components of x and y is symmetric, we also have $y_i = x_i$ for all $1 \leq i \leq n$. But this just means that $y = x$, by Definition 3.5.6. Thus, the definition of equality for ordered n-tuples is symmetric.

Third, suppose that $x = y$ and $y = z$. Then, by Definition 3.5.6, $x_i = y_i$ and $y_i = z_i$ for all $1 \leq i \leq n$. By the transitivity of equality for the components of x, y, z, we have $x_i = z_i$ for all $1 \leq i \leq n$. So, $x = z$ according to Definition 3.5.6. We see that the definition of equality for ordered n-tuples is transitive.

Exercise 3.5.4

Let A, B, C be sets. We will prove the following three equalities:

$$A \times (B \cup C) = (A \times B) \cup (A \times C),$$

$$A \times (B \cap C) = (A \times B) \cap (A \times C),$$

and

$$A \times (B \setminus C) = (A \times B) \setminus (A \times C).$$

We distinguish between the case where all sets are non-empty and the cases where at least one of the sets is empty. For any object x,

First, assume that A, B, and C are all non-empty.

Let x be an object. Suppose that $x \in (A \times B) \cup (A \times C)$. Then, $x \in A \times B$ or $x \in A \times C$, i.e., $x = (x_1, x_2)$ for some $x_1 \in A$ and $x_2 \in B$ or $x = (x_1, x_2)$ for some $x_1 \in A$ and $x_2 \in C$. In either case, $x = (x_1, x_2)$ for some $x_1 \in A$ and $x_2 \in B \cup C$. That is, $x \in A \times (B \cup C)$. Hence, $(A \times B) \cup (A \times C) \subseteq A \times (B \cup C)$. Now, suppose instead that $x \in A \times (B \cup C)$. Then, $x = (x_1, x_2)$ for some $x_1 \in A$ and $x_2 \in B \cup C$. Either $x_2 \in B$ or $x_2 \in C$. If $x_2 \in B$, then $x \in A \times B$; if $x_2 \in C$, then $x \in A \times C$. So, $x \in (A \times B) \cup (A \times C)$. Thus, $A \times (B \cup C) \subseteq (A \times B) \cup (A \times C)$. We deduce that

$$A \times (B \cup C) = (A \times B) \cup (A \times C).$$

Let y be an object. Suppose first that $y \in A \times (B \cap C)$. Then, $y = (y_1, y_2)$ for some $y_1 \in A$ and $y_2 \in B \cap C$. Since $y_2 \in B$ and $y_2 \in C$, we also have $y \in A \times B$ and $y \in A \times C$. Hence, $y \in (A \times B) \cap (A \times C)$. So, $A \times (B \cap C) \subseteq (A \times B) \cap (A \times C)$. Second, suppose that $y \in (A \times B) \cap (A \times C)$. As $y \in A \times B$, $y = (y_1, y_2)$ for some $y_1 \in A$ and $y_2 \in B$. Since $y \in A \times C$, we also have $y_2 \in C$, and hence $y_2 \in B \cap C$. So, $y \in A \times (B \cap C)$. Thus, $(A \times B) \cap (A \times C) \subseteq A \times (B \cap C)$. It follows that

$$A \times (B \cap C) = (A \times B) \cap (A \times C).$$

Let z_1, z_2 be objects and consider the ordered pair $z := (x_1, x_2)$. On the one side, suppose that $z \in A \times (B \setminus C)$. Then, $z_1 \in A$ and $z_2 \in B \setminus C$. This implies that $z_1 \in A$ and $z_2 \in B$, and thus $x \in A \times B$. Also, $z_2 \notin C$, and thus $z \notin A \times C$. As a result, $z \in (A \times B) \setminus (A \times C)$. Hence, $A \times (B \setminus C) \subseteq (A \times B) \setminus (A \times C)$. On the other side, suppose that $z \in (A \times B) \setminus (A \times C)$. Then, $z \in A \times B$ and so $z_1 \in A$ and $z_2 \in B$. Also, $z \notin A \times C$, so $z_1 \notin A$ or $z_2 \notin C$. If $z_1 \notin A$, we have a contradiction, hence, $z_2 \notin C$. Thus, $z_1 \in A$ and $z_2 \in B \setminus C$. This means that $z \in A \times (B \setminus C)$. Hence, $(A \times B) \setminus (A \times C) \subseteq A \times (B \setminus C)$. Therefore,

$$A \times (B \setminus C) = (A \times B) \setminus (A \times C).$$

Second, assume that A is empty. Observe that for any set X,

$$A \times X = \{(s, t) : s \in \emptyset, t \in X\} = \emptyset.$$

Hence,

$$\begin{aligned} A \times (B \cup C) &= \emptyset, \\ (A \times B) \cup (A \times C) &= \emptyset \cup \emptyset = \emptyset, \\ A \times (B \cap C) &= \emptyset, \\ (A \times B) \cap (A \times C) &= \emptyset \cap \emptyset = \emptyset, \\ A \times (B \setminus C) &= \emptyset, \\ (A \times B) \setminus (A \times C) &= \emptyset. \end{aligned}$$

We see that the three equalities hold in this case.

Third, assume that B is empty. For any set X,

$$X \times B = \{(s,t) : s \in X, t \in \emptyset\} = \emptyset.$$

So,

$$\begin{aligned} A \times (B \cup C) &= A \times C, \\ (A \times B) \cup (A \times C) &= \emptyset \cup (A \times C) = A \times C, \\ A \times (B \cap C) &= A \times B = \emptyset, \\ (A \times B) \cap (A \times C) &= \emptyset \cap (A \times C) = \emptyset, \\ A \times (B \setminus C) &= A \times \emptyset = \emptyset, \\ (A \times B) \setminus (A \times C) &= \emptyset \setminus (A \times C) = \emptyset. \end{aligned}$$

Therefore, the equalities hold in this case as well.

Finally, assume that C is empty. For set X, $X \times C = \emptyset$. From the latter case ($B = \emptyset$) and by the commutativity of the union and the intersection of sets, we deduce that the first two equalities also hold in this case. For the last equality, observe that $A \times (B \setminus C) = A \times B$ and $(A \times B) \setminus (A \times C) = (A \times B) \setminus \emptyset = A \times B$, and thus we also have $A \times (B \setminus C) = (A \times B) \setminus (A \times C)$.

In conclusion, the three equalities are true regardless of whether any of the sets A, B, or C is empty.

Exercise 3.5.5

Let A, B, C, D be sets.

Assume that one of these sets is empty. Note that $X \times \emptyset = \emptyset \times X = \emptyset \cap X = X \cap \emptyset = \emptyset$ for any set X. So,

$$(A \times B) \cap (CD) = \emptyset = (A \cap C) \times (B \cap D).$$

Now, assume that each of the sets A, B, C and D is non-empty.

First, suppose that $(A \times B) \cap (C \times D) \neq \emptyset$. Let $x \in (A \times B) \cap (C \times D)$. Since $x \in A \times B$, $x = (x_1, x_2)$ for some $x_1 \in A$ and $x_2 \in B$. Plus, since $x \in C \times D$, $x_1 \in C$ and $x_2 \in D$. So, $x_1 \in A \cap C$ and $x_2 \in B \cap D$, and hence, $x \in (A \cap C) \times (B \cap D)$. Thus, $(A \times B) \cap (C \times D) \subseteq (A \cap C) \times (B \cap D)$.

Second, if $(A \times B) \cap (C \times D) = \emptyset$, then $(A \times B) \cap (C \times D) \subseteq (A \cap C) \times (B \times D)$.

Third, suppose that $(A \cap C) \times (B \cap D) \neq \emptyset$. Let $x \in (A \cap C) \times (B \cap D)$. Then, $x = (x_1, x_2)$ for some $x_1 \in A \times C$ and $x_2 \in B \cap B$. Since $x_1 \in A$ and $x_2 \in B$, we have $x \in A \times B$. Similarly, $x_1 \in C$ and $x_2 \in D$ imply that $x \in C \times D$. Hence, $x \in (A \times B) \cap (B \times D)$, and so $(A \cap C) \times (B \cap D) \subseteq (A \times B) \cap (B \times D)$.

Fourthly, if $(A \cap C) \times (B \cap D) = \emptyset$, we have $(A \cap C) \times (B \cap D) \subseteq (A \times B) \cap (C \times D)$.

We see that we have

$$(A \times B) \cap (C \times D) \subseteq (A \cap C) \times (B \times D)$$

and

$$(A \cap C) \times (B \cap D) \subseteq (A \times B) \cap (C \times D)$$

whether any of these two sets is empty or not. We conclude that

$$(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D).$$

It is not a rule that $(A \times B) \cup (C \times D) = (A \cup C) \times (B \cup D)$. To see this, suppose that $A = \emptyset$, $B = \{11\}$, $C = \mathbf{N}$ and $D = \{22\}$. Then,

$$A = \emptyset \implies A \times B = \emptyset \implies (A \times B) \cup (C \times D) = C \times D$$
$$A = \emptyset \implies A \cup C = C \implies (A \cup C) \times (B \cup D) = C \times (B \cup D)$$

We have $(0, 11) \in C \times (B \cup D)$ but $(0, 11) \notin C \times D$. So, the sets $(A \times B) \cup (C \times D)$ and $(A \cup C) \times (B \cup D)$ are distinct.

It is also not generally true that $(A \times B) \setminus (C \times D) = (A \setminus C) \times (B \setminus D)$. To illustrate, suppose that $C = \emptyset$, A and B are both non-empty, and $D = B$. Then

$$(A \times B) \setminus (C \times D) = (A \times B) \setminus (\emptyset \times D) = (A \times B) \setminus \emptyset = A \times B \neq \emptyset,$$

but

$$(A \setminus C) \times (B \setminus D) = (A \setminus \emptyset) \times (B \setminus B) = A \times \emptyset = \emptyset.$$

Exercise 3.5.6

Let A, B, C, D be non-empty sets.

Let us prove that $A \times B \subseteq C \times D$ if and only if $A \subseteq C$ and $B \subseteq D$.

First, suppose that $A \times B \subseteq C \times D$. Since A and B are non-empty, $A \times B$ is also non-empty. Indeed, we can select an element x in A and an element y in B by the single choice lemma (Lemma 3.1.5), and form the ordered pair (x, y) which belongs to $A \times B$. Take such an element (x, y) in $A \times B$. Since $A \times B \subseteq C \times D$, we have $(x, y) \in C \times D$. Hence, $x \in C$ and $y \in D$. Therefore, $A \subseteq C$ and $B \subseteq D$.

Second, suppose that $A \subseteq C$ and $B \subseteq D$. As A and B are non-empty, we can form an ordered pair (a, b), made of some element a in A and some element b in B (Lemma 3.5.11). Thus, $A \times B$ is non-empty as well. Let $t \in A \times B$. There is some $x \in A$ and $y \in B$ such that $t = (x, y)$. Since $A \subseteq C$ and $B \subseteq D$, we also have $x \in C$ and $y \in D$. Thus, $t \in C \times D$. Therefore, $A \times B \subseteq C \times D$.

In conclusion,

$$A \times B \subseteq C \times D \iff A \subseteq C \text{ and } B \subseteq D.$$

Our second task is to show that $A \times B = C \times D$ if and only if $A = C$ and $B = D$. We simply observe that:

$$\begin{aligned} A \times B = C \times D &\iff A \times B \subseteq C \times D \text{ and } C \times D \subseteq A \times B \\ &\iff (A \subseteq C \text{ and } B \subseteq D) \text{ and } (C \subseteq A \text{ and } D \subseteq B) \\ &\iff A \subseteq C \text{ and } B \subseteq D \text{ and } C \subseteq A \text{ and } D \subseteq B \\ &\iff (A \subseteq C \text{ and } C \subseteq A) \text{ and } (B \subseteq D \text{ and } D \subseteq B) \\ &\iff A = C \text{ and } B = D, \end{aligned}$$

where the second equivalence holds because of the result we proved earlier in this exercise.

Our final task is to examine what happens when some or all of the hypotheses that A, B, C, D are non-empty are removed. We will look at each of the two equivalences separately.

Starting with the first one, note that for any sets A, B, C, and D,

$$A \subseteq C \text{ and } B \subseteq D \implies A \times B \subseteq C \times D.$$

For, if A and/or B is empty, then $A \times B$ is empty as well and the two statements in this implication are true; and if A and B are non-empty, then $A \times B$ is non-empty as well, and any element of $A \times B$ is an ordered pair whose first component belongs to A and

also C, and whose second component belongs to B and also D, so that this ordered pair belongs to $C \times D$. So, we just have to examine the statement:

$$A \times B \subseteq C \times D \implies A \subseteq C \text{ and } B \subseteq D. \qquad (\star)$$

If $B = \emptyset$, then "$A \times B \subseteq C \times D$" and "$B \subseteq D$" are always true irrespectively of whether the other sets are empty (since $A \times B = \emptyset$ and $\emptyset \subseteq S$ for any set S). However, "$A \subseteq C$" may not be true (for example, if $A = \{0\}$ and $C = \{1\}$). So, this implication is false when B is allowed to be empty. The same goes for when A is allowed to be empty by the symmetry of the situation. If $C = \emptyset$, then $C \times D = \emptyset$ and "$A \times B \subseteq C \times D$" is true only when $A \times B = \emptyset$, i.e., only when A or B is empty. If A is empty, then the implication is not true, for even though $A \subseteq C$, we may have $B \nsubseteq D$. If B is empty, the implication is still not true, for we may not have $A \subseteq C$ even though $B \subseteq D$. So, the implication is false when C is allowed to be empty. The same goes for D by symmetry. Therefore, the implication $(\star)$ is false. This shows that the equivalence in the first task is false when any of the hypotheses that A, B, C, and D are non-empty is removed.

Now, let's look at the second equivalence. Given that for any set A, B, C, D,

$$A \subseteq C \text{ and } B \subseteq D \implies A \times B \subseteq C \times D$$

and

$$C \subseteq A \text{ and } D \subseteq B \implies C \times D \subseteq A \times B,$$

we have:

$$A = C \text{ and } B = D \implies A \times B = C \times D.$$

So, we just need to examine the implication

$$A \times B = C \times D \implies A = C \text{ and } B = D. \qquad (\star\star)$$

In the case where $A = \emptyset$, the left-hand side of $(\star\star)$ is true only when $C \times D = \emptyset$, i.e., $C = \emptyset$ or $D = \emptyset$.[12] Hence, the left-hand side of $(\star\star)$ is false if $C \neq \emptyset$ and $D \neq \emptyset$.[13] So, the implication $(\star\star)$ is true if $C \neq \emptyset$ and $D \neq \emptyset$.[14] This results holds regardless of the emptiness or nonemptiness of B. By symmetry, the implication $(\star\star)$ also holds if $C \neq \emptyset$ and $D \neq \emptyset$ in the case where $B = \emptyset$, regardless of whether A is empty or not. Therefore, the implication $(\star\star)$ is true when we allow A and/or B to be empty (keeping the nonemptiness of C and D). Given the symmetry of the equality of sets, it

[12] Given two mathematical statements p and q, the expression "p is true only when q is true" is another way to say that p implies q.

[13] We used the contrapositive here: if p true implies q true, then q false implies p false

[14] Remember that if p is a false statement, then the implication $p \implies q$ is true for any statement q.

also follows that the implication $(\star\star)$ is true when we allow C and/or D to be empty (keeping the nonemptiness of A and B). In conclusion, the equivalence of the second task holds if the nonemptiness hypotheses are removed for A and/or B only, or if the nonemptiness hypotheses are removed for C and/or D only.

Exercise 3.5.7

Let X, Y, Z be sets and denote by $\pi_{X\times Y\to X} : X \times Y \to X$ and $\pi_{X\times Y\to Y} : X \times Y \to Y$ the co-ordinates functions on $X \times Y$. Let $f : Z \to X$ and $g : Z \to Y$ be two functions.

Define the function $h : Z \to X \times Y$ by:

$$h(z) = (f(z), g(z))$$

for any $z \in Z$. Given that f and g are well-defined functions, for any $z \in Z$, $f(z)$ and $g(z)$ are unique, and so $(f(z), g(z))$ is unique. Plus, $(f(z), g(z)) \in X \times Y$ for any $z \in Z$. Thus, h is a well-defined functions.

Now, the functions $\pi_{X\times Y\to X} \circ h$ and $\pi_{X\times Y\to Y} \circ h$ are defined from Z to X and from Z to Y, respectively. Furthermore, for any $z \in Z$, we have:

$$\pi_{X\times Y\to X} \circ h(z) = \pi_{X\times Y\to X}(f(z), g(z)) = f(z)$$

and

$$\pi_{X\times Y\to Y} \circ h(z) = \pi_{X\times Y\to Y}(f(z), g(z)) = g(z).$$

So:

$$\pi_{X\times Y\to X} \circ h = f \quad \text{and} \quad \pi_{X\times Y\to Y} \circ h = g.$$

There is no other function with this property. To see this, suppose that $h' : Z \to X \times Y$ was another function with this property. This means that for any $z \in Z$,

$$\pi_{X\times Y\to X} \circ h'(z) = f(z) \quad \text{and} \quad \pi_{X\times Y\to Y} \circ h'(z) = g(z). \qquad (\star)$$

Given that $\pi_{X\times Y\to X}$ and $\pi_{X\times Y\to Y}$ are the two co-ordinate functions on $X \times Y$ and $h'(z) \in X \times Y$, $(\star)$ implies that

$$h'(z) = (f(z), g(z)).$$

So,

$$h = h'.$$

Therefore, the function h is unique.

Exercise 3.5.8

Let $X_1, \ldots, X_n$ be sets.

On the one hand, the finite choice lemma (Lemma 3.5.11) says that:

$$(X_i \neq \emptyset \text{ for all } 1 \leq i \leq n) \implies \prod_{i=1}^{n} X_i \neq \emptyset.$$

Therefore, the contrapositive is true, that is:

$$\prod_{i=1}^{n} X_i = \emptyset \implies (X_i = \emptyset \text{ for some } 1 \leq i \leq n).$$

On the other hand, suppose that $X_j = \emptyset$ for some $1 \leq j \leq n$. Let $t_1, \ldots, t_n$ be n objects and consider the ordered n-tuple $t = (t_1, \ldots, t_n)$. If $t \in \prod_{i=1}^{n} X_i$, then $t_j \in X_j$ by Definition 3.5.6. But this is a contradiction since X_j is empty. Hence, $t \notin \prod_{i=1}^{n} X_i$. Therefore, no ordered n-tuple belongs in $\prod_{i=1}^{n} X_i$, which is a set defined to be a collection of n-tuples (Definition 3.5.6). So, $\prod_{i=1}^{n} X_i$ is empty.

In summary,

$$\prod_{i=1}^{n} X_i = \emptyset \iff (X_i = \emptyset \text{ for some } 1 \leq i \leq n).$$

Exercise 3.5.9

Let I and J be two sets. For all $\alpha \in I$, let A_α be a set, and for all $\beta \in J$, let B_β be a set.

Let x be an object. Suppose that $x \in \left(\bigcup_{\alpha \in I} A_\alpha\right) \cap \left(\bigcup_{\beta \in J} B_\beta\right)$. Then, $x \in A_{\alpha_0}$ for some $\alpha_0 \in I$ and $x \in B_{\beta_0}$ for some $\beta_0 \in J$. Note that $x \in A_{\alpha_0} \cap B_{\beta_0}$ and $(\alpha_0, \beta_0) \in I \times J$. Thus, $x \in \bigcup_{(\alpha,\beta) \in I \times J} (A_\alpha \cap B_\beta)$. Hence,

$$\left(\bigcup_{\alpha \in I} A_\alpha\right) \cap \left(\bigcup_{\beta \in J} B_\beta\right) \subseteq \bigcup_{(\alpha,\beta) \in I \times J} (A_\alpha \cap B_\beta).$$

Suppose that $x \in \bigcup_{(\alpha,\beta) \in I \times J} (A_\alpha \cap B_\beta)$. Then, $x \in A_{\alpha_0} \cap B_{\beta_0}$ for some $(\alpha_0, \beta_0) \in I \times J$. In particular, $x \in A_{\alpha_0}$ and $\alpha_0 \in I$, and thus $x \in \bigcup_{\alpha \in I} A_\alpha$. Similarly, we have $x \in \bigcup_{\beta \in J} A_\beta$

because $x \in B_{\beta_0}$ and $\beta_0 \in J$. Thus,

$$\bigcup_{(\alpha,\beta)\in I\times J}(A_\alpha \cap B_\beta) \subseteq \left(\bigcup_{\alpha\in I} A_\alpha\right) \cap \left(\bigcup_{\beta\in J} B_\beta\right).$$

We deduce that

$$\left(\bigcup_{\alpha\in I} A_\alpha\right) \cap \left(\bigcup_{\beta\in J} B_\beta\right) = \bigcup_{(\alpha,\beta)\in I\times J}(A_\alpha \cap B_\beta).$$

This equality still holds when one interchanges all the union and intersection symbols, provided that I and J are both non-empty. Note that:

$$\begin{aligned} x \in \left(\bigcap_{\alpha\in I} A_\alpha\right) \cup \left(\bigcap_{\beta\in J} B_\beta\right) &\iff x \in \bigcap_{\alpha\in I} A_\alpha \text{ or } x \in \bigcap_{\beta\in J} B_\beta \\ &\iff (x \in A_\alpha \text{ for all } \alpha \in I) \text{ or } (x \in B_\beta \text{ for all } \beta \in J) \end{aligned} \tag{1}$$

and

$$\begin{aligned} x \in \bigcap_{(\alpha,\beta)\in I\times J}(A_\alpha \cup B_\beta) &\iff x \in A_\alpha \cup B_\beta \text{ for all } (\alpha,\beta) \in I\times J \\ &\iff (x \in A_\alpha \text{ or } x \in B_\beta) \text{ for all } \alpha \in I \text{ and } \beta \in J. \end{aligned} \tag{2}$$

Suppose that statement (1) is true. Then, either x belongs in all the sets A_α or x belongs in all the sets B_β. Without loss of generality, let's assume that $x \in A_\alpha$ for all $\alpha \in I$. Then, for any $\alpha \in I$ and $\beta \in J$, the statement "$x \in A_\alpha$ or $x \in B_\beta$" is true because $x \in A_\alpha$. This means that (1) implies (2). Now, suppose that (2) is true. If $x \notin A_a$ for some $a \in I$, then $x \in B_\beta$ for all $\beta \in J$ since "$x \in A_a$ or $x \in B_\beta$" is true. Otherwise, $x \in A_\alpha$ for all $\alpha \in I$. We see that (2) implies (1). Therefore, statements (1) and (2) are equivalent to each other. It follows that

$$x \in \left(\bigcap_{\alpha\in I} A_\alpha\right) \cup \left(\bigcap_{\beta\in J} B_\beta\right) \iff x \in \bigcap_{(\alpha,\beta)\in I\times J}(A_\alpha \cup B_\beta).$$

In conclusion,

$$\left(\bigcap_{\alpha\in I} A_\alpha\right) \cup \left(\bigcap_{\beta\in J} B_\beta\right) = \bigcap_{(\alpha,\beta)\in I\times J}(A_\alpha \cup B_\beta).$$

Exercise 3.5.10

(i) Let $f : X \to Y$ and $\tilde{f} : X \to Y$ be two functions. We will prove that these two functions are equal if and only if they have the same graph.

First, note that:

$$\begin{aligned} f = \tilde{f} &\implies f(x) = \tilde{f}(x) \text{ for all } x \in X \\ &\implies (x, f(x)) = (x, \tilde{f}(x)) \text{ for all } x \in X \\ &\implies \{(x, f(x)) : x \in X\} = \{(x, \tilde{f}(x)) : x \in X\}, \end{aligned}$$

where the last implication holds because any element of $\{(x, f(x)) : x \in X\}$ is also an element of $\{(x, \tilde{f}(x)) : x \in X\}$ and vice versa.

Second, suppose that $\{(x, f(x)) : x \in X\} = \{(x, \tilde{f}(x)) : x \in X\}$. Then for any object $x \in X$, $(x, \tilde{f}(x))$ belongs in the graph of f; hence there exists an object w in X such that $(x, \tilde{f}(x)) = (w, f(w))$. As two ordered pairs are equal only if their components are equal, we have $x = w$ and $\tilde{f}(x) = f(w)$. By applying the axiom of substitution on the last equality, we see that $\tilde{f}(x) = f(x)$. Hence, $f(x) = \tilde{f}(x)$ for all objects x in X. Therefore, $f = \tilde{f}$.

In conclusion,

$$f = \tilde{f} \iff \{(x, f(x)) : x \in X\} = \{(x, \tilde{f}(x)) : x \in X\}.$$

(ii) Let G be a subset of $X \times Y$ with the property that for each $x \in X$, the set $\{y \in Y : (x, y) \in G\}$ has exactly one element.

Let $f : X \to Y$ be the function that maps each x in X to the unique element of the set $\{y \in Y : (x, y) \in G\}$. Then, for any $x \in X$, $(x, f(x)) \in G$. In addition, for any $(a, b) \in G$, $a \in X$ and $b = f(a)$ (by definition of f). Therefore, $G = \{(x, f(x)) : x \in X\}$. In other words, G is the graph of f. Moreover, we showed in part (i) that two functions are equal if and only if they have the same graph. Thus, f is only function whose graph is equal to G.

(iii) Let X, Y be sets and $P(x, y)$ be a property pertaining to an object $x \in X$ and an object $y \in Y$ that obeys the vertical line test. Let $G := \{(x, y) \in X \times Y : P(x, y)\}$. By its definition, G is a set that obeys the vertical line test. The ordered triple $f := (X, Y, G)$ is a function (following the newly introduced definition).

Note that for every $x \in X$, there exists exactly one $y \in Y$ such that $y = f(x)$; this object y is the only object in Y such that $(x, y) \in G$, i.e., such that $P(x, y)$ is true. Hence, for any $x \in X$ and $y \in Y$,

$$y = f(x) \iff P(x, y) \text{ is true.}$$

Therefore, this definition is compatible with Definition 3.3.1.

Now, let $g = (A, B, F)$ be another function (defined according to the new definition). Since g and f are essentially ordered 3-tuples, $f = g$ if and only if $X = A$, $Y = B$ and $G = F$ (Definition 3.5.6). That is to say, the functions f and g are equal if and only if their domain and codomain agree (i.e., $X = A$ and $Y = B$) and $f(x) = g(x)$ for all $x \in X$ (i.e., $G = \{(x, y) \in X \times Y : P(x, y)\} = F$). This shows that the new definition is also compatible with Definition 3.3.8

Exercise 3.5.11

See Exercise 3.4.6 part (ii). (Note that our solution over there does not follow the hint given here, but is still valid. Besides, it is easy to see how one can implement the hinted solution from that one.)

Exercise 3.5.12

(i) We believe that there is a mistake in the prompt of this exercise. It seems that the set X is not an arbitrary set, but rather the set $\mathbf{N}$. We think so because a is supposed to be a unique function that is defined for all natural numbers, and its domain is X. Besides, the version of this exercise in the third edition of the book uses $\mathbf{N}$ instead of X. If X is really not $\mathbf{N}$, then the mistake is in the definition of a. In this case, the domain of a should be $\mathbf{N}$ and not X. Hence, we consider that the function a we have to find is defined from $\mathbf{N}$ to X in this exercise.

We have the function $f : \mathbf{N} \times X \to X$ and the natural number c.

First, let us show that for every natural number $N \in \mathbf{N}$, there exists a unique function $a_N : \{n \in \mathbf{N} : n \leq N\} \to X$ such that $a_N(0) = c$ and $a_N(n{+}{+}) = f(n, a(n))$ for all $n \in \mathbf{N}$ such that $n < N$. We will do this inductively.

For the base case ($N = 0$), note that $\{n \in \mathbf{N} : n \leq 0\} = \{0\}$ and let a_0 be the function defined from $\{n \in \mathbf{N} : n \leq 0\}$ to X that assigns c to 0. It is vacuously true that $a_0(n{+}{+}) = f(n, a(n))$ for all $n \in \mathbf{N}$ such that $n < 0$ (since there are no such natural numbers n). Hence, a_0 is a function that satisfies the requirement for $N = 0$. Furthermore, a_0 is the only one to do it, for if there was another function b_0 that satisfied the requirement for $N = 0$, then $b_0(0) = c = a_0(0)$, which would imply that $b_0 = a_0$. Therefore, the base case is completed.

For the inductive step, assume that for some $k \in \mathbf{N}$ there is a unique function $a_k : \{n \in \mathbf{N} : n \leq k\} \to X$ such that $a_k(0) = c$ and $a_k(n{+}{+}) = f(n, a(n))$ for all $n < k$. Now, let $a_{k{+}{+}} : \{n \in \mathbf{N} : n \leq k{+}{+}\} \to X$ be the function defined by $a_{k{+}{+}}(n) = a_k(n)$ for $n \leq k$ and $a_{k{+}{+}}(k{+}{+}) = f(k, a(k))$. It is well-defined because each element n of the domain is associated to exactly one element of the range given that a_k and f are well-defined functions. For all $n < k$, we have $a_{k{+}{+}}(n{+}{+}) = a_k(n{+}{+}) = f(n, a(n))$ by the induction hypothesis. Hence, $a_{k{+}{+}}(n{+}{+}) = f(n, a(n))$ for all $n < k{+}{+}$. Moreover, the function $a_{k{+}{+}}$ is unique. Indeed, if $b{+}{+} : \{n \in \mathbf{N} : n \leq k{+}{+}\} \to X$ is a function such that $b_{k{+}{+}}(0) = 0$ and $b_{k{+}{+}}(n{+}{+}) = f(n, a(n))$ for all $n < k{+}{+}$, it is clear that $b_{k{+}{+}}(n) = a_{k{+}{+}}(n)$ for all $n \in \{n \in \mathbf{N} : n \leq k{+}{+}\}$, and so $b_{k{+}{+}} = a_{k{+}{+}}$. This closes the induction.

Now, let $a : \mathbf{N} \to X$ be the function defined by

$$a(n) = a_n(n) \text{ for all } n \in \mathbf{N}.$$

This is a well-defined function because $a_n(n)$ exists and is unique for each n in $\mathbf{N}$ (since a_n is a well-defined function whose domain contains n). Observe that:

$$a(0) = a_0(0) = c.$$

Furthermore, for any $n \in \mathbf{N}$,

$$a(n{+}{+}) = a_{n{+}{+}}(n{+}{+}) = f(n, a_{n{+}{+}}(n)) = f(n, a_n(n)) = f(n, a(n)),$$

where the penultimate equality holds because $a_{n{+}{+}}(n) = a_n(n)$ since $n \leq n$ (see the definition of $a_{k{+}{+}}$ in the inductive step above).

Moreover, this function a is unique. To see this, suppose that there is a function $b : \mathbf{N} \to X$ such that $b(0) = c$ and $b(n{+}{+}) = f(n, b(n))$ for all $n \in \mathbf{N}$. Let us show that $b(n) = a(n)$ for all $n \in \mathbf{N}$ by induction. For $n = 0$, we readily have $b(0) = c = a(0)$. Next, supposing inductively that $b(k) = a(k)$ for some natural number k, we have $b(k{+}{+}) = f(k, b(k)) = f(k, a(k)) = a(k{+}{+})$. This closes the induction. Given that $a(n) = b(n)$ for all $n \in \mathbf{N}$, the functions a and b are equal. Therefore, the function a is indeed unique.

(ii) Here, will construct another proof without using any properties of the natural numbers other than the Peano axioms directly.

First of all, we will show inductively that for every natural number N, there exists a unique pair A_N, B_N of subsets of $\mathbf{N}$ such that:

(a) $A_N \cap B_N = \emptyset$

(b) $A_N \cup B_N = \mathbf{N}$

(c) $0 \in A_N$

(d) $N{+}{+} \in B_N$

(e) Whenever $n \in B_N$, we have $n{+}{+} \in B_N$

(f) Whenever $n \in A_N$ and $n \neq N$, we have $n{+}{+} \in A_N$.

Let us start with the base case $(N = 0)$. Let

$$A_0 := \{0\} \quad \text{and} \quad B_0 := \mathbf{N} \setminus \{0\}.$$

Note that $A_0 \cap B_0 = \emptyset$, $A_0 \cup B_0 = \mathbf{N}$, $0 \in A_0$, and $0{+}{+} \in B_0$ (since $0{+}{+}$ is a natural number different from 0 by Axiom 2.3). Besides, whenever n is a natural number distinct from 0 (i.e., $n \in B_0$), its successor is also a natural number distinct from 0 (i.e., $n{+}{+} \in B_0$ as well), according to Axiom 2.3. Finally, the statement (f) for $N = 0$ is vacuously true because the only element in A_0 is 0. Therefore, the sets A_0 and B_0 satisfy properties (a) – (f) for $N = 0$. Let C_0 and D_0 be any subsets of $\mathbf{N}$ that satisfy the properties (a) – (f). We can very easily prove by induction on n that $n{+}{+} \in D_0$ for all $n \in \mathbf{N}$, by using property (d) in the base case and property (e) in the inductive step. By applying Axiom 2.5 with the property $P(n)$: "n is either 0 or the successor of some natural number", it is straightforward, to see that every natural number is either equal to 0 or is the successor of a natural number.[15] As D_0 contains all the natural numbers that are successors, we have $\mathbf{N} \setminus \{0\} \subseteq D_0$. Given that $0 \in C_0$ and $C_0 \cap D_0 = \emptyset$, we have $C_0 \cap (\mathbf{N} \setminus \{0\}) = \emptyset$, and so $C_0 = \{0\}$. Since $C_0 \cup D_0 = \mathbf{N}$ (i.e., $\{0\} \cup D_0 = \mathbf{N}$), $0 \notin D_0$ (as $C_0 \cap D_0 = \emptyset$), and $\mathbf{N} \setminus \{0\} \subseteq D_0$, we have $D_0 = \mathbf{N} \setminus \{0\}$. Therefore, $A_0 = C_0$ and $B_0 = D_0$. This shows the uniqueness of A_0 and B_0, and concludes the base step.

For the inductive step, suppose that there exists a unique pair A_k, B_k of subsets of $\mathbf{N}$ that satisfy the properties (a) – (f) for $N = k$, where k is some natural number. Now, let

$$A_{k++} := A_k \cup \{k{+}{+}\} \quad \text{and} \quad B_{k++} := B_k \setminus \{k{+}{+}\}.$$

First, we have:

$$\begin{aligned} A_{k++} \cap B_{k++} &= (A_k \cup \{k{+}{+}\}) \cap (B_k \setminus \{k{+}{+}\}) \\ &= (A_k \cap (B_k \setminus \{k{+}{+}\})) \cup (\{k{+}{+}\} \cap (B_k \setminus \{k{+}{+}\})) \\ &= (A_k \cap (B_k \setminus \{k{+}{+}\})) \cup \emptyset \\ &= A_k \cap (B_k \setminus \{k{+}{+}\}). \end{aligned}$$

[15] We do not invoke Lemma 2.2.10 here because we have to use the Peano axioms directly.

Since $B_k \setminus \{k{+}{+}\} \subseteq B_k$, $A_k \cap (B_k \setminus \{k{+}{+}\}) \subseteq A_k \cap B_k$. But $A_k \cap B_k = \emptyset$ by induction hypothesis. So, $A_k \cap (B_k \setminus \{k{+}{+}\}) = \emptyset$. Hence,

$$A_{k{+}{+}} \cap B_{k{+}{+}} = \emptyset.$$

We see that $A_{k{+}{+}}$ and $B_{k{+}{+}}$ satisfy property (a) for $N = k{+}{+}$. Second, note that:

$$\begin{aligned} A_{k{+}{+}} \cup B_{k{+}{+}} &= (A_k \cup \{k{+}{+}\}) \cup (B_k \setminus \{k{+}{+}\}) \\ &= A_k \cup (\{k{+}{+}\} \cup (B_k \setminus \{k{+}{+}\})) \\ &= A_k \cup B_k && \text{since } k{+}{+} \in B_k \\ &= \mathbf{N} && \text{by induction hypothesis.} \end{aligned}$$

So, $A_{k{+}{+}}$ and $B_{k{+}{+}}$ satisfy property (b) for $N = k{+}{+}$. Third, observe that $0 \in A_k$ (induction hypothesis) implies that $0 \in A_k \cup \{k{+}{+}\}$, i.e., $0 \in A_{k{+}{+}}$. Property (c) is therefore satisfied as well. Fourth, we have $k{+}{+} \in B_k$ and $(k{+}{+}){+}{+} \in B_k$ by induction hypothesis. Since $(k{+}{+}){+}{+} \neq k{+}{+}$ by Axiom 2.4, $(k{+}{+}){+}{+} \in B_k \setminus \{k{+}{+}\}$, that is $(k{+}{+}){+}{+} \in B_{k{+}{+}}$. So, property (d) is satisfied. Fifth, for any natural number n,

$$\begin{aligned} n \in A_{k{+}{+}} \text{ and } n \neq k{+}{+} &\implies n \in A_k \cup \{k{+}{+}\} \text{ and } n \notin \{k{+}{+}\} \\ &\implies n \in (A_k \cup \{k{+}{+}\}) \setminus \{k{+}{+}\} \\ &\implies n \in A_k, \end{aligned}$$

where the last implication holds because $k{+}{+} \notin A_k$ (since $k{+}{+} \in B_k$ and, by induction hypothesis, $A_k \cap B_k = \emptyset$). Now, if $n \neq k$,

$$\begin{aligned} n \in A_{k{+}{+}} \text{ and } n \neq k{+}{+} &\implies n \in A_k \\ &\implies n{+}{+} \in A_k && \text{by induction hypothesis} \\ &\implies n{+}{+} \in A_k \cup \{k{+}{+}\} \\ &\implies n{+}{+} \in A_{k{+}{+}}. \end{aligned}$$

If $n = k$, then $n{+}{+} = k{+}{+}$ and $n{+}{+} \in A_k \cup \{k{+}{+}\}$, i.e., $n{+}{+} \in A_{k{+}{+}}$. Therefore, whenever $n \in A_{k{+}{+}}$ and $n \neq k{+}{+}$, $n{+}{+} \in A_{k{+}{+}}$. This means that the sets $A_{k{+}{+}}$ and $B_{k{+}{+}}$ satisfy Property (f) for $N = k{+}{+}$. Lastly, for any natural number n,

$$\begin{aligned} n \in B_{k{+}{+}} &\implies n \in B_k \setminus \{k{+}{+}\} \\ &\implies n \in B_k \text{ and } n \neq k{+}{+} \\ &\implies n \in B_k \\ &\implies n{+}{+} \in B_k && \text{by induction hypothesis.} \end{aligned}$$

If $n \neq k$, then $n{+}{+} \neq k{+}{+}$ and

$$\begin{aligned} n \in B_{k++} &\implies n{+}{+} \in B_k \\ &\implies n{+}{+} \in B_k \setminus \{k{+}{+}\} \\ &\implies n{+}{+} \in B_{k++}. \end{aligned}$$

Now, we will show by contradiction that $k \notin B_{k++}$. So, suppose for the sake contradiction that $k \in B_{k++}$. Then, $k \in B_k$. Property (f) for $N = k$ states that:

$$n \in A_k \text{ and } n \neq k \implies n{+}{+} \in A_k.$$

As $B_k \cap A_k = \emptyset$ (by induction hypothesis) and $k \in B_k$, we have $k \notin A_k$. Thus, property (f) for $N = k$ is equivalent to:

$$n \in A_k \implies n{+}{+} \in A_k. \tag{f2}$$

With a straightforward induction, we can easily show that every natural number belongs in A_k by using property (c) for $N = k$ in the base case and (f2) in the inductive step. However, this implies that $k{+}{+} \in A_k$, a contradiction (since $k{+}{+} \in B_k$ and $A_k \cap B_k = \emptyset$ by induction hypothesis). Hence, $k \notin B_{k++}$. Earlier we showed that if n is a natural number different from k, then $n \in B_{k++}$ implies that $n{+}{+} \in B_{k++}$, for all natural numbers n. As $k \notin B_{k++}$, it is also true that

$$n \in B_{k++} \implies n{+}{+} \in B_{k++}$$

for any natural number n. Therefore, property (e) is true for $N = k{+}{+}$. In summary, the sets A_{k++} and B_{k++} satisfy properties (a) – (f) for $N = k{+}{+}$. Now, we just need to show that they are unique. For this, let C_{k++}, D_{k++} be a pair of sets that satisfy properties (a) – (f) for $N = k{+}{+}$. Let C_k and D_k be the sets defined by:

$$C_k := C_{k++} \setminus \{k{+}{+}\} \text{ and } D_k := D_{k++} \cup \{k{+}{+}\}.$$

Observe first that

$$\begin{aligned} C_k \cap D_k &= (C_{k++} \setminus \{k{+}{+}\}) \cap (D_{k++} \cup \{k{+}{+}\}) \\ &= ((C_{k++} \setminus \{k{+}{+}\}) \cap D_{k++}) \cup ((C_{k++} \setminus \{k{+}{+}\}) \cap \{k{+}{+}\}) \\ &= ((C_{k++} \setminus \{k{+}{+}\}) \cap D_{k++}) \cup \emptyset \\ &= (C_{k++} \setminus \{k{+}{+}\}) \cap D_{k++} \\ &= \emptyset \qquad \text{because } C_{k++} \cap D_{k++} = \emptyset. \end{aligned}$$

Second,

$$\begin{aligned}
C_k \cup D_k &= (C_{k++} \setminus \{k++\}) \cup (D_{k++} \cup \{k++\}) \\
&= (C_{k++} \setminus \{k++\}) \cup D_{k++} \cup \{k++\} \\
&= (C_{k++} \setminus \{k++\}) \cup \{k++\} \cup D_{k++} \\
&= \big((C_{k++} \setminus \{k++\}) \cup \{k++\}\big) \cup D_{k++} \\
&= C_{k++} \cup \{k++\} \cup D_{k++} \\
&= C_{k++} \cup D_{k++} \cup \{k++\} \\
&= \mathbf{N} \cup \{k++\} && \text{because } C_{k++} \cup D_{k++} = \mathbf{N} \\
&= \mathbf{N}.
\end{aligned}$$

Third, as $k++ \neq 0$ (Axiom 2.3), we have

$$0 \in C_{k++} \implies 0 \in C_k.$$

Fourth,

$$k++ \in D_k.$$

Fifth, for any natural number n,

$$\begin{aligned}
n \in D_k &\implies n \in D_{k++} \cup \{k++\} \\
&\implies n \in D_{k++} \text{ or } n = k++ \\
&\implies n++ \in D_{k++} \text{ or } n++ = (k++)++ \\
&\implies n++ \in D_{k++} \\
&\implies n++ \in D_k.
\end{aligned}$$

Lastly, for any natural number n,

$$\begin{aligned}
n \in C_k \text{ and } n \neq k &\implies n \in C_{k++} \setminus \{k++\} \text{ and } n \neq k \\
&\implies n++ \in C_{k++} \text{ and } n++ \neq k++ \quad \text{by property (f) and Axiom 2.4} \\
&\implies n++ \in C_{k++} \setminus \{k++\} \\
&\implies n++ \in C_k.
\end{aligned}$$

We see that, just like A_k and B_k, the sets C_k and D_k satisfy properties (a) – (f) for $N = k$. However, A_k and B_k are unique by induction hypothesis. Hence, $C_k = A_k$ and $D_k = B_k$. So,

$$A_k = C_{k++} \setminus \{k++\} \quad \text{and} \quad B_k = D_{k++} \cup \{k++\}.$$

Remember that we proved earlier that $k \notin B_{k++}$ by showing that $k \in B_{k++}$ implies that $k \in B_k$, which yields a contradiction. We can similarly show that $k{+}{+} \in D_{k++}$ yields a contradiction. Thus, $k{+}{+} \notin D_{k++}$ or, equivalently, $k{+}{+} \in C_{k++}$. We have:

$$A_{k++} = A_k \cup \{k{+}{+}\} = (C_{k++} \setminus \{k{+}{+}\}) \cup \{k{+}{+}\} = C_{k++}$$

and

$$B_{k++} = B_k \setminus \{k{+}{+}\} = (D_{k++} \cup \{k{+}{+}\}) \setminus \{k{+}{+}\} = D_{k++}.$$

We conclude that the sets A_{k++} and B_{k++} are the unique sets that satisfy properties (a) – (f) for $N = k{+}{+}$, and conclude the induction.

Now, we can directly prove that given a function $f : \mathbf{N} \times X \to X$ and an element c of X, there exists a unique function $a : \mathbf{N} \to X$ such that $a(0) = c$ and $a(n{+}{+}) = f(n, a(n))$ for all $n \in \mathbf{N}$.

First of all, we show by induction that for every natural number N, there exists a unique function $a_N : A_N \to \mathbf{N}$ such that $a_N(0) = c$ and $a_N(n{+}{+}) = f(n, a(n))$ for all $n \in \mathbf{N}$ such that $n \in A_N \setminus \{N\}$, where A_N is the first set of the unique pair of sets that satisfy the properties (a) – (f) for N. For the base case, note that $A_0 = \{0\}$ and $A_0 \setminus \{0\} = \emptyset$, so we can use the same line of reasoning we used in part(i) to complete the base case. In the inductive step, we can also use the same line of reasoning of the proof of part (i), but by replacing "$n < N$" with "$n \in A_N \setminus \{N\}$" and "$\{n \in \mathbf{N} : n \leq N\}$" with "$A_N$".

Now, let $a : N \to X$ be the function defined by

$$a(n) = a_n(n) \text{ for all } n \in \mathbf{N}.$$

The same argument used to prove that a is a well-defined and unique function such that $a(n{+}{+}) = f(n, a(n))$ in part (i) can be used here by replacing "$n \leq n$" with "$n \in A_n$".

Exercise 3.5.13

Suppose that we have a set $\mathbf{N}'$ of alternative natural numbers with an alternative zero $0'$, and an alternative increment operation $+{+}'$ such that the Peano axioms all hold with $\mathbf{N}$, 0, and $+{+}$ replace by their alternative counterparts.

Let $g : \mathbf{N} \times \mathbf{N}' \to \mathbf{N}'$ be the function defined by $g(n, n') := n'+{+}'$ for every $n \in \mathbf{N}$ and $n' \in \mathbf{N}'$. By the results of Exercise 3.5.12, there exists a function $f : \mathbf{N} \to \mathbf{N}'$ such that:

$$f(0) = 0'$$

and

$$f(n{+}{+}) = g(n, f(n)) \text{ for all } n \in \mathbf{N}.$$

Note that:

$$f(n{+}{+}) = f(n){+}{+}' \text{ for all } n \in \mathbf{N}.$$

We are going to show that f is surjective by proving inductively that for any $n' \in \mathbf{N}'$, there exists $n \in \mathbf{N}$ such that $f(n) = n'$. The base case is already given: $0 \in \mathbf{N}$ and $f(0) = 0'$. For the inductive step, let $k' \in \mathbf{N}'$. Suppose that there is some $k \in \mathbf{N}$ such that $f(k) = k'$. We have $f(k{+}{+}) = f(k){+}{+}' = k'{+}{+}'$. As $k{+}{+} \in \mathbf{N}$, the proposition is true for $k'{+}{+}'$ as well. This ends the induction. Therefore, f is a surjective function.

Next, let us show that a is also injective. To do this, we will prove that

$$f(x) = f(y) \implies x = y$$

for any natural numbers x and y, by inducting on x (keeping y fixed). In the base case, $x = 0$. Suppose that $f(0) = f(y)$, for some $y \in \mathbf{N}$. Then, $f(y) = 0'$. If, $y \neq 0$, then $y = m{+}{+}$ for some $m \in \mathbf{N}$ (see Definition 2.2.7 and Lemma 2.2.10). Thus, $f(y) = f(m{+}{+}) = f(m){+}{+}' = 0'$. But this is a contradiction by Axiom 2.3. Hence, $y = 0$, and the base case is done. For the inductive step, let $k \in \mathbf{N}$ and assume that:

$$f(k) = f(y) \implies k = y \text{ for all } y \in \mathbf{N}.$$

Let $y \in \mathbf{N}$. Suppose that $f(k{+}{+}) = f(y)$. If $y = 0$, then $f(y) = f(0)$ and, by the base case, $k{+}{+} = 0$. But this is contradictory to Axiom 2.3, so $y \neq 0$. Therefore, $y = m{+}{+}$ for some $m \in \mathbf{N}$. We have

$$f(k{+}{+}) = f(k){+}{+}' \quad \text{and} \quad f(y) = f(m{+}{+}) = f(m){+}{+}'.$$

Now,

$$\begin{aligned}
f(k{+}{+}) = f(y) &\implies f(k){+}{+}' = f(m){+}{+}' \\
&\implies f(k) = f(m) && \text{by Axiom 2.4} \\
&\implies k = m && \text{by induction hypothesis} \\
&\implies k{+}{+} = m{+}{+} && \text{by Axiom 2.4} \\
&\implies k{+}{+} = y.
\end{aligned}$$

This closes our induction. The function f is thus injective.

In conclusion, the function f is a bijection from the natural numbers to the alternative natural numbers.

3.6 Cardinality of Sets

Exercise 3.6.1

We are going to prove that the notion of having equal cardinality is an equivalence relation. Let X, Y, Z be sets.

Define the function $f : X \to X$ by $f(x) := x$. This function is injective (because if x_1, x_2 are distinct elements of X, then $f(x_1)$ and $f(x_2)$ are distinct as well) and surjective (because for any $x \in X$, $f^{-1}(\{x\})$ is non-empty as $x \in f^{-1}(\{x\})$). Hence, f is bijective. By Definition 3.6.1, X has equal cardinality with X. The reflexivity of this relation is thus proven.

Suppose that X has equal cardinality with Y. Definition 3.6.1 guarantees the existence of a bijection $g : X \to Y$. Its inverse $g^{-1} : Y \to X$ is also a bijection (see Exercise 3.3.6). Hence, Y has equal cardinality with X as well by Definition 3.6.1. We see that the notion of having equal cardinality is also symmetric.

Now, suppose that X has equal cardinality with Y and Y has equal cardinality with Z. There exist bijective functions $h_1 : X \to Y$ and $h_2 : Y \to Z$ by Definition 3.6.1. Now, consider their composition $h_2 \circ h_1$. This function is defined from X to Z and, as shown in Exercise 3.3.7, is a bijection. Therefore, X has equal cardinality with Z according to Definition 3.6.1. So, this relation is also transitive.

Exercise 3.6.2

Let X be a set. By Definition 3.6.5, X has cardinality 0 if and only if X has equal cardinality with the empty set (note that $\{i \in \mathbf{N} : 1 \leq i \leq 0\} = \emptyset$). Since the notion of having equal cardinality is symmetric (Propostion 3.6.4), X has equal cardinality with the empty set if and only the empty set has equal cardinality with X. By Definition 3.6.1, the empty set has equal cardinality with X if and only if there exists a bijection f from the empty set to X. But, we saw in Exercise 3.3.3 that the empty function is bijective if and only if its codomain is the empty set. Thus, there exists a bijection f from the empty set to X if and only if $X = \emptyset$. In summary,

$$\#(X) = 0 \iff X = \emptyset.$$

Exercise 3.6.3

Consider the proposition $P(n)$: "for any function $f : \{i \in \mathbf{N} : 1 \leq i \leq n\} \to \mathbf{N}$, there exists a natural number M such that $f(i) \leq M$ for all $1 \leq i \leq n$" that pertains to a natural number n. We will show that $P(n)$ is true for all $n \in \mathbf{N}$ by induction.

The base case is at $n = 0$. Since $\{i \in \mathbf{N} : 1 \leq i \leq 0\} = \emptyset$, $P(0)$ is vacuously true.

For the inductive step, suppose that $P(k)$ is true for some natural number k. Now, let $f : \{i \in \mathbf{N} : 1 \leq i \leq k{+}{+}\} \to \mathbf{N}$ be a function. Let $g : \{i \in \mathbf{N} : 1 \leq i \leq k\} \to \mathbf{N}$ be the function defined by $g(i) := f(i)$ for all $1 \leq i \leq k$. By induction hypothesis, there exists a natural number M such that $g(i) \leq M$ for all $1 \leq i \leq k$. Thus, $f(i) \leq M$ for all $1 \leq i \leq k$. This implies that $f(i) \leq M + f(k{+}{+})$ for all $1 \leq i \leq k{+}{+}$. Hence, $P(k{+}{+})$ is also true.

In conclusion, $P(n)$ is true for all $n \in \mathbf{N}$.

We can use this to prove that the set of natural numbers $\mathbf{N}$ is infinite (Theorem 3.6.12). We proceed by contradiction. Suppose that $\mathbf{N}$ is finite. By Definition 3.6.10, this means that $\mathbf{N}$ has cardinality n for some natural number n. Hence, there exists a bijection f from $\{i \in \mathbf{N} : 1 \leq i \leq n\}$ to $\mathbf{N}$, according to Definition 3.6.1. As f is bijective,

$$f(\{i \in \mathbf{N} : 1 \leq i \leq n\}) = \mathbf{N},$$

i.e.,

$$\{f(i) : 1 \leq i \leq n\} = \mathbf{N}.$$

Since $P(n)$ is true, there exists a natural number M such that $f(i) \leq M$ for all $1 \leq i \leq n$. So, $M{+}{+} \notin \{f(i) : 1 \leq i \leq n\}$ since $M{+}{+} > M$. However, $M{+}{+} \in \mathbf{N}$. This contradiction shows that $\mathbf{N}$ is infinite.

Exercise 3.6.4

(a) Let X be a finite set, and let x be an object which is not an element of X.

Given that X is finite, it has cardinality n for some $n \in \mathbf{N}$. This means that there is a bijection f from X to $\{i \in \mathbf{N} : 1 \leq i \leq n\}$. Consider the function

$$g : X \cup \{x\} \to \{i \in \mathbf{N} : 1 \leq i \leq n+1\}$$

defined by

$$g(y) := \begin{cases} f(y), & y \in X \\ n+1, & y \in \{x\} \end{cases}.$$

Since $g(X) = \{i \in \mathbf{N} : 1 \leq i \leq n\}$ and $g(\{x\}) = \{n+1\}$,

$$g(X \cup \{x\}) = g(X) \cup g(\{x\}) = \{i \in \mathbf{N} : 1 \leq i \leq n+1\},$$

and so g is surjective.

Suppose that there exist two objects x_1 and x_2 in $X \cup \{x\}$ such that $g(x_1) = g(x_2)$. Since $g(X) \cap g(\{x\}) = \emptyset$, $g(x_1) = g(x_2)$ only if $x_1, x_2 \in X$ or $x_1, x_2 \in \{x\}$. In the latter case, it is clear that $x_1 = x_2$. In the former case, $g(x_1) = g(x_2)$ implies that $f(x_1) = f(x_2)$, which in turn implies that $x_1 = x_2$ because f is injective. Therefore, g is an injective function.

Given that g is both injective and surjective, it is bijective. We deduce that $X \cup \{x\}$ is finite and has cardinality $n+1$. We have:

$$\#(X \cup \{x\}) = \#(X) + 1.$$

(b) Let X and Y be finite sets. Denote by n the cardinality of Y.

We will show by induction on n that $\#(X \cup Y) \leq \#(X) + n$, i.e., $\#(X \cup Y) \leq \#(X) + \#(Y)$.

In the base case, $n = 0$. So, $Y = \emptyset$ (see Exercise 3.6.2) and $X \cup Y = X$. Hence,

$$\#(X \cup Y) = \#(X),$$

and so we have

$$\#(X \cup Y) \leq \#(X) + \#(Y).$$

This completes the base case.

For the inductive step, suppose that the proposition is true for $n = k$. Now, suppose that $n = k{+}{+}$, i.e., $\#(Y) = k{+}{+}$. As $k{+}{+} \neq 0$, Y is not empty and we can select an element y_0 in Y. We have: $Y = (Y \setminus \{y_0\}) \cup \{y_0\}$, and thus:

$$X \cup Y = (X \cup (Y \setminus \{y_0\})) \cup \{y_0\} = ((Y \setminus \{y_0\}) \cup X) \cup \{y_0\}.$$

This implies that:

$$\#(X \cup Y) = \#\big(((Y \setminus \{y_0\}) \cup X) \cup \{y_0\}\big).$$

There are two cases to consider. First, if $y_0 \in X$, then $y_0 \in (Y \setminus \{y_0\}) \cup X$, and so $((Y \setminus \{y_0\}) \cup X) \cup \{y_0\})$ is just equal to $(Y \setminus \{y_0\}) \cup X$. In this case, we have $\#(X \cup Y) = \#((Y \setminus \{y_0\}) \cup X)$. In the second case, $y_0 \notin X$. Here, $y_0 \notin (Y \setminus \{y_0\}) \cup X$

and so $\#\big(((Y \setminus \{y_0\}) \cup X) \cup \{y_0\}\big)$ is equal to $\#((Y \setminus \{y_0\}) \cup X) + 1$, by Proposition 3.6.14(a). Thus, $\#(X \cup Y) = \#((Y \setminus \{y_0\}) \cup X) + 1$. We see that in any case,

$$\#(X \cup Y) \leq \#((Y \setminus \{y_0\}) \cup X) + 1. \qquad (\star)$$

Now, since $y_0 \notin Y \setminus \{y_0\}$, $\#((Y \setminus \{y_0\}) \cup \{y_0\}) = \#(Y \setminus \{y_0\}) + 1$ by Proposition 3.6.14(a). Note that:

$$\begin{aligned} \#(Y) = k{+}{+} &\implies \#((Y \setminus \{y_0\}) \cup \{y_0\}) = k{+}{+} \\ &\implies \#(Y \setminus \{y_0\}) + 1 = k{+}{+} \\ &\implies \#(Y \setminus \{y_0\}){+}{+} = k{+}{+} \\ &\implies \#(Y \setminus \{y_0\}) = k.^{16} \end{aligned}$$

By induction hypothesis, $\#(X \cup (Y \setminus \{y_0\})) \leq \#(X) + \#(Y \setminus \{y_0\})$, and so

$$\#((Y \setminus \{y_0\}) \cup X) \leq \#(X) + k.$$

Hence, $(\star)$ yields

$$\#(X \cup Y) \leq \#(X) + k + 1,$$

that is

$$\#(X \cup Y) \leq \#(X) + \#(Y).$$

This closes the induction.

In the case where $X \cap Y = \emptyset$, we can prove that $\#(X \cup Y) = \#(X) + \#(Y)$ by following this same line of argument with a little change. The base case is still straightforward since $\#(\emptyset) = 0$ leads to $\#(X \cup \emptyset) = \#(X) + \#(\emptyset)$. In the inductive step, we simply observe that $y_0 \notin X \cup (Y \setminus \{y_0\})$ for any element y_0 in Y because X and Y are disjoint in this case. It then follows that

$$\begin{aligned} \#(X \cup Y) &= \#(X \cup (Y \setminus \{y_0\})) + 1 \\ &= \#(X) + \#(Y \setminus \{y_0\}) + 1 \qquad \text{by the new induction hypothesis} \\ &= \#(X) + k + 1 \\ &= \#(X) + \#(Y) \end{aligned}$$

for any element y_0 in Y.

[16] We could have also used Lemma 3.6.9, but it makes use of subtraction even though it has not been introduced yet. We insists on only using results that have already been established.

(c) Let X be a finite set, and Y be a subset of X.

Since X is finite, it contains exactly n elements for some natural number n. Suppose, for the sake of contradiction that Y is infinite. Then, Y has more than n elements (otherwise, it would be finite). Hence, we can find a subset S of Y that contains $n+1$ elements.[17] Since $S \subseteq Y$ and $Y \subseteq X$, $S \subseteq X$. So, all the $n+1$ elements of S belong to X. This is contradictory because by Proposition 3.6.8, X has a unique cardinality, and so X cannot have more than n elements. Therefore, Y is finite.

Now, let $W := X \setminus Y$. We know from Proposition 3.1.27(g) that $W \subseteq X$, $W \cup Y = X$ and $W \cap Y = \emptyset$. By Proposition 3.6.14(b), we have:

$$\#(X) = \#(W \cup Y) = \#(W) + \#(Y).$$

Thus,

$$\#(Y) \leq \#(X).$$

If, in addition, Y is a proper subset of X, then W is not empty, and so $\#(W) > 0$. In this case, $\#(X) = \#(W) + \#(Y)$ implies that:

$$\#(Y) < \#(X).$$

(d) Let X be a finite set and $f : X \to Y$ be a function.

Since X is finite, there exists a bijection $g : X \to \{i \in \mathbf{N} : 1 \leq i \leq n\}$ for some natural number n. The set X has n elements. Suppose, for the sake of contradiction, that $f(X)$ has more than n elements. Then, it contains at least $n+1$ elements. Let $y_1, \ldots, y_{n+1}$ be $n+1$ distinct elements of $f(X)$. We have $f^{-1}(\{y_1, \ldots, y_{n+1}\}) \subseteq X$. So,

$$f^{-1}(\{y_1\}) \cup \cdots \cup f^{-1}(\{y_{n+1}\}) \subseteq X.$$

For all $1 \leq i \leq n+1$, $y_i \in f(X)$, so there exists an element in X whose image under f equals to y_i, and thus $f^{-1}(\{y_i\})$ is non-empty. For each $1 \leq i \leq n+1$, we select an element x_i in $f^{-1}(\{y_i\})$ with the single choice lemma (Lemma 3.1.5). Note that the sets $f^{-1}(\{y_1\}), \ldots, f^{-1}(\{y_{n+1}\})$ are pairwise disjoint. This follows because if $f^{-1}(\{y_j\}) \cap f^{-1}(\{y_k\}) \neq \emptyset$ for some $j \neq k$ with $1 \leq j, k \leq n+1$, then there exists

[17] Technically, we have to prove the claim that for any m, there is a subset S of Y with m elements. But this is very straightforward with an induction on the cardinality m of S. For the base case, $\emptyset \subseteq Y$. For the inductive step, if U is some subset of Y with m elements, then there exists $y \in Y$ such that $y \notin U$ (for otherwise, U contains all the elements of Y so that $U = Y$; a contradiction, since Y is infinite and U is not), and the set $U \cup \{y\}$ has $m+1$ elements (Proposition 3.6.14(a)).

some $x \in X$ such that $f(x) = y_j$ and $f(x) = y_k$, which is contradictory because $y_j \neq y_k$ and f is a well-defined function. Hence,

$$x_j \neq x_k \text{ for all } 1 \leq j, k \leq n+1, \text{ with } j \neq k.$$

Thus, the function $\phi : \{i \in \mathbf{N} : 1 \leq i \leq n+1\} \to \{x_1, \ldots, x_{n+1}\}$ defined by $\phi(i) := x_i$ is bijective, and so $\#(\{x_1, \ldots, x_{n+1}\}) = n+1$. In addition,

$$x_i \in f^{-1}(\{y_1\}) \cup \cdots \cup f^{-1}(\{y_{n+1}\}) \text{ for all } 1 \leq i \leq n+1,$$

and hence

$$\{x_1, \ldots, x_{n+1}\} \subseteq f^{-1}(\{y_1\}) \cup \cdots \cup f^{-1}(\{y_{n+1}\}).$$

It follows from Proposition 3.6.14(c) that

$$\#(f^{-1}(\{y_1\}) \cup \cdots \cup f^{-1}(\{y_{n+1}\})) \geq n+1.$$

By the same proposition, we have

$$\#(f^{-1}(\{y_1\}) \cup \cdots \cup f^{-1}(\{y_{n+1}\})) \leq \#(X).$$

Thus,

$$\#(X) \geq n+1.$$

This contradiction shows that $f(X)$ must contain no more than n elements, i.e., $f(X)$ is finite and $\#f(X) \leq \#(X)$.

Now, we will show that

$$f \text{ is injective } \iff \#f(X) = \#(X).$$

Denote by $a_1, \ldots, a_n$ the n elements of X. First, assume that f is injective. Then, $f(a_j) \neq f(a_k)$ for all $1 \leq j, k \leq n$ such that $j \neq k$. Thus, $f(X) = \{f(a_1), \ldots, f(a_n)\}$ contains exactly n elements. We see that $\#(f(X)) = \#(X)$. Second, assume that $\#(f(X)) = \#(X)$. Then, $f(X) = \{f(a_1), \ldots, f(a_n)\}$ contains n elements. Hence, $f(a_1), \ldots, f(a_n)$ are pairwise distinct. So if x and x' are distinct elements of X, then $f(x) \neq f(x')$, given that $x = a_j$ and $x' = a_k$ for some $1 \leq j, k \leq n$ with $j \neq k$. This means that f is injective.

(e) Let X be a finite set.

We will show by induction on the natural number n that for any set Y of cardinality n, the set $X \times Y$ is finite and $\#(X \times Y) = \#(X) + \#(Y)$.

For the base case ($n = 0$), let Y be a set of cardinality 0. It follows from Exercise 3.6.2 that Y is the empty set. Hence, $X \times Y$ is also the empty set, and thus finite. We have $\#(X \times Y) = \#(\emptyset) = 0$ and $\#(X) \times \#(Y) = \#(X) \times \#(\emptyset) = \#(X) \times 0 = 0$. This proves the base case.

For the inductive step, suppose that the claim holds for a natural number k. Let Y be a set of cardinality $k+1$. In particular, Y is non-empty and so we can select an element y_0 in Y with the single choice lemma (Lemma 3.1.5). By Proposition 3.6.14(b), $\#(Y) = \#(Y \setminus \{y_0\}) + \#\{y_0\}$, that is $k+1 = \#(Y \setminus \{y_0\}) + 1$, and so $\#(Y \setminus \{y_0\}) = k$. Hence, $X \times (Y \setminus \{y_0\})$ is finite and $\#(X \times (Y \setminus \{y_0\})) = \#(X) \times \#(Y \setminus \{y_0\})$ by the induction hypothesis. We have

$$X \times Y = X \times \big((Y \setminus \{y_0\}) \cup \{y_0\}\big) = \big((X \times (Y \setminus \{y_0\})\big) \cup (X \times \{y_0\}).$$

Note that the function $f : X \to X \times \{y_0\}$ defined by $f(x) := (x, y_0)$ is bijective, and thus $X \times \{y_0\}$ and X have equal cardinality. In particular, $X \times \{y_0\}$ is finite and $\#(X \times \{y_0\}) = \#(X)$. Next, observe that $X \times (Y \setminus \{y_0\})$ and $X \times \{y_0\}$ are disjoint because any ordered pair in the former does not have y_0 as its second component while every ordered pair in the latter does. Hence, $(X \times (Y \setminus \{y_0\})) \cup (X \times \{y_0\})$ is finite and

$$\#\big((X \times (Y \setminus \{y_0\})) \cup (X \times \{y_0\})\big) = \#(X \times (Y \setminus \{y_0\})) + \#(X \times \{y_0\})$$

by Proposition 3.6.14(b). It follows that $X \times Y$ is finite and

$$\begin{aligned}
\#(X \times Y) &= \#(X \times (Y \setminus \{y_0\})) + \#(X \times \{y_0\}) \\
&= \#(X) \times \#(Y \setminus \{y_0\}) + \#(X) \\
&= \#(X) \times \big(\#(Y \setminus \{y_0\}) + 1\big) \\
&= \#(X) \times (k+1) \\
&= \#(X) \times \#(Y).
\end{aligned}$$

Hence, the claim also holds for $k+1$, which closes the induction.

We conclude that for any finite set Y, $X \times Y$ is finite and $\#(X \times Y) = \#(X) \times \#(Y)$.

(f) Let Y be a finite set.

We will prove by induction on the natural number n that for any set X of cardinality n, the set Y^X is finite and $\#(Y^X) = \#(Y)^{\#(X)}$.

At the base, $n = 0$. Let X be a set of cardinality 0. We know from Exercise 3.6.2 that X is the empty set. Hence, Y^X contains exactly one element: the unique empty function

with codomain Y. Thus, Y^X is finite and $\#(Y^X) = 1$. Also, $\#(Y)^{\#(X)} = \#(Y)^0 = 1$. So, $\#(Y^X) = \#(Y)^{\#(X)}$ and the base case is done.

Now, suppose inductively that the claim is true for some natural number k. Let X be a set of cardinality $k+1$. It follows from Exercise 3.6.2 that X is non-empty. So, there exists an element $x_0 \in X$. Observe that if $Y = \emptyset$, then $Y^X = \emptyset$ because there is no (well-defined) function from a non-empty set to the empty set; in addition, $\#(Y)^{\#(X)} = \#(\emptyset)^{\#(X)} = 0^{\#(X)} = 0^{k+1} = 0 = \#(Y^X)$, and we are done. Now assume that Y is non-empty. According Proposition 3.6.14(b), $\#(X) = \#(X \setminus \{x_0\}) + \#(\{x_0\})$, that is $k+1 = \#(X \setminus \{x_0\}) + 1$, and so $\#(X \setminus \{x_0\}) = k$. Thus, $Y^{X\setminus\{x_0\}}$ is finite and

$$\#\big(Y^{X\setminus\{x_0\}}\big) = \#(Y)^{\#(X\setminus\{x_0\})}$$

by the induction hypothesis. Let $\varphi : Y^X \to Y^{X\setminus\{x_0\}}$ be the function that assigns to each function f in Y^X the function $\varphi_f : X \setminus \{x_0\} \to Y$ defined by $\varphi_f(x) := f(x)$. (In other words, φ_f is the restriction of f to $X \setminus \{x_0\}$.) Next, let $\Phi : Y^X \to Y \times Y^{X\setminus\{x_0\}}$ be the function defined by $\Phi(f) := (f(x_0), \varphi_f)$. Clearly, the function Φ is surjective because for any (y, f) in $Y \times Y^{X\setminus\{x_0\}}$, the function $g : X \to Y$ defined by $g(x) := f(x)$ for all $x \in X \setminus \{x_0\}$ and $g(x_0) := y$ is well-defined and belongs to Y^X, and we have $\Phi(g) = (y, f)$. The function Φ is also injective. In fact, if g_1, $g_2 \in Y^X$ such that $\Phi(g_1) = \Phi(g_2)$, then $g_1(x_0) = g_2(x_0)$; moreover, $\varphi_{g_1} = \varphi_{g_2}$ implies that $g_1(x) = g_2(x)$ for all $x \in X \setminus \{x_0\}$, and hence $g_1 = g_2$. Therefore, Φ is a bijective function. It follows that Y^X and $Y \times Y^{X\setminus\{x_0\}}$ have equal cardinality. By Proposition 3.6.14(e), $Y \times Y^{X\setminus\{x_0\}}$ is finite and $\#\big(Y \times Y^{X\setminus\{x_0\}}\big) = \#(Y) \times \#\big(Y^{X\setminus\{x_0\}}\big)$. Observe that

$$\#(Y) \times \#\big(Y^{X\setminus\{x_0\}}\big) = \#(Y) \times \#(Y)^{\#(X\setminus\{x_0\})} = \#(Y) \times (Y)^k = \#(Y)^{k+1} = \#(Y)^{\#(X)}.$$

Thus, Y^X is also finite and has cardinality $\#(Y)^{\#(X)}$, which closes the induction.

We conclude that for any finite set X, the set Y^X is finite and $\#(Y^X) = \#(Y)^{\#(X)}$.

Exercise 3.6.5

Let A and B be sets. Let $f : A \times B \to B \times A$ be the function defined by

$$f(x, y) := (y, x).$$

Clearly, f is a bijection. On the one hand, we see that f is injective because if $(x, y), (x', y')$ are distinct elements of $A \times B$, then $x \neq x'$ or $y \neq y'$, and so $(y, x) \neq (y', x')$, i.e., $f(x, y) \neq f(x', y')$. On the other hand, f is surjective because for each

$(y,x) \in B \times A$, $(x,y) \in A \times B$ and $f(x,y) = (y,x)$. Given that f is bijective, the sets $A \times B$ and $B \times A$ have equal cardinality.

Now, let m, n be natural numbers. Consider the sets $X := \{i \in \mathbf{N} : 1 \leq i \leq n\}$ and $Y := \{i \in \mathbf{N} : 1 \leq i \leq m\}$. Then, X has cardinality n and Y has cardinality m. By Proposition 3.6.14(e),

$$\#(X \times Y) = \#(X) \times \#(Y) = n \times m$$

and

$$\#(Y \times X) = \#(Y) \times \#(X) = m \times n.$$

Given that $X \times Y$ and $Y \times X$ have equal cardinality, we have:

$$n \times m = m \times n,$$

as stated in Lemma 2.3.2.

Exercise 3.6.6

Let A, B, and C be sets.

For any function f in $(A^B)^C$ and any z in C, we write $f_z := f(z)$ for convenience.

Now, let $\mathcal{F} : (A^B)^C \to A^{B \times C}$ be the function defined by $\mathcal{F}(f) = g$, where g is the function that maps any ordered pair (y,z) of $B \times C$ to $f_z(y)$, i.e.,

$$\begin{aligned} g : B \times C &\to A \\ (y,z) &\mapsto f_z(y). \end{aligned}$$

Suppose that $f, \tilde{f}$ are two functions in $(A^B)^C$ such that $\mathcal{F}(f) = \mathcal{F}(\tilde{f})$. Denote $\mathcal{F}(f)$ by g and $\mathcal{F}(\tilde{f})$ by $\tilde{g}$. Then,

$$\begin{aligned} \mathcal{F}(f) = \mathcal{F}(\tilde{f}) &\implies g = \tilde{g} \\ &\implies g(y,z) = \tilde{g}(y,z) \text{ for all } (y,z) \in B \times C \\ &\implies f_z(y) = \tilde{f}_z(y) \text{ for all } (y,z) \in B \times C \\ &\implies f_z = \tilde{f}_z \text{ for all } z \in C \\ &\implies f(z) = \tilde{f}(z) \text{ for all } z \in C \\ &\implies f = \tilde{f}. \end{aligned}$$

Hence, $\mathcal{F}$ is injective.

Let $h \in A^{B\times C}$. Consider the function t in $(A^B)^C$ that maps each z in C to the function t_z in A^B defined by $t_z(y) := h(y,z)$. We have $\mathcal{F}(t) = h$. Thus, $\mathcal{F}$ is surjective as well.

Therefore, $\mathcal{F}$ is a bijection. This shows that $(A^B)^C$ and $A^{B\times C}$ have equal cardinality.

Let a, b, and c be natural numbers. Let X, Y, and Z be the sets defined by:

$$X := \{i \in \mathbf{N} : 1 \le i \le a\}, \quad Y := \{i \in \mathbf{N} : 1 \le i \le b\}, \text{ and } Z := \{i \in \mathbf{N} : 1 \le i \le c\}.$$

We have:

$$\begin{aligned} \#((X^Y)^Z) &= \#(X^Y)^{\#(Z)} && \text{by Proposition 3.6.14(f)} \\ &= (\#(X)^{\#(Y)})^{\#(Z)} && \text{by Proposition 3.6.14(f)} \\ &= (a^b)^c \end{aligned}$$

and

$$\begin{aligned} \#(X^{Y\times Z}) &= \#(X)^{\#(Y\times Z)} && \text{by Proposition 3.6.14(f)} \\ &= \#(X)^{\#(Y)\times\#(Z)} && \text{by Proposition 3.6.14(e)} \\ &= a^{bc}. \end{aligned}$$

Given that $\#((X^Y)^Z) = \#(X^{Y\times Z})$, we have:

$$(a^b)^c = a^{bc}.$$

Now, suppose that B and C are disjoint., i.e., $B \cap C = \emptyset$. Let $\mathcal{G} : A^B \times A^C \to A^{B\cup C}$ be the function defined by:

$$\mathcal{G}(f_1, f_2) = g$$

where g is the function

$$\begin{aligned} g : B \cup C &\to A \\ w &\mapsto \begin{cases} f_1(w), & w \in B \\ f_2(w), & w \in C \end{cases}. \end{aligned}$$

The function $g = \mathcal{G}(f_1, f_2)$ is well-defined because the domains B and C of f_1 and f_2 are disjoint for all $f_1 \in A^B$ and $f_2 \in A^C$.

Let $(f_1, f_2), (\tilde{f}_1, \tilde{f}_2) \in A^B \times A^C$ and set $g := \mathcal{G}(f_1, f_2)$, $\tilde{g} := \mathcal{G}(\tilde{f}_1, \tilde{f}_2)$. Observe that:

$$\begin{aligned} g = \tilde{g} &\implies g(w) = \tilde{g}(w) \text{ for all } w \in B \cup C \\ &\implies \begin{cases} f_1(w) = \tilde{f}_1(w) & \text{for all } w \in B \\ f_2(w) = \tilde{f}_2(w) & \text{for all } w \in C \end{cases} \\ &\implies f_1 = \tilde{f}_1 \text{ and } f_2 = \tilde{f}_2 \\ &\implies (f_1, f_2) = (\tilde{f}_1, \tilde{f}_2). \end{aligned}$$

Hence, $\mathcal{G}$ is injective.

Let $u \in A^{B \cup C}$. Define the functions:

$$\begin{array}{lll} h_1 : B \to A & \text{and} & h_2 : C \to A \\ \quad y \mapsto u(y) & & \quad z \mapsto u(z). \end{array}$$

We have:

$$u(w) = \begin{cases} h_1(w) & w \in B \\ h_2(w) & w \in C \end{cases}, \text{ for all } w \in B \cup C$$

and

$$\mathcal{G}(h_1, h_2) = u.$$

So, $\mathcal{G}$ is a surjective function. Therefore, $\mathcal{G}$ is bijective. This implies that $A^B \times A^C$ and $A^{B \cup C}$ have equal cardinality.

Now, suppose that the sets Y and Z defined earlier are disjoint. We have:

$$\begin{aligned} \#(X^Y \times X^Z) &= \#(X^Y) \times \#(X^Z) && \text{by Proposition 3.6.14(e)} \\ &= \#(X)^{\#(Y)} \times \#(X)^{\#(Z)} && \text{by Proposition 3.6.14(f)} \\ &= a^b \times a^c \end{aligned}$$

and

$$\begin{aligned} \#(X^{Y \cup Z}) &= \#(X)^{\#(Y \cup Z)} && \text{by Proposition 3.6.14(e)} \\ &= \#(X)^{\#(Y)+\#(Z)} && \text{by Proposition 3.6.14(b)} \\ &= a^{b+c}. \end{aligned}$$

So,

$$a^b + a^c = a^{b+c}.$$

Exercise 3.6.7

Let A and B be two finite sets.

On the one hand, suppose that A has lesser or equal cardinality to B. Then, there exists an injection $f : A \to B$ from A to B. By Proposition 3.6.14(d), $f(A)$ is finite and $\#(f(A)) = \#(A)$. Since $f(A) \subseteq B$, we have $\#(f(A)) \leq \#(B)$ by Proposition 3.6.14(c). Hence, $\#(A) \leq \#(B)$.

On the other hand, suppose that $\#(A) \leq \#(B)$. There exists a bijection f defined from A to $\{i \in \mathbf{N} : 1 \leq i \leq \#(A)\}$ and a bijection g defined from B to $\{i \in \mathbf{N} : 1 \leq i \leq \#(B)\}$. Let $\tilde{g}$ be the restriction of g^{-1} to $\{i \in \mathbf{N} : 1 \leq i \leq \#(A)\}$, that is let $\tilde{g}$ be the function:

$$\begin{aligned} \tilde{g} : \{i \in \mathbf{N} : 1 \leq i \leq \#(A)\} &\to B \\ i &\mapsto g^{-1}(i). \end{aligned}$$

Note that $\tilde{g}$ is injective because g^{-1} is injective. Now, consider the function $\tilde{g} \circ f$. This function is defined from A to B and it is injective because both $\tilde{g}$ and f are injective (see Exercise 3.3.2). Therefore, A has lesser or equal cardinality to B.

We conclude that A has lesser or equal cardinality to B if and only if $\#(A) \leq \#(B)$.

Exercise 3.6.8

Let A and B be sets and $f : A \to B$ an injection. Assume that A is non-empty.

There exists an object a that belongs to A by the single choice lemma (Lemma 3.1.5). The function $\tilde{f} : A \to f(A)$ defined by $\tilde{f}(x) := f(x)$ is clearly a bijection. Let $g : B \to A$ be the function defined by:

$$g(y) := \begin{cases} \tilde{f}^{-1}(y), & y \in f(A) \\ a, & \text{otherwise} \end{cases}.$$

Note that

$$g(B) \subseteq A.$$

Next, we have

$$g(f(A)) = \{g(y) : y \in f(A)\} = \{\tilde{f}(y) : y \in f(A)\} = \tilde{f}^{-1}(f(A)),$$

where the second equality holds because $g(y) = \tilde{f}^{-1}(y)$ for all $y \in f(A)$. Observe that

$$\tilde{f}^{-1}(f(A)) = \tilde{f}^{-1}(\{x \in A : x \in f(x) \in A\}) = \tilde{f}^{-1}(\{x \in A : \tilde{f}(x) \in A\}) = \tilde{f}^{-1}(\tilde{f}(A)) = A,$$

where the second equality holds because $f(x) = \tilde{f}(x)$ for all $x \in A$, and the last equality holds because $\tilde{f}$ is injective (see Exercise 3.4.5). Therefore, $g(f(A)) = A$. Since $f(A) \subseteq B$, we have $g(f(A)) \subseteq g(B)$, and so

$$A \subseteq g(B).$$

Thus,

$$g(B) = A.$$

This shows that the function g is a surjection.

Exercise 3.6.9

Let A and B be finite sets.

By Proposition 3.6.14(b), $A \cup B$ is finite. By Proposition 3.6.14(c), $A \cap B$ is also finite since $A \cap B \subseteq A \cup B$.

We have $A \cup B = (A \setminus B) \cup (B \setminus A) \cup (A \cap B)$, as shown in Exercise 3.1.10. We also proved in that same exercise that the sets $A \setminus B$, $B \setminus A$, and $A \cap B$ are pairwise disjoint. It follows from Proposition 3.6.14(b) that

$$\#(A \cup B) = \#((A \setminus B) \cup (B \setminus A)) + \#(A \cap B) = \#(A \setminus B) + \#(B \setminus A) + \#(A \cap B).$$

We know that $A = (A \setminus B) \cup (A \cap B)$ and $(A \setminus B) \cap (A \cap B) = \emptyset$. So,

$$\#(A) = \#(A \setminus B) + \#(A \cap B)$$

by Proposition 3.6.14(b). Similarly,

$$\#(B) = \#(B \setminus A) + \#(A \cap B).$$

Hence,

$$\#(A) + \#(B) = \#(A \setminus B) + \#(A \cap B) + \#(B \setminus A) + \#(A \cap B).$$

Therefore,

$$\#(A) + \#(B) = \#(A \cup B) + \#(A \cap B).$$

Exercise 3.6.10

Let $A_1, \ldots, A_n$ be finite sets such that $\#(\bigcup_{i \in \{1,\ldots,n\}} A_i) > n$.

Suppose that for every $1 \leq i \leq n$, $\#(A_i) \leq 1$. By a generalization of Proposition 3.6.14(b) that can be easily proven by induction, we have:

$$\#\left(\bigcup_{i \in \{1,\ldots,n\}} A_i \right) \leq \#(A_1) + \cdots + \#(A_n),$$

and so

$$\#\left(\bigcup_{i\in\{1,\ldots,n\}} A_i\right) \leq n.$$

But this is contradictory. Hence, there must exist an index $1 \leq i \leq n$ for which $\#(A_i) > 1$, that is $\#(A_i) \geq 2$.

Exercise 3.6.11

Let $f : X \to Y$ be a function between two sets X, Y. Regardless of the cardinality of X or Y, we observe the following equivalences.

The function f is injective if and only if $f(x_1) \neq f(x_2)$ for any distinct elements x_1, x_2 in X.

Furthermore, $f(x_1) \neq f(x_2)$ for any distinct elements x_1, x_2 in X if and only if $\{f(x_1), f(x_2)\}$ is a set of exactly two elements for any distinct x_1, x_2 in X.

Moreover, $\{f(x_1), f(x_2)\}$ is a set of exactly two elements for any distinct x_1, x_2 in X if and only if $f(\{x_1, x_2\})$ is a set of exactly two elements for any distinct x_1, x_2 in X.

Finally, $f(\{x_1, x_2\})$ is a set of exactly two elements for any distinct x_1, x_2 in X if and only if $f(E)$ contains exactly two elements for any subset E of X that contains exactly two elements.

In summary, f is injective if and only if $\#(f(E)) = 2$ for any subset E of X such that $\#(E) = 2$.

Exercise 3.6.12

For convenience, we define $D_n := \{i \in \mathbf{N} : 1 \leq i \leq n\}$ for each $n \in \mathbf{N}$ throughout this exercise.

(i) Note that for any natural number k, S_k is a subset of ${D_k}^{D_k}$.

Let $n \in \mathbf{N}$. Given that D_n is finite, ${D_n}^{D_n}$ is also finite according to Proposition 3.6.14(f). Since $S_n \subseteq {D_n}^{D_n}$, S_n is also finite by Proposition 3.6.14(c).

Now, consider S_{n++}. This set consists of bijections from D_{n++} to itself. Each of these bijections assign to n++ exactly one of the n++ elements of D_{n++}. So, we can partition S_{n++} into n++ subsets, such that every bijection in a particular subset assigns to n++ the same element. For each $1 \leq i \leq n$++, define the set

$$T_i := \{f \in S_{n++} : f(n++) = i\}.$$

For each $1 \leq i \leq n{+}{+}$, let $\mathcal{F}_i : T_i \to S_n$ be the function that maps each function in T_i to its restriction to D_n. That is, for any $f \in T_i$, $\mathcal{F}_i(f) = g$ if and only if $g \in S_n$ and $g(j) := f(j)$ for all $j \in D_n$. Note that the functions $\mathcal{F}_i$ are injective, for if two functions f_1, f_2 in T_i have the same restriction to D_n (i.e., $\mathcal{F}_i(f_1) = \mathcal{F}_i(f_2)$), then they are also equal since $f_1(n{+}{+}) = f_2(n{+}{+}) = i$. The functions $\mathcal{F}_i$ are also surjective because for any bijection g in S_n, the function $f : D_{n++} \to D_{n++}$ defined by $f(j) = g(j)$ if $j \in D_n$ and $f(n{+}{+}) = i$ belongs to T_i and we have $\mathcal{F}_i(f) = g$. Hence, the functions $\mathcal{F}_i$ are bijective. Therefore, the sets T_i and the set S_n have equal cardinality, i.e., the sets T_i are finite and

$$\#(T_1) = \cdots = \#(T_{n++}) = \#(S_n).$$

We have:

$$S_{n++} = T_1 \cup \cdots \cup T_{n++}$$

and

$$T_i \cap T_j = \emptyset \text{ for all } 1 \leq i, j \leq n{+}{+} \text{ such that } i \neq j.$$

By a generalization of the second claim of Proposition 3.14.6(b) (that can be easily proved with a simple induction), the set S_{n++} is finite and

$$\#(S_{n++}) = \#(T_1) + \cdots + \#(T_{n++}).$$

We deduce that

$$\#(S_{n++}) = (n{+}{+}) \times \#(S_n).$$

(ii) Note that $D_0 = \emptyset$ and so ${D_0}^{D_0}$ contains only the function defined from the empty set to the empty set. Since this function is a bijection (see Exercise 3.3.3) and $S_0 = {D_0}^{D_0}$, we have $\#(S_0) = 1$.

Given that $\#(S_0) = 0!$ and that $\#(S_{n++})$ is recursively defined from $\#(S_n)$ by the same rule that recursively defines $(n{+}{+})!$ from $n!$ for all natural numbers n, we have $\#(S_n) = n!$ for all natural numbers n.[18]

[18] We believe that this argument is sufficient and that rolling out an explicit induction (which would be straightforward) is not needed.

Chapter 4
Integers and Rationals

4.1 The Integers

Exercise 4.1.1

Let $a—b$ and $c—d$ be two integers.

Since a and b are natural numbers, $a+b$ is a natural number. Since the equality for natural numbers is well-defined, we have: $a+b=a+b$.[1] By Definition 4.1.1, this means that $a—b=a—b$. So, the equality of integers is reflexive.

Suppose that $a—b=c—d$. Then, $a+d=c+b$ by Definition 4.1.1, which implies that $c+b=a+d$ since $c+b$ and $a+b$ are natural numbers and the equality of natural numbers is well-defined[2]. But, $c+b=a+d$ means that $c—d=a—b$ according to Definition 4.1.1. We see that this definition is also symmetric.

[1] You can even show this explicitly if you want. Suppose that $N \neq N$ for some $N \in \mathbf{N}$. Then a simple backwards induction (see Exercise 2.2.6) yield $n \neq n$ for all natural numbers $n \leq N$. In particular, $0 \neq 0$. So, by Definition 2.2.7 and Lemma 2.2.10, there exists a natural number m for which $0 = m{+}{+}$, which contradicts Axiom 2.3.

[2] I also believe that the equality of natural numbers is well-defined simply because we constructed them axiomatically without making use of any prior mathematical objects. But I am not sure about this. If you know something, please contact me at keumoe.fozeu.r@gmail.com or open an issue on this manual's GitHub.

Exercise 4.1.2

Let $a—b$ and $a'—b'$ be two integers such that $(a—b) = (a'—b')$. By Definition 4.1.1., $a+b' = a'+b$. Now,

$$\begin{aligned} a+b' = a'+b &\implies a'+b = a+b' \\ &\implies b+a' = b'+a \\ &\implies (b—a) = (b'—a') && \text{by Definition 4.1.1} \\ &\implies -(a—b) = -(a'—b') && \text{by Definition 4.1.4.} \end{aligned}$$

Therefore, the definition of negation on the integers is well-defined.

Exercise 4.1.3

Let a be an integer. By Definition 4.1.1, there exists natural numbers x and y such that $a = x—y$. We have:

$$\begin{aligned} (-1) \times a &= (0—1) \times (x—y) \\ &= (0 \times x + 1 \times y)—(0 \times y + 1 \times x) \\ &= y—x \\ &= -(x—y) \\ &= -a \end{aligned}$$

as desired.

Exercise 4.1.4

Let $x, y, z \in \mathbf{Z}$. There exist $a, b, c, d, e, f \in \mathbf{N}$ such that $x = a—b$, $y = c—d$, and $z = e—f$.

Note that:

$$\begin{aligned} x+y &= (a—b)+(c—d) \\ &= (a+c)—(b+d) \\ &= (c+a)—(d+b) \\ &= (c—d)+(a—b) \\ &= y+x. \end{aligned} \tag{1}$$

Note that:

$$\begin{aligned}(x+y)+z &= ((a—b)+(c—d))+(e—f)\\ &= (a+c)—(b+d)+(e—f)\\ &= ((a+c)+e)—((b+d)+f)\\ &= (a+(c+e))—(b+(d+f))\\ &= (a—b)+(c+e)—(d+f)\\ &= (a—b)+((c—d)+(e—f))\\ &= x+(y+z).\end{aligned}$$

Note that

$$\begin{aligned}x+0 &= (a—b)+(0—0)\\ &= (a+0)—(b+0)\\ &= a—b\\ &= x.\end{aligned}$$

By (1), we have:

$$x+0 = 0+x = x.$$

By (1), we have $x+(-x) = (-x)+x$, given that $-x$ is also an integer. Plus,

$$\begin{aligned}(-x)+x &= -(a—b)+(a—b)\\ &= (b—a)+(a—b)\\ &= (b+a)—(a+b)\\ &= (a+b)—(a+b).\end{aligned}$$

Given that $(a+b)+0 = 0+(a+b)$, we have $(a+b)—(a+b) = 0—0$. Hence, $(-x)+x = 0—0$, so that

$$x+(-x) = (-x)+x = 0.$$

Note that:

$$\begin{aligned}xy &= (a—b)\times(c—d)\\ &= (ac+bd)—(ad+bc)\\ &= (ca+db)—(da+cb)\\ &= (c—d)\times(a—b)\\ &= yx.\end{aligned} \tag{5}$$

We have $x1 = 1x$ by (5). Plus,

$$\begin{aligned} 1x &= (1—0) \times (a—b) \\ &= (1 \times a + 0 \times b)—(1 \times b + 0 \times a) \\ &= a \times b \\ &= x. \end{aligned}$$

Hence,

$$x1 = 1x = x.$$

Note that:

$$\begin{aligned} x(y+z) &= (a—b) \times ((c—d) + (e—f)) \\ &= (a—b) \times ((c+e)—(d+f)) \\ &= (a(c+e) + b(d+f))—(a(d+f) + b(c+e)) \\ &= (ac+bd+ae+bf)—(ad+bc+af+be) \\ &= (ac+bd)—(ad+bc) + (ae+bf)—(af+be) \\ &= (a—b) \times (c—d) + (a—b) \times (e—f) \\ &= xy + xz. \end{aligned} \tag{8}$$

Finally, as $y+z$ is also an integer, (8) and (5) imply that:

$$(y+z)x = x(y+z) = xy + xz = yx + zx.$$

Exercise 4.1.5

Let a and b be integers such that $ab = 0$.

Suppose, for the sake of contradiction, that $a \neq 0$ and $b \neq 0$. By the trichotomy of integers (Lemma 4.1.5), this implies that a is positive or negative (but not both) and b is positive or negative (but not both).

We know from Lemma 2.2.3 that a and b cannot be simultaneously positive. So, we have three cases left to examine.

If a is positive and b is negative, then $b = -k$ for some positive natural number k,

and so

$$\begin{aligned} ab &= a \times (-k) \\ &= a \times ((-1) \times k) && \text{(see Exercise 4.1.3)} \\ &= (a \times (-1)) \times k && \text{by Proposition 4.1.6} \\ &= ((-1) \times a) \times k && \text{by Proposition 4.1.6} \\ &= (-1) \times (a \times k) && \text{by Proposition 4.1.6} \\ &= (-1) \times ak \\ &= -ak && \text{(see Exercise 4.1.3).} \end{aligned}$$

Since a and k are positive natural numbers, so is ak, thus $-ak$ is a negative integer, and so $-ak \neq 0$. Hence, $ab \neq 0$. By the commutativity of the multiplication of integers, it is clear that we also have $ab \neq 0$ in the case where a is negative and b is positive.

If both a and b are negative, then $a = -p$ and $b = -q$ for some positive natural numbers p and q. Thus,

$$\begin{aligned} ab &= (-p) \times (-q) \\ &= (-1) \times p \times (-1) \times q && \text{(see Exercise 4.1.3)} \\ &= (-1) \times (-1) \times p \times q && \text{by Proposition 4.1.6} \\ &= ((-1) \times (-1)) \times pq. \end{aligned}$$

Observe that

$$(-1) \times (-1) = (0-1) \times (0-1) = (0 \times 0 + 1 \times 1) - (0 \times 1 - 1 \times 0) = 1 - 0 = 1.$$

Hence,

$$ab = 1 \times pq = pq,$$

by Proposition 4.1.6. As pq is a positive integer, $pq \neq 0$, and so $ab \neq 0$.

We see that in any case, $a \neq 0$ and $b \neq 0$ yield the contradiction $ab \neq 0$. We conclude that either $a = 0$ or $b = 0$.

Exercise 4.1.6

Let a, b, c be integers such that $ac = bc$ and $c \neq 0$. Observe that

$$\begin{aligned} ac + (-bc) &= ac + (-1)(bc) && \text{(see Exercise 4.1.3)} \\ &= ac + ((-1)(b))c && \text{by Proposition 4.1.6} \\ &= ac + (-b)c && \text{(see Exercise 4.1.3)} \\ &= (a + (-b))c && \text{by Proposition 4.1.6} \\ &= (a-b)c. \end{aligned}$$

It follows from Proposition 4.1.8 that $a - b = 0$ or $c = 0$. But, $c \neq 0$. Hence, we must have $a - b = 0$, that is $a + 0 = 0 + b$, that is $a = b$.

Exercise 4.1.7

Let a, b, $c \in \mathbf{Z}$. We are going to prove the following six statements. In the proofs, Proposition 4.1.6 will be freely used without explicit mention as it should be clear at this point.

(a) $a > b$ if and only if $a - b$ is a positive natural number.

By Definition 4.1.10, $a \geq b$ if and only if $a = b + m$ for some natural number m. We have:

$$\begin{aligned} a = b + m &\iff a + (-b) = b + m + (-b) \\ &\iff a - b = m + b + (-b) \\ &\iff a - b = m + 0 \\ &\iff a - b = m. \end{aligned}$$

Hence, $a \geq b$ if and only if $a - b = m$ for some natural number m. Definition 4.1.10 also states that $a > b$ if and only if $a \geq b$ and $a \neq b$. Thus,

$$a > b \iff a - b = m \text{ for some natural number } m \text{ and } a \neq b.$$

Given that $a = b$ if and only $a - b = 0$, we have $a - b \neq 0$ if and only if $a \neq b$. So:

$$\begin{aligned} &(a - b = m \text{ for some natural number } m \text{ and } a \neq b) \\ &\qquad \iff (a - b = m \text{ for some positive natural number } m). \end{aligned}$$

Therefore,

$$a > b \iff a - b \text{ is a positive natural number.}$$

(b) If $a > b$, then $a+c > b+c$.

As shown in (a), $a > b$ if and only if $a-b$ is a positive natural number. Observe that:

$$\begin{aligned} a-b &= a-b+0 \\ &= a-b+c-c \\ &= a+(-b)+c+(-c) \\ &= a+c+(-b)+(-c) \\ &= a+c+(-1)\times b+(-1)\times c \\ &= a+c+(-1)\times(b+c) \\ &= a+c+(-(b+c)) \\ &= (a+c)-(b+c). \end{aligned}$$

Hence, $a > b$ if and only if $(a+c)-(b+c)$ is a positive natural number. By (a), $(a+c)-(b+c)$ is a positive natural number if and only if $a+c > b+c$. Thus, $a > b$ if and only if $a+c > b+c$.

(c) If $a > b$ and c is positive, then $ac > bc$.

Suppose that $a > b$ and c is positive. From (a), we know that $a-b$ is positive. By Lemma 2.3.3, $(a-b)c$ is a positive natural number, that is to say $ac-bc$ is a positive natural number. By part (a), this means that $ac > bc$.

(d) If $a > b$, then $-a < -b$.

Note that:

$$\begin{aligned} a-b &= a+(-b) \\ &= (-b)+a \\ &= (-b)+1a \\ &= (-b)+(-1)\times(-1)\times a && \text{see Exercise 4.1.5} \\ &= (-b)+(-(-a)) && \text{see Exercise 4.1.3} \\ &= (-b)-(-a). \end{aligned}$$

So,

$$\begin{aligned} a > b &\implies a-b \text{ is a positive integer} && \text{by (a)} \\ &\implies (-b)-(-a) \text{ is a positive integer} \\ &\implies -b > -a && \text{by (a)} \\ &\implies -a < -b. \end{aligned}$$

(e) If $a > b$ and $b > c$, then $a > c$.

Suppose that $a > b$ and $b > c$. By (a), this implies that $a - b$ and $b - c$ are positive integers. So, their sum is also a positive integer (Proposition 2.2.8). We have:

$$\begin{aligned}(a-b)+(b-c) &= a+(-b)+b+(-c)\\ &= a+((-b)+b)+(-c)\\ &= a+0+(-c)\\ &= a+(-c)\\ &= a-c.\end{aligned}$$

Hence, $a - c$ is a positive integer. By (a), this means that $a > c$.

(f) Exactly one of the statements $a > b$, $a < b$, or $a = b$ is true.

Consider the integer $a - b$. By the trichotomy of integers (Lemma 4.1.5), exactly one of the following statement is true: (i) $a - b = 0$, (ii) $a - b$ is a positive natural number, (iii) $a - b$ is the negation of a natural number.

Statement (i) is equivalent to $a + 0 = 0 + b$, so that we have

$$\text{(i)} \iff a = b.$$

By (a),

$$\text{(ii)} \iff a > b.$$

Statement (iii) is equivalent to $a - b = -n$, where n is a positive natural number. But,

$$\begin{aligned}a-b=-n &\iff a+(-b)=-n\\ &\iff a+(-b)+b=(-n)+b\\ &\iff a=(-n)+b\\ &\iff a+n=(-n)+b+n\\ &\iff a+n=b.\end{aligned}$$

So, (iii) is equivalent to $b = a + n$ for some positive natural number n. By (a), we have

$$\text{(iii)} \iff a < b.$$

As exactly one of the three statements (i), (ii), (iii) is true, we conclude that exactly one of the statements $a > b$, $a < b$, $a = b$ is true.

Exercise 4.1.8

Let $P(n)$: "n is a natural number" be a proposition that pertains to an integer n.

By Axiom 2.1, $P(0)$ is true. By Axiom 2.2, if $P(n)$ is true for some natural number n, then $P(n{+}{+})$ is also true. However, $P(-1)$ is false even though -1 is an integer. This shows that Axiom 2.5 does not apply directly to the integers.

Exercise 4.1.9

Let n be an integer.

If $n = 0$, then $n^2 = n \times n = 0 \times 0 = 0$.

If n is positive, then $n^2 = n \times n$ is also a positive natural number by Lemma 2.3.3.

If n is negative, then $-n$ is positive. Recall from Exercise 4.1.5 that $(-1) \times (-1) = 1$. So, $(-1) \times (-n) = (-1) \times (-1) \times n = 1 \times n = n$ in light of Exercise 4.1.3. We have:

$$n \times n = ((-1) \times (-n)) \times ((-1) \times (-n)) = (-1) \times (-1) \times (-n) \times (-n) = (-n) \times (-n).$$

So, $n^2 = (-n) \times (-n)$ is a positive natural number by Lemma 2.3.3.

We see that in any case,

$$n^2 \geq 0.$$

4.2 The Rationals

Exercise 4.2.1

Let x, y, z be three rational numbers. By Definition 4.2.1, there exist integers a, c, e and non-zero integers b, d, f such that $x = a//b$, $y = c//d$ and $z = e//f$.

Since a and b are integers, ab is also an integer and we have $ab = ab$ because the equality of integers is well-defined. But $ab = ab$ means that $a//b = a//b$, i.e., $x = x$. Hence, the equality of rational numbers is reflexive.

Suppose that $x = y$, i.e., $ad = cb$. Then, $cb = ad$ because the equality of integers is well-defined. So, $c//d = a//b$, i.e., $y = x$. Hence, the equality of rational numbers is symmetric.

Suppose that $x = y$ and $y = z$, i.e., $ad = cb$ and $cf = ed$. Observe that:

$$\begin{aligned}
cf = ed &\implies bcf = bed \\
&\implies adf = bed && \text{because } bc = cb = ad \\
&\implies afd = bed \\
&\implies af = be && \text{by Corollary 4.1.9} \\
&\implies a//b = e//f \\
&\implies x = z.
\end{aligned}$$

So, the equality of rational numbers is transitive.

Exercise 4.2.2

Let $a'//b'$ and $a//b$ be rationals such that $a'//b' = a//b$. Note that $a'b = ab'$.

Let $c//d$ be a rational. By Definition 4.2.2, we have:

$$(a//b) \times (c//d) = (ac)//(bd)$$

and

$$(a'//b') \times (c//d) = (a'c)//(b'd).$$

Note that $(ac) \times (b'd) = ab'cd = a'bcd = (a'c) \times (bd)$. So,

$$(a//b) \times (c//d) = (a'//b') \times (c//d).$$

We similarly show that if $(c'//d')$ is a rational equal to $(c//d)$ then

$$(a//b) \times (c'//d') = (a//b) \times (c//d).$$

Now, we have

$$-(a//b) = (-a)//b$$

and

$$-(a'//b') = (-a')//b'$$

by Definition 4.2.2. Observe that

$$(-a) \times b' = -(ab') = (-a'b) = (-a') \times b.$$

So, $(-a)//b = (-a')//b'$, i.e.,

$$-(a//b) = -(a'//b').$$

Exercise 4.2.3

Let x, y, z be rational numbers. By Definition 4.2.1, there exist integers a, c, e and non-zero integers b, d, f such that $x = a//b$, $y = c//d$ and $z = e//f$. At any point in this exercise, we will use previously established identities without explicitly citing them each time.

By Definition 4.2.2,

$$x+y = (ad+bc)//(bd)$$

and

$$y+x = (cb+da)//(db) = (bc+ad)//(bd) = (ad+bc)//(bd),$$

so

$$x+y = y+x.$$

As 0 is a rational, we have $x+0 = 0+x$. Now,

$$0+x = 0//1+a//b = (0b+1a)//(1b) = a//b = x.$$

Hence,

$$x+0 = 0+x = x.$$

We have $-x = (-a)//b$ by Definition 4.2.2. Observe that

$$x+(-x) = a//b+(-a)//b = (ab+b(-a))//(bb) = (ab-ab)//(b^2) = 0//(b^2) = 0.$$

So,

$$x+(-x)=(-x)+x=0.$$

By Definition 4.2.2,

$$xy=(a//b)\times(c//d)=(ac)//(bd)=(ca)//(db)=(c//d)\times(a//b)=yx.$$

Note that

$$\begin{aligned}(xy)z&=((a//b)\times(c//d))\times(e//f)\\&=(ac)//(bd)\times(e//f)\\&=(a(ce))//(b(df))\\&=(a//b)\times((ce)//(df))\\&=(a//b)\times((c//d)\times(e//f))\\&=x(yz).\end{aligned}$$

Observe that

$$1x=(1//1)\times(a//b)=(1a)//(1b)=a//b=x.$$

So,

$$x1=1x=x.$$

We have

$$\begin{aligned}x(y+z)&=(a//b)\times((c//d)+(e//f))\\&=(a//b)\times((cf+de)//(df))\\&=(a(cf+de))//(b(df))\\&=(acf+ade)//(bdf).\end{aligned}$$

Note that

$$\begin{aligned}xy+xz&=(ac)//(bd)+(ae)//(bf)\\&=(acbf+aebd)//(bdbf)\\&=(b(acf+ade))//(b(bdf))\\&=(b//b)\times(acf+ade)//(bdf),\end{aligned}$$

and given that $b1=1b$, we have $b//b=1//1=1$, which leads to

$$\begin{aligned}xy+xz&=1\times(acf+ade)//(bdf)\\&=1\times(x(y+z))\\&=x(y+z).\end{aligned}$$

It is then straightforward to see that

$$(y+z)x = x(y+z) = xy+xz = yx+zx.$$

Finally, if $x \neq 0$, then x^{-1} exists and x^{-1} is a rational. In this case,

$$\begin{aligned} xx^{-1} &= (a//b) \times (b//a) \\ &= (ab)//(ba) \\ &= 1//1 && \text{because } (ab)1 = 1(ba) \\ &= 1, \end{aligned}$$

and so

$$xx^{-1} = x^{-1}x = 1.$$

Exercise 4.2.4

Let x be a rational number, so that there exist an integer a and a non-zero integer b such that $x = a/b$. Since a and b are integers, so is their product ab. By the trichotomy of integers (Lemma 4.1.5), exactly one of the following three statements is true: (i) $ab = 0$, (ii) $ab > 0$, (iii) $ab < 0$. We will examine each case separately.

We start with (i). Given that $b \neq 0$, $ab = 0$ if and only if $a = 0$ by Proposition 4.1.8. But, $a = 0$ if and only if $x = 0$. Hence, $ab = 0$ if and only if $x = 0$. That is to say,

$$\text{(i)} \iff x = 0.$$

Before examining (ii) and (iii), recall that ab is negative if and only a is positive and b is negative or vice versa, and ab is positive if and only if both a and b are positive or both a and b are negative, as shown in the solution of Exercise 4.1.5.

Now, we look at (ii). On the one side, note that

$$\begin{aligned} ab > 0 &\implies (a > 0 \text{ and } b > 0) \text{ or } (a < 0 \text{ and } b < 0) \\ &\implies (a > 0 \text{ and } b > 0) \text{ or } (-a > 0 \text{ and } -b > 0) && \text{by Lemma 4.1.11(d)} \\ &\implies a/b \text{ is positive or } (-a)/(-b) \text{ is positive} && \text{by Definition 4.2.6} \\ &\implies a/b \text{ is positive} \\ &\implies x \text{ is positive,} \end{aligned}$$

where the penultimate implication holds because $(-a)/(-b) = a/b$, since we have $(-a)b = a(-b)$. On the other side, suppose that x is positive. Then, a/b is positive

so that $a/b = c/d$ for some positive integers c and d by Definition 4.2.6. We have $ad = bc$. So, either both a and b are positive or both a and b are negative, for otherwise we would have a positive integer equal a negative integer. In either case, $ab > 0$. Therefore,

$$\text{(ii)} \iff x \text{ is positive.}$$

Finally, we focus on (iii). On the one side, we have

$$\begin{aligned} ab < 0 &\implies (a > 0 \text{ and } b < 0) \text{ or } (a < 0 \text{ and } b > 0) \\ &\implies (a > 0 \text{ and } -b > 0) \text{ or } (-a > 0 \text{ and } b > 0) \\ &\implies a/(-b) \text{ is positive or } (-a)/b \text{ is positive} \\ &\implies (-a)/b \text{ is positive} \\ &\implies a/b \text{ is negative} \\ &\implies x \text{ is negative,} \end{aligned}$$

where the fourth implication holds because $a/(-b) = (-a)/b$, as $ab = (-b)(-a)$. On the other side, suppose that x is negative. Then, $x = a/b = (-c)/d$ for some positive integers c and d by Definition 4.2.6. We have $ad = b(-c)$. So, either a is positive and b is negative, or a is negative and b is positive, for otherwise we would have a positive integer equal a negative integer. In both cases, $ab < 0$. Hence,

$$\text{(iii)} \iff x \text{ is negative.}$$

Given that exactly one of the three statements (i), (ii), or (iii) is true, the trichotomy of rationals is established.

Exercise 4.2.5

Let $x, y, z \in \mathbf{Q}$. Let us prove the following five statements.

(a) Exactly one of the three statements $x = y$, $x < y$ or $x > y$ is true.

Consider the rational $x - y$. By the trichotomy of rationals (Lemma 4.2.7), exactly one of the statements (i) $x - y = 0$, (ii) $x - y$ is a positive rational number, or (iii) $x - y$ is a negative rational number, is true.

Note that

$$\text{(i)} \iff (x-y)+y = 0+y \iff x+(-y)+y = y \iff x+0 = y \iff x = y,$$

$$\text{(ii)} \iff x > y \qquad \text{by Definition 4.2.8,}$$

and

$$\text{(iii)} \iff x < y \qquad \text{by Definition 4.2.8.}$$

Hence, exactly one of the statements $x = y$, $x < y$ or $x > y$ is true.

(b) One has $x < y$ if and only if $y > x$.

By definition,

$$x < y \iff x - y \text{ is a negative rational number.}$$

Before moving on, note that for any rational number s/t, we have

$$(-1) \times (s/t) = (-1/1) \times (s/t) = ((-1)s)/(1t) = (-s)/t = -(s/t),$$

and so

$$-(-(s/t)) = (-1) \times ((-1) \times (s/t)) = ((-1) \times (-1)) \times (s/t) = 1 \times (s/t) = s/t.$$

Now,

$$\begin{aligned} x - y &= -(-x) + (-y) \\ &= (-1) \times (-x) + (-1) \times (y) \\ &= (-1) \times (-x + y) \\ &= -(-x + y) \\ &= -(y + (-x)) \\ &= -(y - x). \end{aligned}$$

We see that $x - y$ is a negative rational number if and only if $y - x$ is a positive rational number by Definition 4.2.6. Given that $y - x$ is a positive rational number if and only if $y > x$ by Definition 4.2.8, we conclude that

$$x < y \iff y > x.$$

(c) If $x < y$ and $y < z$, then $x < z$.

Suppose that $x < y$ and $y < z$. By Definition 4.2.8, this implies that $x - y$ and $y - z$ are negative rationals, so that, by Definition 4.2.6, there exist positive integers a, b, c, d such that $x - y = (-a)/b$ and $y - z = (-c)/d$. Now,

$$(x - y) + (y - z) = (-ad - bc)/(bd) = -(ad + bc)/(bd).$$

Since $ad+bc$ and bd are positive integers, it follows from Definition 4.2.6 that $(ad+bc)/(bd)$ is a positive rational, and so $(x-y)+(y-z)$ is a negative rational. But,

$$(x-y)+(y-z)=x-z.$$

Thus, $x-z$ is a negative rational, which means that $x<z$ according to Definition 4.2.8.

(d) If $x<y$, then $x+z<y+z$.

In light of Definition 4.2.8, note that

$$\begin{aligned} x<y &\implies x-y \text{ is a negative rational} \\ &\implies x-y+z-z \text{ is a negative rational} \\ &\implies x+z-y-z \text{ is a negative rational} \\ &\implies (x+z)-(y+z) \text{ is a negative rational} \\ &\implies x+z<y+z. \end{aligned}$$

(e) If $x<y$ and z is positive, then $xz<yz$.

Suppose that $x<y$ and z is positive.

According Definition 4.2.8, $x-y$ is a negative rational because $x<y$. By Definition 4.2.6, there exist positive integers a, b, c, d such that $x-y=(-a)/b$ and $z=c/d$. Now,

$$xz-yz=(x-y)z=((-a)/b)\times(c/d)=((-a)c)/(bd)=(-(ac))/(bd).$$

Since ac and bd are positive integers, $xz-yz$ is a negative rational by Definition 4.2.6, which means that

$$xz<yz$$

according to Definition 4.2.8.

Exercise 4.2.6

Let x, y, $z\in\mathbf{Q}$ such that $x<y$ and z is negative.

We want to show that $xz>yz$. To properly do this, we need three preliminary results. The first one is that for any rational q, we have $-q=(-1)q$, which we already established in Exercise 4.2.5. The next one is that for any rational q, we have $0q=0$. To establish this, let q be rational so that there exist $a\in\mathbf{Z}$ and $b\in\mathbf{Z}\setminus\{0\}$ such that

$q = a/b$, and observe that $0q = (0/1) \times (a/b) = (0a/1b) = (0/b) = 0$, as desired. The last preliminary is that for any rational q, q is positive if and only if $q > 0$. Note that for any rational q, $q - 0 = q + (-0) = q + ((-1)0) = q + 0 = q$, and so q is positive if and only if $q > 0$ by Definition 4.2.8.[3]

Now, since z is negative, $z = -w$ for some positive rational w by Definition 4.2.6. Note that

$$xz - yz = xz + (-yz).$$

So,

$$\begin{aligned} xz - yz &= xz + ((-1)y)z \\ &= (x + (-1)y)z \\ &= ((-1)y + x)z. \end{aligned}$$

It follows that

$$\begin{aligned} xz - yz &= ((-1)y + x)(-w) \\ &= ((-1)y + x)(-1)w \\ &= ((-1)y(-1) + x(-1))w \\ &= (y + (-x))w \\ &= (y - x)w. \end{aligned}$$

Since $x < y$, we have $y > x$ by Proposition 4.2.9(b), and so $y - x$ is positive according to Definition 4.2.8. We also have $w > 0$. So by Proposition 4.2.9(e), $(y - x)w > (y - w)0$, that is $(y - x)w > 0$. In other words, $xz - yz$ is positive. Therefore,

$$xz > yz$$

according to Definition 4.2.8.

[3] The three preliminaries will be used without explicit mention. Besides, observe that the third preliminary implies that a rational q is negative if and only if $q < 0$. This fourth result will also be used without explicit mention.

4.3 Absolute Value and Exponentiation

Exercise 4.3.1

Let x, y, and z be rational numbers. We will prove the following 7 statements:

(a) We have $|x| \geq 0$. Also, $|x| = 0$ if and only if x is 0.

If $x > 0$, by Definition 4.3.1, $|x| = x$, and so $|x| > 0$.

If $x = 0$, by Definition 4.3.1, $|x| = 0$.

If $x < 0$, by Definition 4.3.1, $|x| = -x$, and so $|x| > 0$ (see Exercise 4.2.6).

We see that in any case,

$$|x| \geq 0.$$

Furthermore, $|x| \neq 0$ if $x < 0$ or $x > 0$, so that if $|x| = 0$ then $x = 0$ (by taking the contrapositive). Therefore,

$$|x| = 0 \iff x = 0.$$

(b) We have $|x+y| \leq |x| + |y|$.

By the trichotomy of rationals, each of x and y is either positive, negative, or zero. Grouping these possibilities, we distinguish four exhaustive cases.

Case 1: Both x and y are positive.

Applying Definition 4.3.1, we have $|x| = x$ and $|y| = y$, and so $|x| + |y| = x + y$. In this case, $x + y$ is positive as well since $x > 0$ implies that $x + y > y$, and then $y > 0$ implies that $x + y > 0$. So $|x+y| = x+y$ by Definition 4.3.1. Since $x + y \leq x + y$, we have:

$$|x+y| \leq |x| + |y|.$$

Case 2: Both x and y are negative.

Here, $x + y$ is also negative because $x < 0$ implies that $x + y < y$, and then $y < 0$ implies that $x + y < 0$. So by Definition 4.3.1, $|x+y| = -(x+y) = -x-y$. The same definition also yields $|x| = -x$ and $|y| = -y$. Given that $-x-y \leq -x-y$, we see that:

$$|x+y| \leq |x| + |y|.$$

Case 3: One of the rationals is zero.

Without loss of generality, assume that $x = 0$. We have $|x+y| = |0+y| = |y|$ and $|x| + |y| = |0| + |y| = 0 + |y| = |y|$. As $|y|$ is a rational, we have $|y| \leq |y|$. So:

$$|x+y| \leq |x| + |y|.$$

Case 4: One of the rationals is negative and the other is positive.

Without loss of generality, assume that x is negative while y is positive. Here, $|x| = -x$ and $|y| = y$ by Definition 4.3.1. So, $|x| + |y| = -x + y$. Plus, by Definition 4.3.1,

$$|x+y| = \begin{cases} x+y, & x+y > 0 \\ 0, & x+y = 0 \\ -x-y, & x+y < 0 \end{cases}.$$

If $x+y > 0$, then $|x+y| = x+y$. Since x is negative, $-x$ is positive, and thus $x \leq 0 \leq -x$, so that $x \leq -x$. Hence, $x+y \leq -x+y$, and so $|x+y| \leq |x| + |y|$.

If $x+y = 0$, then $|x+y| = 0$. By Proposition 4.3.3(a), $|x| \geq 0$ and $|y| \geq 0$, so $|x| + |y| \geq 0$, and thus $|x+y| \leq |x| + |y|$.

If $x+y < 0$, then $|x+y| = -x-y$. Given that y is positive, $-y$ is negative, and thus $-y \leq 0 \leq y$, so that $-y \leq y$. This implies that $-x-y \leq -x+y$, that is $|x+y| \leq |x| + |y|$. In sum, we have

$$|x+y| \leq |x| + |y|$$

in this case.

Conclusion: $|x+y| \leq |x| + |y|$.

(c) We have $-y \leq x \leq y$ if and only if $y \geq |x|$. In particular, we have $-|x| \leq x \leq |x|$.

We distinguish three cases based on the trichotomy of rationals.

Case 1: $x > 0$.

Then $|x| = x$ by Definition 4.3.1. So,

$$-y \leq x \leq y \implies x \leq y \implies |x| \leq y.$$

Now, if $|x| \leq y$, then $x \leq y$. In addition, given that $x \geq 0$, $x \leq y$ implies that $y \geq 0$, so that $-y \leq 0$. It follows that that $-y \leq x \leq y$. Hence,

$$|x| \leq y \implies -y \leq x \leq y.$$

So,

$$|x| \leq y \iff -y \leq x \leq y.$$

Case 2: $x = 0$.

Then, $|x| = 0$ by Definition 4.3.1 and it is straightforward to see that:

$$|x| \leq y \iff -y \leq x \leq y.$$

Case 3: $x < 0$.

Then $|x| = -x$ by Definition 4.3.1. Thus,

$$-y \leq x \leq y \implies -y \leq -x \leq y \implies -x \leq y \implies |x| \leq y.$$

Now, if $|x| \leq y$, then $-x \leq y$ and so $-y \leq x$. Since $x \leq 0$, $-y \leq x$ implies that $-y \leq 0$, which in turn implies that $y \geq 0$, so that $y \geq 0 \geq x \geq -y$, and so $-y \leq x \leq y$. Hence,

$$|x| \leq y \implies -y \leq x \leq y.$$

So we have

$$|x| \leq y \iff -y \leq x \leq y.$$

Conclusion: $-y \leq x \leq y$ if and only if $|x| \leq y$.

In particular, given that $|x| \leq |x|$, we have:

$$-|x| \leq x \leq |x|.$$

(d) We have $|xy| = |x||y|$. In particular, $|-x| = |x|$.

In light of the trichotomy of rationals, there are four exhaustive cases to consider.

Case 1: x and y are both positive.

In this case, xy is positive as well and, by Definition 4.3.1, we have $|xy| = xy$, $|x| = x$, and $|y| = y$. Since $xy = xy$,

$$|xy| = |x||y|.$$

Case 2: x and y are both negative.

Here, xy is positive. By Definition 4.3.1, $|xy| = xy$, $|x| = -x$ and $|y| = -y$. We have $|x||y| = (-x)(-y) = xy$. Thus,

$$|xy| = |x||y|.$$

Case 3: One of the rationals is 0.

Without loss of generality, assume that $x = 0$. We have

$$xy = 0y = 0 \implies |xy| = |0| = 0$$

and

$$|x||y| = |0||y| = 0|y| = 0.$$

So,

$$|xy| = |x||y|.$$

Case 4: One of the rationals is negative and the other one is positive.

Without loss of generality, assume that x is negative while y is positive. Here, xy is negative and so $|xy| = -(xy)$. Furthermore, $|x| = -x$ and $|y| = y$ imply that $|x||y| = (-x)y = -(xy)$. Hence,

$$|xy| = |x||y|.$$

Conclusion: $|xy| = |x||y|$.

In particular,

$$|-x| = |(-1)x| = |-1||x| = -(-1)|x| = (-1)(-1)|x| = |x|.$$

(e) We have $d(x, y) \geq 0$. Also, $d(x, y) = 0$ if and only if $x = y$.

By Definition 4.3.2, $d(x, y) = |x-y|$. By Proposition 4.3.3(a), $|x-y| \geq 0$. So,

$$d(x, y) \geq 0.$$

According to the same proposition, $|x-y| = 0$ if and only if $x-y = 0$. But $x-y = 0$ if and only if $x = y$. Hence,

$$d(x, y) = 0 \iff x = y.$$

(f) $d(x, y) = d(y,x)$.

Note that:

$$\begin{aligned} d(x, y) &= |x-y| && \text{by Definition 4.3.2} \\ &= |-(y-x)| \\ &= |y-x| && \text{by Proposition 4.3.3(d)} \\ &= d(y, x) && \text{by Definition 4.3.2.} \end{aligned}$$

(g) $d(x,z) \leq d(x,y) + d(y,z)$

By Definition 4.3.2, $d(x,z) = |x-z|$, $d(x,y) = |x-y|$, and $d(y,z) = |y-z|$. We have $d(x,y)+d(y,z) = |x-y|+|y-z|$. In addition, $|(x-y)+(y-z)| \leq |x-y|+|y-z|$ by Proposition 4.3.3(b), and so $|x-z| \leq |x-y|+|y-z|$. In other words,

$$d(x,z) \leq d(x,y) + d(y,z).$$

Exercise 4.3.2

Let $x, y, z, w \in \mathbf{Q}$. We are to prove the following seven statements.

(a) One has $x = y$ if and only if x is ε-close to y for every $\varepsilon > 0$.

If $x = y$, then $d(x,y) = |x-y| = |x-x| = |0| = 0$. So, for any rational $\varepsilon > 0$, we have $d(x,y) \leq \varepsilon$. In other words, x is ε-close to y for every $\varepsilon > 0$.

If x is ε-close to y for every $\varepsilon > 0$, suppose, for the sake of contradiction, that $d(x,y) \neq 0$. Then, by Proposition 4.3.3(e), $d(x,y) > 0$. This implies that $d(x,y)/2 > 0$ because 2^{-1} is a positive rational. So, x is also $(d(x,y)/2)$-close to y. This means that

$$d(x,y) \leq d(x,y)/2.$$

Hence,

$$d(x,y) - d(x,y)/2 \leq 0,$$

so that

$$d(x,y)/2 \leq 0,$$

so that

$$d(x,y) \leq 0.$$

But this is contradictory. Therefore, $d(x,y) = 0$. That is to say, $x = y$ by Proposition 4.3.3(e).

In conclusion,

$$x = y \iff d(x,y) \leq \varepsilon \text{ for all } \varepsilon > 0.$$

(b) Let $\varepsilon > 0$. If x is ε-close to y, then y is ε-close to x.

Suppose that x is ε-close to y, that is, $d(x,y) \leq \varepsilon$. We have $d(x,y) = d(y,x)$ by Proposition 4.3.3(f). Hence, we also have $d(y,x) \leq \varepsilon$. That is, y is ε-close to x by Definition 4.3.4.

(c) Let $\varepsilon, \delta > 0$. If x is ε-close to y, and y is δ-close to z, then x and z are $(\varepsilon+\delta)$-close.

Suppose that x is ε-close to y and y is δ-close to z, so that $d(x,y) \leq \varepsilon$ and $d(y,z) \leq \delta$ by Definition 4.3.4. We have $d(x,z) \leq d(x,y) + d(y,z)$ by Proposition 4.3.3(g). So, $d(x,z) \leq \varepsilon + \delta$, which means that x and z are $(\varepsilon+\delta)$-close by Definition 4.3.4.

(d) Let ε, $\delta > 0$. If x and y are ε-close, and z and w are δ-close, then $x+z$ and $y+w$ are $(\varepsilon+\delta)$-close, and $x-z$ and $y-w$ are also $(\varepsilon+\delta)$-close.

Suppose that x is ε-close to y and z is δ-close to w, so that $d(x,y) \leq \varepsilon$ and $d(z,w) \leq \delta$ by Definition 4.3.4.

We have $d(x,y) + d(z,w) \leq \varepsilon + \delta$, and so $|x-y| + |z-w| \leq \varepsilon + \delta$. In addition, $|(x-y)+(z-w)| \leq |x-y| + |z-w|$ by Proposition 4.3.3(b). So, $|x-y+z-w| \leq \varepsilon+\delta$, i.e., $|(x+z)-(y+w)| \leq \varepsilon+\delta$. Hence, $x+z$ and $y+w$ are $(\varepsilon+\delta)$-close.

Note that

$$|x-y| + |z-w| = |x-y| + |w-z| \geq |x-y+w-z|,$$

by Proposition 4.3.3(b). So,

$$|x-y| + |z-w| \leq \varepsilon+\delta \implies |x-y+w-z| \leq \varepsilon+\delta \implies |(x-z)-(y-w)| \leq \varepsilon+\delta.$$

Hence, $x-z$ and $y-w$ are also $(\varepsilon+\delta)$-close.

(e) Let $\varepsilon > 0$. If x and y are ε-close, they are also ε'-close for every $\varepsilon' > \varepsilon$.

Suppose that x and y are ε-close so that $d(x,y) \leq \varepsilon$ by Definition 4.3.4. Let $\varepsilon' > \varepsilon$ be a rational. We have:

$$\varepsilon' > \varepsilon \geq d(x,y) \implies \varepsilon' > d(x,y) \implies d(x,y) \leq \varepsilon'.$$

Hence, x and y are ε'-close as well, by Definition 4.3.4.

(f) Let $\varepsilon > 0$. If y and z are both ε-close to x, and w is between y and z, then w is also ε-close to x.

Suppose that both y and z are ε-close to x and w is between y and z. By Definition 4.3.4, $d(x,y) \leq \varepsilon$ and $d(x,z) \leq \varepsilon$, that is $|x-y| \leq \varepsilon$ and $|x-z| \leq \varepsilon$. Define

$$M := \max(|x-y|, |x-z|) = \begin{cases} |x-y|, & |x-y| \geq |x-z| \\ |x-z|, & |x-y| < |x-z| \end{cases}.$$

The rational M is well-defined given the order trichotomy of rationals (Proposition 4.2.9(a)). We have $M \leq \varepsilon$. In addition, we know from Proposition 4.3.3(c) that

$$|x-y| \leq M \implies -M \leq x-y \leq M$$

and

$$|x-z| \leq M \implies -M \leq x-z \leq M.$$

Now, given that w is between y and z, either $y \leq w \leq z$ or $z \leq w \leq y$. We have:

$$y \leq w \leq z \implies x-z \leq x-w \leq x-y \implies -M \leq x-w \leq M$$

and

$$z \leq w \leq y \implies x-y \leq x-w \leq x-z \implies -M \leq x-w \leq M.$$

So, in any case, $-M \leq x-w \leq M$. By Proposition 4.3.3(c), this is equivalent to $|x-w| \leq M$. Hence, $|x-w| \leq \varepsilon$, that is, $d(x,w) \leq \varepsilon$, which means that w is ε-close to x by Definition 4.3.4.

(g) Let $\varepsilon > 0$. If x and y are ε-close, and z is non-zero, then xz and yz are $\varepsilon|z|$-close.

Suppose that x and y are ε-close so that $d(x,y) \leq \varepsilon$ by Definition 4.3.4. Suppose in addition that $z \neq 0$. We have $|x-y| \leq \varepsilon$. Given that $z \neq 0$, $|z| > 0$ by Proposition 4.3.3(a). Hence,

$$|x-y| \leq \varepsilon \implies |x-y||z| \leq \varepsilon|z|.$$

But, $|x-y||z| = |(x-y)z| = |xz-yz|$ by Proposition 4.3.3(d). So, $|xz-yz| \leq \varepsilon|z|$. In other words,

$$d(xz,yz) \leq \varepsilon|z|.$$

This means that xz and yz are $\varepsilon|z|$-close according to Definition 4.3.4.

Exercise 4.3.3

Let $x, y \in \mathbf{Q}$ and $n, m \in \mathbf{N}$. Let us prove the following four statements.

(a) We have $x^n x^m = x^{n+m}$, $(x^n)^m = x^{nm}$, and $(xy)^n = x^n y^n$.

We will proceed by induction on n (keeping m fixed).

Before that, let us establish that $1^a = 1$ for every natural number a, as it will be useful in subsequent exercises. It is easily done with a simple induction that uses Definition 4.3.9 in the two steps. For the base case ($a = 0$), we have $1^0 := 1$; and for

the inductive step, if $1^a = 1$ for some $a \in \mathbf{N}$, then $1^{a+1} := 1^a \times 1 = 1 \times 1 = 1$, which closes the induction.

Now we move on to our main proof.

The base case is at $n = 0$. On the one side

$$x^0 x^m = 1x^m = x^m, \quad (x^0)^m = 1^m = 1, \quad \text{and} \quad (xy)^0 = 1.$$

On the other side,

$$x^{0+m} = x^m, \quad x^{0\times m} = x^0 = 1, \quad \text{and} \quad x^0 y^0 = 1 \times 1 = 1.$$

Hence, $x^0 x^m = x^{0+m}$, $(x^0)^m = x^{0\times m}$, and $(xy)^0 = x^0 y^0$, and we are done with the base case.

For the inductive step, suppose that the proposition is true for $n = k$. We have (1)

$$\begin{aligned}
x^{k+1} \times x^m &= (x^k \times x) \times x^m && \text{by Definition 4.3.9} \\
&= x \times x^k \times x^m \\
&= x \times x^{k+m} && \text{by induction hypothesis} \\
&= x^{k+m} \times x \\
&= x^{k+m+1} && \text{by Definition 4.3.9} \\
&= x^{(k+1)+m},
\end{aligned}$$

and (2)

$$\begin{aligned}
(xy)^{k+1} &= (xy)^k (xy) && \text{by Definition 4.3.9} \\
&= x^k y^k xy && \text{by induction hypothesis} \\
&= x^k x y^k y \\
&= x^{k+1} y^{k+1} && \text{by Definition 4.3.9,}
\end{aligned}$$

and (3)

$$\begin{aligned}
(x^{k+1})^m &= (x^k x)^m && \text{by Definition 4.3.9} \\
&= (x^k)^m x^m && \text{because } (xy)^a = x^a y^a \text{ for all } a \in \mathbf{N} \text{ as shown in (2)} \\
&= x^{km} x^m && \text{by induction hypothesis} \\
&= x^{km+m} && \text{because } x^a x^b = x^{a+b} \text{ for all } a, b \in \mathbf{N} \text{ as shown in (1)} \\
&= x^{(k+1)m}.
\end{aligned}$$

This concludes our induction.[4]

[4] Note that this was a sort of three-branch induction and, after completing the induction steps of the first and second branches, we used them to prove the third branch.

(b) Suppose $n > 0$. Then we have $x^n = 0$ if and only if $x = 0$.

We will use induction starting from the base case $n = 1$ (see Exercise 2.2.7).

In the base case, $n = 1$. We have $x^1 = x^{0+1} = x^0 \times x = 1 \times x = x$. So, it is evident that $x^1 = 0$ if and only if $x = 0$.

Now for the inductive step, suppose that the statement is true for some natural number k, i.e., $x^k = 0$ if and only if $x = 0$. We have

$$x^{k+1} = 0 \iff x^k x = 0.$$

Note that for any rationals p and q, $p = a/b$ and $q = c/d$ for some integers a, b, c, d, and we have

$$\begin{aligned} pq = 0 &\iff (a/b) \times (c/d) = 0 \\ &\iff (ac)/(bd) = 0 \\ &\iff ac = 0 \\ &\iff a = 0 \text{ or } c = 0 \\ &\iff a/b = 0 \text{ or } c/d = 0 \\ &\iff p = 0 \text{ or } q = 0. \end{aligned}$$

Hence,

$$\begin{aligned} x^{k+1} = 0 &\iff x^k x = 0 \\ &\iff x^k = 0 \text{ or } x = 0 \\ &\iff x = 0 \text{ or } x = 0 && \text{by induction hypothesis} \\ &\iff x = 0. \end{aligned}$$

Thus, the statement is true for $n = k + 1$ as well.

We conclude that the statement is true for all integers $n \geq 1$.

(c) If $x \geq y \geq 0$, then $x^n \geq y^n \geq 0$. If $x > y \geq 0$ and $n > 0$, then $x^n > y^n \geq 0$.

We will proceed by induction on n to prove that if $x \geq y \geq 0$ then $x^n \geq y^n \geq 0$ for all natural numbers n.

The base case is at $n = 0$. Since $x^0 = y^0 = 1$ and $1 \geq 0$, it is clear that $x^0 \geq y^0 \geq 1$. So, the implication

$$x \geq y \geq 0 \implies x^0 \geq y^0 \geq 1$$

is true, which validates the base case.

For the inductive step, suppose that the statement is true for some natural number k. Now,

$$\begin{aligned} x \geq y \geq 0 &\implies x^k \geq y^k \geq 0 && \text{by induction hypothesis} \\ &\implies xx^k \geq xy^k \geq x \times 0 && \text{by Proposition 4.2.9(e)} \\ &\implies x^{k+1} \geq xy^k \geq 0 \\ &\implies x^{k+1} \geq yy^k \geq 0 \\ &\implies x^{k+1} \geq y^{k+1} \geq 0, \end{aligned}$$

where the penultimate implication holds because $x \geq y$ implies that $xy^k \geq yy^k$, and since y and y^k are positive, $yy^k \geq 0$.

This closes the induction and our proof.

Now, we will prove by induction on n starting from the base case $n = 1$ (see Exercise 2.2.7) that if $x > y \geq 0$, then $x^n > y^n \geq 0$ for all natural numbers $n > 0$.

The base case ($n = 1$) is quite straightforward. Since $x^1 = x^0 x = x$ and $y^1 = y^0 y = y$, it is clear that

$$x > y \geq 0 \implies x^1 > y^1 \geq 0.$$

For the inductive step, suppose the statement holds for some natural number l. We have:

$$\begin{aligned} x > y \geq 0 &\implies x^l > y^l \geq 0 && \text{by induction hypothesis} \\ &\implies xx^l > xy^l \geq x \times 0 && \text{by Proposition 4.2.9(e)} \\ &\implies x^{l+1} > xy^l \geq 0 \\ &\implies x^{l+1} > yy^l \geq 0 \\ &\implies x^{l+1} > y^{l+1} \geq 0. \end{aligned}$$

We see that the statement is also true for $n = l + 1$, and so the induction ends here.[5]

(d) We have $|x^n| = |x|^n$.

Once again, we will proceed by induction.

The base case is at $n = 0$. Since $|x^0| = |1| = 1$ and $|x|^0 = 1$, we have $|x^0| = |x|^0$.

[5] A simple corollary of Proposition 4.3.10(c) is that for any rational number q and any positive integer n, if $q > 0$, then $q^n > 0$. We will use this fact without further comment from now on.

Now, suppose inductively that $|x^k| = |x|^k$ for some natural number k. Then,

$$\begin{aligned} |x^{k+1}| &= |x^k x| \\ &= |x^k||x| && \text{by Proposition 4.3.3(d)} \\ &= |x|^k|x| && \text{by induction hypothesis} \\ &= |x|^{k+1}. \end{aligned}$$

This closes the induction.

Exercise 4.3.4

Let x, y be non-zero rational numbers, and let n, m be integers. We are going to prove four statements. But before that, we need to establish three preliminaries.

First, we show (by induction) that $(1/p)^k = 1/(p^k)$ for all natural numbers k, where p is an arbitrary non-zero rational number. Starting at the base case ($k = 0$), we have $(1/p)^0 := 1$ and $1/(p^0) = 1/1 = 1$, and so the equality is clear. Next, suppose inductively that $(1/p)^k = 1/(p^k)$ for some $k \in \mathbf{N}$. Then, $(1/p)^{k+1} := (1/p)^k \times 1/p$. We have $(1/p)^k \times 1/p = 1/(p^k) \times 1/p = 1/(p^k p) = 1/(p^{k+1})$ by the induction hypothesis and Proposition 4.3.10(a). So, $(1/p)^{k+1} = 1/(p^{k+1})$ and the induction is done.

The second preliminary result is that for any p, $p' \in \mathbf{Q}$ and q, $q' \in \mathbf{Q} \setminus \{0\}$, we have $(p/q)(p'/q') = (pp')/(qq')$. To show this, let p and p' be rationals and let q and q' be non-zero rationals. There exist integers a, a' and non-zero integers b, b', c, c', d, d' such that $p = a/b$, $p' = a'/b'$, $q = c/d$, and $q' = c'/d'$. We have

$$\begin{aligned} (p/q)(p'/q') &= \big((a/b)/(c/d)\big) \times \big((a'/b')/(c'/d')\big) \\ &= \big((a/b) \times (c/d)^{-1}\big) \times \big((a'/b') \times (c'/d')^{-1}\big). \end{aligned}$$

Observe that $(c/d)(d/c) = 1$, and hence $(c/d) \times (c/d)^{-1} = (c/d) \times (d/c)$. This implies that $(c/d)^{-1} \times (c/d) \times (c/d)^{-1} = (c/d)^{-1} \times (c/d) \times (d/c)$, and as a result, $(c/d)^{-1} = d/c$. Similarly, $(c'/d')^{-1} = d'/c'$. So,

$$\begin{aligned} (p/q)(p'/q') &= \big((a/b) \times (d/c)\big) \times \big((a'/b') \times (d'/c')\big) \\ &= \big((ad)/(bc)\big) \times \big((a'd')/(b'c')\big) \\ &= (aa'dd')/(bb'cc') \\ &= \big((aa')/(bb')\big) \times \big((dd')/(cc')\big). \end{aligned}$$

Just like we showed that $(c/d)^{-1} = d/c$, we get $((cc')/(dd'))^{-1} = (dd')/(cc')$. Hence,

$$\begin{aligned}(p/q)(p'/q') &= \big((aa')/(bb')\big) \times \big((cc')/(dd')\big)^{-1} \\ &= \big((aa')/(bb')\big)/\big((cc')/(dd')\big) \\ &= \big((a/b)(a'/b')\big)/\big((c/d)(c'/d')\big) \\ &= (pp')/(qq'),\end{aligned}$$

as desired.

Finally, we show that for any non-zero rational p, we have $1/(p^{-1}) = p$. So, take a non-zero rational p. By Definition 4.3.11, $1/(p^{-1}) = (p^{-1})^{-1}$. Note that $(p^{-1})^{-1}p^{-1} = 1 = pp^{-1}$, and so $(p^{-1})^{-1}p^{-1}p = pp^{-1}p$, which leads to $(p^{-1})^{-1} = p$. Therefore, $1/(p^{-1}) = p$ as desired.[6]

(a) We have $x^n x^m = x^{n+m}$, $(x^n)^m = x^{nm}$, and $(xy)^n = x^n y^n$.

We distinguish four exhaustive cases: the case where n and m are both non-negative, the two cases where only one of n and m is negative, and the case where both are negative.

Case 1: n and m are non-negative.

In this case, we know from Proposition 4.3.10(a) that these equalities all hold.

Case 2: n and m are both negative.

Here, $-n$ and $-m$ are positive. So first,

$$\begin{aligned}x^n x^m &= 1/(x^{-n}) \times 1/(x^{-m}) && \text{by Definition 4.3.11} \\ &= 1/(x^{-n}x^{-m}) \\ &= 1/(x^{-n-m}) && \text{by Proposition 4.3.10(a)} \\ &= 1/\big(x^{-(n+m)}\big) \\ &= x^{n+m} && \text{by Definition 4.3.11.}\end{aligned}$$

[6] The three preliminary results will be used without explicit mention.

Second,

$$\begin{aligned}
(x^n)^m &= 1/(x^n)^{-m} && \text{by Definition 4.3.11}\\
&= 1/(1/(x^{-n}))^{-m} && \text{by Definition 4.3.11}\\
&= 1/\big(1/(x^{-n})^{-m}\big)\\
&= 1/\big(1/\big(x^{(-n)(-m)}\big)\big) && \text{by Proposition 4.3.10(a)}\\
&= 1/(1/x^{nm})\\
&= 1/(x^{nm})^{-1} && \text{by Definition 4.3.11}\\
&= x^{nm}.
\end{aligned}$$

Third,

$$\begin{aligned}
(xy)^n &= 1/(xy)^{-n} && \text{by Definition 4.3.11}\\
&= 1/(x^{-n}y^{-n}) && \text{by Proposition 4.3.10(a)}\\
&= 1/(x^{-n}) \times 1/(y^{-n})\\
&= x^n y^n && \text{by Definition 4.3.11.}
\end{aligned}$$

Case 3: n is non-negative and m is negative.

We begin by proving by induction that for any natural number k, $x^n x^{-k} = x^{n-k}$.

At the base case ($k = 0$), we have $x^n x^{-0} = x^n x^0 = x^n$ and $x^{n-k} = x^{n-0} = x^n$, and so the claim holds.

Now suppose inductively that $x^n x^{-k} = x^{n-k}$ for some natural number k. It follows from Case 2 that $x^n x^{-k-1} = x^n x^{-k} x^{-1}$, which leads to $x^n x^{-k-1} = x^{n-k} x^{-1}$ by induction hypothesis. If $n-k = 0$, then $x^{n-k}x^{-1} = x^{-1} = x^{n-k-1}$, and so $x^n x^{-k-1} = x^{n-k-1}$. If $n-k$ is positive, then $n-k-1$ is a natural number, and by Definition 4.3.9, we have $x^{n-k} = x^{n-k-1}x$, so that $x^{n-k}x^{-1} = x^{n-k-1}$, that is $x^n x^{-k-1} = x^{n-k-1}$. If $n-k$ is negative, then $x^{n-k}x^{-1} = x^{n-k-1}$ in light of Case 2, and hence $x^n x^{-k-1} = x^{n-k-1}$. We see that the claim also holds for $k+1$, which completes the induction.

As m is negative, $-m$ is positive. Hence, $x^n x^{-(-m)} = x^{n-(-m)}$. That is,

$$x^n x^m = x^{n+m}.$$

For the second equality, note that $-m$ is positive, nm is negative, and observe that

$$\begin{aligned}
(x^n)^m &= 1/((x^n)^{-m}) && \text{by Definition 4.3.11}\\
&= 1/(x^{-nm}) && \text{by Proposition 4.3.10(a)}\\
&= x^{-(-nm)} && \text{by Definition 4.3.11}\\
&= x^{nm}.
\end{aligned}$$

The third equality follows from Case 1.

Case 4: n is negative and m is non-negative.

First, since $x^n x^m = x^m x^n$, we deduce from Case 3 that

$$x^n x^m = x^{n+m}.$$

Second, we have

$$\begin{aligned}
(x^n)^m &= (1/(x^{-n}))^m && \text{by Definition 4.3.11} \\
&= 1/\big((x^{-n})^m\big) \\
&= 1/(x^{-nm}) && \text{by Proposition 4.3.10(a)} \\
&= x^{-(-nm)} && \text{by Definition 4.3.11} \\
&= x^{nm}.
\end{aligned}$$

Lastly, the third equality follows from Case 2.

(b) If $x \geq y > 0$, then $x^n \geq y^n > 0$ if n is positive, and $0 < x^n \leq y^n$ if n is negative.

Suppose that n is positive. By Proposition 4.3.10(c),

$$x \geq y > 0 \implies x \geq y \geq 0 \implies x^n \geq y^n \geq 0.$$

Since $y \neq 0$, Proposition 4.3.10(b) guarantees that $y^n \neq 0$. Hence, $y^n \geq 0$ if and only if $y^n > 0$. Thus,

$$x^n \geq y^n \geq 0 \implies x^n \geq y^n > 0,$$

so that

$$x \geq y > 0 \implies x^n \geq y^n > 0.$$

Suppose that n is negative. Note that if $x > 0$, then $x^{-1} > 0$, because $x^{-1} \leq 0$ would lead to $xx^{-1} \leq 0$, i.e., $1 \leq 0$, a contradiction. Now,

$$\begin{aligned}
x \geq y > 0 &\implies xx^{-1} \geq yx^{-1} > 0x^{-1} && \text{because } x^{-1} > 0 \\
&\implies 1 \geq y/x > 0.
\end{aligned}$$

Since $-n > 0$ and $y/x \neq 0$, $(y/x)^{-n} \neq 0$ by Proposition 4.3.10(b). In addition, $1 \geq y/x$ implies $1^{-n} \geq (y/x)^{-n}$ by Proposition 4.3.10(c). So,

$$1 \geq y/x > 0 \implies 1^{-n} \geq (y/x)^{-n} > 0 \implies 1 \geq (y/x)^{-n} > 0.$$

Let us examine $(y/x)^{-n}$:

$$\begin{aligned}
(y/x)^{-n} &= (y \times x^{-1})^{-n} \\
&= (x^{-1} \times y)^{-n} \\
&= ((1/x) \times y)^{-n} && \text{by Definition 4.3.11} \\
&= (1/x)^{-n} \times y^{-n} && \text{by Proposition 4.3.10(a)} \\
&= (1/x^{-n}) \times y^{-n} \\
&= x^n \times y^{-n} && \text{by Definition 4.3.11} \\
&= x^n \times (y^n)^{-1} && \text{by Proposition 4.3.12(a)} \\
&= x^n/y^n.
\end{aligned}$$

Hence,

$$1 \geq (y/x)^{-n} > 0 \implies 1 \geq x^n/y^n > 0.$$

Notice that since $y > 0$, we have $y^{-n} > 0^{-n}$ by Proposition 4.3.10(c), that is, $y^{-n} > 0$ by Proposition 4.3.10(b). So, $y^n \leq 0$ implies that $y^{-n}y^n \leq 0$, which implies that $1 \leq 0$ by Proposition 4.3.12(a) and Definition 4.3.9, a contradiction. Thus, $y^n > 0$. It follows that

$$1 \geq x^n/y^n > 0 \implies y^n \geq (x^n/y^n)y^n > 0.$$

Observe that

$$(x^n/y^n)y^n = x^n(y^n)^{-1}y^n = x^n.$$

Therefore,

$$y^n \geq (x^n/y^n)y^n > 0 \implies y^n \geq x^n > 0.$$

In conclusion,

$$x \geq y > 0 \implies y^n \geq x^n > 0.$$

(c) If $x, y > 0$, $n \neq 0$, and $x^n = y^n$, then $x = y$.

Suppose that x and y are positive, n is non-zero, and $x^n = y^n$. Now suppose, for the sake of contradiction, that $x \neq y$. Given the trichotomy of rationals, either $x > y > 0$ or $y > x > 0$. Without loss of generality, assume that $x > y > 0$.

If n is positive, then

$$x > y > 0 \implies x > y \geq 0 \implies x^n > y^n \geq 0$$

by Proposition 4.3.10(c). Since $y \neq 0$, we also have $y^n \neq 0$ by Proposition 4.3.10(b). Hence,

$$x > y > 0 \implies x^n > y^n > 0.$$

If n is negative, $-n$ is positive. Given Proposition 4.3.10(c) and that $y^{-n} \neq 0$ by Proposition 4.3.10(b), we have:

$$x > y > 0 \implies x^{-n} > y^{-n} > 0.$$

The rational $x^n y^n$ is positive, for

$$\begin{aligned} x^n y^n < 0 &\implies x^n y^n x^{-n} y^{-n} < 0 \\ &\implies (x^n x^{-n})(y^n y^{-n}) < 0 \\ &\implies (x^0)(y^0) < 0 && \text{by Proposition 4.3.12(a)} \\ &\implies (1)(1) < 0 \\ &\implies 1 < 0. \end{aligned}$$

So, we can preserve order within inequalities by multiplying by $x^n y^n$. Using Proposition 4.3.12(a), we have:

$$x^{-n} > y^{-n} > 0 \implies x^n y^n x^{-n} > x^n y^n y^{-n} > 0 \implies y^n > x^n > 0.$$

Thus,

$$x > y > 0 \implies y^n > x^n > 0.$$

So in either case, we have $x^n \neq y^n$, a contradiction. Therefore, $x = y$.

(d) We have $|x^n| = |x|^n$.

For $n \in \mathbf{N}$, this statement is true by Proposition 4.3.10(d). So, assume that $n < 0$.

We need to establish a lemma first: for any non-zero rational numbers p and q, $pq = 1$ if and only if $q = p^{-1}$. Suppose that $q = p^{-1}$. Then $pq = 1$ Proposition 4.2.4. Now suppose that $pq = 1$. Then, $p^{-1}pq = (p^{-1})(1)$, so that $p^{-1+1}q = p^{-1}$ by Proposition 4.3.12(a), that is $p^0 q = p^{-1}$. It follows that $q = p^{-1}$. This completes the proof of the lemma.

Going back to our main matter, note that:

$$\begin{aligned}
|x^n| &= |x^{(-1)(-n)}| \\
&= |(x^{-1})^{-n}| && \text{by Proposition 4.3.12(a)} \\
&= |x^{-1}|^{-n} && \text{by Proposition 4.3.10(d)} \\
&= |x^{-1}|^{(-1)n} \\
&= (|x^{-1}|^{-1})^n && \text{by Proposition 4.3.12(a)} \\
&= |x|^n,
\end{aligned}$$

where the last equality follows because

$$|x^{-1}||x| = |x^{-1}x| = |1| = 1 \implies |x^{-1}|^{-1} = |x|.$$

Exercise 4.3.5

Let us prove by induction on N starting from the base case $N = 1$ (see Exercise 2.2.7) that $2^N \geq N$ for all positive integers N.

The base case ($N = 1$) is simple since $2^1 = 2$ and $2 \geq 1$ imply that $2^1 \geq 1$.

Now, suppose inductively that $2^k \geq k$ for some positive integer k. Then, we have $2 \times 2^k \geq 2 \times k$, so that $2^{k+1} \geq k + k$. Since $k \geq 1$, $k + k \geq k + 1$. Thus, $2^{k+1} \geq k + 1$. This closes the induction.

4.4 Gaps in the Rational Numbers

Exercise 4.4.1

Let $x \in \mathbf{Q}$. By definition, there exist an integer p and a non-zero integer q such that $x = p/q$. We may assume that q is positive, for if q is negative, we can just write $x = p/q = (-p)/(-q)$ and $-q$ is positive.

Suppose that p is non-negative, so that p is a natural number. By the Euclid's division lemma (Proposition 2.3.9), there exist natural numbers n and r such that $p = nq + r$ and $0 \leq r < q$. So,

$$x = (nq + r)/q = n + r/q.$$

Since $r \geq 0$ and $q > 0$, $r/q \geq 0$ and so $x \geq n$. Given that $r < q$ and $q > 0$, we have $r/q < 1$, and so $x < n + 1$. Therefore,

$$n \leq x < n + 1.$$

Suppose that p is negative. We have $x = -((-p)/q)$. Since $(-p)$ is positive, $(-p)/q$ is positive. It follows from the previous case that there exists a natural number m such that $m \leq (-p)/q < m + 1$. This implies that $-m - 1 < -((-p)/q) \leq -m$, which is equivalent to

$$-m - 1 < x \leq -m. \qquad (\star)$$

Now, we distinguish two subcases. In the first one, $x \in \mathbf{Z}$. Here, $(\star)$ implies that

$$-m \leq x < -m + 1.$$

In the second subcase, $x \notin \mathbf{Z}$. So, $(\star)$ implies that $-m - 1 < x < -m$, which leads to

$$-m - 1 \leq x < -m.$$

We see that in any case, there exists a natural number n such that $n \leq x < n + 1$.

Now, suppose that there exists another natural number k distinct from n such that $k \leq x < k + 1$. Since $n \neq k$, either $n < k$ or $n > k$. If $n < k$, then $n + 1 \leq k$, and so $x < n + 1$ implies that $x < k$, a contradiction. If $n > k$, then $n \geq k + 1$, and so $x < k + 1$ implies that $x < n$, another contradiction. These contradictions show that we cannot have another natural number that satisfies the property. In conclusion, there exists a unique natural number n such that $n \leq x < n + 1$.

Lastly, we see that for any integer N such that $N > n+1$, we have $N > x$.

Exercise 4.4.2

(a) Let $a_0, a_1, \ldots$ be a sequence of natural numbers. Suppose, for the sake of contradiction, that this sequence is in infinite descent, so that $a_n > a_{n+1}$ for all $n \in \mathbf{N}$. Note that for all $n \in \mathbf{N}$, $a_n \geq 0$ given that $a_n \in \mathbf{N}$.

We claim that $a_n \geq k$ for any natural numbers n, k, and prove it by induction.

The base case $(k = 0)$ is readily validated as we have $a_n \geq 0$ for all $n \in \mathbf{N}$.

For the inductive step, suppose that $a_n \geq k$ for some natural number k and all natural numbers n. In particular, we have $a_{n+1} \geq k$ for all $n \in \mathbf{N}$. Since $a_n > a_{n+1}$, this implies that $a_n > k$, that is, $a_n \geq k+1$ for all $n \in \mathbf{N}$. So, the claim also holds for $k+1$. Thus, the claim holds for all natural numbers n by induction.

Given that $a_0 + 1$ is a natural number, we have $a_0 \geq a_0 + 1$ by the claim. This contradiction shows that it is impossible for a sequence $a_0, a_1, \ldots$ of natural numbers to be in infinite descent.

(b) The principle of infinite descent does not work if the sequence is allowed to take integer values instead of natural numbers. This is because for any integer m, there exist an integer m' such that $m' < m$ (such an integer is $m-1$). An example of a sequence of integers in infinite descent is $0, -1, -2, -3, -4, \ldots$.

The principle does not work if we have positive rational numbers instead of natural numbers, either. This is because for any positive rational number x, there is always another positive rational number x' such that $x > x'$ (such a rational number is $x/2$). An example of a sequence of positive rationals in infinite descent is $1, 1/2, 1/4, 1/8, 1/16, \ldots$.

Exercise 4.4.3

Let us complete the gaps in the proof of Proposition 4.4.4 by showing

1. Why every natural number is either even or odd, but not both.

First, there exist even natural numbers such as 2 $(2 = 2 \times 1)$ and 14 $(14 = 2 \times 7)$, and there exist odd natural numbers such as 3 $(3 = 2 \times 1 + 1)$ and 11 $(11 = 2 \times 5 + 1)$.

Now, suppose that there exists some natural number m that is both odd and even. Then, there exist natural numbers p and q such that $m = 2p$ and $m = 2q+1$. So, $2p = 2q+1$, so that $2p-2q = 1$, so that $2(p-q) = 1$. If $p = q$, then $p-q = 0$ and so $2(p-q) = 0$, a contradiction since $0 \neq 1$. If $p > q$, then $p \geq q+1$, so $p-q \geq 1$, and thus $2(p-q) \geq 2$, i.e., $1 \geq 2$, a contradiction. If $p < q$, then $p-q < 0$, and so $2(p-q) < 0$, i.e., $1 < 0$, another contradiction. By the trichotomy of order for natural numbers, one of the statements $p = q$, $p > q$, and $p < q$ is true. Yet, each one leads to a contradiction. So, the natural number m does not exist. Therefore, a natural number cannot be both odd and even.

To summarize, we have proven thus far that among the natural numbers, some are odd, some are even, and none are both odd and even. Now, we just need to show that no natural number can be neither odd nor even. There are many ways to do this, but we will use induction because it is fast. The base case is readily validated because $0 = 2 \times 0$ is even. For the inductive step, suppose that a natural number n is even or odd. If n is even, then $n = 2k$ for some $k \in \mathbf{N}$ and $n+1 = 2k+1$ is odd. If n is odd, then $n = 2k+1$ for some $k \in \mathbf{N}$ and $n+1 = 2(k+1)$ is even. Hence, $n+1$ is either even or odd. This closes the induction.

In conclusion, every natural number is either even or odd, but not both.

2. Why p odd implies p^2 odd.

If the natural number p is odd, then $p = 2k+1$ for some $k \in \mathbf{N}$. We have:

$$p^2 = (2k+1)^2 = 4k^2+4k+1 = 2(2k^2+2k)+1 = 2K+1,$$

where $K := 2k^2+2k$. Since K is also a natural number, p^2 is odd.

3. Why $p^2 = 2q^2$ implies $q < p$.

Suppose that $p^2 = 2q^2$. By the order trichotomy of natural numbers, exactly one of the following statements is true: $p < q$, $p = q$, and $p > q$.

If $p < q$, then $p^2 < q^2$ (since both p and q are positive), and so $p^2 < q^2+q^2$ (because $q^2 > 0$). Thus $p^2 < 2q^2$, which contradicts $p^2 = 2q^2$. Hence, $p \not< q$.

If $p = q$, then $p^2 = q^2$. As $q \neq 0$, $q^2 \neq 0$ (note that $q^2 = 0$ implies $q = 0$ and take the contrapositive.) So, $q^2+q^2 \neq q^2$ (note that $q^2+q^2 = q^2$ implies $q^2 = 0$ and take the contrapositive), i.e., $2q^2 \neq q^2$, i.e., $p^2 \neq q^2$. This contradiction shows that $p \neq q$.

Therefore, we must have $p > q$, that is $q < p$.

We do not yet know what the axiom of choice is. But since we did not need it to complete the proof, we conclude that it is not required to establish this proposition.

Chapter 5
The Real Numbers

5.1 Cauchy Sequences

Exercise 5.1.1

Let $(a_n)_{n=1}^{\infty}$ be a Cauchy sequence of rationals.

Since $(a_n)_{n=1}^{\infty}$ is Cauchy, it is eventually 1-steady by Definition 5.1.8. So, there exist a natural number $N \geq 1$ such that $(a_n)_{n=N}^{\infty}$ is 1-steady. Let us split the sequence $(a_n)_{n=1}^{\infty}$ into $a_1, \ldots, a_N$ and $(a_n)_{n=N+1}^{\infty}$.

Since $a_1, \ldots, a_N$ is a finite sequence, it is bounded by some positive rational M_1, according to Lemma 5.1.14.

Now we focus on $(a_n)_{n=N+1}^{\infty}$. By Definition 5.1.3, we have $|a_j - a_N| \leq 1$ for all $j \geq N$ since $(a_n)_{n=N}^{\infty}$ is 1-steady. Because $j \geq N+1$ implies $j \geq N$, we also have $|a_j - a_N| \leq 1$ for all $j \geq N+1$. In light of the triangle inequality for the absolute value, note that:

$$|a_j| = |a_j - a_N + a_N| \leq |a_j - a_N| + |a_N| \leq 1 + |a_N|$$

for all $j \geq N+1$.

Define $M := M_1 + 1 + |a_N|$. Since $|a_i| \leq M_1$ for all $1 \leq i \leq N$ and $|a_i| \leq 1 + |a_N|$ for all $i \geq N+1$, we have

$$|a_i| \leq M \text{ for all } i \geq 1.$$

This means that $(a_n)_{n=1}^{\infty}$ is bounded (by M) according to Definition 5.1.12.

Exercise 5.1.2

Let $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ be bounded sequences, so that, by Definition 5.1.12, there exist non-negative rationals A and B such that

$$|a_i| \leq A \quad \text{and} \quad |b_i| \leq B$$

for all $i \geq 1$.

By the triangle inequality for the absolute value, we have $|a_i + b_i| \leq |a_i| + |b_i|$, and thus

$$|a_i + b_i| \leq A + B$$

for all $i \geq 1$.

This means that $(a_n + b_n)_{n=1}^{\infty}$ is bounded (by $A + B$) by Definition 5.1.12.

Note that for all $i \geq 1$,

$$|a_i - b_i| = |a_i + (-b_i)| \leq |a_i| + |-b_i| = |a_i| + |b_i| \leq A + B.$$

With this observation, it is clear that $(a_n - b_n)_{n=1}^{\infty}$ is also a bounded sequence (bounded by $A + B$) according to Definition 5.1.12.

Finally, note that for all $i \geq 1$, $|a_i b_i| = |a_i||b_i|$. Given that $|a_i|$, $|b_i|$, A, and B are all positive, this yields

$$|a_i b_i| \leq AB$$

for all $i \geq 1$. Therefore, $(a_n b_n)_{n=1}^{\infty}$ is bounded (by AB), by Definition 5.1.12.

5.2 Equivalent Cauchy Sequences

Exercise 5.2.1

Let $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ be two equivalent sequences of rationals.

Suppose that $(a_n)_{n=1}^{\infty}$ is a Cauchy sequence. Let $\varepsilon > 0$ be an arbitrary positive rational.

As $\varepsilon > 0$, $\varepsilon/3 > 0$. By Definition 5.2.6, since $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ are equivalent, there exists a natural number N_1 for which $(a_n)_{n=N_1}^{\infty}$ and $(b_n)_{n=N_1}^{\infty}$ are $\varepsilon/3$ close, that is $|a_n - b_n| \leq \varepsilon/3$ for all $n \geq N_1$. Since $(a_n)_{n=1}^{\infty}$ is Cauchy, there exists a natural number N_2 for which $(a_n)_{n=N_2}^{\infty}$ is $(\varepsilon/3)$-steady, i.e., $|a_j - a_k| \leq \varepsilon/3$ for all $j, k \geq N_2$.

Now, let $N := N_1 + N_2$. For any $j, k \geq N$, we have:

$$|a_j - b_j| \leq \varepsilon/3, \quad |a_k - b_k| \leq \varepsilon/3, \quad \text{and} \quad |a_j - a_k| \leq \varepsilon/3,$$

thus

$$|a_j - b_j| + |a_k - b_k| + |a_j - a_k| \leq \varepsilon.$$

Since

$$\begin{aligned} |a_j - b_j| + |a_k - b_k| + |a_j - a_k| &= |-(a_j - b_j)| + |a_k - b_k| + |a_j - a_k| \\ &\geq |-(a_j - b_j) + a_k - b_k + a_j - a_k| \\ &= |b_j - b_k|, \end{aligned}$$

we have

$$|b_j - b_k| \leq \varepsilon$$

for all $j, k \geq N$. Therefore, $(b_n)_{n=1}^{\infty}$ is eventually ε-steady. Since ε is arbitrary, we conclude that $(b_n)_{n=1}^{\infty}$ is a Cauchy sequence.

By symmetry, we can similarly show that if $(b_n)_{n=1}^{\infty}$ is a Cauchy sequence, then $(a_n)_{n=1}^{\infty}$ is also a Cauchy sequence. This establishes the equivalence.

Exercise 5.2.2

Let $\varepsilon > 0$. Let $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ be two eventually ε-close sequences, so that by Definition 5.2.3, there exist a natural number $N \geq 1$ such that $|a_n - b_n| \leq \varepsilon$ for all $n \geq N$.

We will prove that $(a_n)_{n=1}^{\infty}$ is bounded if and only if $(b_n)_{n=1}^{\infty}$ is bounded. Given the symmetry of this statement and that nothing particularly distinguishes $(a_n)_{n=1}^{\infty}$ from $(b_n)_{n=1}^{\infty}$, it is sufficient to show that if $(a_n)_{n=1}^{\infty}$ is bounded, then $(b_n)_{n=1}^{\infty}$ is bounded.

Suppose that $(a_n)_{n=1}^{\infty}$ is bounded. Then, there exists a positive rational M such that $|a_i| \leq M$ for all $i \geq 1$. Suppose for the sake of contraction that $(b_n)_{n=1}^{\infty}$ is unbounded. Then, there exists a natural number $j \geq N$ such that $|b_j| > M + \varepsilon$ (because otherwise $(b_n)_{n=N}^{\infty}$ would be bounded by $M+\varepsilon$ which would imply that $(b_n)_{n=1}^{\infty}$ is bounded, given that $(b_n)_{n=1}^{N}$ is already bounded by Lemma 5.1.4). Now, $|b_j| > M + \varepsilon$ implies that $b_j > M + \varepsilon$ or $-b_j > M + \varepsilon$. Note that:

$$\begin{aligned} b_j > M+\varepsilon &\implies b_j > a_j + \varepsilon && \text{because } a_j \leq |a_j| \leq M \\ &\implies b_j - a_j > \varepsilon \\ &\implies |b_j - a_j| > \varepsilon && \text{because } bj - a_j \leq |b_j - a_j| \\ &\implies |a_j - b_j| > \varepsilon \end{aligned}$$

and

$$\begin{aligned} -b_j > M+\varepsilon &\implies a_j - b_j > -M + M + \varepsilon && \text{because } a_j \geq -|a_j| \geq -M \\ &\implies a_j - b_j > \varepsilon \\ &\implies |a_j - b_j| > \varepsilon && \text{because } aj - b_j \leq |b_j - a_j|. \end{aligned}$$

In sum, $|b_j| > M + \varepsilon$ implies that $|a_j - b_j| > \varepsilon$. But this is contradictory because $|a_n - b_n| \leq \varepsilon$ for all $n \geq N$. Hence, $(b_n)_{n=1}^{\infty}$ is a bounded sequence.

5.3 The Construction of the Real Numbers

Exercise 5.3.1

Let $x = \text{LIM}_{n\to\infty} a_n$, $y = \text{LIM}_{n\to\infty} b_n$, and $z = \text{LIM}_{n\to\infty} c_n$ be real numbers.

First, since a_n is a rational, we have $|a_n - a_n| = |0| = 0$ for every $n \geq 1$. Hence, for any rational $\varepsilon > 0$, we have $|a_n - a_n| \leq \varepsilon$ for all $n \geq 1$. Thus, $(a_n)_{n=1}^{\infty}$ is equivalent to $(a_n)_{n=1}^{\infty}$. That is, $x = x$.

Next, suppose that $x = y$. By definition, $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ are equivalent. In other words, for any rational $\varepsilon > 0$, there exists an integer $N \geq 1$ such that $|a_n - b_n| \leq \varepsilon$ for all $n \geq N$. But $|a_n - b_n| \leq \varepsilon$ if and only if $|b_n - a_n| \leq \varepsilon$. So, $(b_n)_{n=1}^{\infty}$ and $(a_n)_{n=1}^{\infty}$ are equivalent as well, i.e., $y = x$.

Finally, suppose that $x = y$ and $y = z$. Then by definition, $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ are equivalent and $(b_n)_{n=1}^{\infty}$ and $(c_n)_{n=1}^{\infty}$ are equivalent. Take an arbitrary positive rational ε. There is an integer $N_1 \geq 1$ such that a_n and b_n are $(\varepsilon/2)$-close for all $n \geq N_1$, and there exists an integer N_2 such that b_n and c_n are $(\varepsilon/2)$-close for all $n \geq N_2$. Define $N := N_1 + N_2$. Then, a_n and b_n are $(\varepsilon/2)$-close, and b_n and c_n are $(\varepsilon/2)$-close for all $n \geq N$. It follows from Proposition 4.3.7(c) that a_n and c_n are ε-close for all $n \geq N$. Thus, $(a_n)_{n=1}^{\infty}$ and $(c_n)_{n=1}^{\infty}$ are eventually ε-close. As ε is arbitrary, this means that $(a_n)_{n=1}^{\infty}$ and $(c_n)_{n=1}^{\infty}$ are equivalent. In other words, $x = z$.

Exercise 5.3.2

Let $x = \text{LIM}_{n\to\infty} a_n$, $y = \text{LIM}_{n\to\infty} b_n$, and $x' = \text{LIM}_{n\to\infty} a'_n$ be real numbers.

Let $\varepsilon > 0$ be a rational number. Since $x, y \in \mathbf{R}$, $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ are Cauchy sequences of rationals by Definition 5.3.1. Therefore, they are bounded sequences (Lemma 5.1.15). This means that there exist non-negative rationals M_1 and M_2 such that $|a_n| \leq M_1$ and $|b_n| \leq M_2$ for all integers $n \geq 1$. Set $M := M_1 + M_2 + \varepsilon + 1$. Then, $\varepsilon \leq M$, plus $|a_n| \leq M$ and $|b_n| \leq M$ for all $n \geq 1$.

Note that $\varepsilon/(3M)$ is a positive rational. Since $(a_n)_{n=1}^{\infty}$ is a Cauchy sequence of rationals, there exists a natural number $N_1 \geq 1$ such that $|a_j - a_k| \leq \varepsilon/(3M)$ for all $j, k \geq N_1$. Similarly, $(b_n)_{n=1}^{\infty}$ is a Cauchy sequence and so there exists a natural number $N_2 \geq 1$ such that $|b_j - b_k| \leq \varepsilon/(3M)$ for all $j, k \geq N_2$. Let $N := N_1 + N_2$. Then, for all $j, k \geq N$, we have $|a_j - a_k| \leq \varepsilon/(3M)$ and $|b_j - b_k| \leq \varepsilon/(3M)$.

Now, by Proposition 4.3.7(h), we have

$$|a_j b_j - a_k b_k| \leq (\varepsilon/(3M))|b_j| + (\varepsilon/(3M))|a_j| + (\varepsilon^2/(9M^2)),$$

which leads to

$$|a_j b_j - a_k b_k| \leq (\varepsilon/(3M))M + (\varepsilon/(3M))M + (\varepsilon^2/(9M^2)),$$

and hence

$$|a_j b_j - a_k b_k| \leq \varepsilon/3 + \varepsilon/3 + (\varepsilon^2/(9M^2))$$

for all $j, k \geq N$. The right-hand side simplifies to $\varepsilon(2/3 + \varepsilon/(9M^2))$. Observe that:

$$\begin{aligned}
M \geq 1 \text{ and } M \geq \varepsilon &\implies M^2 \geq \varepsilon \\
&\implies 9M^2 \geq 9\varepsilon \\
&\implies 9M^2(1/(9\varepsilon)) \geq 1 \\
&\implies 1/(9\varepsilon) \geq 1/(9M^2) \\
&\implies 1/(9M^2) \leq 1/(9\varepsilon) \\
&\implies \varepsilon/(9M^2) \leq 1/9 \\
&\implies 2/3 + \varepsilon/(9M^2) \leq 7/9 \\
&\implies 2/3 + \varepsilon/(9M^2) \leq 1 \\
&\implies \varepsilon(2/3 + \varepsilon/(9M^2)) \leq \varepsilon.
\end{aligned}$$

Thus, for all $j, k \geq N$,

$$|a_j b_j - a_k b_k| \leq \varepsilon.$$

So, the sequence $(a_n b_n)_{n=1}^{\infty}$ is eventually ε-steady. Since ε is an arbitrary positive rational, we deduce that $(a_n b_n)_{n=1}^{\infty}$ is a Cauchy sequence. Therefore, $xy := \mathrm{LIM}_{n\to\infty} a_n b_n$ is a real number according to Definition 5.3.1.

Suppose that $x = x'$. By Definition 5.3.1, $(a_n)_{n=1}^{\infty}$ and $(a'_n)_{n=1}^{\infty}$ are equivalent sequences. We have $xy := \mathrm{LIM}_{n\to\infty} a_n b_n$ and $x'y := \mathrm{LIM}_{n\to\infty} a'_n b_n$.

Take an arbitrary rational $\gamma > 0$. The rational γ/M is positive, and so the sequences $(a_n)_{n=1}^{\infty}$ and $(a'_n)_{n=1}^{\infty}$ are eventually (γ/M)-close. That is, there exists a positive integer K such that $|a_n - a'_n| \leq \gamma/M$ for all $n \geq K$. This implies that $|a_n - a'_n||b_n| \leq (\gamma/M)M$, i.e., $|a_n b_n - a'_n b_n| \leq \gamma$, for all $n \geq K$. Hence, the sequences $(a_n b_n)_{n=1}^{\infty}$ and $(a'_n b_n)_{n=1}^{\infty}$ are eventually γ-close. Therefore, $(a_n b_n)_{n=1}^{\infty}$ and $(a'_n b_n)_{n=1}^{\infty}$ are equivalent Cauchy sequences. By Definition 5.3.1, this means that $xy = x'y$.

Exercise 5.3.3

Let a and b be rational numbers. Note that the sequences $a, a, a, \ldots$ and $b, b, b, \ldots$ are Cauchy sequences because each is eventually ε-steady for any $\varepsilon > 0$ (indeed, $a - a = b - b = 0$).

Suppose first that $a = b$. Then, the sequences $a, a, a, \ldots$ and $b, b, b, \ldots$ are ε-close for any $\varepsilon > 0$ because $|a - b| = |a - a| = |0| = 0$. Hence, these two Cauchy sequences are equivalent. By Definition 5.3.1, we have $\text{LIM}_{n\to\infty}\, a = \text{LIM}_{n\to\infty}\, b$.

Second, suppose that $\text{LIM}_{n\to\infty}\, a = \text{LIM}_{n\to\infty}\, b$. Then, the sequences $a, a, a, \ldots$ and $b, b, b, \ldots$ are equivalent by Definition 5.3.1. That is, the two sequences $(a)_{n=1}^{\infty}$ and $(b)_{n=1}^{\infty}$ are eventually ε-close for any $\varepsilon > 0$. This implies that a and b are ε-close for any $\varepsilon > 0$. Therefore, $a = b$ by Proposition 4.3.7(a).

In conclusion,

$$a = b \iff \text{LIM}_{n\to\infty}\, a = \text{LIM}_{n\to\infty}\, b.$$

Exercise 5.3.4

Let $(a_n)_{n=1}^{\infty}$ be a sequence of rationals that is bounded. Let $(b_n)_{n=1}^{\infty}$ be a sequence of rationals that is equivalent to $(a_n)_{n=1}^{\infty}$. This means that $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ are eventually ε-close for every $\varepsilon > 0$. Given that $(a_n)_{n=1}^{\infty}$ is bounded, this implies that $(b_n)_{n=1}^{\infty}$ is bounded as well (see Exercise 5.2.2).

Exercise 5.3.5

We are going to prove that $\text{LIM}_{n\to\infty}\, 1/n = 0$. It makes sense to write $\text{LIM}_{n\to\infty}\, 1/n$ because $(1/n)_{n=1}^{\infty}$ is a Cauchy sequence of rationals (cf. Proposition 5.1.11).

Note that $0 = \text{LIM}_{n\to\infty}\, 0$. By Definition 5.3.1, $\text{LIM}_{n\to\infty}\, 1/n = 0$ if and only if $(1/n)_{n=1}^{\infty}$ and $(0)_{n=1}^{\infty}$ are equivalent sequences.

Take an arbitrary rational $\varepsilon > 0$. We need to show that $(1/n)_{n=1}^{\infty}$ and $(0)_{n=1}^{\infty}$ are eventually ε-close, i.e., that there exists a positive natural number N such that $|1/n| \leq \varepsilon$ for all $n \geq N$. But $n \geq N > 0$ implies that $0 < 1/n \leq 1/N$, which implies that $|1/n| \leq |1/N|$. Hence, we just need to find some N such that $|1/N| \leq \varepsilon$. But since $1/N$ is positive,

$$|1/N| \leq \varepsilon \iff 1/N \leq \varepsilon \iff N \geq 1/\varepsilon.$$

So, all we really need is a positive integer N such that $N \geq 1/\varepsilon$. But, since $1/\varepsilon$ is a rational, Proposition 4.4.1 guarantees that we can find such a positive integer N (just

take $N = \lfloor 1/\varepsilon \rfloor + 1$, for example). So, $(1/n)_{n=1}^{\infty}$ and $(0)_{n=1}^{\infty}$ are eventually ε-close. Therefore, $(1/n)_{n=1}^{\infty}$ and $(0)_{n=1}^{\infty}$ are equivalent Cauchy sequences.

In conclusion,

$$\mathrm{LIM}_{n\to\infty}\, 1/n = 0.$$

5.4 Ordering the Reals

Exercise 5.4.1

Let x and y be real numbers.

Since the equality of real numbers is well-defined, x cannot be zero and non-zero simultaneously. So, either $x = 0$ or $x \neq 0$ but not both.

Suppose that $x = 0$. If x is positive or negative, then by Definition 5.4.3, x is the formal limit of a Cauchy sequence that is positively bounded away from zero or negatively bounded away from zero. In either case, x is the formal limit of a Cauchy sequence bounded away from zero (as any sequence positively bounded away from zero or negatively bounded away from zero is clearly bounded away from zero)[1] . Hence, there exists a positive rational c such that the absolute value of any term of this sequence is greater than or equal to c, and thus this sequence and the zero sequence are not eventually $(c/2)$-close, which implies that $x \neq 0$. So, x is neither positive nor negative when $x = 0$.

Now suppose that $x \neq 0$. Then, there exists a Cauchy sequence $(a_n)_{n=1}^{\infty}$, bounded away from zero, such that $x = \mathrm{LIM}_{n\to\infty} a_n$. There is a positive rational c such that $|a_n| \geq c$ for all $n \geq 1$. Since $(a_n)_{n=1}^{\infty}$ is Cauchy, it is eventually c-steady; that is, there is an integer $N \geq 1$ such that $|a_j - a_k| \leq c$ for all $j, k \geq N$. We claim that $(a_n)_{n=N}^{\infty}$ is either a sequence of positive rationals or a sequence of negative rationals (but of course not both, since a rational cannot be both positive and negative). To show this, suppose that there exist $j, k \geq N$ such that $a_j > 0$ and $a_k < 0$. Then, $|a_j| \geq c > c/2$ implies that $a_j > c/2$ and $|a_k| \geq c > c/2$ implies that $-a_k > c/2$. Thus, $a_j - a_k > c$, and so $|a_j - a_k| > c$. But this is a contradiction; hence, our claim is correct. Let us now look at the two cases separately.

If $(a_n)_{n=N}^{\infty}$ is a sequence of positive rationals, let $(b_n)_{n=1}^{\infty}$ be the sequence defined by $b_n := c$ if $1 \leq n \leq N-1$ and $b_n := a_n$ if $n \geq N$. Clearly, the sequences $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ are eventually ε-close for any rational $\varepsilon > 0$ since they are eventually the same. This means that the sequences $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ are equivalent, and so $x = \mathrm{LIM}_{n\to\infty} b_n$. We have $b_n \geq c$ for all $n \geq 1$. Thus, $(b_n)_{n=1}^{\infty}$ is a sequence positively bounded away from zero according Definition 5.4.1. By Definition 5.4.3, this means that x is a positive real number.

If $(a_n)_{n=N}^{\infty}$ is a sequence of negative rationals, let $(b_n)_{n=1}^{\infty}$ be the sequence defined

[1] Clearly, if $(x_n)_{n=1}^{\infty}$ is a sequence of rationals, r is a positive rational, and we have either $x_n \geq r$ for all integers $n \geq 1$ or $x_n \leq -r$ for all integers $n \geq 1$, then $|x_n| \geq r$ for all integers $n \geq 1$.

by $b_n := -c$ if $1 \leq n \leq N-1$ and $b_n := a_n$ if $n \geq N$. Clearly, the sequences $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ are eventually ε-close for any rational $\varepsilon > 0$ since they are eventually the same. This means that the sequences $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ are equivalent, and so $x = \text{LIM}_{n\to\infty}\, b_n$. We have $b_n \leq -c$ for all $n \geq 1$. Hence, $(b_n)_{n=1}^{\infty}$ is a sequence negatively bounded away from zero, according Definition 5.4.1. Therefore, x is a negative real number by Definition 5.4.3.

So far, we have shown that either $x = 0$ or $x \neq 0$ but not both; that if $x = 0$, then x is neither positive nor negative; and that if $x \neq 0$, then x is either positive or negative. It remains to show that when $x \neq 0$, x cannot be simultaneously positive and negative. Suppose toward a contradiction that x is both a positive and a negative real number. Then, by Definition 5.4.3, $x = \text{LIM}_{n\to\infty}\, c_n$ and $x = \text{LIM}_{n\to\infty}\, d_n$ for some Cauchy sequence $(c_n)_{n=1}^{\infty}$ positively bounded away from zero and some Cauchy sequence $(d_n)_{n=1}^{\infty}$ negatively bounded away from zero. Given that $\text{LIM}_{n\to\infty}\, c_n = \text{LIM}_{n\to\infty}\, d_n$, the sequences $(c_n)_{n=1}^{\infty}$ and $(d_n)_{n=1}^{\infty}$ are equivalent. By Definition 5.4.1, there exist positive rationals p and q such that $c_n \geq p$ and $d_n \leq -q$ for all $n \geq 1$. Thus, for all $n \geq 1$, $c_n - d_n \geq p + q$, so that $|c_n - d_n| > p$ because $p + q > p$. But this implies that the sequences $(c_n)_{n=1}^{\infty}$ and $(d_n)_{n=1}^{\infty}$ are not eventually p-close. Thus, $(c_n)_{n=1}^{\infty}$ and $(d_n)_{n=1}^{\infty}$ are not equivalent, a contradiction. Therefore, x cannot be both positive and negative.

In summary, exactly one of the following three statements is true: x is zero; x is positive; x is negative.

Suppose that x is negative. Then, $x = \text{LIM}_{n\to\infty}\, a_n$ where $(a_n)_{n=1}^{\infty}$ is a Cauchy sequence negatively bounded away from zero by Definition 5.4.3. Thus, there exists a positive rational c such that $a_n \leq -c$ for all $n \geq 1$, by Definition 5.4.1. It follows that $-a_n \geq c > 0$ for all $n \geq 1$. Hence, $(-a_n)_{n=1}^{\infty}$ is a sequence positively bounded away from zero by Definition 5.4.1. As $-x = \text{LIM}_{n\to\infty}\, (-a_n)$, we see that $-x$ is positive following Definition 5.4.3.

Suppose that x and y are positive. According to Definition 5.4.3, $x = \text{LIM}_{n\to\infty}\, (a_n)_{n=1}^{\infty}$ and $y = \text{LIM}_{n\to\infty}\, (b_n)_{n=1}^{\infty}$, for some Cauchy sequences $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ positively bounded away from zero. By Definition 5.4.1, there exist positive rationals c_1 and c_2 such that $a_n \geq c_1$ and $b_n \geq c_2$ for all $n \geq 1$. We have $x + y := \text{LIM}_{n\to\infty}\, a_n + b_n$ and $xy := \text{LIM}_{n\to\infty}\, a_n b_n$. Now, $a_n + b_n \geq c_1 + c_2 > 0$ for all $n \geq 1$. Hence, $(a_n + b_n)_{n=1}^{\infty}$ is positively bounded away from zero by Definition 5.4.1. Similarly, given that a_n and b_n are positive, we have $a_n b_n \geq c_1 c_2 > 0$ for all $n \geq 1$, and hence $(a_n b_n)_{n=1}^{\infty}$ is also positively bounded away from zero. Therefore, by Definition 5.4.3, $x + y$ and xy are positive real numbers.

Exercise 5.4.2

Let x, y, and z be real numbers. We are to prove the following four statements.

(a) Exactly one of the three statements $x = y$, $x < y$, or $x > y$ is true.

Note that $x - y$ is a real number. Proposition 5.4.4 tells us that exactly one of the following statements is true: (i) $x - y$ is zero; (ii) $x - y$ is negative; (iii) $x - y$ is positive. Now,

$$\begin{aligned}
&\text{(i)}: \ x-y=0 \iff x-y+y=0+y \iff x+0=y \iff x=y, \\
&\text{(ii)}: \ x-y \text{ is negative} \iff x<y && \text{by Definition 5.4.6,} \\
&\text{(iii)}: \ x-y \text{ is positive} \iff x>y && \text{by Definition 5.4.6.}
\end{aligned}$$

This concludes the proof.

(b) One has $x < y$ if and only if $y > x$.

By Definition 5.4.6, $x < y$ if and only if $x - y$ is negative. By Proposition 5.4.4, $x - y$ is negative if and only if $-(x - y)$ is positive. But $-(x - y) = -x + y = y - x$. So, $-(x - y)$ is positive if and only if $y - x$ is positive. By Definition 5.4.6, $y - x$ is positive if and only if $y > x$. Therefore, $x < y$ if and only if $y > x$.

(c) If $x < y$ and $y < z$, then $x < z$.

By Proposition 5.4.7(b), $x < y$ is equivalent to $y > x$, that is to say $y - x$ is positive (according to Definition 5.4.6). Similarly, $y < z$ if and only if $z > y$ which is equivalent to $z - y$ is positive. Thus, $x < y$ and $y < z$ imply that $y - x$ and $z - y$ are positive. By Proposition 5.4.4, this in turn implies that $y - x + z - y$ is positive, i.e., $z - x$ is positive. But $z - x$ is positive if and only if $z > x$ by Definition 5.4.6, which is equivalent to $x < z$ by Proposition 5.4.7(b). In summary, $x < y$ and $y < z$ imply that $x < z$.

(d) If $x < y$, then $x + z < y + z$.

We have:

$$\begin{aligned}
x<y &\implies x-y \text{ is a negative real} && \text{by Definition 5.4.6} \\
&\implies x-y+0 \text{ is a negative real} \\
&\implies x-y+z-z \text{ is a negative real} \\
&\implies (x+z)-(y+z) \text{ is a negative real} \\
&\implies x+z<y+z && \text{by Definition 5.4.6,}
\end{aligned}$$

as desired.

Exercise 5.4.3

Let $x \in \mathbf{R}$. We are to show that there is exactly one integer N such that $N \leq x < N+1$. By Proposition 4.4.1, this statement is true if $x \in \mathbf{Q}$. So, we may now assume that $x \notin \mathbf{Q}$. In this case, x is non-zero (as $0 \in \mathbf{Q}$), and thus x is either positive or negative but not both (by Proposition 5.4.4).

Suppose that x is positive. Then, $x = \mathrm{LIM}_{n\to\infty} a_n$ where $(a_n)_{n=1}^{\infty}$ is a Cauchy sequence of rationals which is positively bounded away from zero by Definition 5.4.3. There exists a positive rational c such that $a_n \geq c > 0$ for all $n \geq 1$ by Definition 5.4.1. Since $(a_n)_{n=1}^{\infty}$ is a Cauchy sequence, it is eventually 1-steady. So, there is an integer $K \geq 1$ such that $(a_n)_{n=K}^{\infty}$ is 1-steady. That is, $|a_n - a_K| \leq 1$ for all $n \geq K$. As $a_K \in \mathbf{Q}$, there exists a unique integer M such that $M \leq a_K < M+1$ by Proposition 4.4.1. Since a_K is positive, we also have $M \leq |a_K| < M+1$. With these inequalities in mind, we use the triangle inequality for the absolute value to see that:

$$\begin{aligned} |a_n| \leq |a_K| + |a_n - a_K| &\implies |a_n| < M+2 \\ |a_K| \leq |a_n| + |a_n - a_K| &\implies M \leq |a_n| + 1 \implies |a_n| \geq M-1, \end{aligned}$$

for all $n \geq K$. Therefore, $M-1 \leq |a_n| < M+2$ for all $n \geq K$.

Now, let $(b_n)_{n=1}^{\infty}$ be the sequence defined by $b_n := M+1$ if $1 \leq n \leq K-1$ and $b_n := a_n$ if $n \geq K$. Clearly, $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ are equivalent, and so $x = \mathrm{LIM}_{n\to\infty} b_n$. For all $n \geq 1$, we have $M-1 \leq |b_n| < M+2$. Note that $M+1 > 0$ and $a_n > 0$ for all $n \geq K$. Thus, for all $n \geq 1$, $|b_n| = b_n$ because $b_n > 0$. Hence, $M-1 \leq b_n < M+2$ for all $n \geq 1$. By Corollary 5.4.10, $\mathrm{LIM}_{n\to\infty}(M-1) \leq \mathrm{LIM}_{n\to\infty} b_n \leq \mathrm{LIM}_{n\to\infty}(M+2)$. As a result, $M-1 \leq x \leq M+2$.

Note that $x \neq M-1$, $x \neq M$, $x \neq M+1$ and $x \neq M+2$ because $x \notin \mathbf{Q}$. So, we have

$$M-1 < x < M+2.$$

Using the order trichotomy of real numbers (Proposition 5.4.7), we see that exactly one of the following statements is true: (a) $x < M$; (b) $x > M$ and $x < M+1$; (c) $x > M+1$. If (a) $x < M$, then

$$M-1 \leq x < M.$$

If (b) $x > M$ and $x < M+1$, then

$$M \leq x < M+1.$$

If (c) $x > M+1$, then

$$M+1 \leq x < M+2.$$

So in any case, we can always find an integer N such that

$$N \leq x < N+1.$$

To show that this integer N is unique, suppose that N' is an integer such that $N' \leq x < N'+1$. Then, $-N'-1 < -x \leq -N'$. Now,

$$\begin{aligned}
&\begin{cases} -N'-1 < -x \leq -N' \\ N \leq x < N+1 \end{cases} && \Longrightarrow N-N'-1 < x-x < N+1-N' \\
&&& \Longrightarrow N-N'-1 < 0 < N-N'+1 \\
&&& \Longrightarrow -1 < N-N' < 1 \\
&&& \Longrightarrow N-N' = 0 \qquad \text{since } N-N' \in \mathbf{Z} \\
&&& \Longrightarrow N' = N.
\end{aligned}$$

So, the integer N for which $N \leq x < N+1$ is indeed unique.

Suppose that x is negative. By Proposition 5.4.4, $-x$ is positive. We just showed that this implies the existence of a unique integer L such that $L \leq -x < L+1$. This leads to $-L-1 < x \leq -L$, which implies that $-L-1 \leq x \leq -L$. Since $x \notin \mathbf{Q}$, $x \neq -L$. So, $-L-1 \leq x < -L$. The uniqueness of the integer satisfying the latter equality as $-L$ does follows by the same argument used above.

In conclusion, whether x is zero, positive, or negative, there is exactly one integer N such that

$$N \leq x < N+1.$$

Exercise 5.4.4

Let x be a positive real number. Since $x > 0$ and $1 > 0$, by the Archimedean property of real numbers (Corollary 5.4.13), there exists a positive integer N such that $Nx > 1 > 0$. Now, we multiply by the positive rational N^{-1} to get

$$x > 1/N > 0.$$

Exercise 5.4.5

Let $x < y$ be two real numbers.

First, suppose that $x > 0$, so that $y > x > 0$. Then, $y - x$ is positive by Definition 5.4.6. Hence, $(y-x)/2$ is positive by Proposition 5.4.4. Given the result of Exercise 5.4.4, we know that there exists a positive integer N such that $(y-x)/2 > 1/N$. Thus

$$2 + Nx < Ny.$$

As Nx is a real number, there exists a unique integer m such that $m \leq Nx < m+1$ (cf. Exercise 5.4.3). We have:

$$Nx < m+1 \leq Nx+1 < m+2 \leq Nx+2.$$

So, $Nx < m+1 < Nx+2$, so that $Nx < m+1 < Ny$. This implies that:

$$x < (m+1)/N < y.$$

Note that $(m+1)/N$ is a rational number.

Second, suppose that $x = 0$. Then, $x < y$ means that $y > 0$ by Proposition 5.4.7. Thus, there exists a positive natural number N such that $y > 1/N > 0$ (cf. Exercise 5.4.4). In other words, $x < 1/N < y$. Note that $1/N$ is a rational number.

Third, suppose that $x < 0$ and $y = 0$. From Proposition 5.4.7 and Exercise 4.2.6, we deduce that $-x > 0$. Hence, there exists a positive natural number N such that $-x > 1/N > 0$ (cf. Exercise 5.4.4). Using Proposition 5.4.7 and the result of Exercise 4.2.6 again leads to $x < -1/N < 0$. That is, $x < -1/N < y$. Note that $-1/N$ is a rational number.

Fourth, suppose that $x < 0$ and $y > 0$. Clearly, $x < 0 < y$.

Finally, suppose that $x < 0$ and $y < 0$. It follows from Proposition 5.4.7 and Exercise 4.2.6 that $-x > 0$, $-y > 0$ and $-y < -x$. From the first case, we deduce that there exists some rational p such that $-y < p < -x$. Applying Proposition 5.4.7 and the result of Exercise 4.2.6, we have $x < -p < y$. Note that $-p$ is a rational number.

In conclusion, there exists a rational q such that

$$x < q < y.$$

Exercise 5.4.6

Let x, y be real numbers and let $\varepsilon > 0$ be a positive real.

We begin by noting that for any positive real r, if $x-y<0$, we have $x-y<r$ because $r>0$, and if $x-y\geq 0$, then $x-y>-r$ because $-r<0$.

Now, we have:

$$\begin{aligned}
|x-y|<\varepsilon &\iff \begin{cases} x-y<\varepsilon, & x-y\geq 0 \\ -(x-y)<\varepsilon, & x-y<0 \end{cases} \\
&\iff \begin{cases} x-y<\varepsilon, & x-y\geq 0 \\ x-y>-\varepsilon, & x-y<0 \end{cases} \\
&\iff \begin{cases} x-y<\varepsilon \text{ and } x-y>-\varepsilon, & x-y\geq 0 \\ x-y>-\varepsilon \text{ and } x-y<\varepsilon, & x-y<0 \end{cases} \qquad \text{because } \varepsilon>0 \\
&\iff -\varepsilon<x-y<\varepsilon \\
&\iff y-\varepsilon<x<y+\varepsilon.
\end{aligned}$$

Next, we have $|x-y|\leq\varepsilon$ if and only if $-\varepsilon\leq x-y\leq\varepsilon$ by Proposition 4.3.3(c) applied to reals. Therefore,

$$|x-y|\leq\varepsilon \iff -\varepsilon\leq x-y\leq\varepsilon \iff y-\varepsilon\leq x\leq y+\varepsilon.$$

Exercise 5.4.7

Let $x, y\in\mathbf{R}$.

Note that for any real number $\varepsilon>0$, $y+\varepsilon>y+0$, i.e., $y+\varepsilon>y$. So,

$$x\leq y \implies x\leq y+\varepsilon$$

for any real $\varepsilon>0$.

Conversely, suppose that $x\leq y+\varepsilon$ for all $\varepsilon>0$. If $x>y$, then $1>2^{-1}>0$ leads to $x-y>(x-y)/2>0$, and so $x>y+(x-y)/2$, a contradiction. Hence, $x\leq y$.

In summary, $x\leq y+\varepsilon$ for all positive real numbers ε if and only if $x\leq y$.

By Proposition 4.3.7(a) applied to real numbers[2], $|x-y|\leq\varepsilon$ for all real numbers $\varepsilon>0$ if and only if $x=y$.

Exercise 5.4.8

Let $(a_n)_{n=1}^{\infty}$ be a Cauchy sequence of rationals, and let x be a real number.

[2] See text after Remark 5.4.11

Suppose that $a_n \leq x$ for all $n \geq 1$. For the sake of contradiction, suppose that $\text{LIM}_{n\to\infty} a_n > x$. Then, by Proposition 5.4.14, there is a rational number q such that $x < q < \text{LIM}_{n\to\infty} a_n$. So, $a_n \leq x$ implies that $a_n < q$ for all $n \geq 1$. Given that $(q)_{n=1}^{\infty}$ is also a Cauchy sequence of rationals, we have: $\text{LIM}_{n\to\infty} q \geq \text{LIM}_{n\to\infty} a_n$ by Corollary 5.4.10. In other words, $q \geq \text{LIM}_{n\to\infty} a_n$. But this contradicts $q < \text{LIM}_{n\to\infty} a_n$. Hence, $\text{LIM}_{n\to\infty} a_n \leq x$.

Suppose that $a_n \geq x$ for all $n \geq 1$. Then, $-a_n \leq -x$ for all $n \geq 1$. Given that $(-a_n)_{n=1}^{\infty}$ is also a Cauchy sequence of rationals, we can use the result of the above paragraph to deduce that $\text{LIM}_{n\to\infty} (-a_n) \leq -x$. So, $-\text{LIM}_{n\to\infty} (-a_n) \geq -(-x)$, i.e., $\text{LIM}_{n\to\infty} (-(-a_n)) \geq x$, i.e., $\text{LIM}_{n\to\infty} a_n \geq x$.

Exercise 5.4.9

(i) Let x and y be real numbers.

Given that

$$\min(-x, -y) := \begin{cases} -x, & -x \leq -y \\ -y, & -x > -y \end{cases} = \begin{cases} -x, & x \geq y \\ -y, & x < y \end{cases}$$

and

$$\max(-x, -y) := \begin{cases} -x, & -x \geq -y \\ -y, & -x < -y \end{cases} = \begin{cases} -x, & x \leq y \\ -y, & x > y \end{cases},$$

it is clear that

$$-\min(-x - y) = \max(x, y) \quad \text{and} \quad -\max(-x, -y) = \min(x, y).$$

(ii) Let x, y, and z be real numbers.

To start with,

$$\begin{aligned}
\max(x,y) &:= \begin{cases} x, & x \geq y \\ y, & x < y \end{cases} \\
&= \begin{cases} x, & x > y \text{ or } x = y \\ y, & x < y \end{cases} \\
&= \begin{cases} x, & x > y \\ x, & x = y \\ y, & x < y \end{cases} \\
&= \begin{cases} x, & x > y \\ y, & x = y \\ y, & x < y \end{cases} \\
&= \begin{cases} x, & x > y \\ y, & x = y \text{ or } x < y \end{cases} \\
&= \begin{cases} y, & y \geq x \\ x, & y < x \end{cases} \\
&= \max(y,x).
\end{aligned}$$

Next, we clearly have

$$\max(x,x) := \begin{cases} x, & x \geq x \\ x, & x < x \end{cases} = x$$

Now, note that

$$\begin{aligned}
\max(x,y)+z &:= \begin{cases} x+z, & x \geq y \\ y+z, & x < y \end{cases} \\
&= \begin{cases} x+z, & x+z \geq y+z \\ y+z, & x+z < y+z \end{cases} \\
&:= \max(x+z, y+z).
\end{aligned}$$

Here, the second equality follows because $x \geq y$ if and only if $x+z \geq y+z$ and $x < y$ if and only if $x+z < y+z$.

If z is positive, then

$$\begin{aligned}\max(xz, yz) &:= \begin{cases} xz, & xz \geq yz \\ yz, & xz < yz \end{cases} \\ &= \begin{cases} xz, & x \geq y \\ yz, & x < y \end{cases} \\ &= \max(x, y)z,\end{aligned}$$

where the second equality holds because, since $z > 0$, we have $xz \geq yz$ if and only if $x \geq y$, and we have $xz < yz$ if and only if $x < y$, by Proposition 5.4.7. If z is zero, then

$$\max(xz, yz) := \max(0, 0) = 0 = 0\max(x, y).$$

Thus, whenever z is non-negative, we have

$$\max(xz, yz) = z\max(x, y).$$

But if z is negative, then

$$\begin{aligned}\max(xz, yz) &:= \begin{cases} xz, & xz \geq yz \\ yz, & xz < yz \end{cases} \\ &= \begin{cases} xz, & x \leq y \\ yz, & x > y \end{cases} \\ &\text{because here } xz \geq yz \iff x \leq y \text{ and } xz < yz \iff x > y \\ &= \min(x, y)z.\end{aligned}$$

(iii) Let x, y, and z be real numbers.[3]

First,

$$\min(x, y) = -\max(-x, -y) = -\max(-y, -x) = \min(y, x).$$

Second,

$$\min(x, x) = -\max(-x, -x) = -(-x) = x.$$

[3] Here, we will freely use the results established in part (i) and (ii) without explicitly mentioning, but it should be very clear at this point.

Third,

$$\begin{aligned}\min(x+z, y+z) &= -\max(-(x+z), -(y+z)) \\ &= -\max(-x-z, -y-z) \\ &= -(\max(-x, -y)-z) \\ &= -\max(-x, -y)+z \\ &= \min(x, y)+z.\end{aligned}$$

Fourth, if z is non-negative, then

$$\begin{aligned}\min(xz, yz) &= -\max(-xz, -yz) \\ &= -z\max(-x, -y) \\ &= z(-\max(-x, -y)) \\ &= z\min(x, y).\end{aligned}$$

Finally, if z is negative, then

$$\begin{aligned}\min(xz, yz) &= -\max(-xz, -yz) \\ &= -(z\min(-x, -y)) \\ &= z(-\min(-x, -y)) \\ &= z\max(x,y).\end{aligned}$$

(iv) Let x and y be positive real numbers.

By Propositions 5.4.8 and 5.4.7,

$$x > y \iff x^{-1} < y^{-1} \quad \text{and} \quad x < y \iff x^{-1} > y^{-1}.$$

Furthermore, we have $x = y$ if and only if $x^{-1} = y^{-1}$ because $x = y$ implies $x^{-1} = y^{-1}$ as the reciprocal for non-zero reals is well defined, and $x^{-1} = y^{-1}$ yields $x = y$ by multiplying both sides by xy and using the field axioms (Proposition 4.2.4 for reals). It follows that

$$x \geq y \iff x^{-1} \leq y^{-1} \quad \text{and} \quad x \leq y \iff x^{-1} \geq y^{-1}.$$

Using these equivalences, we see that

$$\max(x, y)^{-1} := \begin{cases} x^{-1}, & x \geq y \\ y^{-1}, & x < y \end{cases} = \begin{cases} x^{-1}, & x^{-1} \leq y^{-1} \\ y^{-1}, & x^{-1} > y^{-1} \end{cases} := \min(x^{-1}, y^{-1}),$$

and

$$\min(x, y)^{-1} := \begin{cases} x^{-1}, & x \leq y \\ y^{-1}, & x > y \end{cases} = \begin{cases} x^{-1}, & x^{-1} \geq y^{-1} \\ y^{-1}, & x^{-1} < y^{-1} \end{cases} := \max(x^{-1}, y^{-1}).$$

5.5 The Least Upper Bound Property

Exercise 5.5.1

Let E be a subset of $\mathbf{R}$, and suppose that E has a least upper bound M that is a real number, i.e., $M = \sup(E)$. Let $-E$ be the set

$$-E := \{-x : x \in E\}.$$

Since the least upper bound of E is real, E must be non-empty (cf. Definition 5.5.10). Consequently, $-E$ must also be non-empty.

Let y_0 be an element of $-E$. By the definition of $-E$, $-y_0 \in E$. Given that $-y_0 \leq M$ (since M is an upper bound for E, see Definition 5.5.1), we have $y_0 \geq -M$. Hence, $-M$ is a lower bound for $-E$.

Now, let N be a lower bound for $-E$. Then, $y \geq N$ for any $y \in -E$. In other words, $-y \geq N$ for any $y \in E$ (because $y \in -E$ if and only if $-y \in E$)[4]. So, $y \leq -N$ for any $y \in E$. It means that $-N$ is an upper bound for E. Since M is the least upper bound of E, we thus have $-N \geq M$, that is, $-M \geq N$.

We have showed that $-M$ is a lower bound for $-E$ that is greater than or equal to any other lower bound. Therefore, $-M$ is the greatest lower bound for $-E$. That is to say,

$$-M = \inf(-E).$$

Exercise 5.5.2

Let E be a non-empty subset of $\mathbf{R}$, let $n \geq 1$ be an integer, and let $L < K$ be integers. Suppose that K/n is an upper bound for E and L/n is not an upper bound for E.

Suppose, for the sake of contradiction, that there is no integer $L < m \leq K$ such that m/n is a an upper bound for E and $(m-1)/n$ is not an upper bound for E. In other words, suppose that for all $L < m \leq K$, the statement $Q(m)$: "either m/n is not an upper bound for E or $(m-1)/n$ is an upper bound for E" is true.

Consider the statement $P(x)$: "$(K-x)/n$ is not an upper bound for E" that pertains to a natural number x. Using the principle of backwards induction (see Exercise 2.2.6), we will prove that $P(x)$ is true for all $0 \leq x \leq K-L$.

[4] Here is an alternative argument that yields the same result: For any $y \in E$, $-y \in -E$, and so $-y \geq N$.

First, note that for $x = K - L$, $(K - x)/n = L/n$. As L/n is not an upper bound for E, $P(K - L)$ is true.

Second, suppose inductively that $P(x+1)$ is true for some natural number x with $0 \leq x \leq K - L$. This means that $(K - x - 1)/n$ is not an upper bound for E. If $0 \leq x < K - L$, then $L < K - x \leq K$, and so $(K - x)/n$ is not an upper bound for E by $Q(K - x)$. If $x = K - L$, then $K - x = L$ and we already know that $(K - x)/n$ is not an upper bound for E. Thus, $P(x)$ is also true.

We conclude that $P(x)$ is true for all $0 \leq x \leq K - L$.

In particular, $P(0)$: "K/n is not an upper bound for E" is true. But this is contradictory because K/n is an upper bound for E. This contradiction shows that there must exist an integer m between L and K such that m/n is an upper bound for E while $(m-1)/n$ is not.

Exercise 5.5.3

Let E be a non-empty subset of $\mathbf{R}$, let $n \geq 1$ be an integer, and let m, m' be integers with the properties that m/n and m'/n are upper bounds for E, but $(m-1)/n$ and $(m'-1)/n$ are not upper bounds for E.

Given that neither $(m-1)/n$ nor $(m'-1)/n$ is an upper bound for E and that E is non-empty, there exist elements $x, x' \in E$ such that $(m-1)/n < x$ and $(m'-1)/n < x'$, by Definition 5.5.1 (note that we might have $x = x'$, but it does not matter here). The same definition tells us that $x \leq m'/n$ and $x' \leq m/n$ because m/n and m'/n are upper bounds for E. Using the transitivity of order, we have:

$$(m-1)/n < m'/n \quad \text{and} \quad (m'-1)/n < m/n.$$

This implies that

$$m - 1 < m' \quad \text{and} \quad m' - 1 < m,$$

so that

$$m \leq m' \quad \text{and} \quad m' \leq m.$$

Therefore,

$$m = m'.$$

Exercise 5.5.4

Let $(q_n)_{n=1}^{\infty}$ be a sequence of rational numbers with the property that $|q_n - q'_n| \leq 1/M$

whenever $M \geq 1$ is an integer and $n, n' \geq M$.

Let $\varepsilon > 0$ be a positive rational. From Exercise 5.4.4, we know that there exists a positive integer P such that $\varepsilon > 1/P$. We have $|q_n - q'_n| \leq 1/P$ for all $n, n' \geq P$, by the property of $(q_n)_{n=1}^{\infty}$. Hence, $|q_n - q'_n| < \varepsilon$ for all $n, n' \geq P$. We see that $(q_n)_{n=1}^{\infty}$ is eventually ε-steady. Therefore, $(q_n)_{n=1}^{\infty}$ is a Cauchy sequence.

Define the real number $S := \mathrm{LIM}_{n\to\infty} q_n$. Let M be a positive integer. For every $n \geq M$, we have $|q_M - q_n| \leq 1/M$, and so $-1/M \leq q_M - q_n \leq 1/M$.

Now, $q_M - q_n \leq 1/M$ implies that $q_M - 1/M \leq q_n$. Let $(a_n)_{n=1}^{\infty}$ be the sequence defined by $a_n = q_M - 1/M$ if $n < M$ and $a_n = q_n$ if $n \geq M$. Then, $(q_n)_{n=1}^{\infty}$ and $(a_n)_{n=1}^{\infty}$ are equivalent since they eventually are the same. Thus, $S = \mathrm{LIM}_{n\to\infty} a_n$. Furthermore, we have $q_M - 1/M \leq a_n$ for all $n \geq 1$. Using Exercise 5.4.8, we see that $q_M - 1/M \leq \mathrm{LIM}_{n\to\infty} a_n$, i.e., $q_M - 1/M \leq S$. Thus,

$$q_M - S \leq 1/M. \tag{1}$$

Similarly, since $q_M - q_n \geq -1/M$, we have $q_M + 1/M \geq q_n$ for all $n \geq M$. Using an argument similar to the one above, we see that this leads to $q_M + 1/M \geq S$. So,

$$-1/M \leq q_M - S. \tag{2}$$

From (1) and (2), we conclude that

$$|q_M - S| \leq 1/M.$$

Exercise 5.5.5

By Proposition 5.5.12, there exists a real number k such that $k^2 = 2$. From Proposition 4.4.4, we know that k is an irrational number. Note that

$$k^2 = 2 \implies k^2/2^2 = 1/2 \implies (k/2)(k/2) = 1/2 \implies (k/2)(k/2) < 1.$$

If $k/2 \geq 1$, then $(k/2)(k/2) \geq 1$. Hence, $k/2 < 1$.

Let r be an arbitrary positive rational number. By the results of Exercise 5.4.4, there exists a positive integer N_r such that

$$0 < 1/N_r < r.$$

Since $k/2 < 1$, we have $k/(2N_r) < 1/N_r$, so that

$$0 < k/(2N_r) < r.$$

Note that $k/(2N_r)$ is an irrational number, for if $k/(2N_r) = q$ for some rational number q, then $k = 2N_r q$, which is contradictory since $2N_r q$ is rational while k is not. We have just proven that there is always an irrational number strictly between 0 and any positive real.

Let $x < y$ be two real numbers. The number $y - x$ is positive. Consequently, there exists an irrational number z such that $0 < z < y - x$. By Proposition 5.4.14, there is a rational number p such that $0 < p < z$. We have:

$$x < p + x < z + x < y.$$

If x is a rational number, then $z + x$ is an irrational number, for otherwise z would be rational since $z = (z + x) - x$ and the subtraction of two rationals is a rational. If x is an irrational number, then $p + x$ is an irrational number, for otherwise x would be rational since $x = (p + x) - p$ and the subtraction of rationals is a rational.

We see that in any case there exists an irrational number i such that

$$x < i < y.$$

5.6 Real Exponentiation, Part I

Exercise 5.6.1

Let x, y be non-negative reals and let n, m be positive integers. Define the sets

$$E := \{z \in \mathbf{R} : z \geq 0 \text{ and } z^n \leq x\} \text{ and } F := \{z \in \mathbf{R} : z \geq 0 \text{ and } z^n \leq y\}.$$

We will prove the seven statements of Lemma 5.6.6. But before that, we must establish the binomial theorem, which states that for any real numbers a and b and any natural number k, we have

$$(a+b)^k = \sum_{i=0}^{k} a^i b^{k-i} \frac{k!}{i!(k-i)!}.$$

But, stated like this, the theorem uses the sigma notation, which we will only see in Chapter 7. Given that we insist on not using concepts that will only be introduced in later parts of the text,[5] we will rewrite the binomial theorem in a way that avoids using the sigma notation or any other concepts that will be defined in later chapters. We defined the factorial $k!$ of a natural number k in Exercise 3.6.12, so we can safely use that notion.

Now, for convenience, we define the notation:

$$\binom{p}{q} := \frac{p!}{q!(p-q)!}$$

for any natural numbers $p \geq q$. It is easy to verify that this definition is valid by showing that $\binom{p}{q} = \binom{p'}{q} = \binom{p}{q'}$ for any natural numbers $p \geq q$ and natural numbers p' and q' such that $p = p'$ and $q = q'$. Note that:

$$\begin{aligned}
\binom{p+1}{q} &= \frac{(p+1)!}{q!(p+1-q)!} \\
&= \frac{(p+1)p!}{q!(p+1-q)(p-q)!} \\
&= \frac{(p+1-q)p!}{q!(p+1-q)(p-q)!} + \frac{(q)p!}{q!(p+1-q)(p-q)!} \\
&= \frac{p!}{q!(p-q)!} + \frac{p!}{(q-1)!(p-(q-1))!} \\
&= \binom{p}{q} + \binom{p}{q-1},
\end{aligned}$$

[5] The only exception is within Exercise 3.4.6(ii) where we believe that Axiom 3.12, which would only be introduced at a later point, was needed to solve the exercise. Of course, if you found a way to solve it without Axiom 3.12, please let us know at keumoe.fozeu.r@gmail.com.

for any positive natural numbers $p \geq q \geq 1$.

Let $a, b \in \mathbf{R}$. For any $k \in \mathbf{N}$, define the function $f_k : \{i \in \mathbf{N} : i \leq k\} \to \mathbf{R}$ by:

$$f_k(i) := a^i b^{k-i} \binom{k}{i}.$$

We will prove by induction that for any natural number k, we have:

$$(a+b)^k = f_k(0) + \cdots + f_k(k),$$

where $f_k(0) + \cdots + f_k(k)$ is the result of the addition of the k real numbers $f_k(0), \ldots, f_k(k)$. This is an equivalent formulation of the binomial theorem.[6]

Let's start with the base case: at $k = 0$. We have:

$$(a+b)^0 = 0 \text{ and } f_0(0) = a^0 b^0 \binom{0}{0} = 1,$$

and the base case is completed.

Next, note that

$$(a+b)^1 = a+b \text{ and } f_1(0) + f_1(1) = b+a,$$

and hence the binomial theorem holds for $k = 1$. Also,

$$(a+b)^2 = a^2 + 2ab + b^2 \text{ and } f_2(0) + f_2(1) + f_2(2) = b^2 + 2ab + a^2,$$

and so the binomial theorem equally holds for $k = 2$. Hence, we can start our induction step from $k \geq 2$. So, suppose inductively that $(a+b)^k = f_k(0) + \cdots + f_k(k)$ for some natural number $k \geq 2$. Given that $k+1 \geq 3$, we can write:

$$f_{k+1}(0) + f_{k+1}(1) + \cdots + f_{k+1}(k) + f_{k+1}(k+1) = b^{k+1} + f_{k+1}(1) + \cdots + f_{k+1}(k) + a^{k+1}.$$

Define the functions g and h from $\{i \in \mathbf{N} : 1 \leq i \leq k\}$ to $\mathbf{R}$ by

$$g(i) := a^i b^{k+1-i} \binom{k}{i} \text{ and } h(i) := a^i b^{k+1-i} \binom{k}{i-1}.$$

It is easy to see that:

$$f_{k+1}(i) = g(i) + h(i), \quad g(i) = b f_k(i), \quad \text{and} \quad h(i) = a f_k(i-1)$$

[6] Is there perhaps a way to solve this exercise without using the binomial theorem (and any other concept not yet seen)? Tell me at keumoe.fozeu.r@gmail.com or on GitHub @KeumoeFozeu. Note that in any case, the proofs in this exercise are correct because the definition of the sigma notation does not depend on anything we proved here.

for all $1 \leq i \leq k$. We have:

$$\begin{aligned} f_{k+1}(0)+f_{k+1}(1)+\cdots+f_{k+1}(k)+f_{k+1}(k+1) &= b^{k+1}+(g(1)+h(1))+\ldots \\ &\quad +(g(k)+h(k))+a^{k+1} \\ &= b^{k+1}+g(1)+\cdots+g(k)+h(1)+\ldots \\ &\quad +h(k)+a^{k+1}. \end{aligned}$$

Using the induction hypothesis, we see that:

$$\begin{aligned} g(1)+\cdots+g(k) &= b(f_k(1)+\cdots+f_k(k)) \\ &= b(f_k(0)+f_k(1)+\cdots+f_k(k)-f_k(0)) \\ &= b((a+b)^k-b^k) \\ &= b(a+b)^k-b^{k+1} \end{aligned}$$

and

$$\begin{aligned} h(1)+\cdots+h(k) &= a(f_k(0)+\ldots f_k(k-1)) \\ &= a(f_k(0)+\ldots f_k(k-1)+f_k(k)-f_k(k)) \\ &= a((a+b)^k-a^k) \\ &= a(a+b)^k-a^{k+1}. \end{aligned}$$

So,

$$\begin{aligned} f_{k+1}(0)+f_{k+1}(1)+\cdots+f_{k+1}(k)+f_{k+1}(k+1) &= b^{k+1}+b(a+b)^k-b^{k+1} \\ &\quad +a(a+b)^k-a^{k+1}+a^{k+1} \\ &= (b+a)(a+b)^k \\ &= (a+b)^{k+1}. \end{aligned}$$

This closes the induction.[7]

Having established the binomial theorem, we are now in a position to prove Lemma 5.6.6.

(a) If $y = x^{1/n}$, then $y^n = x$.

By Definition 5.6.4,

$$x^{1/1} = \sup\{z \in \mathbf{R} : z \geq 0 \text{ and } z^1 \leq x\} = \sup\{z \in \mathbf{R} : 0 \leq z \leq x\}.$$

[7] A more direct way of proving the binomial theorem is presented in Exercise 7.1.4.

Since any element of $\{z \in \mathbf{R} : 0 \leq z \leq x\}$ is less than or equal to x, x is an upper bound of this set. Plus, any other upper bound is greater than or equal to x because x belongs to the set. So, x is the least upper bound of $\{z \in \mathbf{R} : 0 \leq z \leq x\}$. That is to say, $x = \sup\{z \in \mathbf{R} : 0 \leq z \leq x\}$. Using this fact, we see that when $n = 1$,

$$y = x^{1/n} \implies y = x^{1/1} \implies y = x \implies y^1 = x.$$

Hence, statement (a) is true when $n = 1$. From now on, assume that $n > 1$.

Suppose that $y = x^{1/n}$. By the trichotomy of order, exactly one of the following three statement is true: (i) $y^n < x$, (ii) $y^n > x$, (iii) $y^n = x$.

First, suppose that (i) $y^n < x$. Let ε be a positive real less than 1. Applying the binomial theorem, we have:

$$(y+\varepsilon)^n = \binom{n}{0} y^n + \binom{n}{1} y^{n-1}\varepsilon + \cdots + \binom{n}{n}\varepsilon^n.$$

Since $\varepsilon^k \leq \varepsilon \leq 1$ for all integers $k \geq 1$, we have $\binom{n}{k} y^{k-1}\varepsilon^k \leq \binom{n}{k} y^{k-1}\varepsilon$ for all $1 \leq k \leq n$. So:

$$\begin{aligned}(y+\varepsilon)^n &\leq \binom{n}{0} y^n + \binom{n}{1} y^{n-1}\varepsilon + \cdots + \binom{n}{n}\varepsilon \\ &= y^n + \varepsilon\left[\binom{n}{1} y^{n-1} + \cdots + \binom{n}{n}\right].\end{aligned}$$

Given that

$$\frac{x - y^n}{\binom{n}{1} y^{n-1} + \cdots + \binom{n}{n}} > 0,$$

we can choose ε such that

$$\varepsilon < \frac{x - y^n}{\binom{n}{1} y^{n-1} + \cdots + \binom{n}{n}}.$$

In this case, $y^n + \varepsilon\left[\binom{n}{1} y^{n-1} + \cdots + \binom{n}{n}\right] < x$, and thus $(y+\varepsilon)^n < x$. This implies that $y + \varepsilon \in E$. But that is contradictory because $y + \varepsilon > y$ and $y = x^{1/n} = \sup(E)$. Therefore, statement (i) $y^n < x$ is false.

Second, suppose that (ii) $y^n > x$. Then, $y \neq 0$ since $0^n = 0$ and $0 \leq x$. Thus, $y > 0$. Let δ be a non-negative real less than 1 and less than y (i.e., $\delta \geq 0$, $\delta < 1$, $\delta < y$). Note that $y > y - \delta > 0$. By the binomial theorem, we have:

$$(y-\delta)^n = y^n + \binom{n}{1} y^{n-1}(-\delta) + \cdots + \binom{n}{n}(-\delta)^n.$$

For all integers $k \geq 1$, we have $(-1)^k = \pm 1$ (this is easily established with Proposition 5.6.3 and a straightforward induction), and hence $(-\delta)^k = \pm\delta^k \geq -\delta^k$ by Proposition 5.6.3. It follows that

$$(y-\delta)^n \geq y^n - \binom{n}{1} y^{n-1}\delta - \cdots - \binom{n}{n}\delta^n.$$

For any integer $k \geq 1$, $0 < \delta^k \leq \delta < 1$ implies that $-\delta^k \geq -\delta$. So,

$$\begin{aligned}(y-\delta)^n &\geq y^n - \binom{n}{1} y^{n-1}\delta - \cdots - \binom{n}{n}\delta \\ &= y^n - \delta\left[\binom{n}{1} y^{n-1} + \cdots + \binom{n}{n}\right].\end{aligned}$$

As

$$\frac{y^n - x}{\binom{n}{1} y^{n-1} + \cdots + \binom{n}{n}} > 0,$$

we can choose δ such that

$$\delta < \frac{y^n - x}{\binom{n}{1} y^{n-1} + \cdots + \binom{n}{n}}.$$

In this case, $y^n - \delta\left[\binom{n}{1} y^{n-1} + \cdots + \binom{n}{n}\right] > x$ leads to $(y-\delta)^n > x$. For any $z_0 \in E$, if $z_0 > y - \delta$, then ${z_0}^n > (y-\delta)^n$ by Proposition 4.3.10(c) applied to reals (see Proposition 5.6.3), and so $x > (y-\delta)^n$, a contradiction. Thus, $y - \delta$ is greater than or equal to every element of E. But this makes $y - \delta$ an upper bound for E, which contradicts the fact that y is the least upper bound of E. This contradiction shows that the statement (ii) $y^n > x$ is false.

Therefore, statement (iii) $y^n = x$ must be true.

(b) Conversely, if $y^n = x$, then $y = x^{1/n}$.

Suppose that $x = y^n$. Then,

$$z^n \leq x \iff z^n \leq y^n$$

for all $z \in \mathbf{R}$. So, $E = \{z \in \mathbf{R} : z \geq 0 \text{ and } z^n \leq y^n\}$. For any $z \in E$, since $z, y \geq 0$ and $z^n \leq y^n$, we have $z \leq y$ by Proposition 5.6.3 (consider the contrapositive of Proposition 4.3.10(c) applied to reals). Thus, y is an upper bound for E. Also, note that $y \in E$ because $y \geq 0$ and $y^n \leq y^n$. So, any upper bound for E is greater that or equal to y. Therefore, y is the least upper bound of E. That is, $y = x^{1/n}$.

(c) $x^{1/n}$ is a non-negative real number, and is positive if and only if x is positive.

By Lemma 5.6.5, $x^{1/n}$ is a real number. Given that $0 \geq 0$ and $0^n = 0 \leq x$, $0 \in E$. As $x^{1/n} = \sup(E)$, we have $x^{1/n} \geq 0$.

If $x > 0$, then $x^{1/n} = 0$ implies that $x = 0^n$ by (a), which yields $x = 0$, a contradiction. So, if $x > 0$, then $x^{1/n} > 0$. Conversely, if $x^{1/n} > 0$, then $x = 0$ implies that $x = 0^n$, which implies that $x^{1/n} = 0$ by (b), a contradiction. So, if $x^{1/n} > 0$, then $x > 0$.

In summary, $x^{1/n}$ is a non-negative real number. Moreover, $x^{1/n}$ is positive if and only if x is positive.

(d) We have $x > y$ if and only if $x^{1/n} > y^{1/n}$.

On the one hand, suppose that $x > y$. Then for any $z \in F$, we have $z^n \leq y \leq x$, and so $z \in E$. Hence, $F \subseteq E$. Therefore, any upper bound for E is also an upper bound for F. Indeed, if M_E is an upper bound for E, then $M_E \geq z$ for all $z \in E$; in particular $M_E \geq z$ for all $z \in F$ since $F \subseteq E$. As $x^{1/n}$ is an upper bound for E and $y^{1/n}$ is the least upper bound for F, we have $x^{1/n} \geq y^{1/n}$. Suppose that $x^{1/n} = y^{1/n}$. Then we have $(x^{1/n})^n = y$ and $(y^{1/n})^n = x$ by part (a), and also $(x^{1/n})^n = (y^{1/n})^n$. This yields $x = y$, a contradiction. Thus, $x^{1/n} \neq y^{1/n}$. Therefore, $x^{1/n} > y^{1/n}$.

On the other hand, suppose that $x^{1/n} > y^{1/n}$. First, it is clear that $x \neq y$. Indeed, $x = y$ implies that $x^{1/n} = y^{1/n}$, since the nth root of a non-negative real is well-defined (as n is a positive integer). Next, if $x < y$, then $E \subseteq F$ (by an argument analogous to the one which was used to show that $F \subseteq E$ above), which implies that $x^{1/n} \leq y^{1/n}$ (by an argument similar to the one used to show that $x^{1/n} \geq y^{1/n}$ in the previous paragraph). But this is a contradiction. Therefore, $x > y$.

In conclusion, $x > y$ is equivalent to $x^{1/n} > y^{1/n}$.

(e) If $x > 1$, then $x^{1/k}$ is a decreasing function of k, where k ranges over the positive integers. If $0 < x < 1$, then $x^{1/k}$ is an increasing function of k. If $x = 1$, then $x^{1/k} = 1$ for all k.

Before we start, define for any positive integer k the set E_k by

$$E_k := \{z \in \mathbf{R} : z \geq 0 \text{ and } z^k \leq x\}.$$

Now, suppose that $x > 1$. Then, for any positive integer k, $1 \in E_k$ as $1^k = 1 < x$. Hence, $x^{1/k} \geq 1$ since $x^{1/k}$ is an upper bound for E_k. Moreover, $x^{1/k} \neq 1$ because $1^k \neq x$ (see part(a)). Thus, $x^{1/k} > 1$ for any positive integer k. Let $s < t$ be two positive integers. We have $x^{1/s} = \sup(E_s)$ and $x^{1/t} = \sup(E_t)$. Since $t - s > 0$ and

$x^{1/t} > 1$, $(x^{1/t})^{t-s} > 1$. So, $(x^{1/t})^t = (x^{1/t})^{t-s}(x^{1/t})^s > (x^{1/t})^s$. But, $(x^{1/t})^t = x$ by part (a). Hence, $(x^{1/t})^s < x$. It follows that $x^{1/t} \in E_s$, and thus $x^{1/t} \leq x^{1/s}$. Since $(x^{1/s})^s = x$ by part (a) while $(x^{1/t})^s \neq x$, we have $x^{1/t} \neq x^{1/s}$. So, $x^{1/t} < x^{1/s}$. This shows that $x^{1/k}$ is a decreasing function of k, where k ranges over the positive integers.

Next, suppose that $0 < x < 1$. Then for any positive integer k and for all $z \in E_k$, $0 \leq z^k \leq x < 1 = 1^k$ implies that $0 \leq z^k < 1^k$, which leads to $z < 1$ by Proposition 5.6.3. Hence, 1 is an upper bound for E_k, for any positive integer k. Furthermore, $1 \neq x^{1/k}$ because $1^k = 1 \neq x$ (see part (a)). Thus, $x^{1/k} < 1$ for all positive integers k. Let $s < t$ be two positive integers. Since $s < t$ and $0 < x^{1/s} < 1$, we have $(x^{1/s})^t < (x^{1/s})^s$ (to see this, note that $(x^{1/s})^t = (x^{1/s})^{t-s}(x^{1/s})^s$ and that $(x^{1/s})^{t-s} < 1$ because $t - s > 0$ and $0 < x^{1/s} < 1$). As $(x^{1/s})^s = x$ by part (a), $(x^{1/s})^t < x$. So, $x^{1/s} \in E_t$, and thus $x^{1/s} \leq x^{1/t}$. Moreover, $(x^{1/t})^t = x$ by part (a) while $(x^{1/s})^t \neq x$, and hence $x^{1/s} \neq x^{1/t}$. Thus, $x^{1/s} < x^{1/t}$. We deduce that $x^{1/k}$ is an increasing function of k, where k ranges over the positive integers.

Lastly, suppose that $x = 1$. Assume toward contradiction that $1^{1/k} \neq 1$ for some positive integer k. Then by part (b), $1 \neq 1^k$, a clear contradiction. So, $1^{1/k} = 1$ for any positive integer k.

(f) We have $(xy)^{1/n} = x^{1/n}y^{1/n}$.

We proceed by contradiction. Suppose that $(xy)^{1/n} \neq x^{1/n}y^{1/n}$. This implies that $xy \neq (x^{1/n}y^{1/n})^n$ by part (b). So, $xy \neq (x^{1/n})^n(y^{1/n})^n$ by Proposition 5.6.3. By (a) and (b), we have $(x^{1/n})^n = x$ and $(y^{1/n})^n = y$.[8] Hence, $xy \neq xy$. This contradiction shows that $(xy)^{1/n} = x^{1/n}y^{1/n}$.

(g) We have $(x^{1/n})^{1/m} = x^{1/nm}$.

Suppose for the sake of contradiction that $(x^{1/n})^{1/m} \neq x^{1/nm}$. Then, $x^{1/n} \neq (x^{1/nm})^m$ by part (b). Either $x^{1/n} > (x^{1/nm})^m$ or $x^{1/n} < (x^{1/nm})^m$. In either case, it follows from Proposition 5.6.3 that $(x^{1/n})^n \neq \left((x^{1/nm})^m\right)^n$ because $x^{1/n} \geq 0$ and $(x^{1/nm})^m \geq 0$ by part (c). So, $x \neq \left((x^{1/nm})^m\right)^n$. Thus, $x \neq (x^{1/nm})^{nm}$ by Proposition 5.6.3. Hence, $x \neq x$, which is a contradiction. So, $(x^{1/n})^{1/m} = x^{1/nm}$.

Exercise 5.6.2

Let $x, y > 0$ be positive reals, and let q, r be rationals. There are integers a, c and

[8] From now on, we will use the fact that $(x^{1/n})^n = x$ for any non-negative real x and positive integer n, without further comment.

positive integers b, d such that $q = a/b$ and $r = c/d$. We will prove the following six statements.

(a) x^q is a positive real.

By Definition 5.6.7, $x^q := (x^{1/b})^a$. By Lemma 5.6.6(c), $x^{1/b}$ is a positive real number because x is positive. If $a > 0$ or $a < 0$, then $(x^{1/b})^a$ is positive by Proposition 5.6.3. If $a = 0$, then $(x^{1/b})^a = 1$, and so $(x^{1/b})^a$ is positive. In any case, $(x^{1/b})^a$ is a positive real. That is, x^q is a positive real.

(b) $x^{q+r} = x^q x^r$ and $(x^q)^r = x^{qr}$.

We have:

$$\begin{aligned}
x^{q+r} &= x^{(ad+bc)/bd} \\
&= (x^{1/bd})^{ad+bc} && \text{by Definition 5.6.7} \\
&= (x^{1/bd})^{ad}(x^{1/bd})^{bc} && \text{by Proposition 5.6.3} \\
&= x^{ad/bd}x^{bc/bd} && \text{by Definition 5.6.7} \\
&= x^{a/b}x^{c/d} \\
&= x^q x^r.
\end{aligned}$$

Before proving the second equality, note that for any integer n and any positive integer m, we have $x^{n/m} = (x^{1/m})^n = (x^n)^{1/m}$. To see this, suppose toward a contradiction that $(x^{1/m})^n \neq (x^n)^{1/m}$. Then by Lemma 5.6.6(b), $((x^{1/m})^n)^m \neq x^n$, i.e., $(x^{1/m})^{nm} \neq x^n$. But, $(x^{1/m})^{nm} := x^{nm/m} = x^n$ by Definition 5.6.7. This contradiction shows that $(x^{1/m})^n = (x^n)^{1/m}$. Now, we have

$$\begin{aligned}
(x^q)^r &= (x^q)^{c/d} \\
&= ((x^q)^{1/d})^c && \text{by Definition 5.6.7} \\
&= \left(((x^{1/b})^a)^{1/d}\right)^c && \text{by Definition 5.6.7} \\
&= \left(((x^{1/b})^{1/d})^a\right)^c && \text{by the result we just established}[9] \\
&= ((x^{1/bd})^a)^c && \text{by Lemma 5.6.6(g)} \\
&= (x^{1/bd})^{ac} && \text{by Proposition 5.6.3} \\
&= x^{ac/bd} && \text{by Definition 5.6.7} \\
&= x^{qr},
\end{aligned}$$

as desired.

[9] This result will be used without further comment from now on.

(c) $x^{-q} = 1/x^q$.

Note that for any integer n, $1/x^{-n} = 1 \times (x^{-n})^{-1}$ by the definition of division in $\mathbf{R}$; and since $(x^{-n})^{-1} = x^n$ by Proposition 5.6.3, it follows that $1/x^{-n} = x^n$. This result will be used without further comment. Now, We have:

$$\begin{aligned} x^{-q} &= x^{(-a)/b} \\ &= (x^{1/b})^{(-a)} && \text{by Definition 5.6.7} \\ &= 1/(x^{1/b})^a \\ &= 1/x^{a/b} && \text{by Definition 5.6.7} \\ &= 1/x^q. \end{aligned}$$

(d) If $q > 0$, then $x > y$ if and only if $x^q > y^q$.

Suppose that $q > 0$ so that $a > 0$. By Lemma 5.6.6(d), $x > y$ if and only if $x^{1/b} > y^{1/b}$. By Proposition 4.3.10(c) applied to reals[10], $x^{1/b} > y^{1/b}$ if and only if $(x^{1/b})^a > (y^{1/b})^a$. By Definition 5.6.7, $(x^{1/b})^a > (y^{1/b})^a$ means that $x^{a/b} > y^{a/b}$. In sum, $x > y$ if and only if $x^q > y^q$.

(e) If $x > 1$, then $x^q > x^r$ if and only if $q > r$. If $x < 1$, then $x^q > x^r$ if and only if $q < r$.

First, suppose that $x > 1$. We have

$$\begin{aligned} x^q > x^r &\iff x^q x^{-r} > x^r x^{-r} && \text{because } x^{-r} > 0 \text{ by (a)} \\ &\iff x^{q-r} > x^0 && \text{by (b)} \\ &\iff x^{q-r} > 1 \\ &\iff (x^{ad-bc})^{1/bd} > 1 && \text{by Definition 5.6.7} \\ &\iff (x^{ad-bc})^{1/bd} > 1^{1/bd} && \text{by Lemma 5.6.6(e)} \\ &\iff x^{ad-bc} > 1 && \text{by Lemma 5.6.6(d)} \\ &\iff x^{ad-bc} > 1^{ad-bc}. \end{aligned}$$

Given that $x > 1 > 0$, $x^{ad-bc} > 1^{ad-bc}$ if and only $ad - bc > 0$ by Proposition 5.6.3.

[10] The proposition states two implications, but it is straightforward to see that the second one can be upgraded to an equivalence by considering the contrapositive of the first one. Observe that when n is positive, $x \leq y$ implies that $x^n \leq y^n$, and $x > y$ implies that $x^n > y^n$; the contrapositive of the former implication is: $x^n > y^n$ implies that $x > y$; so $x > y$ if and only if $x^n > y^n$.

So

$$\begin{aligned} x^q > x^r &\iff ad - bc > 0 \\ &\iff (ad - bc)/(bd) > 0 \\ &\iff q - r > 0 \\ &\iff q > r. \end{aligned}$$

Second, suppose that $x < 1$. In this case,

$$\begin{aligned} x^q > x^r &\iff x^q x^{-r} > x^r x^{-r} && \text{because } x^{-r} > 0 \text{ by (a)} \\ &\iff x^{q-r} > x^0 && \text{by (b)} \\ &\iff x^{q-r} > 1 \\ &\iff (x^{ad-bc})^{1/bd} > 1 && \text{by Definition 5.6.7} \\ &\iff (x^{ad-bc})^{1/bd} > 1^{1/bd} && \text{by Lemma 5.6.6(e)} \\ &\iff x^{ad-bc} > 1 && \text{by Lemma 5.6.6(d)} \\ &\iff x^{ad-bc} > 1^{ad-bc}. \end{aligned}$$

Given that $0 < 1 < x$, $x^{ad-bc} > 1^{ad-bc}$ if and only $ad - bc < 0$ by Proposition 5.6.3. So

$$\begin{aligned} x^q > x^r &\iff ad - bc < 0 && \text{by Proposition 5.6.3 since } 0 < 1 < x \\ &\iff (ad - bc)/(bd) < 0 \\ &\iff q - r < 0 \\ &\iff q < r. \end{aligned}$$

(f) $(xy)^q = x^q y^q$.

We have:

$$\begin{aligned} x^q y^q &= (x^{1/b})^a (y^{1/b})^a && \text{by Definition 5.6.7} \\ &= (x^{1/b} y^{1/b})^a && \text{by Proposition 5.6.3} \\ &= ((xy)^{1/b})^a && \text{by Lemma 5.6.6(f)} \\ &= (xy)^{a/b} && \text{by Definition 5.6.7} \\ &= (xy)^q \end{aligned}$$

as desired.

Exercise 5.6.3

Let x be a real number and n an even natural number. There exists a natural number m such that $n = 2m$. Thus, $x^n = x^{2m}$. By Proposition 5.6.3, $x^n = (x^2)^m$.

If $x \leq 0$, then $-x \geq 0$. So, $(-x)(-x) \geq 0$, so that $x^2 \geq 0$. If $x \geq 0$, then $x^2 \geq 0$. So, in any case, $x^2 \geq 0$. By Proposition 5.6.3, this implies that $(x^2)^m \geq 0$.

Therefore, $x^n \geq 0$.

Exercise 5.6.4

Let $x \in \mathbf{R}$. Recall from Exercise 5.6.3 that $x^2 \geq 0$, and so $(x^2)^{1/2}$ is well-defined. We distinguish three cases.

First, suppose that $x > 0$. Then, $|x| = x$. Moreover, we have

$$(x^2)^{1/2} = x^{2/2} = x^1 = x$$

by Definition 5.6.7. Therefore, $|x| = (x^2)^{1/2}$.

Second, suppose that $x = 0$. Then, $|x| = 0$. We have

$$(x^2)^{1/2} = (0^2)^{1/2} = 0^{1/2} = \sup\{z \in \mathbf{R} : z \geq 0 \text{ and } z^2 \leq 0\}.$$

For any real $z \geq 0$, $z^2 \leq 0$ yields $z = 0$, because if $z > 0$, then $z^2 > 0$ by Proposition 5.6.3. It follows that $\{z \in \mathbf{R} : z \geq 0 \text{ and } z^2 \leq 0\} = \{0\}$, so that $0^{1/2} = \sup\{0\} = 0$. Hence, $|x| = (x^2)^{1/2}$.

Third, suppose that $x < 0$. Then, $|x| = -x$. In addition, we see with Lemma 5.6.9(b) that

$$(x^2)^{1/2} = \left((-|x|)^2\right)^{1/2} = \left((-1)^2|x|^2\right)^{1/2} = (|x|^2)^{1/2} = |x|^{2/2} = |x|^1 = |x|,$$

as desired.

So in all cases, $|x| = (x^2)^{1/2}$.

Exercise 5.6.5

Let x, y be positive reals and q a positive rational. By considering the contrapositive of Lemma 5.6.9(d), it is easy to see that

$$x^q \leq y^q \iff x \leq y,\ x^q < y^q \iff x < y,\ x^q \geq y^q \iff x \geq y, \text{ and } x^q > y^q \iff x > y.$$

In light of these equivalences, we have:

$$\max(x^q, y^q) := \begin{cases} x^q, & x^q \geq y^q \\ y^q, & x^q < y^q \end{cases} = \begin{cases} x^q, & x \geq y \\ y^q, & x < y \end{cases} = \max(x, y)^q,$$

and

$$\min(x^q, y^q) := \begin{cases} x^q, & x^q \leq y^q \\ y^q, & x^q > y^q \end{cases} = \begin{cases} x^q, & x \leq y \\ y^q, & x > y \end{cases} = \min(x, y)^q.$$

At no point did we assume that $q \geq 1$; therefore, these results hold whether $q \geq 1$ or $q < 1$.

Chapter 6
Limits of Sequences

6.1 Convergence and Limit Laws

Exercise 6.1.1

Let $(a_n)_{n=0}^{\infty}$ be a sequence of real numbers such that $a_{n+1} > a_n$ for each natural number n.

Let k be a natural number. We will show by induction on the natural number j starting at the base case $k+1$ that $a_j > a_k$ for all natural numbers $j > k$. The base case ($j = k+1$) is readily validated because $a_{k+1} > a_k$. Now, suppose inductively that $a_j > a_k$ for some natural number $j > k$. We know that $a_{j+1} > a_j$. Hence, $a_{j+1} > a_k$, and the induction is completed.

We conclude that $a_m > a_n$ for any natural numbers $m > n$.

Exercise 6.1.2

Let $(a_n)_{n=m}^{\infty}$ be a sequence of real numbers, and let L be a real number.

The sequence $(a_n)_{n=m}^{\infty}$ converges to L if and only if it is eventually ε-close to L for any real $\varepsilon > 0$ (cf. Definition 6.1.5). Saying that $(a_n)_{n=m}^{\infty}$ is eventually ε-close to L for any real $\varepsilon > 0$ is equivalent to saying that: for any real $\varepsilon > 0$, $(a_n)_{n=N}^{\infty}$ is ε-close to L for some $N \geq m$ (cf. Definition 6.1.5). Finally, this just means that for any $\varepsilon > 0$,

$|a_n - L| \leq \varepsilon$ for all $n \geq N$ for some $N \geq m$ (cf. Definition 6.1.5).

In summary, the sequence $(a_n)_{n=m}^{\infty}$ converges to L if and only if, for any real $\varepsilon > 0$, there is an integer $N \geq m$ such that $|a_n - L| \leq \varepsilon$ for all $n \geq N$.

Exercise 6.1.3

Let $(a_n)_{n=m}^{\infty}$ be a sequence of real numbers, let c be a real number, and let $m' \geq m$ be an integer.

Suppose that $(a_n)_{n=m}^{\infty}$ converges to c. Take an arbitrary real $\varepsilon > 0$. By Definition 6.1.5 (also see Exercise 6.1.2), there exists an integer $N \geq m$ such that $|a_n - c| \leq \varepsilon$ for all $n \geq N$. Let $M := \max(N, m')$. Then, $M \geq m'$ and $|a_n - c| \leq \varepsilon$ for all $n \geq M$. Given that ε is arbitrary, this means that $(a_n)_{n=m'}^{\infty}$ converges to c by Definition 6.1.5.

Similarly, suppose that $(a_n)_{n=m'}^{\infty}$ converges to c and take an arbitrary real $\varepsilon > 0$. By Definition 6.1.5, there exists an integer $N \geq m'$ such that $|a_n - c| \leq \varepsilon$ for all $n \geq N$. Since we also have $N \geq m$ because $m' \geq m$, this implies that $(a_n)_{n=m}^{\infty}$ also converges to c.

In conclusion, $(a_n)_{n=m}^{\infty}$ converges to c if and only if $(a_n)_{n=m'}^{\infty}$ converges to c.

Exercise 6.1.4

Let $(a_n)_{n=m}^{\infty}$ be a sequence of real numbers, let c be a real number, and let $k \geq 0$ be a non-negative integer.

On the one hand, suppose that $(a_{n+k})_{n=m}^{\infty}$ converges to c. Let ε be an arbitrary positive real. By Definition 6.1.5, there is an integer $N \geq m$ such that $|a_{n+k} - c| \leq \varepsilon$ for all $n \geq N$. It follows that $|a_n - c| \leq \varepsilon$ for all $n \geq N + k$. Let $M := N + k$. Then, $M \geq m$ and $|a_n - c| \leq \varepsilon$ for all $n \geq M$. Since ε is arbitrary, we deduce that $(a_n)_{n=m}^{\infty}$ converges to c by Definition 6.1.5.

On the other hand, suppose that $(a_n)_{n=m}^{\infty}$ converges to c. Take an arbitrary real $\varepsilon > 0$. By Definition 6.1.5, there is an integer $N \geq m$ such that $|a_n - c| \leq \varepsilon$ for all $n \geq N$. Note that for any $n \geq N$, $n + k \geq N$. So, we also have $|a_{n+k} - c| \leq \varepsilon$ for all $n \geq N$. Hence, $(a_{n+k})_{n=m}^{\infty}$ converges to c according to Definition 6.1.5.

Therefore, $(a_n)_{n=m}^{\infty}$ converges to c if and only if $(a_{n+k})_{n=m}^{\infty}$ converges to c.

Exercise 6.1.5

Let $(a_n)_{n=m}^{\infty}$ be a convergent sequence of real numbers. Define $L := \lim_{n\to\infty} a_n$.

Let ε be an arbitrary positive real. There is an integer $N \geq m$ such that $|a_n - L| \leq \varepsilon/2$ for all $n \geq N$, since $(a_n)_{n=m}^{\infty}$ and L are eventually $(\varepsilon/2)$-close by Definition 6.1.5. For any $j, k \geq N$,

$$\begin{aligned} |a_j - L| \leq \varepsilon/2 \text{ and } |a_k - L| \leq \varepsilon/2 &\implies |a_j - L| + |a_k - L| \leq \varepsilon \\ &\implies |a_j - a_k| \leq \varepsilon \end{aligned}$$

by the triangle inequality. Hence, $(a_n)_{n=m}^{\infty}$ is eventually ε-steady, We conclude that $(a_n)_{n=m}^{\infty}$ is a Cauchy sequence according to Definition 6.1.3.

Exercise 6.1.6

Let $(a_n)_{n=1}^{\infty}$ be a Cauchy sequence of rationals, and write $L := \mathrm{LIM}_{n\to\infty} a_n$. We will prove that $(a_n)_{n=1}^{\infty}$ converges to L.

Let ε be a positive real number. Suppose for sake of contradiction that the sequence $(a_n)_{n=1}^{\infty}$ is not eventually ε-close to L.

Since $(a_n)_{n=1}^{\infty}$ is a Cauchy sequence, it is eventually $(\varepsilon/2)$-steady; that is, there exists an integer $N \geq 1$ such that $|a_j - a_k| \leq \varepsilon/2$ for all $j, k \geq N$ (cf. Definition 6.1.3)[1]. Since $(a_n)_{n=1}^{\infty}$ is not eventually ε-close to L, there exists an integer $m \geq N$ such that $|a_m - L| > \varepsilon$. Using the triangular inequality, we have $|a_n - L| + |a_m - a_n| \geq |a_m - L|$ for all $n \geq N$. Now, for all $n \geq N$,

$$\begin{aligned} |a_n - L| + |a_m - a_n| \geq |a_m - L| &\implies |a_n - L| \geq |a_m - L| - |a_m - a_n| \\ &\implies |a_n - L| > \varepsilon - \varepsilon/2 \\ &\implies |a_n - L| > \varepsilon/2 \\ &\implies a_n > L + \varepsilon/2 \text{ or } a_n < L - \varepsilon/2. \end{aligned}$$

Observe that if there are some distinct integers $j, k \geq N$ such that $a_j > L + \varepsilon/2$ and $a_k < L - \varepsilon/2$, then $|a_j - a_k| \geq a_j - a_k > \varepsilon > \varepsilon/2$, a contradiction. So, either $a_n > L + \varepsilon/2$ for all $n \geq N$, or $a_n < L - \varepsilon/2$ for all $n \geq N$.

If $a_n > L + \varepsilon/2$ for all $n \geq N$, define the sequence $(b_n)_{n=1}^{\infty}$ by $b_n := L + \varepsilon/2$ if $n < N$ and $b_n := a_n$ if $n \geq N$. Then, $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ are clearly equivalent, and so $\mathrm{LIM}_{n\to\infty} b_n = \mathrm{LIM}_{n\to\infty} a_n = L$. We have $b_n \geq L + \varepsilon/2$ for all $n \geq 1$. Using Exercise

[1] This is the last time this definition is invoked explicitly. It should be clear from now on.

5.4.8, we see that $\mathrm{LIM}_{n\to\infty} b_n \geq L+\varepsilon/2$, i.e., $L \geq L+\varepsilon/2$. But that is a contradiction since $\varepsilon/2 > 0$. If instead $a_n < L-\varepsilon/2$ for all $n \geq N$, define the sequence $(c_n)_{n=1}^{\infty}$ by $c_n := L-\varepsilon/2$ if $n < N$ and $c_n := a_n$ if $n \geq N$. Clearly, $(a_n)_{n=1}^{\infty}$ and $(c_n)_{n=1}^{\infty}$ are equivalent. Thus, $\mathrm{LIM}_{n\to\infty} c_n = \mathrm{LIM}_{n\to\infty} a_n = L$. Given that $c_n \leq L-\varepsilon/2$ for all $n \geq 1$, we can use Exercise 5.4.8 to deduce that $\mathrm{LIM}_{n\to\infty} c_n \leq L-\varepsilon/2$, that is $L \leq L-\varepsilon/2$. But this is contradictory as $-\varepsilon/2 < 0$.

These two contradictions show that our supposition is false. Therefore, $(a_n)_{n=1}^{\infty}$ is eventually ε-close to L. Since ε is arbitrary, it means that $(a_n)_{n=1}^{\infty}$ converges to L by Definition 6.1.5. In other words,

$$\lim_{n\to\infty} a_n = \mathrm{LIM}_{n\to\infty} a_n.$$

Exercise 6.1.7

Let $(a_n)_{n=m}^{\infty}$ be a sequence of rationals bounded by a rational M, in the sense of Definition 5.1.12. By Definition 5.1.12,

$$|a_n| \leq M \text{ for all } n \geq m. \qquad (\star)$$

As $\mathbf{Q} \subseteq \mathbf{R}$, M is also real. Hence, $(\star)$ shows that $(a_n)_{n=m}^{\infty}$ is also bounded by M in the sense of Definition 6.1.16.

Let $(b_n)_{n=m}^{\infty}$ be a sequence of reals bounded by a real N, in the sense of Definition 6.1.16. We have $|b_n| \leq N$ for all $n \geq m$. By Proposition 5.4.12 (bounding of reals by rationals), there exists an integer K such that $N \leq K$. Hence,

$$|b_n| \leq K \text{ for all } n \geq m.$$

As K is a rational number, $(b_n)_{n=m}^{\infty}$ is also bounded by K in the sense of Definition 5.1.12.

In sum, Definition 6.1.16 is consistent with Definition 5.1.12.

Exercise 6.1.8

Let $(a_n)_{n=m}^{\infty}$ and $(b_n)_{n=m}^{\infty}$ be convergent sequences of real numbers, and let x, y be the real numbers $x := \lim_{n\to\infty} a_n$ and $y := \lim_{n\to\infty} b_n$. We have to prove the following eight statements.

(a) The sequence $(a_n+b_n)_{n=m}^{\infty}$ converges to $x+y$.

Let ε be a positive real number. By Definition 6.1.5[2], $(a_n)_{n=m}^{\infty}$ and x are eventually $(\varepsilon/2)$-close, and $(b_n)_{n=m}^{\infty}$ and y are eventually $(\varepsilon/2)$-close. So, there exist integers N_1 and N_2 such that $(a_n)_{n=N_1}^{\infty}$ and x are $(\varepsilon/2)$-close, and $(b_n)_{n=N_2}^{\infty}$ and y are $(\varepsilon/2)$-close. Let $N := \max(N_1, N_2)$. For any $n \geq N$,

$$|a_n - x| \leq \varepsilon/2 \quad \text{and} \quad |b_n - y| \leq \varepsilon/2,$$

so

$$|a_n - x + b_n - y| \leq |a_n - x| + |b_n - y| \leq \varepsilon,$$

that is

$$|(a_n + b_n) - (x+y)| \leq \varepsilon.$$

Thus, $(a_n+b_n)_{n=m}^{\infty}$ is eventually ε-close to $x+y$. As ε is an arbitrary positive real, we conclude that $(a_n+b_n)_{n=m}^{\infty}$ converges to $x+y$. In other words,

$$\lim_{n\to\infty}(a_n+b_n) = \lim_{n\to\infty} a_n + \lim_{n\to\infty} b_n.$$

(b) The sequence $(a_n b_n)_{n=m}^{\infty}$ converges to xy.

Consider an arbitrary positive real ε. As $(a_n)_{n=m}^{\infty}$ and $(b_n)_{n=m}^{\infty}$ are convergent sequences, they are bounded by some positive real numbers X and Y respectively (see Corollary 6.1.17). Define $M := X + Y + \varepsilon + 1$. Then, $(a_n)_{n=m}^{\infty}$ and $(b_n)_{n=m}^{\infty}$ are both bounded by M.

Since $(a_n)_{n=m}^{\infty}$ converges to x, there exists an integer $N_1 \geq m$ such that $(a_n)_{n=N_1}^{\infty}$ is $(\varepsilon/(3M))$-close to x. Similarly, there is an integer $N_2 \geq m$ such that $(b_n)_{n=N_2}^{\infty}$ is $(\varepsilon/(3M))$-close to y. Let $N := \max(N_1, N_2)$. Then, for all $n \geq N$, $|a_j - x| \leq \varepsilon/(3M)$ and $|b_j - y| \leq \varepsilon/(3M)$. Now, by Proposition 4.3.7(h) applied to reals, we have:

$$|a_j b_j - xy| \leq (\varepsilon/(3M))|b_j| + (\varepsilon/(3M))|a_j| + (\varepsilon^2/(9M^2))$$

so that

$$|a_j b_j - xy| \leq (\varepsilon/(3M))M + (\varepsilon/(3M))M + (\varepsilon^2/(9M^2))$$

so that

$$|a_j b_j - xy| \leq \varepsilon/3 + \varepsilon/3 + (\varepsilon^2/(9M^2))$$

[2] This definition will be used without explicit mention from now on.

for all $j, k \geq N$. The right-hand side simplifies to $\varepsilon(2/3 + \varepsilon/(9M^2))$. Observe that:

$$\begin{aligned} M \geq 1 \text{ and } M \geq \varepsilon &\implies M^2 \geq \varepsilon \\ &\implies 9M^2 \geq 9\varepsilon \\ &\implies 1/(9M^2) \leq 1/(9\varepsilon) \\ &\implies \varepsilon/(9M^2) \leq 1/9 \\ &\implies 2/3 + \varepsilon/(9M^2) \leq 7/9 \\ &\implies 2/3 + \varepsilon/(9M^2) \leq 1 \\ &\implies \varepsilon(2/3 + \varepsilon/(9M^2)) \leq \varepsilon. \end{aligned}$$

Thus, for all $j, k \geq N$,

$$|a_j b_j - xy| \leq \varepsilon.$$

So, $(a_n b_n)_{n=1}^{\infty}$ is eventually ε-close to xy.

Therefore, $(a_n b_n)_{n=1}^{\infty}$ converges to xy. That is to say,

$$\lim_{n\to\infty}(a_n b_n) = \left(\lim_{n\to\infty} a_n\right)\left(\lim_{n\to\infty} b_n\right).$$

(c) For any real number c, the sequence $(ca_n)_{n=m}^{\infty}$ converges to cx.

Let c be a real number. Define the sequence $(c_n)_{n=1}^{\infty}$ by $c_n := c$ for all $n \geq m$. As $|c_n - c| = 0$ for all $n \geq m$, $(c_n)_{n=1}^{\infty}$ is ε-close to c for any real $\varepsilon > 0$. This means that $\lim_{n\to\infty} c_n = c$. By (b),

$$\lim_{n\to\infty}(c_n a_n) = \left(\lim_{n\to\infty} c_n\right)\left(\lim_{n\to\infty} a_n\right).$$

So,

$$\lim_{n\to\infty}(ca_n) = c \lim_{n\to\infty} a_n.$$

(d) The sequence $(a_n - b_n)_{n=m}^{\infty}$ converges to $x - y$.

By (c), the sequence $(-b_n)_{n=1}^{\infty}$ converges, and it converges to $-y$. By (a), the sequence $(a_n + (-b_n))_{n=1}^{\infty}$ converges, and it converges to $x + (-y)$. In other words,

$$\lim_{n\to\infty}(a_n + (-b_n)) = \lim_{n\to\infty} a_n + (-\lim_{n\to\infty} b_n),$$

which simplifies to

$$\lim_{n\to\infty}(a_n - b_n) = \lim_{n\to\infty} a_n - \lim_{n\to\infty} b_n.$$

(e) Suppose that $y \neq 0$, and that $b_n \neq 0$ for all $n \geq m$. Then the sequence $(b_n^{-1})_{n=m}^{\infty}$ converges to y^{-1}.

Suppose that $y \neq 0$ and $b_n \neq 0$ for all $n \geq m$. Then for any $n \geq m$, b_n^{-1} is a real number, and so $(b_n^{-1})_{n=m}^{\infty}$ is a sequence of real numbers.

Since $y \neq 0$, $(b_n)_{n=m}^{\infty}$ is not eventually ε-close to 0 for some $\varepsilon > 0$. Let us fix that ε. Given that $(b_n)_{n=m}^{\infty}$ is convergent, it is also a Cauchy sequence by Proposition 6.1.12. Hence, $(b_n)_{n=m}^{\infty}$ is eventually $(\varepsilon/2)$-steady. That is, there exists an integer $N \geq m$ such that for all $j, k \geq N$, we have $|b_j - b_k| \leq \varepsilon/2$. However, there is some $n_0 \geq N$ such that $|b_{n_0}| > \varepsilon$; otherwise $(b_n)_{n=N}^{\infty}$ would be contradictorily ε-close to 0. For all $n \geq N$, $|b_n - b_{n_0}| \leq \varepsilon/2$. By the triangle inequality, $|b_{n_0}| \leq |b_n - b_{n_0}| + |b_n|$, which leads to $|b_n| \geq |b_{n_0}| - |b_n - b_{n_0}|$. As $|b_{n_0}| - |b_n - b_{n_0}| \geq \varepsilon - \varepsilon/2$, we have $|b_n| \geq \varepsilon/2$ for all $n \geq N$.

Now, let $\gamma > 0$ be an arbitrary positive real. Since $y = \lim_{n\to\infty} b_n$, the sequence $(b_n)_{n=m}^{\infty}$ is eventually $(|y|\gamma\varepsilon/2)$-close to y (note that $|y| > 0$ because $y \neq 0$). Thus, there exists an integer $M \geq m$ such that $|b_n - y| \leq |y|\gamma\varepsilon/2$ for all $n \geq M$. Let $P := \max(N, M)$. For all $n \geq P$,

$$|b_n^{-1} - y^{-1}| = \left|\frac{1}{b_n} - \frac{1}{y}\right| = \frac{|b_n - y|}{|b_n y|};$$

next,

$$\frac{|b_n - y|}{|b_n y|} \leq \frac{|y|\gamma\varepsilon/2}{|b_n||y|},$$

and since $1/|b_n| \leq 1/(\varepsilon/2)$, we have

$$\frac{|y|\gamma\varepsilon/2}{|b_n||y|} \leq \frac{|y|\gamma\varepsilon/2}{(\varepsilon/2)|y|},$$

hence

$$\frac{|y|\gamma\varepsilon/2}{|b_n||y|} \leq \gamma,$$

so that

$$|b_n^{-1} - y^{-1}| \leq \gamma.$$

Thus, $(b_n^{-1})_{n=m}^{\infty}$ and y^{-1} are eventually γ-close. Since γ is arbitrary, this means that $(b_n^{-1})_{n=m}^{\infty}$ converges to y^{-1}. In other words,

$$\lim_{n\to\infty} b_n^{-1} = \left(\lim_{n\to\infty} b_n\right)^{-1}.$$

(f) Suppose that $y \neq 0$, and that $b_n \neq 0$ for all $n \geq m$. Then the sequence $(a_n/b_n)_{n=m}^{\infty}$ converges to x/y.

Suppose that $y \neq 0$ and $b_n \neq 0$ for all $n \geq m$. Then for any $n \geq m$, a_n/b_n is a real number, and so $(a_n b_n^{-1})_{n=m}^{\infty}$ is a sequence of real numbers. By (e), the sequence $(b_n^{-1})_{n=m}^{\infty}$ converges, and it converges to y^{-1}. By (b), the sequence $(a_n b_n^{-1})_{n=m}^{\infty}$ converges, and it converges to xy^{-1}. That is, the sequence $(a_n/b_n)_{n=m}^{\infty}$ converges and it converges to x/y. That is to say,

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \frac{\lim_{n\to\infty} a_n}{\lim_{n\to\infty} b_n}.$$

(g) The sequence $(\max(a_n, b_n))_{n=m}^{\infty}$ converges to $\max(x, y)$.

Let ε be an arbitrary positive real. Since $(a_n)_{n=m}^{\infty}$ and $(b_n)_{n=m}^{\infty}$ converge respectively to x and y, there exist integers N_1 and N_2 such that $|a_n - x| \leq \varepsilon$ for all $n \geq N_1$ and $|b_n - y| \leq \varepsilon$ for all $n \geq N_2$. Let $N := \max(N_1, N_2)$. For all $n \geq N$,

$$-\varepsilon + x \leq a_n \leq \varepsilon + x \quad \text{and} \quad -\varepsilon + y \leq b_n \leq \varepsilon + y. \qquad (\star)$$

Consider an integer $n \geq N$. We have:

$$\max(a_n, b_n) = \begin{cases} a_n, & a_n \geq b_n \\ b_n, & a_n < b_n \end{cases}.$$

Thus, it follows from $(\star)$ that $-\varepsilon + x \leq \max(a_n, b_n) \leq \varepsilon + x$ if $a_n \geq b_n$, and also that $-\varepsilon + y \leq \max(a_n, b_n) \leq \varepsilon + y$ if $a_n < b_n$. If $a_n \geq b_n$, then $\max(a_n, b_n) = a_n \leq \varepsilon + x$ implies that $\max(a_n, b_n) \leq \max(\varepsilon + x, \varepsilon + y)$; also, $\max(a_n, b_n) = a_n$ implies that $\max(a_n, b_n) \geq -\varepsilon + x$ and $\max(a_n, b_n) \geq b_n$ implies that $\max(a_n, b_n) \geq -\varepsilon + y$, and so $\max(a_n, b_n) \geq \max(-\varepsilon + x, -\varepsilon + y)$. If $a_n < b_n$, then $\max(a_n, b_n) = b_n \leq \varepsilon + y$ implies that $\max(a_n, b_n) \leq \max(\varepsilon + x, \varepsilon + y)$; also, $\max(a_n, b_n) = b_n$ implies that $\max(a_n, b_n) \geq -\varepsilon + y$ and $\max(a_n, b_n) \geq a_n$ implies that $\max(a_n, b_n) \geq -\varepsilon + x$, so that $\max(a_n, b_n) \geq \max(-\varepsilon + x, -\varepsilon + y)$. So, we have:

$$\max(-\varepsilon + x, -\varepsilon + y) \leq \max(a_n, b_n) \leq \max(\varepsilon + x, \varepsilon + y).$$

It follows that $-\varepsilon + \max(x, y) \leq \max(a_n, b_n) \leq \varepsilon + \max(x, y)$ (see Exercise 5.4.9(ii)). Thus, $-\varepsilon \leq \max(a_n, b_n) - \max(x, y) \leq \varepsilon$, so that

$$|\max(a_n, b_n) - \max(x, y)| \leq \varepsilon.$$

Hence, $\max(a_n, b_n)$ is ε-close to $\max(x, y)$. Therefore, the sequence $(\max(a_n, b_n))_{n=m}^{\infty}$ is eventually ε-close to $\max(x, y)$. Given that ε is arbitrary, this means that

$$\lim_{n\to\infty} \max(a_n, b_n) = \max(x, y).$$

(h) The sequence $(\min(a_n, b_n))_{n=m}^{\infty}$ converges to $\min(x, y)$.

We can go through an argument analogous to the one in (g) to obtain

$$\lim_{n\to\infty} \min(a_n, b_n) = \min(x, y).$$

We can also observe with Exercise 5.4.9 that $\min(a_n, b_n) = -\max(-a_n, -b_n)$ for all $n \geq m$. By (c), the sequences $(-a_n)_{n=m}^{\infty}$ and $(-b_n)_{n=m}^{\infty}$ are both convergent, each one converting to $-x$ and $-y$ respectively. So, the sequence $(\max(-a_n, -b_n))_{n=m}^{\infty}$ converges to $\max(-x, -y)$ by (g). It follows from (c) again that $(-\max(-a_n, -b_n))_{n=m}^{\infty}$ converges to $-\max(-x, -y)$. In other words, the sequence $(\min(a_n, b_n))_{n=m}^{\infty}$ converges to $\min(x, y)$.

Exercise 6.1.9

When the limit of the denominator is 0, the right-hand side of Theorem 6.1.19(f) is undefined since the division of a real number by 0 is undefined.

Exercise 6.1.10

Let $(a_n)_{n=m}^{\infty}$ and $(b_n)_{n=m}^{\infty}$ be sequences of real numbers.

Suppose that $(a_n)_{n=m}^{\infty}$ and $(b_n)_{n=m}^{\infty}$ are eventually δ-close for every rational $\delta > 0$. Let $\varepsilon > 0$ be a positive real. By Proposition 5.4.12, there exists a positive rational number q such that $q \leq \varepsilon$. As $(a_n)_{n=m}^{\infty}$ and $(b_n)_{n=m}^{\infty}$ are eventually q-close, there exists an integer $N \geq 0$ such that $|a_n - b_n| \leq q$ for all $n \geq N$ by Definition 5.2.6. Hence, $|a_n - b_n| \leq \varepsilon$ for all $n \geq N$. So, $(a_n)_{n=m}^{\infty}$ and $(b_n)_{n=m}^{\infty}$ are eventually ε-close for any real $\varepsilon > 0$.

Conversely, suppose that $(a_n)_{n=m}^{\infty}$ and $(b_n)_{n=m}^{\infty}$ are eventually ε-close for any real $\varepsilon > 0$. Then as $\mathbf{Q} \subseteq \mathbf{R}$, $(a_n)_{n=m}^{\infty}$ and $(b_n)_{n=m}^{\infty}$ are also eventually ε-close for any rational $\varepsilon > 0$.

In conclusion, the concept of equivalent sequence, as defined in Definition 5.2.6, does not change if ε is required to be a positive real instead of a positive rational.

6.2 The Extended Real Number System

Exercise 6.2.1

Let $x, y, z \in \mathbf{R}^*$. We will prove the four following statements.

(a) We have $x \leq x$.

If $x \in \mathbf{R}$, then $x = x$ as equality in $\mathbf{R}$ is reflexive (Proposition 5.3.3), and so $x \leq x$ (Definition 5.4.6). If $x = +\infty$, then $x \leq x$ by Definition 6.2.3. If $x = -\infty$, then $x \leq x$ by Definition 6.2.3. So in any case,

$$x \leq x.$$

(b) Exactly one of the statements $x < y$, $x = y$, or $x > y$ is true.

There are three cases to consider.

First, suppose that both x and y are real numbers. By the order trichotomy of real numbers (Proposition 5.4.7), exactly one of the following statements is true: $x < y$, $x = y$, $x > y$.

Second, suppose that one of x and y, say x, is $+\infty$. Then by Definition 6.2.3, $y \leq x$. If $y = +\infty$, then $y = x$, and by Definition 6.2.3 $y \not< x$ and $y \not> x$ (since $y = x$). If $y \neq +\infty$, then $y \neq x$ and so $y < x$ by Definition 6.2.3; plus, $y \not> x$ by Definition 6.2.3 (since no extended real is strictly greater than $+\infty$). So, when one of x and y is $+\infty$, exactly one of the statements $x < y$, $x > y$ and $x = y$ holds.

Third, suppose that one of x and y, say x, is $-\infty$. Then by Definition 6.2.3, $y \geq x$. If $y = -\infty$, then $y = x$, and by Definition 6.2.3 $y \not< x$ and $y \not> x$ (since $y = x$). If $y \neq -\infty$, then $y \neq x$ and so $y > x$ by Definition 6.2.3; plus, $y \not< x$ by Definition 6.2.3 (since no extended real is strictly less than $-\infty$). So, when one of x and y is $-\infty$, exactly one of the statements $x < y$, $x > y$ and $x = y$ holds.

Therefore, it is always true that exactly one of the following statements holds: $x < y$, $x = y$, and $x > y$.

(c) If $x \leq y$ and $y \leq z$, then $x \leq z$.

Suppose that $x \leq y$ and $y \leq z$. We distinguish five exhaustive cases.

If x, y, z are finite real numbers, then $x \leq z$ by the transitivity of order in $\mathbf{R}$ (Proposition 5.4.7).

If $z = +\infty$, then $x \leq z$ by Definition 6.2.3.

If $x = -\infty$, then $x \leq z$ by Definition 6.2.3.

If $z = -\infty$, then $y \leq z$ implies that $y = -\infty$ by Definition 6.2.3 (note that if y is finite or $+\infty$, then $y > -\infty$). Similarly, $x \leq y$ leads to $x = -\infty$ since $y = -\infty$; and we already know that $x = -\infty$ implies that $x \leq z$.

If $x = +\infty$, then $x \leq y$ implies that $y = +\infty$ by Definition 6.2.3 (note that if y is finite or $-\infty$, then $y < +\infty$), and then $y \leq z$ implies that $z = +\infty$, which yields $x \leq z$ by Definition 6.2.3.

Therefore, we always have $x \leq z$ when $x \leq y$ and $y \leq z$.

(d) If $x \leq y$, then $-y \leq -x$.

Suppose that $x \leq y$. There are five cases to consider.

If x and y are finite reals, then we know that $-x \geq -y$ (see Proposition 5.4.7 and Exercise 4.2.6).

If $x = -\infty$, then $-x = +\infty$ by Definition 6.2.2, and we have $-x \geq -y$ by Definition 6.2.3.

If $x = +\infty$, then it follows from Definition 6.2.3 that $x \leq y$ if and only if $y = +\infty$, and hence $-y = -\infty$. So by Definition 6.2.3, $-x \geq -y$.

If $y = -\infty$, then $x \leq y$ if and only if $x = -\infty$ by Definition 6.2.3. Thus, $-x = +\infty$, and so $-x \geq -y$ by Definition 6.2.3.

If $y = +\infty$, then $-y = -\infty$, and so $-x \geq -y$ by Definition 6.2.3.

We see that in any case, $-x \geq -y$, that is $-y \leq -x$ by Definition 6.2.3.

Exercise 6.2.2

Let E be a subset of $\mathbf{R}^*$.

Suppose that E is empty. Then it is vacuously true that $x \leq \sup(E)$ and $x \geq \inf(E)$ for all $x \in E$. In addition, we know that $\sup(E) = -\infty$ and $\inf(E) = +\infty$ (see Example 6.2.10[3]). So if $M \in \mathbf{R}^*$ is an upper bound for E, then $\sup(E) \leq M$ by Definition 6.2.3. Similarly, if $M \in \mathbf{R}^*$ is a lower bound for E, then $\inf(E) \geq M$ by Definition 6.2.3. For the rest of this exercise, assume that E is non-empty.

[3] We know from Definition 5.5.10 that $\sup(\emptyset) = -\infty$. From Exercise 5.5.1 and Definition 6.2.2, we deduce that $\inf(\emptyset) = +\infty$.

Suppose that $+\infty \in E$. Then, $\sup(E) = +\infty$ by Definition 6.2.6. Also, the only upper bound for E is $+\infty$ as $r < +\infty$ for any other extended real r and $+\infty \in E$. Hence, if $M \in \mathbf{R}^*$ is an upper bound for E, then $M = +\infty$ and hence $\sup(E) \leq M$. In addition, $x \leq \sup(E)$ for every $x \in E$ by Definition 6.2.3.

Suppose that $+\infty \notin E$ and $-\infty \notin E$. Then, E is a non-empty subset of $\mathbf{R}$. Here, $\sup(E)$ is either $+\infty$ or a real number by Definition 5.5.10; plus, $-\infty$ is not an upper bound for E since $x > -\infty$ for any $x \in E$. There are two sub-cases to consider. In the first one, $\sup(E)$ is $+\infty$. Here, E has no finite upper bound (cf. Definition 5.5.10) and so the only upper bound of E is $+\infty$ (note that $+\infty$ is an upper bound for any subset of $\mathbf{R}^*$; see Definition 6.2.3). Thus, if $M \in \mathbf{R}^*$ is an upper bound for E, then $M = +\infty$ and so $\sup(E) \leq M$. Besides, we have $x \leq \sup(E)$ for every $x \in E$, according to Definition 6.2.3. In the second sub-case, $\sup(E)$ is a real number. This means that E has some finite upper bound, and $\sup(E)$ is the least upper bound of E (cf. Definition 5.5.10). So, $x \leq \sup(E)$ for all $x \in E$. In addition, $\sup(E) \leq M$ for any finite upper bound $M \in \mathbf{R}$ by Definition 5.5.5. If M is infinite, then M is an upper bound for E if and only if $M = +\infty$ (since E is non-empty and $-\infty < x$ for all $x \in E$), and we also have $\sup(E) \leq M$ by Definition 6.2.3.

Suppose that $+\infty \notin E$ and $-\infty \in E$. Then, $\sup(E) = \sup(E \setminus \{-\infty\})$ by Definition 6.2.6. If $E \setminus \{-\infty\} = \emptyset$, then $E = \{-\infty\}$ and $\sup(E) = -\infty$. It is then easy to see that $x \leq \sup(E)$ for any $x \in E$ and $x \leq M$ for any upper bound $M \in \mathbf{R}^*$ for E ($-\infty$ is less than or equal to any extended real number; see Definition 6.2.3). If $E \setminus \{-\infty\} \neq \emptyset$, then $E \setminus \{-\infty\}$ is a non-empty subset of $\mathbf{R}$. Consequently, $x \leq \sup(E \setminus \{-\infty\})$ for all $x \in E \setminus \{-\infty\}$ and $\sup(E \setminus \{-\infty\}) \leq M$ for any upper bound $M \in \mathbf{R}^*$ of $E \setminus \{-\infty\}$, according to the previous paragraph. Since $\sup(E \setminus \{-\infty\}) = \sup(E)$ and $-\infty$ is less than or equal to any extended real, we also have: $x \leq \sup(E)$ for all $x \in E$. Furthermore, any upper bound for E is an upper bound for $E \setminus \{-\infty\}$, so we also have: $\sup(E) \leq M$ for any upper bound $M \in \mathbf{R}^*$ of E.

We see that if E is non-empty, it is always true that $x \leq \sup(E)$ for all $x \in E$ and $\sup(E) \leq M$ for any upper bound $M \in \mathbf{R}^*$. Now, let $M \in \mathbf{R}^*$ be a lower bound for E. Then for all $x \in E$, $M \leq x$, so that $-M \geq -x$. Hence, $-M$ is an upper bound for $-E$. It follows that $\sup(-E) \leq -M$, using the same argument we employed to show that $\sup(E)$ is less than or equal to any upper bound for E. So, $-\sup(-E) \geq M$. That is, $\inf(E) \geq M$ by Definition 6.2.6. Apart from this, note that for any $-x \in -E$, we have $-x \leq \sup(-E)$ (using the same argument we employed for E in the previous paragraph), and so $x \geq -\sup(-E)$, i.e., $x \geq \inf(E)$.

In conclusion, it is always true that

(a) For every $x \in E$ we have $x \leq \sup(E)$ and $x \geq \inf(E)$.

(b) We have $\sup(E) \leq M$ for any upper bound $M \in \mathbf{R}^*$.

(c) We have $\inf(E) \geq M$ for any lower bound $M \in \mathbf{R}^*$.

6.3 Suprema and Infima of Sequences

Exercise 6.3.1

Let $(a_n)_{n=1}^{\infty}$ be the sequence defined by $a_n = 1/n$. So, $(a_n)_{n=1}^{\infty}$ is the sequence $1, 1/2, 1/3, 1/4, \ldots$. Let A be the set $A := \{a_n : n \geq 1\}$, so $A = \{1, 1/2, 1/3, 1/4, \ldots\}$. Note that $\sup(a_n)_{n=1}^{\infty} = \sup(A)$ and $\inf(a_n)_{n=1}^{\infty} = \inf(A)$ by Definition 6.3.1.

For any integer $n \geq 1$, we have $1/n \leq 1$. Hence, 1 is an upper bound for A. Furthermore, since 1 is in A, 1 is less than or equal to any upper bound for A. Consequently, 1 is the least upper bound for A, that is $\sup(A) = 1$.

Since $x > 0$ for all $x \in A$, 0 is a lower bound for A. Hence, $\inf(A) \geq 0$. Suppose for the sake of contradiction that $\inf(A) > 0$. If $\inf(A)$ is infinite, then $\inf(A) = +\infty$. But this is contradictory because $1 \geq \inf(A)$ as $1 \in A$. If $\inf(A)$ is finite, then there exists a positive integer N such that $1/N < \inf(A)$ (by the Archimedean property). But $1/N \in A$ since $a_N \in A$, and so $1/N \geq \inf(A)$, a contradiction. These two contradictions show that $\inf(A) \not> 0$. Therefore, $\inf(A) = 0$.

In conclusion,

$$\sup(a_n)_{n=1}^{\infty} = 1 \quad \text{and} \quad \inf(a_n)_{n=1}^{\infty} = 0.$$

Exercise 6.3.2

Let $(a_n)_{n=m}^{\infty}$ be a sequence of real numbers, and let x be the extended real number $x := \sup(a_n)_{n=m}^{\infty}$.

By Definition 6.3.1, $x = \sup\{a_n : n \geq m\}$. For all $n \geq m$, $a_n \leq \sup\{a_n : n \geq m\}$ by Theorem 6.2.11(a); in other words, $a_n \leq x$.

Let $M \in \mathbf{R}^*$. Suppose that M is an upper bound for $(a_n)_{n=m}^{\infty}$. Then $a_n \leq M$ for all $n \geq m$. This also means that M is an upper bound for the set $\{a_n : n \geq m\}$. Thus, $\sup\{a_n : n \leq m\} \leq M$ by Theorem 6.2.11(b). That is, $x \leq M$.

Let y be an extended real number for which $y < x$. Note that such a number always exists because $x \neq -\infty$ (since $\{a_n : n \geq m\}$ is a non-empty subset of $\mathbf{R}$), and for any extended real number different from $-\infty$, we can always find another extended real number (such as $-\infty$) that is strictly less than it. Since x is the supremum of $\{a_n : n \geq m\}$ and $y < x$, y is not an upper bound for $\{a_n : n \geq m\}$ by Theorem 6.2.11(b). Hence, there exists some $n \geq m$ for which $y < a_n$. We know that $a_n \leq x$. Therefore, $y < a_n \leq x$.

Exercise 6.3.3

Let $(a_n)_{n=m}^{\infty}$ be a sequence of real numbers which has some finite upper bound $M \in \mathbf{R}$, and which is also increasing. Let $x := \sup(a_n)_{n=m}^{\infty}$. Observe that $x \neq -\infty$ because $\{a_n : n \geq m\}$ is a non-empty subset of $\mathbf{R}$. In addition, $x \leq M$ by Proposition 6.3.6, and so $x \neq +\infty$. Hence, x is a finite real number.

Let ε be a positive real number. By Proposition 6.3.6, there exists an integer N for which $x - \varepsilon < a_N \leq x$. For all $n \geq N$, $a_n \leq x$ by Proposition 6.3.6, and $a_N \leq a_n$ since $(a_n)_{n=m}^{\infty}$ is increasing[4]. Thus, $x - \varepsilon < a_n \leq x$ for all $n \geq N$. This implies that for all $n \geq N$, $0 \leq x - a_n < \varepsilon$, so that $|a_n - x| \leq \varepsilon$.

So, the sequence $(a_n)_{n=m}^{\infty}$ is eventually ε-close to x. Since ε is arbitrary, this means that $(a_n)_{n=m}^{\infty}$ converges to x. In other words,

$$\lim_{n\to\infty} a_n = \sup(a_n)_{n=m}^{\infty}.$$

Exercise 6.3.4

Proposition 6.3.10 fails when $x > 1$ because the sequence $(x^n)_{n=1}^{\infty}$ diverges when $x > 1$. To see this, assume that $x > 1$ and suppose, for the sake of contradiction, that $(x^n)_{n=1}^{\infty}$ converges to some limit L. We have $0 < 1/x < 1$. By Proposition 6.3.10, $\lim_{n\to\infty}(1/x)^n = 0$. Using Theorem 6.1.19(b), we have

$$\lim_{n\to\infty} x^n(1/x)^n = L \times 0 = 0. \qquad (\star)$$

But $x^n(1/x)^n = x^n/x^n = 1$. So $(\star)$ implies that

$$\lim_{n\to\infty} 1 = 0,$$

which is contradictory since $\lim_{n\to\infty} 1 = 1 \neq 0$. So, the sequence $(x^n)_{n=1}^{\infty}$ diverges when $x > 1$.

There are two flaws with Example 1.2.3. The first is assuming that the limit of x^n exists by writing $L = \lim_{n\to\infty} x^n$ (we had to first establish convergence, or that the limit is $+\infty$ or $-\infty$; some sequences have no limit at all). The second flaw is assuming that the limit L of x^n as n tends to $+\infty$ (even if it exists) is real, when they applied Theorem 6.1.19(c) by writing $\lim_{m+1\to\infty} x \times x^m = x \lim_{m+1\to\infty} x^m$. Theorem 6.1.19(c) requires that $(x^n)_{n=1}^{\infty}$ converges, i.e., it requires that L exists and is a real number.

[4] The proof of this claim is a slightly edited version of the proof in Exercise 6.1.1 (just use $\geq$ instead of $>$).

6.4 Limsup, Liminf, and Limit Points

Exercise 6.4.1

Let $(a_n)_{n=m}^{\infty}$ be a sequence that converges to a real number c.

Take an arbitrary positive real ε. As $(a_n)_{n=m}^{\infty}$ converges to c, there exists an integer $M \geq m$ such that $|a_n - c| \leq \varepsilon$ for all $n \geq M$. Thus, it is clear that c is ε-adherent to the sequence $(a_n)_{n=N}^{\infty}$ for all $N \geq M$. In addition, if N is an integer such that $m \leq N < M$, then a_M is an element of $(a_n)_{n=N}^{\infty}$, and since $|a_M - c| \leq \varepsilon$, c is ε-adherent to this sequence. We see that for any integer $N \geq m$, c is ε-adherent to the sequence $(a_n)_{n=N}^{\infty}$. Therefore, c is a limit point of $(a_n)_{n=m}^{\infty}$.

Suppose that there is another limit point denoted c'. Let $\delta > 0$. As $(a_n)_{n=m}^{\infty}$ converges to c, there exists an integer $N \geq m$ such that $|a_n - c| \leq \delta/2$ for all $n \geq N$. As c' is continually δ-adherent to $(a_n)_{n=m}^{\infty}$, there exists an integer $n' \geq N$ such that $|a_{n'} - c'| \leq \delta/2$. By the triangle inequality, $|c - c'| \leq \delta$. Hence, c and c' are δ-close. Since δ is an arbitrary positive real, we deduce that $c = c'$ (by Proposition 4.3.7(a) applied to reals). In conclusion, c is the only limit point of $(a_n)_{n=m}^{\infty}$.

Exercise 6.4.2

Let $(a_n)_{n=m}^{\infty}$ be a sequence of real numbers. Let x be a real number and let ε be a positive real.

Let $m' \geq m$ be an integer.

Suppose that x is continually ε-adherent to $(a_n)_{n=m}^{\infty}$. Then for every $N \geq m$, there exists an $n \geq N$ such that $|a_n - x| \leq \varepsilon$. So for every $N \geq m'$, there exists an $n \geq N$ such that $|a_n - x| \leq \varepsilon$ (since $N \geq m'$ implies $N \geq m$); which means that x is continually ε-adherent to $(a_n)_{n=m'}^{\infty}$. Conversely, suppose that x is continually ε-adherent to $(a_n)_{n=m'}^{\infty}$. Then for every $N \geq m'$, there exists an $n \geq N$ such that $|a_n - x| \leq \varepsilon$. For every $m \leq N < m'$, there also exists an $n \geq N$ such that $|a_n - x| \leq \varepsilon$, because there exists an $n \geq m'$ such that $|a_n - x| \leq \varepsilon$ (as x is ε-adherent to $(a_n)_{n=m'}^{\infty}$) and $n \geq m'$ implies $n \geq N$. We see that x is continually ε-adherent to $(a_n)_{n=m}^{\infty}$. Given that x is continually ε-adherent to $(a_n)_{n=m}^{\infty}$ if and only if x is continually ε-adherent to $(a_n)_{n=m'}^{\infty}$, it is clear that x is a limit point of $(a_n)_{n=m}^{\infty}$ if and only if x is a limit point of $(a_n)_{n=m'}^{\infty}$.

Let S_1 be the limit superior of $(a_n)_{n=m}^{\infty}$ and S_2 be limit superior of $(a_n)_{n=m'}^{\infty}$. In other

words,

$$S_1 := \inf\{\sup(a_n)_{n=N}^{\infty} : N \geq m\} \text{ and } S_2 := \inf\{\sup(a_n)_{n=N}^{\infty} : N \geq m'\}.$$

Take an integer $m \leq M \leq m'$. We have $\sup(a_n)_{n=M}^{\infty} \geq a_i$ for all $i \geq m'$, since $i \geq m'$ implies $i \geq M$. Thus for every $N \geq m'$, we have $\sup(a_n)_{n=M}^{\infty} \geq a_i$ for all $i \geq N$, and so $\sup(a_n)_{n=M}^{\infty}$ is an upper bound for $(a_n)_{n=N}^{\infty}$. Consequently,

$$\sup(a_n)_{n=M}^{\infty} \geq \sup(a_n)_{n=N}^{\infty} \text{ for all } N \geq m'.$$

We can use this to deduce that since $S_2 \leq \sup(a_n)_{n=N}^{\infty}$ for all $N \geq m'$, we also have

$$S_2 \leq \sup(a_n)_{n=N}^{\infty} \text{ for all } N \geq m.$$

Thus, S_2 is a lower bound for $\{\sup(a_n)_{n=N}^{\infty} : N \geq m\}$, and so $S_2 \leq S_1$. Next, we have $S_1 \leq \sup(a_n)_{n=N}^{\infty}$ for all $N \geq m$. In particular,

$$S_1 \leq \sup(a_n)_{n=N}^{\infty} \text{ for all } N \geq m'.$$

Thus, S_1 is a lower bound for $\{\sup(a_n)_{n=N}^{\infty} : N \geq m'\}$. Hence, $S_1 \leq S_2$. Therefore $S_1 = S_2$, which means that $(a_n)_{n=m}^{\infty}$ and $(a_n)_{n=m'}^{\infty}$ have the same limit superior.

We can follow an analogous argument to show that the limit inferior of $(a_n)_{n=m}^{\infty}$ is equal to the limit inferior of $(a_n)_{n=m'}^{\infty}$.

Let $k \geq 0$ be a non-negative integer. Consider the sequences $(a_{n+k})_{n=m}^{\infty}$ and $(a_n)_{n=m+k}^{\infty}$.

Since $m+k \geq m$, x is continually ε-adherent to $(a_n)_{n=m}^{\infty}$ if and only if it is continually ε-adherent to $(a_n)_{n=m+k}^{\infty}$ (as we proved earlier in this exercise). Now, suppose that x is continually ε-adherent to $(a_n)_{n=m+k}^{\infty}$. This means that:

$$\text{for all } N \geq m+k, \text{ there exists an integer } n \geq N \text{ such that } |a_n - x| \leq \varepsilon.$$

We also see that:

$$\text{for all } m \leq N \leq m+k, \text{ there exists an integer } n \geq N \text{ such that } |a_n - x| \leq \varepsilon$$

because we can find an integer $n \geq m+k \geq N$ that satisfies the condition, given that x is ε-adherent to $(a_n)_{n=m+k}^{\infty}$. So:

$$\text{for all } N \geq m, \text{ there exists an integer } n \geq N \text{ such that } n \geq m+k \text{ and } |a_n - x| \leq \varepsilon. \quad (\star)$$

Note that if i is an integer greater than or equal to $m+k$, then $i = n+k$ for some integer $n \geq m$. Therefore, we can deduce from $(\star)$ that:

for all $N \geq m$, there is an integer of the form $n+k$ such that $n+k \geq N$ and $|a_{n+k} - x| \leq \varepsilon$. $(\star\star)$

Now, take an integer $N \geq m$. Since $N+k \geq m$, there exists an integer $n+k \geq N+k$ such that $|a_{n+k} - x| \leq \varepsilon$ by $(\star\star)$. Note that $n \geq N$. Hence there exists an integer $n \geq N$ such that $|a_{n+k} - x| \leq \varepsilon$. Given that N is arbitrary, this means that x is continually ε-adherent to $(a_{n+k})_{n=m}^{\infty}$. Conversely, suppose that x is continually ε-adherent to $(a_{n+k})_{n=m}^{\infty}$. Then:

for all $N \geq m$, there exists an integer $n \geq N$ such that $|a_{n+k} - x| \leq \varepsilon$. $(\star\star\star)$

Take an arbitrary integer $N \geq m+k$. Since $N \geq m$, there is an integer $n_0 \geq N$ such that $|a_{n_0+k} - x| \leq \varepsilon$ by $(\star\star\star)$. Let $n := n_0 + k$. Observe that $n \geq N$ and $|a_n - k| \leq \varepsilon$. Thus, x is continually ε-adherent to $(a_n)_{n=m+k}^{\infty}$. We have established that x is continually ε-adherent to $(a_n)_{n=m+k}^{\infty}$ if and only if it is continually ε-adherent to $(a_{n+k})_{n=m}^{\infty}$. Hence, x is continually ε-adherent to $(a_n)_{n=m}^{\infty}$ if and only if it is continually ε-adherent to $(a_{n+k})_{n=m}^{\infty}$. In conclusion, x is a limit point of $(a_n)_{n=m}^{\infty}$ if and only if x is a limit point of $(a_{n+k})_{n=m}^{\infty}$.

Observe that for any $N \geq m$,

$$\{a_{n+k} : n \geq N\} = \{a_n : n \geq N+k\},$$

and thus

$$\sup\{a_{n+k} : n \geq N\} = \sup\{a_n : n \geq N+k\},$$

that is

$$\sup(a_{n+k})_{n=N}^{\infty} = \sup(a_n)_{n=N+k}^{\infty}.$$

So,

$$\{\sup(a_{n+k})_{n=N}^{\infty} : N \geq m\} = \{\sup(a_n)_{n=N+k}^{\infty} : N \geq m\}.$$

With a change of variable (from $N+k$ to N), we see that

$$\{\sup(a_n)_{n=N+k}^{\infty} : N \geq m\} = \{\sup(a_n)_{n=N}^{\infty} : N \geq m+k\}.$$

Thus,

$$\{\sup(a_{n+k})_{n=N}^{\infty} : N \geq m\} = \{\sup(a_n)_{n=N}^{\infty} : N \geq m+k\}.$$

Therefore,

$$\inf\{\sup(a_{n+k})_{n=N}^{\infty} : N \geq m\} = \inf\{\sup(a_n)_{n=N}^{\infty} : N \geq m+k\}.$$

This means that the limit superior of $(a_{n+k})_{n=m}^{\infty}$ is equal to the limit superior of $(a_n)_{n=m+k}^{\infty}$. Given that $m+k \geq m$, we know that the limit superior of $(a_n)_{n=m+k}^{\infty}$ is equal to that of $(a_n)_{n=m}^{\infty}$ (as shown earlier). Therefore, the limit of superior of $(a_{n+k})_{n=m}^{\infty}$ is equal to the limit superior of $(a_n)_{n=m}^{\infty}$.

We use an analogous line of reasoning to show that $(a_{n+k})_{n=m}^{\infty}$ and $(a_n)_{n=m}^{\infty}$ have the same limit inferior.

Exercise 6.4.3

Let $(a_n)_{n=m}^{\infty}$ be a sequence of real numbers, and let L^- and L^+ be the limit inferior and limit superior of $(a_n)_{n=m}^{\infty}$ respectively. We will prove the four remaining statements of Proposition 6.4.12.

(c) We have $\inf(a_n)_{n=m}^{\infty} \leq L^- \leq L^+ \leq \sup(a_n)_{n=m}^{\infty}$.

By Definition 6.4.6, $L^- = \sup\{\inf(a_n)_{n=N}^{\infty} : N \geq m\}$. Given that $\inf(a_n)_{n=m}^{\infty}$ is an element of $\{\inf(a_n)_{n=N}^{\infty} : N \geq m\}$, we have

$$L^- \geq \inf(a_n)_{n=m}^{\infty}.$$

Similarly, $L^+ = \inf\{\sup(a_n)_{n=N}^{\infty} : N \geq m\}$ and $\sup(a_n)_{n=m}^{\infty} \in \{\sup(a_n)_{n=N}^{\infty} : N \geq m\}$ imply that

$$L^+ \leq \sup(a_n)_{n=m}^{\infty}.$$

Now, suppose that $L^- > L^+$. Then, there exists an integer $M \geq m$ such that $a_n < L^-$ for all $n \geq M$ by Proposition 6.4.12(a). Thus, L^- is an upper bound for the set $\{a_n : n \geq M\}$, and so $L^- \geq \sup(a_n)_{n=M}^{\infty}$. It can be easily proven that for any integers k and k' such that $k' > k \geq m$, we have $\sup(a_n)_{n=k}^{\infty} \geq \sup(a_n)_{n=k'}^{\infty}$ by using the fact that $\{a_n : n \geq k'\} \subseteq \{a_n : n \geq k\}$. From this, we see that

$$L^- \geq \sup(a_n)_{n=N}^{\infty} \text{ for all } N \geq M.$$

Assume that $L^- > \sup(a_n)_{n=N_0}^{\infty}$ for some $N_0 \geq M$. Then, there exists an integer $P \geq m$ such that

$$a_i > \sup(a_n)_{n=N_0}^{\infty} \text{ for all } i \geq P \qquad (\star)$$

according to Proposition 6.4.12(a). Let $K := \max(N_0, P)$. Given that $K \geq N_0$, we have $\sup(a_n)_{n=N_0}^{\infty} \geq \sup(a_n)_{n=K}^{\infty}$. So, $(\star)$ implies that $a_i > \sup(a_n)_{n=K}^{\infty}$ for all $i \geq K$. But this contradicts Proposition 6.3.6. Thus,

$$L^- = \sup(a_n)_{n=N}^{\infty} \text{ for all } N \geq M.$$

Now, we know from Exercise 6.4.2 that the limit superior of $(a_n)_{n=m}^{\infty}$ is also the limit superior of $(a_n)_{n=M}^{\infty}$. That is,

$$L^+ = \inf\{\sup(a_n)_{n=N}^{\infty} : N \geq m\} = \inf\{\sup(a_n)_{n=N}^{\infty} : N \geq M\}.$$

But since $\{\sup(a_n)_{n=N}^{\infty} : N \geq M\} = \{L^-\}$, this yields $L^+ = L^-$, a contradiction. Therefore, $L^- \not> L^+$. In other words, $L^- \leq L^+$.

In summary, we have:

$$\inf(a_n)_{n=m}^{\infty} \leq L^- \leq L^+ \leq \sup(a_n)_{n=m}^{\infty}.$$

(d) If c is a limit point of $(a_n)_{n=m}^{\infty}$, then $L^- \leq c \leq L^+$.

Let c be a real number. Suppose that c is a limit point of $(a_n)_{n=m}^{\infty}$. Now, suppose for the sake of contraction that $c > L^+$. Then, by Proposition 6.4.12(a), there exists an integer $M \geq m$ such that $a_n < c$ for all $n \geq M$.

Since $c \in \mathbf{R}$ and $c > L^+$, $L^+ \neq +\infty$. Furthermore, we claim that $L^+ \neq -\infty$ and prove it by contradiction[5]. Assume that $L^+ = -\infty$. Let $\delta \in \mathbf{R}$. Since $-|c| - |\delta| > L^+$, there is an integer $M_\delta \geq m$ such that $a_n < -|c| - |\delta|$ for all $n \geq M_\delta$ by Proposition 6.4.12(a). Since c is a limit point of $(a_n)_{n=m}^{\infty}$, there is an integer $n_\delta \geq \max(M, M_\delta)$ such that $|a_{n_\delta} - c| \leq |\delta|$. As $n_\delta \geq M$, we have $a_{n_\delta} < c$, and so $c - a_{n_\delta} \leq |\delta|$. Thus, $a_{n_\delta} \geq c - |\delta|$, and hence $-|c| - |\delta| > c - |\delta|$, so that $-|c| > c$. This contradiction shows that $L^+ \neq -\infty$. Hence, $L^+ \in \mathbf{R}$.

Given that c is a limit point of $(a_n)_{n=m}^{\infty}$, there exists an integer $n_1 \geq M$ such that $|a_{n_1} - c| \leq 1$. For the same reason, there exist an integer $n_2 > n_1$ such that $|a_{n_2} - c| \leq 1/2$, and an integer $n_3 > n_2$ such that $|a_{n_3} - c| \leq 1/3$. Continuing in this way, we build a sequence $(a_{n_k})_{k=1}^{\infty}$ with the property that

$$|a_{n_k} - c| \leq 1/k$$

for all integers $k \geq 1$, where $n_i < n_j$ for any positive integers $i < j$. Given an arbitrary positive real ε, the Archimedean property of real numbers guarantees the existence of an integer t such that $1/t \leq \varepsilon$; so we have $|a_{n_k} - c| \leq \varepsilon$ for all $k \geq t$. This means that

$$\lim_{k \to \infty} a_{n_k} = c.$$

[5] It is important to establish this here because we will later perform arithmetic with L^+.

For all $k \geq 1$, given that $n_k \geq M$, we have $a_{n_k} < c$, so that $|a_{n_k} - c| = c - a_{n_k}$. Since $L^+ < c$, we have

$$L^+ - a_{n_k} < c - a_{n_k} \leq 1/k$$

for all $k \geq 1$. Suppose that $a_{n_k} \leq L^+$ for all $k \geq 1$. Then $|a_{n_k} - L^+| \leq 1/k$ for all $k \geq 1$, and we already saw that this yields $\lim_{k\to\infty} a_{n_k} = L^+$. But, by Property 6.1.7 (uniqueness of limits), this implies that $c = L^+$, a contradiction. So, there exists an integer $q \geq 1$ such that $a_{n_q} > L^+$. By Proposition 6.4.12(a), there is an integer $P \geq m$ such that $a_N < a_{n_q}$ for all $N \geq P$. Given that $n_1 \geq m$ and $n_{k+1} > n_k$ for all $k \geq 1$, we can show with a straightforward induction that $n_k \geq m + k - 1$ for all $k \geq 1$. The base case is given, and for the inductive step, note that if $n_k \geq m + k - 1$ for some $k \geq 1$, then $n_{k+1} > n_k$ yield $n_{k+1} \geq m + k$, i.e, $n_{k+1} \geq m + (k+1) - 1$. Now, let $p = P - m + 1$. We have $n_p \geq P$. Therefore, $a_{n_k} < a_{n_q}$ for all $k > p$. So, $c - a_{n_k} > c - a_{n_q}$ for all $k > p$. But this means that c is not eventually $(c - a_{n_q})$-close to $(a_{n_k})_{k=1}^{\infty}$, even though c is the limit of $(a_{n_k})_{k=1}^{\infty}$. From this contradiction, it follows that $c \not> L^+$, that is,

$$c \leq L^+.$$

We similarly prove that $c < L^-$ yields a contradiction, and thus $c \geq L^-$. Therefore,

$$L^- \leq c \leq L^+.$$

(e) If L^- or L^+ is finite, then it is a limit point of $(a_n)_{n=m}^{\infty}$.

Suppose that L^+ is finite. Let ε be an arbitrary positive real. Then $L^+ - \varepsilon$ and $L^+ + \varepsilon$ are real numbers, and we have $L^+ - \varepsilon < L^+ < L^+ + \varepsilon$. Since $L^+ + \varepsilon > L^+$, there exists an integer $M \geq m$ such that $a_n < L^+ + \varepsilon$ for all $n \geq M$ by Proposition 6.4.12(a).

Now, take an integer $N \geq m$. Let $K := \max(N, M)$. Since $L^+ - \varepsilon < L^+$, there exists an integer $k \geq K \geq N$ such that $a_k > L^+ - \varepsilon$, by Proposition 6.4.12(b). As $k \geq K \geq M$, we also have $a_k < L^+ - \varepsilon$. Thus, $L^+ - \varepsilon < a_k < L^+ + \varepsilon$, and so $|a_k - L^+| \leq \varepsilon$. So, L^+ is ε-adherent to $(a_n)_{n=N}^{\infty}$. Since N is arbitrary, we deduce that L^+ is continually ε-adherent to $(a_n)_{n=m}^{\infty}$. Since ε is arbitrary, this means that L^+ is a limit point of $(a_n)_{n=m}^{\infty}$.

Following an analogous argument, we show that if L^- is finite, then it is a limit point of $(a_n)_{n=m}^{\infty}$.

(f) If c is a real number, $(a_n)_{n=m}^{\infty}$ converges to c if and only if $L^+ = L^- = c$.

Let c be a real number.

Suppose that $(a_n)_{n=m}^{\infty}$ converges to c. Then, $(a_n)_{n=m}^{\infty}$ is a bounded sequence (see Corollary 6.1.17). That is, there exists a positive real number M such that $|a_n| \leq M$ for all $n \geq m$. Since $a_n \leq M$ and $-M \leq a_n$ for all $n \geq m$, we have $M \geq \sup(a_n)_{n=m}^{\infty}$ and $-M \leq \inf(a_n)_{n=m}^{\infty}$. We also know that from Proposition 6.4.12(c) that

$$\sup(a_n)_{n=m}^{\infty} \geq \inf\{\sup(a_n)_{n=N}^{\infty} : N \geq m\} = L^+$$

and

$$\inf(a_n)_{n=m}^{\infty} \leq \sup\{\inf(a_n)_{n=N}^{\infty} : N \geq m\} = L^-.$$

Consequently,

$$M \geq L^+ \quad \text{and} \quad -M \leq L^-.$$

Since $M \in \mathbf{R}$, $M \geq L^+$ implies that $L^+ \neq +\infty$. Since $-M \in \mathbf{R}$, $-M \leq L^-$ implies that $L^- \neq -\infty$. By Proposition 6.4.12(c), we have $L^- \leq L^+$. If $L^+ = -\infty$, then $L^- = -\infty$, a contradiction. If $L^- = +\infty$, then $L^+ = +\infty$, another contradiction. So, L^+ and L^- are both finite. According to Proposition 6.4.12(e), this implies that L^+ and L^- are limit points of $(a_n)_{n=m}^{\infty}$. But by Proposition 6.4.5, c is the only limit point of $(a_n)_{n=m}^{\infty}$. Therefore,

$$L^+ = L^- = c.$$

Conversely, suppose that $L^+ = L^- = c$. Then, L^+ is finite as c is finite. By Proposition 6.4.12(e), L^+ is a limit point of $(a_n)_{n=m}^{\infty}$. So, c is a limit point of $(a_n)_{n=m}^{\infty}$. Let $\varepsilon > 0$. Since $c - \varepsilon < c = L^-$, there exists an integer $M \geq m$ such that $a_n > c - \varepsilon$ for all $n \geq M$, by Proposition 6.4.12(a). Thus, $a_n - c > -\varepsilon$ for all $n \geq M$. Since $c + \varepsilon > c = L^+$, there exists an integer $K \geq m$ such that $a_n < c + \varepsilon$ for all $n \geq K$, by Proposition 6.4.12(a). Thus, $a_n - c < \varepsilon$ for all $n \geq K$. Let $N := \max(K, M)$. For all $n \geq N$, we have $-\varepsilon < a_n - c < \varepsilon$, so that $|a_n - c| \leq \varepsilon$. Therefore, c is eventually ε-close to $(a_n)_{n=m}^{\infty}$. As ε is arbitrary, this means that $(a_n)_{n=m}^{\infty}$ converges to c.

Exercise 6.4.4

Suppose that $(a_n)_{n=m}^{\infty}$ and $(b_n)_{n=m}^{\infty}$ are two sequences of real numbers such that $a_n \leq b_n$ for all $n \geq m$.

We know that $b_i \leq \sup(b_n)_{n=m}^{\infty}$ for all $i \geq m$. Hence, $a_i \leq \sup(b_n)_{n=m}^{\infty}$ for all $i \geq m$. Consequently, $\sup(b_n)_{n=m}^{\infty}$ is an upper bound for $\{a_n : n \geq m\}$, and so

$$\sup(a_n)_{n=m}^{\infty} \leq \sup(b_n)_{n=m}^{\infty}.$$

We know that $\inf(a_n)_{n=m}^{\infty} \leq a_i$ for all $i \geq m$. Hence, $\inf(a_n)_{n=m}^{\infty} \leq b_i$ for all $i \geq m$. Thus, $\inf(a_n)_{n=m}^{\infty}$ is a lower bound for $(b_n)_{n=m}^{\infty}$, and hence

$$\inf(a_n)_{n=m}^{\infty} \leq \inf(b_n)_{n=m}^{\infty}.$$

Consider the sequences $(a_N^+)_{N=m}^{\infty}$ and $(b_N^+)_{N=m}^{\infty}$ defined by $a_N^+ := \sup(a_n)_{n=N}^{\infty}$ and $b_N^+ := \sup(b_n)_{n=N}^{\infty}$ for all $N \geq m$. Note that for all $N \geq m$, $a_n \leq b_n$ for all $n \geq N$, and so $\sup(a_n)_{n=N}^{\infty} \leq \sup(b_n)_{n=N}^{\infty}$ (by the first property established in this exercise), that is $a_N^+ \leq b_N^+$. Using the second property established in this exercise, we deduce that $\inf(a_N^+)_{N=m}^{\infty} \leq \inf(b_N^+)_{N=m}^{\infty}$. In other words,

$$\limsup_{n\to\infty} a_n \leq \limsup_{n\to\infty} b_n.$$

Let $(a_N^-)_{N=m}^{\infty}$ and $(b_N^-)_{N=m}^{\infty}$ be the sequences defined by $a_N^- := \inf(a_n)_{n=N}^{\infty}$ and $b_N^- := \inf(b_n)_{n=N}^{\infty}$ for all $N \geq m$. For all $N \geq m$, we have $a_n \leq b_n$ for all $n \geq N$, and so $\inf(a_n)_{n=N}^{\infty} \leq \inf(b_n)_{n=N}^{\infty}$ (by the second property we proved in this exercise). Hence, $a_N^- \leq b_N^-$ for all $N \geq m$. Given the first property we established in this exercise, it follows that $\sup(a_N^-)_{N=m}^{\infty} \leq \sup(b_N^-)_{N=m}^{\infty}$. That is,

$$\liminf_{n\to\infty} a_n \leq \liminf_{n\to\infty} b_n.$$

Exercise 6.4.5

Let $(a_n)_{n=m}^{\infty}$, $(b_n)_{n=m}^{\infty}$, and $(c_n)_{n=m}^{\infty}$be sequences of real numbers such that $a_n \leq b_n \leq c_n$ for all $n \geq m$. Suppose that $(a_n)_{n=m}^{\infty}$ and $(c_n)_{n=m}^{\infty}$ both converge to the same limit L.

Given that $a_n \leq b_n$ and $b_n \leq c_n$ for all $n \geq m$, we have:

$$\limsup_{n\to\infty} a_n \leq \limsup_{n\to\infty} b_n \leq \limsup_{n\to\infty} c_n$$

and

$$\liminf_{n\to\infty} a_n \leq \liminf_{n\to\infty} b_n \leq \liminf_{n\to\infty} c_n$$

by Lemma 6.4.13. As $(a_n)_{n=m}^{\infty}$ and $(c_n)_{n=m}^{\infty}$ converge to L, Proposition 6.4.12(f) implies that $\limsup_{n\to\infty} a_n$, $\limsup_{n\to\infty} c_n$, $\liminf_{n\to\infty} a_n$, and $\liminf_{n\to\infty} c_n$ are all real numbers and we have $\limsup_{n\to\infty} a_n = \limsup_{n\to\infty} c_n = \liminf_{n\to\infty} a_n = \liminf_{n\to\infty} c_n = L$. So,

$$\limsup_{n\to\infty} a_n \leq \limsup_{n\to\infty} b_n \leq \limsup_{n\to\infty} c_n \implies L \leq \limsup_{n\to\infty} b_n \leq L \implies \limsup_{n\to\infty} b_n = L$$

and

$$\liminf_{n\to\infty} a_n \leq \liminf_{n\to\infty} b_n \leq \liminf_{n\to\infty} c_n \implies L \leq \liminf_{n\to\infty} b_n \leq L \implies \liminf_{n\to\infty} b_n = L.$$

Given that $\limsup_{n\to\infty} b_n = \liminf_{n\to\infty} b_n = L$, the sequence $(b_n)_{n=m}^{\infty}$ converges to L by Proposition 6.4.12(f).

Exercise 6.4.6

Consider the sequences $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ defined by

$$a_n = -\frac{1}{n} \quad \text{and} \quad b_n = -\frac{1}{n+1}$$

for all $n \geq 1$. For any $n \geq 1$, $n+1 > n > 0$, so that $-1/n < -1/(n+1)$, i.e., $a_n < b_n$. Besides, we have $|a_n| = 1/n \leq 1$ and $|b_n| = 1/(n+1) \leq 1$ for all $n \geq 1$; thus, $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ are bounded sequences.

Note that 0 is clearly an upper bound for both $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$. Let x be an arbitrary negative real. Since $-x > 0$, there exists an integer $N \geq 1$ such that $0 < 1/N < -x$, so that $x < -1/N < 0$. Thus, $x < a_N$ and, as $a_N < b_N$, we also have $x < b_N$. This shows that x is not an upper bound for $(a_n)_{n=1}^{\infty}$ or $(b_n)_{n=1}^{\infty}$. Hence, any upper bound for $(a_n)_{n=1}^{\infty}$ or $(b_n)_{n=1}^{\infty}$ is greater than or equal to 0. Therefore, 0 is the least upper bound of $(a_n)_{n=1}^{\infty}$ and the least upper bound of $(b_n)_{n=1}^{\infty}$. In other words, $0 = \sup(a_n)_{n=1}^{\infty} = \sup(b_n)_{n=1}^{\infty}$. So,

$$\sup(a_n)_{n=1}^{\infty} \not< \sup(b_n)_{n=1}^{\infty}.$$

This does not contradict Lemma 6.4.13 because

$$a_n < b_n \implies a_n \leq b_n$$

for all $n \geq 1$, and

$$\sup(a_n)_{n=1}^{\infty} = \sup(b_n)_{n=1}^{\infty} \implies \sup(a_n)_{n=1}^{\infty} \leq \sup(b_n)_{n=1}^{\infty}.$$

Thus, it is indeed true that $a_n \leq b_n$ for all $n \geq 1$, and $\sup(a_n)_{n=1}^{\infty} \leq \sup(b_n)_{n=1}^{\infty}$, as guaranteed by Lemma 6.4.13.

Exercise 6.4.7

Let $(a_n)_{n=m}^{\infty}$ be a sequence of real numbers.

On the one hand, suppose that $(a_n)_{n=m}^{\infty}$ converges to 0. Then, $(a_n)_{n=m}^{\infty}$ is eventually ε-close to 0 for any real $\varepsilon > 0$. This means that for any $\varepsilon > 0$, there exists an integer

$N \geq m$ such that $|a_n| \leq \varepsilon$ for all $n \geq N$. Since $|a_n| = ||a_n||$ for all $n \geq m$, it follows that $(|a_n|)_{n=m}^{\infty}$ is also eventually ε-close to 0 for any real $\varepsilon > 0$. Hence, $(|a_n|)_{n=m}^{\infty}$ converges to 0.

On the other hand, suppose that $(|a_n|)_{n=m}^{\infty}$ converges to 0. By the limit laws, $(-|a_n|)_{n=m}^{\infty}$ also converges to 0. For all $n \geq m$, we have $-|a_n| \leq a_n \leq |a_n|$. By Corollary 6.4.14, this implies that $(a_n)_{n=m}^{\infty}$ is convergent to 0.

In conclusion, the limit $\lim_{n\to\infty} a_n$ exists and is equal to zero if and only if the limit $\lim_{n\to\infty} |a_n|$ exists and is equal to zero.

This statement is not true if we replace 0 by any other real number. To see this, let r be a nonzero real number. Define the sequence $(a_n)_{n=m}^{\infty}$ by $a_n := -|r|$ for all $n \geq m$. Then, $|a_n| = |r|$ for all $n \geq m$. It is clear that $\lim_{n\to\infty} |a_n|$ exists and is equal to $|r|$; but even though $\lim_{n\to\infty} a_n$ also exists, it is equal to $-|r|$ which is different from $|r|$. This shows that when we replace 0 by any other real number, we can always find a sequence that contradicts the statement.[6] Therefore, the statement is not true in that case.

Exercise 6.4.8

Let $(a_n)_{n=M}^{\infty}$ be a sequence of real numbers. Let L^+ and L^- be the limit superior and limit inferior of $(a_n)_{n=M}^{\infty}$, respectively.

If L^+ is finite, then L^+ is a limit point of $(a_n)_{n=M}^{\infty}$ according to Proposition 6.4.12(e). If $L^+ = +\infty$, we have $x < L^+$ for any finite real number x; and so there exists an integer $n \geq M$ such that $a_n > x$ by Proposition 6.4.12(b). Thus, no finite real number is an upper bound for $(a_n)_{n=M}^{\infty}$. Following the new definition, this means that $L^+ = +\infty$ is a limit point of $(a_n)_{n=M}^{\infty}$. If $L^+ = -\infty$, then $x > L^+$ for any finite real number x; and so there exists an integer $N \geq M$ such that $a_n < x$ for all $n \geq N$, by Proposition 6.4.12(a). Hence, no finite real number is a lower bound for $(a_n)_{n=M}^{\infty}$. Following the new definition, this means that $L^+ = -\infty$ is a limit point of $(a_n)_{n=M}^{\infty}$. We have just shown that L^+ is a limit point of $(a_n)_{n=M}^{\infty}$.

Let c be an extended real number. If c is a finite limit point of $(a_n)_{n=M}^{\infty}$, then $c \leq L^+$ by Proposition 6.4.12(d). If $c = +\infty$ and c is a limit point of $(a_n)_{n=M}^{\infty}$, then $(a_n)_{n=M}^{\infty}$ has no finite upper bound according to the new definition. This implies that $\sup(a_n)_{n=N}^{\infty} = +\infty$ for any integer $N \geq M$ for two reasons. First, if $\sup(a_n)_{n=N}^{\infty} \in \mathbf{R}$ for some $N \geq M$, then as the sequence $(a_n)_{n=M}^{N}$ has some finite upper bound A, the real number $\max(A, \sup(a_n)_{n=N}^{\infty})$ is a finite upper bound for $(a_n)_{n=M}^{\infty}$, a contradiction.

[6] If $r > 0$, note that $\lim_{n\to\infty} |a_n|$ exists and is equal to r while $\lim_{n\to\infty} a_n$ exists and is equal to $-r$. If $r < 0$, note that $\lim_{n\to\infty} a_n$ exists and is equal to r while $\lim_{n\to\infty} |a_n|$ exists and is equal to $-r$.

Second, $\sup(a_n)_{n=N}^{\infty} \neq -\infty$ because $\{a_n : n \geq N\}$ is a non-empty subset of $\mathbf{R}$. It follows that $L^+ = \inf\{\sup(a_n)_{n=N}^{\infty} : N \geq M\} = \inf\{+\infty\} = +\infty$. Thus, $c \leq L^+$. If $c = -\infty$ and c is a limit point of $(a_n)_{n=M}^{\infty}$, then $c \leq L^+$, regardless of the value of L^+. We have just shown that if c is a limit point of $(a_n)_{n=M}^{\infty}$, then $c \leq L^+$.

In summary, L^+ is the largest limit point of $(a_n)_{n=M}^{\infty}$.

If L^- is finite, then L^- is a limit point of $(a_n)_{n=M}^{\infty}$ according to Proposition 6.4.12(e). If $L^- = -\infty$, we have $x > L^-$ for any finite real number x; and so there exists an integer $n \geq M$ such that $a_n < x$ by Proposition 6.4.12(b). Thus, no finite real number is a lower bound for $(a_n)_{n=M}^{\infty}$. Following the new definition, this means that $L^- = -\infty$ is a limit point of $(a_n)_{n=M}^{\infty}$. If $L^- = +\infty$, then $x < L^-$ for any finite real number x; and so there exists an integer $N \geq M$ such that $a_n > x$ for all $n \geq N$, by Proposition 6.4.12(a). Hence, no finite real number is an upper bound for $(a_n)_{n=M}^{\infty}$. Following the new definition, this means that $L^- = +\infty$ is a limit point of $(a_n)_{n=M}^{\infty}$.

Let b be an extended real number. If b is a finite limit point of $(a_n)_{n=M}^{\infty}$, then $b \geq L^-$ by Proposition 6.4.12(d). If $b = -\infty$ is a limit point of $(a_n)_{n=M}^{\infty}$, then $(a_n)_{n=M}^{\infty}$ has no finite lower bound according to the new definition. It follows that $\inf(a_n)_{n=N}^{\infty} = -\infty$ for any integer $N \geq M$ for the following two reasons. First, if $\inf(a_n)_{n=N}^{\infty} \in \mathbf{R}$ for some $N \geq M$, then as the sequence $(a_n)_{n=N}^{M}$ has some finite lower bound B, the real number $\min(B, \inf(a_n)_{n=N}^{\infty})$ is a lower bound for $(a_n)_{n=M}^{\infty}$, which is contradictory. Second, $\inf(a_n)_{n=N}^{\infty} \neq +\infty$ because $\{a_n : n \geq N\}$ is a non-empty subset of $\mathbf{R}$. We have $L^- = \sup\{\inf(a_n)_{n=N}^{\infty} : N \geq M\} = \sup\{-\infty\} = -\infty$. Thus, $b \geq L^-$. If $b = +\infty$ is a limit point of $(a_n)_{n=M}^{\infty}$, then $b \geq L^-$, regardless of the value of L^-.

In summary, L^- is the smallest limit point of $(a_n)_{n=M}^{\infty}$.

Exercise 6.4.9

Let $(a_n)_{n=1}^{\infty}$ be the sequence defined by

$$a_{3n-2} := n, \quad a_{3n-1} := -n, \quad \text{and} \quad a_{3n} := 0 \quad \text{for all } n \geq 1.$$

More informally, $(a_n)_{n=1}^{\infty}$ is the sequence

$$1, -1, 0, 2, -2, 0, 3, -3, 0, 4, -4, 0, 5, \ldots.$$

We see that for any natural number $z \geq 1$, $a_{3z-2} = z$ while $a_{3z-1} = -z$. Thus, every integer is a term of the sequence $(a_n)_{n=1}^{\infty}$. Since for any finite real number there always exists an integer greater than it and an integer lower than it, $(a_n)_{n=1}^{\infty}$ has no finite upper bound and no finite lower bound. Hence, $+\infty$ and $-\infty$ are limit points of $(a_n)_{n=1}^{\infty}$ according to the definition in Exercise 6.4.8.

For any real $\varepsilon > 0$ and any integer $N \geq 1$, we have $3N \geq N$ and $|a_{3N} - 0| = 0 \leq \varepsilon$. Thus, 0 is a limit point of $(a_n)_{n=1}^{\infty}$.

Now, let p be a non-zero real number. First, suppose that $p > 0$. We know that there exists a positive integer M such that $M > p$. There also exists a positive real number δ such that $M > p + \delta$. Let us choose a positive real ε such that $\varepsilon < p$ and $\varepsilon < \delta$. For any integer $N \geq M$, we also have $N > p + \varepsilon$, hence $N - p > \varepsilon$, and so $|N - p| > \varepsilon$. In addition, for any integer $N \geq M$, we have $N + p > N - p$, and so $|N + p| > \varepsilon$, i.e., $|-N - p| > \varepsilon$. Furthermore, $|p| > \varepsilon$. Now, note that $M = a_{3M-2}$ and $-M = a_{3M-1}$. So for any $n \geq 3M - 2$,

$$\begin{cases} a_n \geq M, & n \text{ is of the form } 3k-2 \text{ for some integer } k \\ a_n = -m \text{ for some integer } m \geq M, & n \text{ is of the form } 3k-1 \text{ for some integer } k \\ a_n = 0, & n \text{ is of the form } 3k \text{ for some integer } k \end{cases}.$$

In any case, we have $|a_n - p| > \varepsilon$ for all $n \geq 3M - 2$. Therefore, p is not continually ε-adherent to $(a_n)_{n=1}^{\infty}$. Consequently, p is not a limit point of $(a_n)_{n=1}^{\infty}$.

Second, suppose that $p < 0$. Here, $-p > 0$. So, there exists a positive integer K such that $K > -p > 0$. There also exists a positive real number β such that $K > -p + \beta$. Let us choose a positive real γ such that $\gamma < -p$ and $\gamma < \beta$. For any integer $k \geq K$, we have $k > -p + \gamma$, thus $k + p > \gamma$, and so $|-k - p| > \gamma$. Furthermore, for any integer $k \geq K$, we have $k > p + \gamma$ (because $-p > p$), hence $k - p > \gamma$, and so $|k - p| > \gamma$. Therefore, $|a_n - p| > \gamma$ for all $n \geq 3K - 2$. Hence, p is not continually γ-adherent to $(a_n)_{n=1}^{\infty}$. Consequently, p is not a limit point of $(a_n)_{n=1}^{\infty}$.

We deduce that no nonzero real number is a limit point of $(a_n)_{n=1}^{\infty}$.

In conclusion, the sequence $(a_n)_{n=1}^{\infty}$ has exactly three limit points, namely $-\infty$, 0, and $+\infty$.

Exercise 6.4.10

Let $(a_n)_{n=N}^{\infty}$ be a sequence of real numbers, and let $(b_m)_{m=M}^{\infty}$ be another sequence of real numbers such that each b_m is a limit point of $(a_n)_{n=N}^{\infty}$. Let c be a limit point of $(b_m)_{m=M}^{\infty}$.

Consider an arbitrary positive real number ε. Given that c is a limit point of $(b_m)_{m=M}^{\infty}$, it is $(\varepsilon/2)$-adherent to $(b_m)_{m=M}^{\infty}$. That is, there exists an integer $m_0 \geq M$ such that $|b_{m_0} - c| \leq \varepsilon/2$.

Now, take an integer $K \geq N$. Since b_{m_0} is a limit point of $(a_n)_{n=N}^{\infty}$, it is continually $(\varepsilon/2)$-adherent to $(a_n)_{n=N}^{\infty}$. So, there exists an integer $n \geq K$ such that $|a_n - b_{m_0}| \leq \varepsilon/2$.

Therefore, $|a_n - b_{m_0}| + |b_{m_0} - c| \leq \varepsilon$. Using the triangle inequality yields $|a_n - c| \leq \varepsilon$. We see that c is continually ε-adherent to $(a_n)_{n=N}^{\infty}$. Therefore, c is a limit point of $(a_n)_{n=N}^{\infty}$.

6.5 Some Standard Limits

Exercise 6.5.1

Let $q > 0$ be a rational. Since q is positive, there exist positive integers a and b such that $q = a/b$. We have

$$\frac{1}{n^q} = \frac{1}{n^{a/b}} = \left(\frac{1}{n^{1/b}}\right)^a$$

for all integers $n \geq 1$. We know from Corollary 6.5.1 that $\lim_{n\to\infty} 1/n^{1/b} = 0$.

Before proceeding, we need to establish the following lemma: given any converging sequence of positive real numbers $(a_n)_{n=1}^{\infty}$ and a positive integer m, we have $\lim_{n\to\infty}(a_n)^m = (\lim_{n\to\infty}(a_n))^m$.

Let $(a_n)_{n=1}^{\infty}$ be a sequence of positive real numbers that converges, and let the real number L be its limit. We argue by induction. At the base case ($m = 1$), we have $\lim_{n\to\infty}(a_n)^m = \lim_{n\to\infty}(a_n)^1 = \lim_{n\to\infty} a_n = L$ and $L^m = L^1 = L$, so we are done. For the inductive step, suppose that the claim holds for a positive integer m. Then,

$$L^{m+1} = L^m L = \left(\lim_{n\to\infty}(a_n)\right)^m \left(\lim_{n\to\infty} a_n\right) = \left(\lim_{n\to\infty}(a_n)^m\right)\left(\lim_{n\to\infty} a_n\right)$$

by the induction hypothesis. As

$$\left(\lim_{n\to\infty}(a_n)^m\right)\left(\lim_{n\to\infty} a_n\right) = \lim_{n\to\infty}((a_n)^m a_n)$$

by Theorem 6.1.9(b), we have

$$\lim_{n\to\infty}(a_n)^{m+1} = L^{m+1},$$

which closes the induction.

Applying the lemma above yields $\lim_{n\to\infty}\left(\frac{1}{n^{1/b}}\right)^a = 0$. That is,

$$\lim_{n\to\infty}\frac{1}{n^q} = 0.$$

Now, suppose that $\lim_{n\to\infty} n^q$ exists. Then by Theorem 6.1.19(b),

$$\lim_{n\to\infty} n^q(1/n^q) = \left(\lim_{n\to\infty} n^q\right) \times 0 = 0. \qquad (\star)$$

However, $n^q(1/n^q) = 1$ for all integers $n \geq 1$, and so $(\star)$ implies that

$$\lim_{n\to\infty} 1 = 0,$$

a clear contradiction. Hence, $\lim_{n\to\infty} n^q$ does not exist.[7]

Exercise 6.5.2

Let x be a real number.

First, suppose that $|x| < 1$. If $|x| = 0$, then $x = 0$. In this case, $x^n = 0^n = 0$ for all positive integers n. Thus, $\lim_{n\to\infty} x^n$ exists and is equal to 0. If instead $0 < |x| < 1$, then $\lim_{n\to\infty} |x|^n = 0$ by Proposition 6.3.10. Observe that $|x|^n = |x^n|$ for all integers n. So, $\lim_{n\to\infty} |x|^n = 0$ if and only if $\lim_{n\to\infty} |x^n| = 0$. Furthermore, $(|x^n|)_{n=1}^{\infty}$ converges to 0 if and only if $(x^n)_{n=1}^{\infty}$ converges to 0, by Corollary 6.4.17. Therefore, we also have $\lim_{n\to\infty} x^n = 0$ when $0 < |x| < 1$. In sum, x^n converges to zero when $|x| < 1$.

Second, suppose that $x = 1$. Then, since $x^n = 1^n = 1$ for all positive integers n, $\lim_{n\to\infty} x^n$ exists and is equal to 1.

Third, suppose that $x = -1$. Assume toward contradiction that $(x^n)_{n=1}^{\infty}$ converges to a limit c. Then, c is eventually $(1/2)$-close to $(x^n)_{n=1}^{\infty}$. That is, there exists an integer $N \geq 1$ such that $|x^n - c| \leq 1/2$ for all $n \geq N$. Let $m \geq N$. We have $|x^m - c| \leq 1/2$ and $|x^{m+1} - c| \leq 1/2$. Using the triangle inequality, we get

$$|x^m - x^{m+1}| \leq 1,$$

that is

$$|(-1)^m - (-1)^{m+1}| \leq 1,$$

that is

$$2 \leq 1,$$

a contradiction. Therefore, $(x^n)_{n=1}^{\infty}$ diverges when $x = -1$.

Finally, suppose that $|x| > 1$. Then, $x \neq 0$ and $|1/x| < 1$. Hence, $((1/x)^n)_{n=1}^{\infty}$ converges and its limit is 0 (given the result established in the case $|x| < 1$). In other words,

$$\lim_{n\to\infty} \frac{1}{x^n} = 0.$$

[7] If you wish to do this second part of the exercise using the hint provided, you may proceed as follows. Suppose, for the sake of contradiction, that $\lim_{n\to\infty} n^q$ exists. Let $L = \lim_{n\to\infty} n^q$. For any integer $n \geq 1$, we have $n^q \geq 1$ (see Lemma 5.6.9(d)), so $L \geq 1$ (see Exercise 5.4.8). Given that $n^q \neq 0$ for all $n \geq 1$ and $L \neq 0$, it follows from Theorem 6.1.19(e) that $\lim_{n\to\infty} \frac{1}{n^q} = L^{-1}$. But this is contradictory because $\lim_{n\to\infty} \frac{1}{n^q} = 0$. Hence, $\lim_{n\to\infty} n^q$ does not exist.

Suppose, for the sake of contradiction, that $(x^n)_{n=1}^{\infty}$ converges to a limit c. Then, by Theorem 6.1.19(b), the sequence $((1/x^n)x^n)_{n=1}^{\infty}$ converges and we have

$$\lim_{n\to\infty} x^n \frac{1}{x^n} = c \times 0 = 0,$$

that is

$$\lim_{n\to\infty} 1 = 0,$$

which is contradictory. Thus, $(x^n)_{n=1}^{\infty}$ does not converge when $|x| > 1$.

Exercise 6.5.3

Let x be a positive real number.

Let $\delta > 0$ and $M > 0$ be arbitrary positive real numbers. Given that $1 + \delta > 1$, the sequence $((1+\delta)^n)_{n=1}^{\infty}$ diverges by Lemma 6.5.2. Clearly, it is an increasing sequence. If $(1+\delta)^n < M$ for all integers $n \geq 1$, then $((1+\delta)^n)_{n=1}^{\infty}$ is bounded, and so it converges by Proposition 6.3.8, which is contradictory. Thus, there exists some integer $k \geq 1$ for which $(1+\delta)^k \geq M$. This implies that

$$M^{1/k} \leq 1 + \delta. \qquad (\star)$$

Now, let ε be a positive real.

To begin with, suppose that $0 < x < 1$. Then, by Lemma 6.5.2, $(x^n)_{n=1}^{\infty}$ converges to 0. Note that for any positive integer n,

$$x < 1 + \varepsilon \implies x^{1/n} < (1+\varepsilon)^{1/n}$$

by Lemma 5.6.6(d), and

$$x^{1/n} < (1+\varepsilon)^{1/n} \implies x^{1/n} < 1 + \varepsilon$$

by Lemma 5.6.6(e). We also have $1/x > 0$. It follows from $(\star)$ that there exists an integer k such that $(1/x)^{1/k} \leq 1 + \varepsilon$. Observe that

$$\begin{aligned} \left(\frac{1}{x}\right)^{1/k} \leq 1 + \varepsilon &\implies \frac{1}{x^{1/k}} \leq 1 + \varepsilon \\ &\implies 1 \leq x^{1/k} + x^{1/k}\varepsilon, \end{aligned}$$

and given that $x^{1/k} < 1$ (by Lemmas 5.6.6(d) and 5.6.6(e)),

$$\begin{aligned} 1 \leq x^{1/k} + x^{1/k}\varepsilon &\implies 1 \leq x^{1/k} + \varepsilon \\ &\implies x^{1/k} \geq 1 - \varepsilon. \end{aligned}$$

Now, since $x < 1$, we have $x^{1/n} \geq x^{1/k}$ for all integers $n \geq k$ according to Lemma 5.6.6(e). Hence, $x^{1/n} \geq 1 - \varepsilon$ for all $n \geq k$. It follows that $1 - \varepsilon \leq x^{1/n} \leq 1 + \varepsilon$, i.e., $|x^{1/n} - 1| \leq \varepsilon$, for all integers $n \geq k$. Thus, $(x^{1/n})_{n=1}^{\infty}$ is eventually ε-close to 1. We deduce that $(x^{1/n})_{n=1}^{\infty}$ converges to 1.

Next, Suppose that $x > 1$. Then $0 < 1/x < 1$. Since, in addition, $\varepsilon/x > 0$, it follows from Lemmas 5.6.6(d) and 5.6.6(f) that $(1/x)^{1/n} < 1 + \varepsilon/x$ for all positive integers n. We see that

$$\begin{aligned} \frac{1}{x^{1/n}} \leq 1 + \frac{\varepsilon}{x} &\implies \frac{1}{x^{1/n}} \leq 1 + \frac{\varepsilon}{x} \\ &\implies 1 \leq x^{1/n} + \varepsilon\frac{x^{1/n}}{x}, \end{aligned}$$

and since $\frac{x^{1/n}}{x} \leq 1$ (see Lemma 5.6.6(e)),

$$\begin{aligned} 1 \leq x^{1/n} + \varepsilon\frac{x^{1/n}}{x} &\implies 1 \leq x^{1/n} + \varepsilon \\ &\implies x^{1/n} \geq 1 - \varepsilon \end{aligned}$$

for all positive integers n. Since $x > 0$, there exists an integer k such that $x^{1/k} \leq 1 + \varepsilon$ by $(\star)$. But, $x^{1/n} \leq x^{1/k}$ for all $n \geq k$ since $x > 1$ (see Lemma 5.6.6(e)). Thus, $x^{1/n} \leq 1 + \varepsilon$ for all $n \geq k$. For all $n \geq k$, we have $1 - \varepsilon \leq x^{1/n} \leq 1 + \varepsilon$, so that $|x^{1/n} - 1| \leq \varepsilon$. Hence, $(x^{1/n})_{n=1}^{\infty}$ is eventually ε-close to 1. This means that $(x^{1/n})_{n=1}^{\infty}$ converges to 1.

Lastly, suppose that $x = 1$. Then, $x^{1/n} = 1$ for all integers $n \geq 1$ by Lemma 5.6.6(e), and so the sequence $(x^{1/n})_{n=1}^{\infty}$ converges to 1.

In conclusion,

$$\lim_{n\to\infty} x^{1/n} = 1$$

in all cases.

6.6 Subsequences

Exercise 6.6.1

Let $(a_n)_{n=0}^{\infty}$, $(b_n)_{n=0}^{\infty}$, and $(c_n)_{n=0}^{\infty}$ be sequences of real numbers.

Given that the function $f : \mathbf{N} \to \mathbf{N}$ defined by $f(n) := n$ is strictly increasing and $a_n = a_{f(n)}$ for all $n \in \mathbf{N}$, the sequence $(a_n)_{n=0}^{\infty}$ is a subsequence of $(a_n)_{n=0}^{\infty}$ by Definition 6.6.1.

Suppose that $(b_n)_{n=0}^{\infty}$ is a subsequence of $(a_n)_{n=0}^{\infty}$ and $(c_n)_{n=0}^{\infty}$ is a subsequence of $(b_n)_{n=0}^{\infty}$. By Definition 6.6.1, there exist strictly increasing functions $g : \mathbf{N} \to \mathbf{N}$ and $h : \mathbf{N} \to \mathbf{N}$ such that $b_n = a_{g(n)}$ and $c_n = b_{h(n)}$ for all $n \in \mathbf{N}$. We see that $c_n = a_{g \circ h(n)}$ for all $n \in \mathbf{N}$. Now, the function $g \circ h$ is defined from $\mathbf{N}$ to $\mathbf{N}$. Moreover, for any $n \in \mathbf{N}$, $h(n+1) > h(n)$ (because h is strictly increasing), and so $g(h(n+1)) > g(h(n))$ (because g is strictly increasing). Thus, $g \circ h$ is a strictly increasing function. It follows from Definition 6.6.1 that $(c_n)_{n=0}^{\infty}$ is also a subsequence of $(a_n)_{n=0}^{\infty}$.

Exercise 6.6.2

Let $(a_n)_{n=0}^{\infty}$ be the sequence defined by $a_{2n} := 0$ and $a_{2n+1} := 1$ for all $n \in \mathbf{N}$, and let $(b_n)_{n=0}^{\infty}$ be the sequence defined by $b_{2n} := 1$ and $b_{2n+1} := 0$ for all $n \in \mathbf{N}$. More informally, $(a_n)_{n=0}^{\infty}$ is the sequence $0, 1, 0, 1, 0, 1, \ldots$, and $(b_n)_{n=0}^{\infty}$ is the sequence $1, 0, 1, 0, 1, 0, \ldots$.

Since $a_0 = 0 \neq 1 = b_0$, the sequences $(a_n)_{n=0}^{\infty}$ and $(b_n)_{n=0}^{\infty}$ are not the same. Nonetheless, we have:

$$a_n = b_{n+1} \quad \text{and} \quad b_n = a_{n+1}$$

for all $n \in \mathbf{N}$. Note that the function : $n \mapsto n+1$ defined from $\mathbf{N}$ to $\mathbf{N}$ is a strictly increasing function. Consequently, $(a_n)_{n=0}^{\infty}$ is a subsequence of $(b_n)_{n=0}^{\infty}$ and $(b_n)_{n=0}^{\infty}$ is a subsequence of $(a_n)_{n=0}^{\infty}$, according to Definition 6.6.1.

Exercise 6.6.3

Let $(a_n)_{n=0}^{\infty}$ be a sequence that is not bounded.

In particular, $(a_n)_{n=0}^{\infty}$ is not bounded by 0. Thus, there exists a natural number n such that $|a_n| > 0$. Hence, the set $\{n \in \mathbf{N} : |a_n| > 0\}$ is a non-empty subset of $\mathbf{N}$. By

the well-ordering principle, there exists a unique natural number n_0 defined by

$$n_0 := \min\{n \in \mathbf{N} : |a_n| > 0\}.$$

Let $i \geq 1$ be a natural number. Given that $(a_n)_{n=0}^{\infty}$ is unbounded, for any natural number N, there exists a natural number $n > N$ such that $|a_n| \geq i$ (otherwise $(a_n)_{n=N+1}^{\infty}$ would be bounded, and thus $(a_n)_{n=0}^{\infty}$ would be bounded given that $(a_n)_{n=0}^{N}$ is bounded). Hence, the set $\{n \in \mathbf{N} : |a_n| \geq i;\ n > N\}$ is a non-empty subset of $\mathbf{N}$ for any natural number N. It follows from the well-ordering principle that for any natural number N, $\min\{n \in \mathbf{N} : |a_n| \geq i;\ n > N\}$ exists and is a unique natural number.

Now, for any natural number $j \geq 1$, we can recursively define the unique natural number

$$n_j := \min\{n \in \mathbf{N} : |a_n| \geq j;\ n > n_{j-1}\}.$$

Next, define the sequence $(b_n)_{n=0}^{\infty}$ by $b_j := a_{n_j}$ for all $j \in \mathbf{N}$. It is clear that $(b_n)_{n=0}^{\infty}$ is a subsequence of $(a_n)_{n=0}^{\infty}$ because $n_{j+1} > n_j$ for any $j \in \mathbf{N}$.

Note that $b_n \neq 0$ for any natural number n. Furthermore, for any natural number $n \geq 1$, we have $|b_n| \geq n$, so that $1/|b_n| \leq 1/n$. Let $\varepsilon > 0$. There exists a natural number $N \geq 1$ such that $1/N \leq \varepsilon$. For all $n \geq N$, $1/n \leq 1/N \leq \varepsilon$, and so $1/|b_n| \leq \varepsilon$, i.e., $|1/b_n| \leq \varepsilon$. So, $(1/b_n)_{n=0}^{\infty}$ is eventually ε-close to 0. As ε is arbitrary, we conclude that $(1/b_n)_{n=0}^{\infty}$ converges to 0.[8] In other words, $\lim_{n\to\infty} 1/b_n$ exists and is equal to 0.

Exercise 6.6.4

Let $(a_n)_{n=0}^{\infty}$ be a sequence of real numbers and let L be a real number.

On the one hand, suppose that every subsequence of $(a_n)_{n=0}^{\infty}$ converges to L. By Lemma 6.6.4, $(a_n)_{n=0}^{\infty}$ is a subsequence of $(a_n)_{n=0}^{\infty}$. Thus, $(a_n)_{n=0}^{\infty}$ also converges to L.

On the other hand, suppose that $(a_n)_{n=0}^{\infty}$ converges to L. Let $(b_n)_{n=0}^{\infty}$ be a subsequence of $(a_n)_{n=0}^{\infty}$. By Definition 6.6.1, there exists a strictly increasing function $f : \mathbf{N} \to \mathbf{N}$ such that $b_n = a_{f(n)}$ for all $n \in \mathbf{N}$. Now, let ε be an arbitrary positive real. Since $(a_n)_{n=0}^{\infty}$ converges to L, there exists a natural number N such that $|a_n - L| \leq \varepsilon$ for all $n \geq N$. It is easy to prove by induction that $f(n) \geq n$ for all $n \in \mathbf{N}$ by noting that $f(0) \geq 0$ (as $f(0) \in \mathbf{N}$), and that $f(n+1) > f(n)$ and $f(n+1) \in \mathbf{N}$ for all $n \in \mathbf{N}$.[9] So

[8] Another approach would consist of using the squeeze test to show that $(1/|b_n|)_{n=0}^{\infty}$ converges to 0, and then applying Corollary 6.4.7 to conclude that $(1/b_n)_{n=0}^{\infty}$ converges to 0 as well.

[9] From now on, we will use the fact that if $(p_n)_{n=m}^{\infty}$ is a subsequence of $(q_n)_{n=m'}^{\infty}$, then for any $M \geq m$, $p_M = q_N$ for some $N \geq M$, without further comment.

for any $n \geq N$, we also have $f(n) \geq N$. Thus, for all $n \geq N$, we have $|a_{f(n)} - L| \leq \varepsilon$, that is $|b_n - L| \leq \varepsilon$. This shows that $(b_n)_{n=0}^{\infty}$ is eventually ε-close to L. Therefore, $(b_n)_{n=0}^{\infty}$ converges to L.

In conclusion, $(a_n)_{n=0}^{\infty}$ converges to L if and only if every subsequence of $(a_n)_{n=0}^{\infty}$ converges to L.

Exercise 6.6.5

Let $(a_n)_{n=0}^{\infty}$ be a sequence of real numbers and let L be a real number.

On the one hand, suppose that there exists a subsequence $(b_n)_{n=0}^{\infty}$ of $(a_n)_{n=0}^{\infty}$ which converges to L. By Proposition 6.4.5, L is a limit point of $(b_n)_{n=0}^{\infty}$. Therefore, for every real $\varepsilon > 0$ and every natural number N, there exists an integer $m \geq N$ such that $|b_m - L| \leq \varepsilon$; this implies that $|a_n - L| \leq \varepsilon$ for some $n \geq N$ (since $b_m = a_n$ for some integer $n \geq m$). Thus, L is continually ε-adherent to $(a_n)_{n=0}^{\infty}$ for any $\varepsilon > 0$. This means that L is a limit point of $(a_n)_{n=0}^{\infty}$.

On the other hand, suppose that L is a limit point of $(a_n)_{n=0}^{\infty}$. This means that L is continually δ-adherent to $(a_n)_{n=0}^{\infty}$ for any real $\delta > 0$. So, for any real $\delta > 0$ and any natural number N, there exists an integer $n \geq N$ such that $|a_n - L| \leq \delta$; hence, the set $\{n \in \mathbf{N} : |a_n - L| \leq \delta;\ n > N\}$ is always a non-empty subset of $\mathbf{N}$. According to the well-ordering principle, $\min\{n \in \mathbf{N} : |a_n - L| \leq \delta;\ n > N\}$ is a unique natural number for any real $\delta > 0$ and natural number N. So, for any natural number $j \geq 1$, we can recursively define the natural number n_j by:

$$n_j := \min\{n \in \mathbf{N} : |a_n - L| \leq 1/j;\ n > n_{j-1}\},$$

with $n_0 := 0$. Now, let $(b_n)_{n=0}^{\infty}$ be the subsequence of $(a_n)_{n=0}^{\infty}$ defined by $b_j := a_{n_j}$ for all $j \in \mathbf{N}$. Take an arbitrary positive real ε. There exists a natural number $N \geq 1$ such that $1/N \leq \varepsilon$. For all integers $n \geq N$, we have $1/n \leq 1/N$, and so $1/n \leq \varepsilon$. But for all integers $n \geq N$, we have $|b_n - L| \leq 1/n$, and hence $|b_n - L| \leq \varepsilon$. This shows that $(b_n)_{n=0}^{\infty}$ is eventually ε-close to L. As ε is arbitrary, we deduce that the sequence $(b_n)_{n=0}^{\infty}$, a subsequence of $(a_n)_{n=0}^{\infty}$, converges to L.

In conclusion, there exists a subsequence of $(a_n)_{n=0}^{\infty}$ which converges to L if and only if L is a limit point of $(a_n)_{n=0}^{\infty}$.

6.7 Real Exponentiation, Part II

Exercise 6.7.1

Let $x, y > 0$ be positive reals, and let q, r be real numbers. There exist a Cauchy sequence of rationals $(q_n)_{n=1}^{\infty}$ that converges to q and a Cauchy sequence of rationals $(r_n)_{n=1}^{\infty}$ that converges to r. We are to prove the following six statements.

(a) x^q is a positive real.

By Definition 6.7.2,

$$x^q = \lim_{n\to\infty} x^{q_n}.$$

Given that $x > 0$, $x^{q_n} > 0$ for any integer $n \geq 1$ by Lemma 5.6.9(a). Thus, we can apply Lemma 6.4.13 to get $\limsup_{n\to\infty} x^{q_n} \geq \limsup(0)_{n=1}^{\infty}$, that is $\limsup_{n\to\infty} x^{q_n} \geq 0$. But since $(x^{q_n})_{n=1}^{\infty}$ converges, we have $\limsup_{n\to\infty} x^{q_n} = \lim_{n\to\infty} x^{q_n}$ by Proposition 6.4.12(f). Therefore,

$$x^q \geq 0.$$

Suppose that $x^q = 0$. Then $x^q \times z = 0$ for any real number z. In particular, $x^q x^{-q} = 0$. However, $x^q x^{-q} = 0$ implies that $x^{q-q} = 0$ by the first part of Lemma 5.6.9(b) for real numbers (whose proof does not depend on the statement we are currently proving[10]). So, $x^0 = 0$, that is $1 = 0$. This contradictions shows that $x^q \neq 0$. Therefore,

$$x^q > 0.$$

(b) $(x^q)^r = x^{qr}$.[11]

First, assume that $r = 0$. Then clearly, $(x^q)^r = (x^q)^0 = 1$ and $x^{qr} = x^0 = 1$, and thus $(x^q)^r = x^{qr}$.

Second, assume that $r > 0$. By Proposition 5.4.14, there exists a rational q_1^+ such that $q < q_1^+ < q + 1$. It also follows from that proposition that there exists a rational q_2^+ such that $q < q_2^+ < \min(q_1^+, q + 1/2)$. Similarly, there exist a rational q_3^+ such that

[10] It was proved in the textbook, under the statement of Proposition 6.7.3

[11] This is a very long proof with several cases and sub-cases. To help the reader follow the argument, we summarize its structure here:
Case 1: $r = 0$, Case 2: $r > 0$, Case 3: $r < 0$.
Case 2.1: $x = 1$, Case 2.2: $x > 1$, Case 2.3: $x < 1$.
Case 2.2.1: $x^q > 1$, Case 2.2.2: $x^q < 1$, Case 2.2.3: $x^q = 1$.

$q < q_3^+ < \min(q_2^+, q+1/3)$ and a rational q_4^+ such that $q < q_4^+ < \min(q_3^+, q+1/4)$. Continuing in this way, we can build a sequence $(q_n^+)_{n=1}^\infty$ of rationals such that

$$q < q_n^+ < q + \frac{1}{n}$$

for all integers $n \geq 1$ and

$$q_i^+ > q_j^+$$

for all positive integers $i < j$. Applying Corollary 6.4.14 (squeeze test), we see that

$$\lim_{n\to\infty} q_n^+ = q.$$

So, $(q_n^+)_{n=1}^\infty$ is a decreasing sequence of rationals that converges to q. By Proposition 6.3.8, we also have $q = \inf(q_n^+)_{n=1}^\infty$. Using the same logic, we can build a decreasing sequence $(r_n^+)_{n=1}^\infty$ of rationals that converges to r, and thus has r as its infimum. Note that for all $n \geq 1$, we have $r_n^+ > 0$ because $r_n^+ \geq r$ and $r > 0$.

By Proposition 5.4.14 again, there exists a rational q_1^- such that $q-1 < q_1^- < q$. Similarly, there exist a rational q_2^- such that $\max(q_1^-, q-1/2) < q_2^- < q$ and a rational q_3^- such that $\max(q_2^-, q-1/3) < q_3^- < q$. Continuing in this way, we obtain a sequence $(q_n^-)_{n=1}^\infty$ of rationals such that

$$q - \frac{1}{n} < q_n^- < q$$

for all integers $n \geq 1$ and

$$q_i^- < q_j^-$$

for all positive integers $i < j$. By Corollary 6.4.14 and Proposition 6.3.8, $(q_n^-)_{n=1}^\infty$ is an increasing sequence of rationals that converges to its supremum q. Now, since $r > 0$, there exists a positive integer K such that $1/K < r$ by the Archimedean property. Thus, $r - 1/k > 0$ for all positive integers $k \geq K$. According to Proposition 5.4.14, there exists a rational r_1^- such that $r - 1/(K+1) < r_1^- < r$. By the same proposition, there exist a rational r_2^- such that $\max(r_1^-, r-1/(K+2)) < r_2^- < r$ and a rational r_3^- such that $\max(r_2^-, r-1/(K+3)) < r_3^- < r$. Continuing in this way, we build a sequence $(r_n^-)_{n=1}^\infty$ of rationals such that

$$r - \frac{1}{K+n} < r_n^- < r$$

for all integers $n \geq 1$, and

$$r_i^- < r_j^-$$

for all positive integers $i < j$, and

$$r_n^- > 0$$

for all integers $n \geq 1$. By Corollary 6.4.14 and Proposition 6.3.8, $(r_n^-)_{n=1}^\infty$ is an increasing sequence of positive rationals that converges to its supremum r.

Now, there are three cases to consider (still under the assumption that $r > 0$).

In the first case, $x = 1$. We have $(x^q)^r = (1^q)^r = 1^r = 1 = 1^{qr} = x^{qr}$, so the result holds.

In the second case, $x > 1$. By Lemma 5.6.9(e),

$$x^{q_n^+} \geq x^{q_{n+1}^+}$$

for all integers $n \geq 1$. Hence, $(x^{q_n^+})_{n=1}^\infty$ is a decreasing sequence. According to Definition 6.7.2,

$$\lim_{n\to\infty} x^{q_n^+} = x^q.$$

Since $(x^{q_n^+})_{n=1}^\infty$ converges, it is a bounded sequence by Corollary 6.1.17. So according to Proposition 6.3.8, we also have $x^q = \inf(x^{q_n^+})_{n=1}^\infty$. Thus,

$$x^{q_n^+} \geq x^q \qquad (\star)$$

for all $n \geq 1$. Suppose that $x^q > 1$. Then by Lemma 5.6.9(d), $(\star)$ implies that $(x^{q_n^+})^{r_m^+} \geq (x^q)^{r_m^+}$ for all integers $n, m \geq 1$. By Lemma 5.6.9(b), $x^{q_n^+ r_m^+} \geq (x^q)^{r_m^+}$ for all integers $n, m \geq 1$. We have $(x^q)^{r_m^+} \geq (x^q)^{r_{m+1}^+}$ for all integers $m \geq 1$ by Lemma 5.6.9(e). So, $((x^q)^{r_m^+})_{n=1}^\infty$ is a decreasing sequence, and since it converges to $(x^q)^r$ by Definition 6.7.2, we have $\inf((x^q)^{r_m^+})_{n=1}^\infty = (x^q)^r$ by Corollary 6.1.17 and Proposition 6.3.8. Hence for all integers $m \geq 1$, $(x^q)^{r_m^+} \geq (x^q)^r$. Therefore, $x^{q_n^+ r_m^+} \geq (x^q)^r$ for all integers $n, m \geq 1$. In particular,

$$x^{q_n^+ r_n^+} \geq (x^q)^r$$

for all integers $n \geq 1$. So, $\limsup_{n\to\infty} x^{q_n^+ r_n^+} \geq (x^q)^r$ by Lemma 6.4.13. But, $(x^{q_n^+ r_n^+})_{n=1}^\infty$ converges to x^{qr} according to Definition 6.7.2, as $(q_n^+ r_n^+)_{n=1}^\infty$ converges to qr. So, $\limsup_{n\to\infty} x^{q_n^+ r_n^+} = x^{qr}$ by Proposition 6.4.12(f). Hence,

$$x^{qr} \geq (x^q)^r.$$

Suppose that $x^q < 1$. Then by Lemma 5.6.9(d), $(\star)$ implies that $(x^{q_n^+})^{r_m^-} \geq (x^q)^{r_m^-}$ for all integers $n, m \geq 1$. By Lemma 5.6.9(b), $x^{q_n^+ r_m^-} \geq (x^q)^{r_m^-}$ for all integers $n, m \geq 1$. We have $(x^q)^{r_m^-} \geq (x^q)^{r_{m+1}^-}$ for all integers $m \geq 1$ by Lemma 5.6.9(e). So, $((x^q)^{r_m^-})_{n=1}^\infty$ is

a decreasing sequence, and since it converges to $(x^q)^r$ by Definition 6.7.2, we have $\inf((x^q)^{r_m^-})_{n=1}^\infty = (x^q)^r$ by Corollary 6.1.17 and Proposition 6.3.8. Thus, $(x^q)^{r_m^-} \geq (x^q)^r$ so that $x^{q_n^+ r_m^-} \geq (x^q)^r$ for all integers $n, m \geq 1$. In particular,

$$x^{q_n^+ r_n^-} \geq (x^q)^r$$

for all integers $n \geq 1$. So, $\limsup_{n\to\infty} x^{q_n^+ r_n^-} \geq (x^q)^r$ by Lemma 6.4.13. But, $(x^{q_n^+ r_n^-})_{n=1}^\infty$ converges to x^{qr} according to Definition 6.7.2, as $(q_n^+ r_n^-)_{n=1}^\infty$ converges to qr. So, $\limsup_{n\to\infty} x^{q_n^+ r_n^-} = x^{qr}$ by Proposition 6.4.12(f). Hence,

$$x^{qr} \geq (x^q)^r.$$

Suppose that $x^q = 1$. Then by Lemma 5.6.9(b) and Lemma 5.6.9(d), $(\star)$ implies that $x^{q_n^+ r_m^-} \geq (x^q)^{r_m^-}$ for all integers $n, m \geq 1$. Since $x^q = 1$, $(x^q)^{r_m^-} = (x^q)^r = 1$ for all integers $m \geq 1$. So, $x^{q_n^+ r_m^-} \geq (x^q)^r$ for all integers $n, m \geq 1$. In particular,

$$x^{q_n^+ r_n^-} \geq (x^q)^r$$

for all integers $n \geq 1$. So, $\limsup_{n\to\infty} x^{q_n^+ r_n^-} \geq (x^q)^r$ by Lemma 6.4.13. But, $(x^{q_n^+ r_n^-})_{n=1}^\infty$ converges to x^{qr} according to Definition 6.7.2. So, $\limsup_{n\to\infty} x^{q_n^+ r_n^-} = x^{qr}$ by Proposition 6.4.12(f). Hence,

$$x^{qr} \geq (x^q)^r.$$

We have just established that $x^{qr} \geq (x^q)^r$ in the case where $x > 1$ (note that we are still under the assumption that $r > 0$). We now show that $x^{qr} \leq (x^q)^r$ by a similar argument. According to Lemma 5.6.9(e),

$$x^{q_n^-} \leq x^{q_{n+1}^-}$$

for all integers $n \geq 1$. Hence, $(x^{q_n^-})_{n=1}^\infty$ is an increasing sequence. According to Definition 6.7.2,

$$\lim_{n\to\infty} x^{q_n^-} = x^q.$$

So by Proposition 6.3.8 and Corollary 6.1.17, we also have $x^q = \sup(x^{q_n^-})_{n=1}^\infty$. Thus,

$$x^{q_n^-} \leq x^q \qquad (\star\star)$$

for all $n \geq 1$. Suppose that $x^q > 1$. Then by Lemma 5.6.9(d), $(\star\star)$ implies that $(x^{q_n^-})^{r_m^-} \leq (x^q)^{r_m^-}$ for all integers $n, m \geq 1$. By Lemma 5.6.9(b), $x^{q_n^- r_m^-} \leq (x^q)^{r_m^-}$ for all integers $n, m \geq 1$. We have $(x^q)^{r_m^-} \leq (x^q)^{r_{m+1}^-}$ for all integers $m \geq 1$ by Lemma 5.6.9(e). So, $((x^q)^{r_m^-})_{n=1}^\infty$ is an increasing sequence, and since it converges to $(x^q)^r$ by Definition 6.7.2, we have $\sup((x^q)^{r_m^-})_{n=1}^\infty = (x^q)^r$ by Proposition 6.3.8 and Corollary

6.1.17. Hence for all integers $m \geq 1$, $(x^q)^{r_m^-} \leq (x^q)^r$. Therefore, $x^{q_n^- r_m^-} \leq (x^q)^r$ for all integers $n, m \geq 1$. In particular,

$$x^{q_n^- r_n^-} \leq (x^q)^r$$

for all integers $n \geq 1$. So, $\limsup_{n\to\infty} x^{q_n^- r_n^-} \geq (x^q)^r$ by Lemma 6.4.13. But, $(x^{q_n^- r_n^-})_{n=1}^{\infty}$ converges to x^{qr} according to Definition 6.7.2. So, $\limsup_{n\to\infty} x^{q_n^- r_n^-} = x^{qr}$ by Proposition 6.4.12(f). Hence,

$$x^{qr} \leq (x^q)^r.$$

Suppose that $x^q < 1$. Then by Lemma 5.6.9(d), $(\star\star)$ implies that $(x^{q_n^-})^{r_m^+} \leq (x^q)^{r_m^+}$ for all integers $n, m \geq 1$. By Lemma 5.6.9(b), $x^{q_n^- r_m^+} \leq (x^q)^{r_m^+}$ for all integers $n, m \geq 1$. We have $(x^q)^{r_m^+} \leq (x^q)^{r_{m+1}^+}$ for all integers $m \geq 1$ by Lemma 5.6.9(e). So, $((x^q)^{r_m^+})_{n=1}^{\infty}$ is an increasing sequence, and since it converges to $(x^q)^r$ by Definition 6.7.2, we have $\sup((x^q)^{r_m^+})_{n=1}^{\infty} = (x^q)^r$ by Proposition 6.3.8 and Corollary 6.1.17. Thus, $(x^q)^{r_m^+} \leq (x^q)^r$ so that $x^{q_n^- r_m^+} \leq (x^q)^r$ for all integers $n, m \geq 1$. In particular,

$$x^{q_n^- r_n^+} \leq (x^q)^r$$

for all integers $n \geq 1$. So, $\limsup_{n\to\infty} x^{q_n^- r_n^+} \geq (x^q)^r$ by Lemma 6.4.13. But, $(x^{q_n^- r_n^+})_{n=1}^{\infty}$ converges to x^{qr} according to Definition 6.7.2. So, $\limsup_{n\to\infty} x^{q_n^- r_n^+} = x^{qr}$ by Proposition 6.4.12(f). Hence,

$$x^{qr} \leq (x^q)^r.$$

Suppose that $x^q = 1$. Then by Lemma 5.6.9(b) and Lemma 5.6.9(d), $(\star\star)$ implies that $x^{q_n^- r_m^-} \leq (x^q)^{r_m^-}$ for all integers $n, m \geq 1$. Since $x^q = 1$, $(x^q)^{r_m^-} = (x^q)^r = 1$ for all integers $m \geq 1$. So, $x^{q_n^- r_m^-} \leq (x^q)^r$ for all integers $n, m \geq 1$. In particular,

$$x^{q_n^- r_n^-} \leq (x^q)^r$$

for all integers $n \geq 1$. So, $\limsup_{n\to\infty} x^{q_n^- r_n^-} \geq (x^q)^r$ by Lemma 6.4.13. But, $(x^{q_n^- r_n^-})_{n=1}^{\infty}$ converges to x^{qr} according to Definition 6.7.2. So, $\limsup_{n\to\infty} x^{q_n^- r_n^-} = x^{qr}$ by Proposition 6.4.12(f). Hence,

$$x^{qr} \leq (x^q)^r.$$

We have just established that $x^{qr} \leq (x^q)^r$ in the case where $x > 1$. Since we also have $x^{qr} \geq (x^q)^r$, it follows that

$$x^{qr} = (x^q)^r$$

when $x > 1$ (still assuming that $r > 0$).

The third case is $x < 1$. Note that $(x^{-1})^{-1} = x$ by Lemma 5.6.9(b). Here, $x^{-1} > 1$ by Proposition 5.6.3. So using the second case, we have:

$$x^{qr} = \left((x^{-1})^{-1}\right)^{qr} = (x^{-1})^{-qr}$$

and

$$(x^q)^r = \left(\left((x^{-1})^{-1}\right)^q\right)^r = \left((x^{-1})^{-q}\right)^r = (x^{-1})^{-qr}.$$

Therefore,

$$x^{qr} = (x^q)^r,$$

as desired. We have now established that $x^{qr} = (x^q)^r$ whenever $r > 0$. It only remains to address the case where $r < 0$.

Third, assume that $r < 0$. Here, $-r > 0$, and so $x^{q(-r)} = (x^q)^{-r}$ in light of the previous case. That is, $x^{-qr} = (x^q)^{-r}$. With the first part of Lemma 5.6.9(b) applied to reals, observe that:

$$\begin{aligned}
x^{-qr} = (x^q)^{-r} &\implies x^{-qr}(x^q)^r = (x^q)^{-r}(x^q)^r \\
&\implies x^{-qr}(x^q)^r = (x^q)^{-r+r} \\
&\implies x^{-qr}(x^q)^r = (x^q)^0 \\
&\implies x^{-qr}(x^q)^r = 1 \\
&\implies x^{qr}x^{-qr}(x^q)^r = x^{qr} \\
&\implies x^{qr-qr}(x^q)^r = x^{qr} \\
&\implies x^0(x^q)^r = x^{qr} \\
&\implies 1(x^q)^r = x^{qr} \\
&\implies (x^q)^r = x^{qr}.
\end{aligned}$$

In conclusion, we have

$$(x^q)^r = x^{qr}.$$

in any case.

(c) $x^{-q} = 1/x^q$

By Definition 5.6.2, $1/x^q = (x^q)^{-1}$. According to (b), $(x^q)^{-1} = x^{-q}$. Hence,

$$1/x^q = x^{-q}.$$

(d) If $q > 0$, then $x > y$ if and only if $x^q > y^q$.

Assume that $q > 0$. Then, there exists a sequence $(a_n)_{n=1}^{\infty}$ of rationals which is positively bounded away from zero and converges to q. For any integer $n \geq 1$, $a_n > 0$.

On the one hand, suppose that $x > y$. For any integer $n \geq 1$, a_n is a positive rational, and so we have $x^{a_n} > y^{a_n}$ by Lemma 5.6.9(d). According to Lemma 6.4.13, $\liminf_{n\to\infty} x^{a_n} \geq \liminf_{n\to\infty} y^{a_n}$. Given that the sequences $(x^{a_n})_{n=1}^{\infty}$ and $(y^{a_n})_{n=1}^{\infty}$ converge to x^q and y^q respectively, we have

$$\liminf_{n\to\infty}(x^{a_n})_{n=1}^{\infty} = \lim_{n\to\infty}(x^{a_n})_{n=1}^{\infty} = x^q$$

and

$$\liminf_{n\to\infty}(y^{a_n})_{n=1}^{\infty} = \lim_{n\to\infty}(y^{a_n})_{n=1}^{\infty} = y^q$$

by Proposition 6.4.12(f). So,

$$x^q \geq y^q.$$

Suppose toward a contradiction that $x^q = y^q$. Then, $x^q x^{-q} = y^q x^{-q}$, so that $1 = y^q x^{-q}$ by the first part of Lemma 5.6.9(b) applied to reals. We have $x^{-q} = (x^{-1})^q$ by (b). By Lemma 5.6.9(f) for reals (whose proof is independent of the work here), $1 = y^q x^{-q}$ implies that $1 = (yx^{-1})^q$. Next, $1 = (yx^{-1})^q$ implies that $1^{1/q} = ((yx^{-1})^q)^{1/q}$, so that $1 = yx^{-1}$ by (b). But this yields $x = y$, a contradiction. So, $x^q \neq y^q$, and we have

$$x^q > y^q.$$

On the other hand, suppose that $x^q > y^q$. If $x = y$, then clearly $x^q = y^q$, which is contradictory. If $x < y$, then the argument in the previous paragraph shows that we must have $y^q > x^q$, a contradiction. Therefore, $x > y$.

In conclusion, $x > y$ if and only if $x^q > y^q$.

(e) If $x > 1$, then $x^q > x^r$ if and only if $q > r$. If $x < 1$, then $x^q > x^r$ if and only if $q < r$.

Suppose that $x > 1$. If $q > r$, then $q - r > 0$, and so $x^{q-r} > 1^{q-r}$ (by (c)), and so $x^q x^{-r} > 1$ (by Lemma 5.6.9(b) applied to reals), and so $x^q x^{-r} x^r > x^r$, so that $x^q > x^r$ (by Lemma 5.6.9(b) applied to reals). If $x^q > x^r$, then $q = r$ yields the contradiction $x^q = x^r$ and $q < r$ yields the contradiction $x^q < x^r$ (as shown in the previous sentence), so that we must have $q > r$. Therefore, $x^q > x^r$ if and only if $q > r$.

Now suppose that $x < 1$. Here, $x^{-1} > 1$. So, $(x^{-1})^q < (x^{-1})^r$ if and only if $q < r$,

as shown in the previous paragraph. Note that

$$\begin{aligned}
(x^{-1})^q < (x^{-1})^r &\iff x^{-q} < x^{-r} && \text{by Lemma 5.6.9(b) applied to reals} \\
&\iff x^q x^r x^{-q} < x^q x^r x^{-r} \\
&\qquad \text{since } x^q, x^{-q}, x^r, x^{-r} > 0 \text{ by Lemma 5.6.9(a) for reals} \\
&\iff x^r < x^q && \text{by Lemma 5.6.9(b) applied to reals.}
\end{aligned}$$

Therefore, $x^q > x^r$ if and only if $q < r$.

(f) $(xy)^q = x^q y^q$.

By Definition 6.7.2, $(x^{q_n})_{n=1}^{\infty}$ converges to x^q and $(y^{q_n})_{n=1}^{\infty}$ converges to y^q. We have:

$$\begin{aligned}
(xy)^q &= \lim_{n\to\infty} (xy)^{q_n} && \text{by Definition 6.7.2} \\
&= \lim_{n\to\infty} x^{q_n} y^{q_n} && \text{by Lemma 5.6.9(f)} \\
&= \left(\lim_{n\to\infty} x^{q_n}\right)\left(\lim_{n\to\infty} y^{q_n}\right) && \text{by Theorem 6.1.19(b)} \\
&= x^q y^q && \text{by Definition 6.7.2.}
\end{aligned}$$

Chapter 7
Series

7.1 Finite Series

Exercise 7.1.1

This exercise would be much simpler if we restricted our attention to natural numbers. We would just use induction on n or on p, starting at 0 or at 1. But here, we must show that the statements hold for negative integers as well. To do this, we will need a modified version of the principle of induction, one that allows us to start below 0. So before proving the 6 statements of Lemma 7.1.4, let us establish the following:

Corollary 7.1.0: "Let m be an integer. Let $P(n)$ be a property pertaining to an integer n such that, for any integer $n \geq m$, if $P(n)$ is true, then $P(n+1)$ is also true. Suppose that $P(m)$ is true. Then, $P(n)$ is true for all integers $n \geq m$."

Proof of Corollary 7.1.0: Given all the assumptions of Corollary 7.1.0, let $Q(n)$ be a property pertaining to a natural number n defined by $Q(n) \iff P(m+n)$. That is, for any natural number n, $Q(n)$ is true if and only if $P(m+n)$ is true. Note that $Q(0)$ is true because $P(m)$ is true. Furthermore, if k is a natural number such that $Q(k)$ is true, then $P(m+k)$ is true, which implies that $P(m+k+1)$ is true as $m+k \geq m$, and so $Q(k+1)$ is also true. Thus by Axiom 2.5, $Q(n)$ is true for all natural numbers n. In other words, $P(m+n)$ is true for all natural numbers n. With a change of variable, this means that $P(n)$ is true for all integers $n \geq m$.

With the corollary established, we now turn to Lemma 7.1.4.[1]

[1] Corollary 7.1.0 will be used without further comment.

(a) Let m be an integer. We will show by induction on p that the statement

$$A(p): \text{ “For any integer } m \leq n < p \text{ and any real sequence } (a_i)_{i=m}^{p},$$
$$\text{we have } \sum_{i=m}^{n} a_i + \sum_{i=n+1}^{p} a_i = \sum_{i=m}^{p} a_i.\text{”}$$

is true for all integers $p > m$.

Let's start at the base case $p = m+1$. Let n be an integer such that $m \leq n < p$. Then, $n = m$ and $n+1 = m+1 = p$. Let $(a_i)_{i=m}^{p}$ be a sequence of real numbers (note that $(a_i)_{i=m}^{p} = (a_i)_{i=m}^{m+1}$). By Definition 7.1.1., we have

$$\sum_{i=m}^{n} a_i = \sum_{i=m}^{m} a_i = \sum_{i=m}^{m-1} a_i + a_m = a_m,$$

and

$$\sum_{i=n+1}^{p} a_i = \sum_{i=m+1}^{m+1} a_i = \sum_{i=m+1}^{m} a_i + a_{m+1} = a_{m+1},$$

and also

$$\sum_{i=m}^{p} a_i = \sum_{i=m}^{m+1} a_i = \sum_{i=m}^{m} a_i + a_{m+1} = a_m + a_{m+1}.$$

Therefore,

$$\sum_{i=m}^{n} a_i + \sum_{i=n+1}^{p} a_i = \sum_{i=m}^{p} a_i$$

as required.

Now, suppose inductively that $A(p)$ is true for some integer $p \geq m+1$. Let n be an integer such that $m \leq n < p+1$. Let $(a_i)_{i=m}^{p+1}$ be a real sequence. Note that $\sum_{i=n+1}^{p+1} a_i = \sum_{i=n+1}^{p} a_i + a_{p+1}$ by Definition 7.1.1, since $p \geq n$. So,

$$\sum_{i=m}^{n} a_i + \sum_{i=n+1}^{p+1} a_i = \sum_{i=m}^{n} a_i + \sum_{i=n+1}^{p} a_i + a_{p+1}.$$

If $n < p$, we have

$$\sum_{i=m}^{n} a_i + \sum_{i=n+1}^{p} a_i = \sum_{i=m}^{p} a_i$$

by considering the sequence $(a_i)_{i=m}^{p}$ and using the induction hypothesis. If $n = p$, we have

$$\sum_{i=m}^{n} a_i + \sum_{i=n+1}^{p} a_i = \sum_{i=m}^{p} a_i + \sum_{i=p+1}^{p} a_i = \sum_{i=m}^{p} a_i,$$

because $\sum_{i=p+1}^{p} a_i = 0$ by Definition 7.1.1. In any case,

$$\sum_{i=m}^{n} a_i + \sum_{i=n+1}^{p+1} a_i = \sum_{i=m}^{p} a_i + a_{p+1}.$$

As $p \geq m-1$, we have

$$\sum_{i=m}^{p+1} a_i = \sum_{i=m}^{p} a_i + a_{p+1}$$

by Definition 7.1.1, so that

$$\sum_{i=m}^{n} a_i + \sum_{i=n+1}^{p+1} a_i = \sum_{i=m}^{p+1} a_i.$$

Therefore, $A(p+1)$ is also true, and this closes the induction.

(b) Let m be an integer. We will show by induction on the integer n that the statement

$$B(n): \text{ “For any integer } k \text{ and any real sequence } (a_i)_{i=m}^{n},$$
$$\text{we have } \sum_{i=m}^{n} a_i = \sum_{j=m+k}^{n+k} a_{j-k}.\text{”}$$

is true for all integers $n \geq m$.

At the base case, $n = m$. Let $(a_i)_{i=m}^{n}$ be a sequence of real numbers. For any natural number k, we have

$$\sum_{j=m+k}^{n+k} a_{j-k} = \sum_{j=m+k}^{m+k} a_{j-k} = a_{(m+k)-k} = a_m.$$

Also,

$$\sum_{i=m}^{n} a_i = \sum_{i=m}^{m} a_i = a_m.$$

So,

$$\sum_{i=m}^{n} a_i = \sum_{j=m+k}^{n+k} a_{j-k},$$

and the base case is done.

For the inductive step, suppose that $B(n)$ is true for some integer $n \geq m$. Let $(a_i)_{i=m}^{n+1}$ be a sequence of real numbers, and let k be an arbitrary integer. We have:

$$\sum_{j=m+k}^{n+1+k} a_{j-k} = \sum_{j=m+k}^{n+k} a_{j-k} + a_{n+1}$$

by Definition 7.1.1. By the induction hypothesis applied to the sequence $(a_i)_{i=m}^{n}$, we have

$$\sum_{j=m+k}^{n+k} a_{j-k} = \sum_{i=m}^{n} a_i.$$

Thus,

$$\sum_{j=m+k}^{n+1+k} a_{j-k} = \sum_{i=m}^{n} a_i + a_{n+1}.$$

It then follows from Definition 7.1.1 that

$$\sum_{j=m+k}^{n+1+k} a_{j-k} = \sum_{i=m}^{n+1} a_i.$$

Therefore, $B(n+1)$ is true as well. We conclude that $B(n)$ is true for integers $n \geq m$.

(c) Let m be an integer. We will show by induction on n that the statement

$C(n)$: "For any real sequences $(a_i)_{i=m}^{n}$ and $(b_i)_{i=m}^{n}$,

$$\text{we have } \sum_{i=m}^{n} (a_i + b_i) = \left(\sum_{i=m}^{n} a_i\right) + \left(\sum_{i=m}^{n} b_i\right)."$$

is true for all integers $n \geq m$.

For the base case, $n = m$, observe that for any sequences $(a_i)_{i=m}^{n}$ and $(b_i)_{i=m}^{n}$, we have

$$\sum_{i=m}^{n} (a_i + b_i) = \sum_{i=m}^{m} (a_i + b_i) = a_m + b_m = \sum_{i=m}^{m} a_i + \sum_{i=m}^{m} b_i = \sum_{i=m}^{n} a_i + \sum_{i=m}^{n} b_i,$$

and we are done.

Now, suppose inductively that $C(n)$ is true for some integer $n \geq m$. Let $(a_i)_{i=m}^{n+1}$ be a sequence. We have:

$$\sum_{i=m}^{n+1} (a_i + b_i) = \sum_{i=m}^{n} (a_i + b_i) + (a_{n+1} + b_{n+1})$$

by Definition 7.1.1. By considering the sequence $(a_i)_{i=m}^{n}$ and the induction hypothesis, we see that:

$$\sum_{i=m}^{n} (a_i + b_i) = \sum_{i=m}^{n} a_i + \sum_{i=m}^{n} b_i.$$

Thus,

$$\sum_{i=m}^{n+1}(a_i+b_i)=\sum_{i=m}^{n}a_i+\sum_{i=m}^{n}b_i+(a_{n+1}+b_{n+1})=\left(\sum_{i=m}^{n}a_i+a_{n+1}\right)+\left(\sum_{i=m}^{n}b_i+b_{n+1}\right),$$

so that

$$\sum_{i=m}^{n+1}(a_i+b_i)=\sum_{i=m}^{n+1}a_i+\sum_{i=m}^{n+1}b_i,$$

by Definition 7.1.1. Therefore, $C(n+1)$ is also true, and the induction is complete.

(d) Let m be an integer. We will show by induction on n that the statement

$$D(n):\ \text{“For any real number } c \text{ and any real sequence } (a_i)_{i=m}^{n},\ \text{we have } \sum_{i=m}^{n}(ca_i)=c\left(\sum_{i=m}^{n}a_i\right).\text{”}$$

is true for all integers $n \geq m$.

Let's start at the base case $n=m$. Let $(a_i)_{i=m}^{n}$ be a sequence of real numbers and let c be a real number. Observe that

$$\sum_{i=m}^{n}(ca_i)=\sum_{i=m}^{m}(ca_i)=ca_m=c(a_m)=c\left(\sum_{i=m}^{m}a_i\right)=c\left(\sum_{i=m}^{n}a_i\right),$$

and we are done with the base case.

Suppose inductively that $D(n)$ is true for integer $n \geq m$. Let $(a_i)_{i=m}^{n+1}$ be a sequence of reals, and let c be a real. We have

$$\sum_{i=m}^{n+1}(ca_i)=\sum_{i=m}^{n}(ca_i)+ca_{n+1}$$

according Definition 7.1.1. By the induction hypothesis applied on $(a_i)_{i=m}^{n}$,

$$\sum_{i=m}^{n}(ca_i)+ca_{n+1}=c\left(\sum_{i=m}^{n}a_i\right)+c(a_{n+1}).$$

So,

$$\sum_{i=m}^{n+1}(ca_i)=c\left(\sum_{i=m}^{n}a_i\right)+c(a_{n+1})=c\left(\sum_{i=m}^{n}a_i+a_{n+1}\right)=c\left(\sum_{i=m}^{n+1}a_i\right),$$

where the last equality holds by Definition 7.1.1. Therefore, $D(n+1)$ is true, and this closes the induction.

(e) Let m be an integer. We will show by induction on n that the statement

$$E(n): \text{ “For any real sequence } (a_i)_{i=m}^n, \text{ we have } \left|\sum_{i=m}^{n} a_i\right| \leq \sum_{i=m}^{n} |a_i|.\text{”}$$

is true for all $n \geq m$.

Let us start at the base case $n = m$. Let $(a_i)_{i=m}^n$ be a sequence of real numbers. We have:

$$\left|\sum_{i=m}^{n} a_i\right| = \left|\sum_{i=m}^{m} a_i\right| = |a_m| = \sum_{i=m}^{m} |a_i| = \sum_{i=m}^{n} |a_i|.$$

Hence, $E(0)$ is true.

For the inductive step, suppose that $E(n)$ is true for some natural number n. We have

$$\left|\sum_{i=m}^{n+1} a_i\right| = \left|\sum_{i=m}^{n} a_i + a_{n+1}\right|$$

according to Definition 7.1.1. Thus,

$$\left|\sum_{i=m}^{n+1} a_i\right| \leq \left|\sum_{i=m}^{n} a_i\right| + |a_{n+1}|.$$

by the triangle inequality. Observe that

$$\left|\sum_{i=m}^{n} a_i\right| + |a_{n+1}| \leq \sum_{i=m}^{n} |a_i| + |a_{n+1}|$$

by the induction hypothesis applied on $(a_i)_{i=m}^n$. Hence,

$$\left|\sum_{i=m}^{n+1} a_i\right| \leq \sum_{i=m}^{n} |a_i| + |a_{n+1}|,$$

that is,

$$\left|\sum_{i=m}^{n+1} a_i\right| \leq \sum_{i=m}^{n+1} |a_i|$$

by Definition 7.1.1. So, $E(n+1)$ is true. We conclude that $E(n)$ is true for all integers $n \geq m$.

(f) Let m be an integer. We will show by induction on n that the statement

$$F(n): \text{ “For any real sequences } (a_i)_{i=m}^n \text{ and } (b_i)_{i=m}^n \text{ such that } a_i \leq b_i \text{ for all } m \leq i \leq n, \text{ we have } \sum_{i=m}^{n} a_i \leq \sum_{i=m}^{n} b_i.\text{”}$$

is true for all integers $n \geq m$.

The base case, $n = m$, is straightforward. Let $(a_i)_{i=m}^{n}$ and $(b_i)_{i=m}^{n}$ be sequences of real numbers such that $a_i \leq b_i$ for all $m \leq i \leq n$. We have

$$\sum_{i=m}^{n} a_i = \sum_{i=m}^{m} a_i = a_m \leq b_m = \sum_{i=m}^{m} b_i = \sum_{i=n}^{n} b_i,$$

and the base case is done.

Now, suppose inductively that $F(n)$ is true for some integer $n \geq m$. Let $(a_i)_{i=m}^{n+1}$ and $(b_i)_{i=m}^{n+1}$ be sequences of real numbers such that $a_i \leq b_i$ for all $m \leq i \leq n+1$. By Definition 7.1.1,

$$\sum_{i=m}^{n+1} a_i = \sum_{i=m}^{n} a_i + a_{n+1}$$

By the induction hypothesis applied on $(a_i)_{i=m}^{n}$ and $(b_i)_{i=m}^{n}$, we have

$$\sum_{i=m}^{n} a_i + a_{n+1} \leq \sum_{i=m}^{n} b_i + a_{n+1}.$$

This implies that

$$\sum_{i=m}^{n} a_i + a_{n+1} \leq \sum_{i=m}^{n} b_i + b_{n+1}$$

because $a_{n+1} \leq b_{n+1}$. So,

$$\sum_{i=m}^{n+1} a_i \leq \sum_{i=m}^{n} b_i + b_{n+1}.$$

That is to say,

$$\sum_{i=m}^{n+1} a_i \leq \sum_{i=m}^{n+1} b_i$$

according to Definition 7.1.1. Thus, $F(n+1)$ is also true. This closes the induction.

Exercise 7.1.2

(a) Let X be the empty set, and $f : X \to \mathbf{R}$ be a function.

Let $g : \{i \in \mathbf{N} : 1 \leq i \leq 0\} \to X$ be a function. Since the domain and codomain of g are both empty, g is a bijective function (see Exercise 3.3.3). By Definition 7.1.6,

$$\sum_{x \in X} f(x) := \sum_{i=1}^{0} f(g(i)).$$

By Definition 7.1.1, $\sum_{i=1}^{0} f(g(i)) = 0$. Therefore,

$$\sum_{x \in X} f(x) = 0.$$

(b) Let X be a singleton set; that is, let $X = \{x_0\}$, where x_0 is an object. Let $f : X \to \mathbf{R}$ be a function.

Let $g : \{i \in \mathbf{N} : 1 \leq i \leq 1\} \to X$ be the function defined by $g(1) := x_0$. Clearly, g is a bijection, as every element of the codomain is the image through g of exactly one element of the domain. By Definitions 7.1.6 and 7.1.1,

$$\sum_{x \in X} f(x) = \sum_{i=1}^{1} f(g(i)) = f(g(1)) = f(x_0).$$

(c) Let X be a finite set, $f : X \to \mathbf{R}$ be a function, and $g : Y \to X$ be a bijection.

Let $n = \#(X)$. Since g is a bijection, we also have $\#(Y) = n$.

Let $h : \{i \in \mathbf{R} : 1 \leq i \leq n\} \to X$ be a bijection. Note that $g^{-1} \circ h$ is a bijection from the set $\{i \in \mathbf{R} : 1 \leq i \leq n\}$ to the set Y. Hence, by Definition 7.1.6, we have:

$$\sum_{y \in Y} f(g(y)) = \sum_{i=1}^{n} f(g(g^{-1} \circ h(i))).$$

Next,

$$\sum_{i=1}^{n} f(g(g^{-1} \circ h(i))) = \sum_{i=1}^{n} f(h(i)) = \sum_{x \in X} f(x)$$

by Definition 7.1.6. Therefore,

$$\sum_{y \in Y} f(g(y)) = \sum_{x \in X} f(x).$$

(d) Let $n \leq m$ be integers, and let X be the set $X := \{i \in \mathbf{Z} : n \leq i \leq m\}$. Let a_i be a real number assigned to each integer $i \in X$.

Let $f : X \to \mathbf{R}$ be the function defined by $f(i) := a_i$ for all $i \in X$. By Lemma 7.1.4(b),

$$\sum_{i=n}^{m} a_i = \sum_{i=1}^{(m-n)+1} a_{i+n-1}.$$

Given that for any i such that $1 \leq i \leq (m-n)+1$ we have $n \leq i+n-1 \leq m$, we can write

$$\sum_{i=1}^{(m-n)+1} a_{i+n-1} = \sum_{i=1}^{(m-n)+1} f(i+n-1),$$

so that

$$\sum_{i=n}^{m} a_i = \sum_{i=1}^{(m-n)+1} f(i+n-1).$$

Define the function $g : \{i \in \mathbf{N} : 1 \leq i \leq (m-n)+1\} \to X$ by $g(i) := i+n-1$. The function g is injective because for any $1 \leq i_1, i_2 \leq (m-n)+1$ with $i_1 \neq i_2$, we have $i_1+n-1 \neq i_2+n-1$, that is $g(i_1) \neq g(i_2)$. Plus, g is surjective because for any $x \in X$, we have $1 \leq x-n+1 \leq (m-n)+1$ and $g(x-n+1) = x$. Thus, g is a bijection. Now,

$$\sum_{i=n}^{m} a_i = \sum_{i=1}^{(m-n)+1} f(g(i)).$$

Given that g is a bijection from $\{i \in \mathbf{N} : 1 \leq i \leq (m-n)+1\}$ to X and f is a function defined from X to $\mathbf{R}$, it follows from Definition 7.1.6 that

$$\sum_{i=1}^{(m-n)+1} f(g(i)) = \sum_{i \in X} f(i).$$

Therefore,

$$\sum_{i=n}^{m} a_i = \sum_{i \in X} a_i.$$

(e) Let X and Y be disjoint finite sets, and let $f : X \cup Y \to \mathbf{R}$ be a function.

Set $n := \#(X)$ and $m := \#(Y)$. Note that $\#(X \cup Y) = n+m$. There exists a bijection h_x defined from $\{i \in \mathbf{N} : 1 \leq i \leq n\}$ to X. Since $\#(\{i \in \mathbf{N} : n+1 \leq i \leq n+m\}) = m$, there also exists a bijection h_y defined from $\{i \in \mathbf{N} : n+1 \leq i \leq n+m\}$ to Y.

Let $g : \{i \in \mathbf{N} : 1 \leq i \leq n+m\} \to X \cup Y$ be the function defined by

$$g(i) := \begin{cases} h_x(i), & i \in \{i \in \mathbf{N} : 1 \leq i \leq n\} \\ h_y(i), & i \in \{i \in \mathbf{N} : n+1 \leq i \leq n+m\} \end{cases}.$$

The function g is well-defined because X and Y are disjoint and the functions h_x and h_y are well-defined on $\{i \in \mathbf{N} : 1 \leq i \leq n\}$ and $\{i \in \mathbf{N} : n+1 \leq i \leq n+m\}$ respectively. Furthermore, g is clearly a bijection. To see this, take an arbitrary element $z \in X \cup Y$ (assuming $X \cup Y$ isn't empty; if $X \cup Y$ is empty, then g is a bijection from $\emptyset$ to $\emptyset$). Either $z \in X$ or $z \in Y$, but not both. Without loss of generality, assume that $z \in X$ and $z \notin Y$. Then, there exists a unique integer $j \in \{i \in \mathbf{N} : 1 \leq i \leq n\}$ such that $h_x(j) = z$ because h_x is a bijection. Since $X \cap Y = \emptyset$ and $h_y(i) \in Y$ for all $n+1 \leq i \leq n+m$, we see that $g(i) = z$ only if $1 \leq i \leq n$. But if $1 \leq i \leq n$, then $g(i) = h_x(i)$, and since h_x is bijective, $h_x(i) = h_x(j)$ if and only if $i = j$. Therefore, j is the unique element of $\{i \in \mathbf{N} : 1 \leq i \leq n+m\}$ such that $g(j) = z$. Hence, g is bijective.

Now, by Definition 7.1.6,

$$\sum_{z\in X\cup Y} f(z) = \sum_{i=1}^{n+m} f(g(i))$$

We have

$$\sum_{i=1}^{n+m} f(g(i)) = \sum_{i=1}^{n} f(g(i)) + \sum_{i=n+1}^{n+m} f(g(i))$$

by Definition 7.1.1 and Lemma 7.1.4(a).[2] Note that:

$$\sum_{i=1}^{n} f(g(i)) + \sum_{i=n+1}^{n+m} f(g(i)) = \sum_{i=1}^{n} f(h_x(i)) + \sum_{i=n+1}^{n+m} f(h_y(i)).$$

Let $\tilde{h}_y$ be the function defined from $\{i \in \mathbf{N} : 1 \le i \le m\}$ to Y by $\tilde{h}_y(i) := h_y(i+n)$. Clearly, $\tilde{h}$ is a bijection because it is the composition of the bijection h_y and the bijection from the set $\{i \in \mathbf{N} : 1 \le i \le m\}$ to the set $\{i \in \mathbf{N} : n+1 \le i \le n+m\}$ that maps each i to $i+n$. By Lemma 7.1.4(b), we have

$$\sum_{i=n+1}^{n+m} f(h_y(i)) = \sum_{i=1}^{m} f(h_y(i+n)) = \sum_{i=1}^{m} f(\tilde{h}_y(i)).$$

So,

$$\sum_{i=1}^{n} f(g(i)) + \sum_{i=n+1}^{n+m} f(g(i)) = \sum_{i=1}^{n} f(h_x(i)) + \sum_{i=1}^{m} f(\tilde{h}_y(i)).$$

By Definition 7.1.6, we have

$$\sum_{i=1}^{n} f(h_x(i)) + \sum_{i=1}^{m} f(\tilde{h}_y(i)) = \sum_{x\in X} f(x) + \sum_{y\in Y} f(y).$$

Therefore,

$$\sum_{z\in X\cup Y} f(z) = \sum_{x\in X} f(x) + \sum_{y\in Y} f(y).$$

(f) Let X be a finite set, and let $f : X \to \mathbf{R}$ and $g : X \to \mathbf{R}$ be functions.

Let $n := \#(X)$. There exists a bijection $h : \{i \in \mathbf{N} : 1 \le i \le n\} \to X$. By Definition 7.1.6,

$$\sum_{x\in X} f(x) + \sum_{x\in X} g(x) = \sum_{i=1}^{n} f(h(i)) + \sum_{i=1}^{n} g(h(i)).$$

[2] If $1 \le n < n+m$, then the result follows from Lemma 7.1.4(a), otherwise Definition 7.1.1 guarantees that this equality holds.

Using Lemma 7.1.4(c), we have

$$\sum_{i=1}^{n} f(h(i)) + \sum_{i=1}^{n} g(h(i)) = \sum_{i=1}^{n} \big(f(h(i)) + g(h(i))\big) = \sum_{i=1}^{n} (f+g)(h(i)).$$

By Definition 7.1.6 again,

$$\sum_{i=1}^{n} (f+g)(h(i)) = \sum_{x\in X} (f+g)(x) = \sum_{x\in X} \big(f(x)+g(x)\big).$$

In conclusion,

$$\sum_{x\in X} f(x) + \sum_{x\in X} g(x) = \sum_{x\in X} \big(f(x)+g(x)\big).$$

(g) Let X be a finite set, let $f : X \to \mathbf{R}$ be a function, and let c be a real number.

Denote by n the cardinality of X. There is a bijective function g defined from $\{i \in \mathbf{N} : 1 \le i \le n\}$ to X.

Let $h : X \to \mathbf{R}$ be the function defined by $h(x) := cf(x)$ for all $x \in X$. By Definition 7.1.6,

$$\sum_{x\in X} cf(x) = \sum_{x\in X} h(x) = \sum_{i=1}^{n} h(g(i)) = \sum_{i=1}^{n} c(f(g(i)).$$

According to Lemma 7.1.4(d),

$$\sum_{i=1}^{n} c(f(g(i)) = c\sum_{i=1}^{n} f(g(i)).$$

Given that

$$\sum_{i=1}^{n} f(g(i)) = \sum_{x\in X} f(x)$$

by Definition 7.1.6, we have

$$\sum_{x\in X} cf(x) = c\sum_{x\in X} f(x).$$

(h) Let X be a finite set, and let $f : X \to \mathbf{R}$ and $g : X \to \mathbf{R}$ be functions such that $f(x) \le g(x)$ for all $x \in X$.

Let $n := \#(X)$. There is a bijective function h defined from $\{i \in \mathbf{N} : 1 \le i \le n\}$ to X. We have

$$\sum_{x\in X} f(x) = \sum_{i=1}^{n} f(h(i)) \quad \text{and} \quad \sum_{x\in X} g(x) = \sum_{i=1}^{n} g(h(i))$$

by Definition 7.1.6.

Note that for any natural number $1 \le i \le n$, $f(h(i)) \le g(h(i))$ since $h(i) \in X$ and $f(x) \le g(x)$ for all $x \in X$. So according to Lemma 7.1.4(f), we have

$$\sum_{i=1}^{n} f(h(i)) \le \sum_{i=1}^{n} g(h(i)).$$

Hence,

$$\sum_{x \in X} f(x) \le \sum_{x \in X} g(x).$$

(i) Let X be a finite set and let $f : X \to \mathbf{R}$ be a function.

Denote by n the cardinality of X. There is a bijective function g defined from $\{i \in \mathbf{N} : 1 \le i \le n\}$ to X. By Definition 7.1.6,

$$\sum_{x \in X} f(x) = \sum_{i=1}^{n} f(g(i)) \quad \text{and} \quad \sum_{x \in X} |f(x)| = \sum_{i=1}^{n} |f(g(i))|.$$

According to Lemma 7.1.4(e),

$$\left| \sum_{i=1}^{n} f(g(i)) \right| \le \sum_{i=1}^{n} |f(g(i))|.$$

Therefore,

$$\left| \sum_{x \in X} f(x) \right| \le \sum_{x \in X} |f(x)|.$$

Exercise 7.1.3

Given any integers m, n and any finite sequence of real numbers $(a_i)_{i=m}^{n}$ we define the finite product $\prod_{i=m}^{n} a_i$ by the recursive formula[3]

$$\prod_{i=m}^{n} a_i := 1 \text{ whenever } n < m;$$

$$\prod_{i=m}^{n} a_i := \left(\prod_{i=m}^{n-1} a_i \right) \times a_n \text{ whenever } n \ge m.$$

[3] Notice that for the recursive step of the formula, we used $\prod_{i=m}^{n} a_i$ and not $\prod_{i=m}^{n+1} a_i$. We prefer the former because it does not assume the existence of a_{n+1}, unlike the latter.

Now, let X be a finite set with n elements, and let $f : X \to \mathbf{R}$ be a function. We define $\prod_{x\in X} f(x)$ by

$$\prod_{x\in X} f(x) := \prod_{i=1}^{n} f(g(i)),$$

where g is any bijection from $\{i \in \mathbf{N} : 1 \leq i \leq n\}$ to X (such a bijection exists because $\#(X) = n$).

We will now explore the results established for finite sums that have analogue for finite products. In doing so, we will not prove rigorously each statement. We will only provide a meta-proof that indicates the reasoning of the proof based on the proof of the original statement, because all the results for finite sums have already been proven. Corollary 7.1.0 (see Exercise 7.1.1) will be used without explicit mention.

Let us start with Lemma 7.1.4. Take an arbitrary integer m.

The analogue of Lemma 7.1.4(a) for finite products holds. That is, the statement

$$A'(p): \text{ “For any integer } m \leq n < p \text{ and any real sequence } (a_i)_{i=m}^{p},$$
$$\text{we have } \left(\prod_{i=m}^{n} a_i\right) \times \left(\prod_{i=n+1}^{p} a_i\right) = \prod_{i=m}^{p} a_i.\text{”}$$

is true for any natural number $p \geq m$. The proof follows the same structure as the proof of Lemma 7.4.1(a).

The analogue of Lemma 7.1.4(b), which is

$$B'(n): \text{ “For any integer } k \text{ and any real sequence } (a_i)_{i=m}^{n},$$
$$\text{we have } \prod_{i=m}^{n} a_i = \prod_{j=m+k}^{n+k} a_{j-k}.\text{”}$$

is true for all integers $n \geq m$, holds; the proof is just the proof of Lemma 7.1.4(b), with $\sum$ replaced by $\prod$, and addition replaced by multiplication.

Similarly, the analogue of Lemma 7.1.4(c), which says that

$$C'(n): \text{ “For any real sequences } (a_i)_{i=m}^{n} \text{ and } (b_i)_{i=m}^{n},$$
$$\text{we have } \prod_{i=m}^{n} (a_i \times b_i) = \left(\prod_{i=m}^{n} a_i\right) \times \left(\prod_{i=m}^{n} b_i\right).\text{”}$$

is true for all integers $n \geq m$, holds and the proof follows the same structure.

The analogue of Lemma 7.1.4(d) is that the statement

$$D'(n): \text{ “For any real number } c, \text{ and any sequence } (a_i)_{i=m}^{n},$$
$$\text{we have } \prod_{i=m}^{n} (ca_i) = c^n \left(\prod_{i=m}^{n} a_i\right).\text{”}$$

holds for all integers $n \geq m$. The proof is quite simple and very similar to the proof of Lemma 7.1.4(d). For the base case, just note that $c^0 = 0$ for any $c \in \mathbf{R}$; and for the inductive step, just note that for any integer $n \geq m$, $\prod_{i=m}^{n+1}(ca_i) = \left(\prod_{i=m}^{n}(ca_i)\right) \times ca_{n+1}$.

The analogue of Lemma 7.1.4(e) is that the statement

$$E'(n): \text{ “For any real sequence } (a_i)_{i=m}^n \text{, we have } \left|\prod_{i=m}^{n} a_i\right| = \prod_{i=m}^{n} |a_i|\text{.”}$$

is true for all integers $n \geq m$. To prove this, we simply follow the structure of the proof of Lemma 7.1.4(e), while keeping in mind that $|xy| = |x||y|$ for any $x, y \in \mathbf{R}$.

Lemma 7.1.4(f) does not have an analogue for finite products, unless we require $(a_i)_{i=m}^n$ and $(b_i)_{i=m}^n$ to be sequences of non-negative terms. Indeed, the statement

$$F'(n): \text{ “For any real sequences } (a_i)_{i=m}^n \text{ and } (b_i)_{i=m}^n \text{ such that } 0 \leq a_i \leq b_i \text{ for all } m \leq i \leq n \text{ we have } \prod_{i=m}^{n} a_i \leq \prod_{i=m}^{n} b_i\text{.”}$$

is true or any integer $n \geq m$. To prove this, we modify the proof of Lemma 7.1.4(f) while noting that if $x \leq y$, then $xz \leq yz$ for any non-negative reals x, y, and z.

Next, we move on to Proposition 7.1.8. It is clear that given a finite set X of n elements, a function $f : X \to \mathbf{R}$, and two bijections $g : \{i \in \mathbf{N} : 1 \leq i \leq n\} \to X$ and $h : \{i \in \mathbf{N} : 1 \leq i \leq n\} \to X$, we have:

$$\prod_{i=1}^{n} f(g(i)) = \prod_{i=1}^{n} f(h(i)).$$

The proof is just the same, but with $\sum$ replaced by $\prod$, and addition replaced by multiplication.

Now, we are at Proposition 7.1.11.

The analogue of Proposition 7.1.11(a) for finite product is that

$$\prod_{x \in X} f(x) = 1,$$

where $X = \emptyset$ and f is any function from X to $\mathbf{R}$. The proof is really just the same one we made for Proposition 7.1.11(a), but with finite products instead of finite sums.

The only difference between Proposition 7.1.11(b) and its analogue is that $\sum$ is replaced by $\prod$, and the analogue is as simple to prove as the initial proposition.

Similarly, we obtain the analogues of Proposition 7.1.11(c) and Proposition 7.1.11(d) for finite products by just replacing $\sum$ with $\prod$. Also, the proofs of the analogues follow the same structure as the proofs of the initial propositions.

The analogue of Proposition 7.1.11(e) which says that for any finite disjoint sets X, Y and any function $f : X \cup Y \to \mathbf{R}$, we have

$$\prod_{z \in X \cup Y} f(z) = \left(\prod_{x \in X} f(x) \right) \times \left(\prod_{y \in Y} f(y) \right),$$

holds. The proof follows the same logic as that of Proposition 7.1.11(e).

The analogue of Proposition 7.1.11(f) which says that for any finite set X and any functions $f : X \to \mathbf{R}$ and $g : X \to \mathbf{R}$, we have

$$\prod_{x \in X} (f(x) \times g(x)) = \left(\prod_{x \in X} f(x) \right) \times \left(\prod_{x \in X} g(x) \right),$$

holds. The proof follows the same logic as that of Proposition 7.1.11(f).

The analogue for finite products of Proposition 7.1.11(g) is that given a finite set X with n elements, a function $f : X \to \mathbf{R}$ and a real number c, we have

$$\prod_{x \in X} cf(x) = c^n \prod_{x \in X} f(x).$$

It is true and the proof just follows the same structure of the proof of Proposition 7.1.11(g), with the same modifications in the proof of the analogue of Lemma 7.1.4(g).

Proposition 7.1.11(h) does not have an analogue for finite products, unless we require the range of f and g to be a subset of non-negative reals. It is true that for any set X and two functions $f : X \to \{y \in \mathbf{R} : y \geq 0\}$ and $g : X \to \{y \in \mathbf{R} : y \geq 0\}$ such that $f(x) \leq g(x)$ for all $x \in X$, we have

$$\prod_{x \in X} f(x) \leq \prod_{x \in X} g(x).$$

The proof follows the same logic of the proof of Proposition 7.1.11(h) with the modifications in the proof of the analogue of Lemma 7.1.4(f).

The analogue of Proposition 7.1.11(i) is that for any finite set X,

$$\left| \prod_{x \in X} f(x) \right| = \prod_{x \in X} |f(x)|$$

is true. To show this, we simply follow the structure of the proof of Proposition 7.1.11(i), while keeping in mind that $|xy| = |x||y|$ for any $x, y \in \mathbf{R}$.

We now move on to Lemma 7.1.13. Its analogue for finite products is that if X and Y are finite sets and $f : X \times Y \to \mathbf{R}$ is a function, then

$$\prod_{x \in X} \left(\prod_{y \in Y} f(x, y) \right) = \prod_{(x,y) \in X \times Y} f(x, y).$$

This is true and to prove this, you can just follow the same steps in the proof of Lemma 7.1.13, while replacing finite sums by finite products and using the analogues of Propositions 7.1.11(a), 7.1.11(c), and 7.1.11(e) where they are invoked.

Finally, the analogue of Corollary 7.1.14 also holds for finite products. Through a similar proof which uses the analogue of Proposition 7.1.11(c) and the analogue of Lemma 7.1.13, we can easily show that

$$\begin{aligned}\prod_{x\in X}\left(\prod_{y\in Y} f(x,y)\right) &= \prod_{(x,y)\in X\times Y} f(x,y)\\ &= \prod_{(y,x)\in Y\times X} f(x,y)\\ &= \prod_{y\in Y}\left(\prod_{x\in X} f(x,y)\right).\end{aligned}$$

Exercise 7.1.4

Let x and y be two real numbers. We will prove by induction on n that the binomial formula

$$(x+y)^n = \sum_{i=0}^{n} \frac{n!}{j!(n-j)!} x^j y^{n-j}$$

holds for every natural number n.

We start at the base case $n = 0$. On the one hand, $(x+y)^0 = 0$. On the other hand,

$$\sum_{j=0}^{0} \frac{n!}{j!(n-j)!} x^j y^{n-j} = \frac{0!}{0!(0-0)!} x^0 y^{0-0} = 1.$$

Thus, the binomial formula holds for $n = 0$.

In addition, observe that

$$(x+y)^1 = x+y = \frac{1!}{0!(1-0)!} x^1 + \frac{1!}{1!(1-1)!} y^1 = \sum_{j=0}^{1} \frac{n!}{j!(n-j)!} x^j y^{n-j},$$

and so the binomial formula also holds for $n = 1$. Thus, we may proceed to the inductive step for $n \geq 1$. Suppose inductively that the binomial formula is true for some natural number $n \geq 1$. We have

$$\sum_{j=0}^{n+1} \frac{(n+1)!}{j!(n+1-j)!} x^j y^{n+1-j} = y^{n+1} + \sum_{j=1}^{n} \frac{(n+1)!}{j!(n+1-j)!} x^j y^{n+1-j} + x^{n+1}.$$

Note that for any natural number $j \geq 1$,

$$\begin{aligned}
\frac{n!}{j!(n-j)!} + \frac{n!}{(j-1)(n-j+1)!} &= \frac{1}{j}\frac{n!}{(j-1)!(n-j)!} + \frac{1}{n-j+1}\frac{n!}{(j-1)!(n-j)!} \\
&= \frac{n!}{(j-1)!(n-j)!}\left(\frac{1}{j} + \frac{1}{n-j+1}\right) \\
&= \frac{n!}{(j-1)!(n-j)!}\left(\frac{n+1}{j(n-j+1)}\right) \\
&= \frac{n! \times (n+1)}{(j-1)! \times j \times (n-j)! \times (n-j+1)} \\
&= \frac{(n+1)!}{j!(n+1-j)!}.
\end{aligned}$$

So,

$$\begin{aligned}
\sum_{j=1}^{n} \frac{(n+1)!}{j!(n+1-j)!} x^j y^{n+1-j} &= \sum_{j=1}^{n} \left(\frac{n!}{j!(n-j)!} + \frac{n!}{(j-1)(n-j+1)!}\right) x^j y^{n+1-j} \\
&= \sum_{j=1}^{n} \frac{n!}{j!(n-j)!} x^j y^{n+1-j} \\
&\qquad + \sum_{j=1}^{n} \frac{n!}{(j-1)(n-j+1)!} x^j y^{n+1-j} \\
&= y \sum_{j=1}^{n} \frac{n!}{j!(n-j)!} x^j y^{n-j} + \sum_{j=0}^{n-1} \frac{n!}{j(n-j)!} x^{j+1} y^{n-j} \\
&= y \sum_{j=1}^{n} \frac{n!}{j!(n-j)!} x^j y^{n-j} + x \sum_{j=0}^{n-1} \frac{n!}{j(n-j)!} x^j y^{n-j}.
\end{aligned}$$

By the induction hypothesis,

$$\sum_{j=1}^{n} \frac{n!}{j!(n-j)!} x^j y^{n-j} = \sum_{j=0}^{n} \frac{n!}{j!(n-j)!} x^j y^{n-j} - y^n = (x+y)^n - y^n$$

and

$$\sum_{j=0}^{n-1} \frac{n!}{j(n-j)!} x^j y^{n-j} = \sum_{j=0}^{n} \frac{n!}{j(n-j)!} x^j y^{n-j} - x^n = (x+y)^n - x^n.$$

It follows that

$$\begin{aligned}
\sum_{j=1}^{n} \frac{(n+1)!}{j!(n+1-j)!} x^j y^{n+1-j} &= y((x+y)^n - y^n) + x((x+y)^n - x^n) \\
&= (x+y)(x+y)^n - x^{n+1} - y^{n+1} \\
&= (x+y)^{n+1} - x^{n+1} - y^{n+1}.
\end{aligned}$$

Therefore,

$$\sum_{i=0}^{n+1} \frac{(n+1)!}{j!(n+1-j)!} x^j y^{n+1-j} = (x+y)^{n+1}.$$

We see that the binomial formula also holds for $n+1$. This completes the induction.

Exercise 7.1.5

Let X be a finite set, let m be an integer, and for each $x \in X$ let $(a_n(x))_{n=m}^{\infty}$ be a convergent sequence of real numbers. Consider the sequence $\left(\sum_{x\in X} a_n(x)\right)_{n=m}^{\infty}$.

Let k be the cardinality of X. If $k = 0$, then $X = \emptyset$ and $\left(\sum_{x\in X} a_n(x)\right)_{n=m}^{\infty} = (0)_{n=m}^{\infty}$ while $\sum_{x\in X} \lim_{n\to\infty} a_n(x) = 0$ by Proposition 7.1.11(a); hence, $\left(\sum_{x\in X} a_n(x)\right)_{n=m}^{\infty}$ converges to $\sum_{x\in X} \lim_{n\to\infty} a_n(x)$ and we are done. From now on, assume that $k > 0$. There exists a bijection g defined from $\{i \in \mathbf{N} : 1 \le i \le k\}$ to X. For each x in X, there is an integer $1 \le i \le k$ such that $g(i) = x$ because g is surjective. Hence,

$$(a_n(g(i)))_{n=m}^{\infty} \in \{(a_n(x))_{n=m}^{\infty} : x \in X\}$$

for each integer $1 \le i \le k$ and

$$(a_n(x))_{n=m}^{\infty} \in \{(a_n(g(i)))_{n=m}^{\infty} : i \in \mathbf{N}, 1 \le i \le k\}$$

for each $x \in X$. So,

$$\{(a_n(x))_{n=m}^{\infty} : x \in X\} = \{(a_n(g(i)))_{n=m}^{\infty} : i \in \mathbf{N}, 1 \le i \le k\}.$$

For each sequence $(a_n(g(i)))_{n=m}^{\infty}$, $1 \le i \le k$, let L_i be its limit; that is, let

$$L_i := \lim_{n\to\infty} a_n(g(i)).$$

Take an arbitrary positive real ε. For each sequence $(a_n(g(i)))_{n=m}^{\infty}$, $1 \le i \le k$, there exists an integer $N_i \ge m$ such that $|a_n(g(i)) - L_i| \le \varepsilon/k$ for all $n \ge N_i$. Let $N := \sum_{i=1}^{k} N_i$. Then, $N \ge N_i$ for every $1 \le i \le k$. For any integer $n \ge N$, we have $|a_n(g(i)) - L_i| \le \varepsilon/k$ for every $1 \le i \le k$. By Lemma 7.1.4(f), this implies that for any integer $n \ge N$, we have

$$\sum_{i=1}^{k} |a_n(g(i)) - L_i| \le \sum_{i=1}^{k} \varepsilon/k,$$

considering the finite sequences $(|a_n(g(i)) - L_i|)_{i=1}^{k}$ and $(\varepsilon/k)_{i=1}^{k}$. On the left side, applying the triangle inequality for finite series (Lemma 7.1.4(e)) yields

$$\sum_{i=1}^{k} |a_n(g(i)) - L_i| \ge \left|\sum_{i=1}^{k} a_n(g(i)) - L_i\right| = \left|\sum_{i=1}^{k} a_n(g(i)) - \sum_{i=1}^{k} L_i\right|.$$

On the right side, we have[4]

$$\sum_{i=1}^{k} \varepsilon/k = (\varepsilon/k) \sum_{i=1}^{k} 1 = (\varepsilon/k)k = \varepsilon.$$

Hence,

$$\left| \sum_{i=1}^{k} a_n(g(i)) - \sum_{i=1}^{k} L_i \right| \leq \varepsilon$$

for any integer $n \geq N$. Given that ε is arbitrary, we deduce that the sequence $\left(\sum_{i=1}^{k} a_n(g(i))\right)_{n=m}^{\infty}$ converges to $\left(\sum_{i=1}^{k} L_i\right)$. But by Definition 7.1.6, we have:

$$\sum_{i=1}^{k} a_n(g(i)) = \sum_{x \in X} a_n(x)$$

and

$$\sum_{i=1}^{k} L_i = \sum_{i=1}^{k} \lim_{n\to\infty} a_n(g(i)) = \sum_{x \in X} \lim_{n\to\infty} a_n(x).$$

Therefore, the sequence $\left(\sum_{x \in X} a_n(x)\right)_{n=m}^{\infty}$ is convergent and

$$\lim_{n\to\infty} \sum_{x \in X} a_n(x) = \sum_{x \in X} \lim_{n\to\infty} a_n(x).$$

Exercise 7.1.6

We will prove by induction on n that the statement

$P(n)$: "If I is a set with n elements, and for each $i \in I$, there is a finite set E_i such that the sets E_i are pairwise disjoint, then

$$\sum_{x \in \bigcup_{i\in I} E_i} f(x) = \sum_{i \in I} \sum_{x \in E_i} f(x)$$

where $f : \bigcup_{i\in I} E_i \to \mathbf{R}$ is a function."

is true for all natural numbers n.

Let's start at the base case ($n = 0$). Here, $I = \emptyset$. On the one hand,

$$\sum_{i \in I} \sum_{x \in E_i} f(x) = \sum_{i \in \emptyset} \left(\sum_{x \in E_i} f(x) \right) = 0$$

[4] It can be proved with a straightforward induction that $\sum_{i=1}^{n} 1 = n$ for any natural number n.

by Proposition 7.1.11(a). On the other hand,

$$\sum_{x\in\bigcup_{i\in I}E_i} f(x) = \sum_{x\in\bigcup_{i\in\emptyset}E_i} f(x) = \sum_{x\in\emptyset} f(x) = 0$$

by Proposition 7.1.11(a). So, $P(0)$ is true.

For the inductive step, suppose inductively that $P(n)$ is true for some natural number n. Let us see if $P(n+1)$ is also true. Let I be a finite set with $n+1$ elements, and for each $i \in I$, let E_i be a finite set. Suppose that the sets E_i are pairwise disjoint. Let f be a function from $\bigcup_{i\in I} E_i$ to $\mathbf{R}$. Take an element $\iota \in I$ (such an element exists because $\#(I) > 0$). Note that

$$\bigcup_{i\in I} E_i = \left(\bigcup_{i\in I\setminus\{\iota\}} E_i\right) \cup E_\iota \quad \text{and} \quad \left(\bigcup_{i\in I\setminus\{\iota\}} E_i\right) \cap E_\iota = \emptyset,$$

where the second equality holds because the sets E_i, $i \in I$, are pairwise disjoint. So,

$$\sum_{x\in\bigcup_{i\in I}E_i} f(x) = \sum_{x\in\bigcup_{i\in I\setminus\{\iota\}}E_i} f(x) + \sum_{x\in E_\iota} f(x)$$

by Proposition 7.1.11(e). Given that $I \setminus \{\iota\}$ has n elements and the sets E_i, with $i \in I \setminus \{\iota\}$, are pairwise disjoint, we have

$$\sum_{x\in\bigcup_{i\in I\setminus\{\iota\}}E_i} f(x) = \sum_{i\in I\setminus\{\iota\}} \sum_{x\in E_i} f(x)$$

by the induction hypothesis. In addition,

$$\sum_{x\in E_\iota} f(x) = \sum_{i\in\{\iota\}} \sum_{x\in E_i} f(x)$$

by Proposition 7.1.11(b). Hence,

$$\sum_{x\in\bigcup_{i\in I}E_i} f(x) = \sum_{i\in I\setminus\{\iota\}} \sum_{x\in E_i} f(x) + \sum_{i\in\{\iota\}} \sum_{x\in E_i} f(x).$$

Using Proposition 7.1.11(e) once again, we see that

$$\sum_{x\in\bigcup_{i\in I}E_i} f(x) = \sum_{i\in(I\setminus\{\iota\})\cup\{\iota\}} \sum_{x\in E_i} f(x),$$

so that

$$\sum_{x\in\bigcup_{i\in I}E_i} f(x) = \sum_{i\in I} \sum_{x\in E_i} f(x).$$

Therefore, $P(n+1)$ is true and the induction is complete.

Exercise 7.1.7

Let n, m be natural numbers, and for each integer $1 \leq i \leq n$, let a_i be a natural number with $a_i \leq m$. Consider the series

$$\sum_{j=1}^{m} \#(\{1 \leq i \leq n : a_i \geq j\}).$$

Note that if $m = 0$, then $a_i = 0$ for each $1 \leq i \leq n$ and we have

$$\sum_{j=1}^{m} \#(\{1 \leq i \leq n : a_i \geq j\}) = \sum_{j=1}^{0} \#(\{1 \leq i \leq n : a_i \geq j\}) = 0 = \sum_{i=1}^{n} 0 = \sum_{i=1}^{n} a_i.$$

If $n = 0$, then $\{1 \leq i \leq n\} = \{1 \leq i \leq 0\} = \emptyset$, and so

$$\sum_{j=1}^{m} \#(\{1 \leq i \leq n : a_i \geq j\}) = \sum_{j=1}^{m} \#(\emptyset) = \sum_{j=1}^{m} 0 = 0 = \sum_{i=1}^{0} a_i = \sum_{i=1}^{n} a_i.$$

So from now, assume that $n > 0$ and $m > 0$.

For each $1 \leq i \leq n$ and $1 \leq j \leq m$, define the number $c_{i,j}$ by

$$c_{i,j} := \begin{cases} 1, & a_i \geq j \\ 0, & a_i < j \end{cases}.$$

Note that

$$\#(\{1 \leq i \leq n : a_i \geq j\}) = \sum_{i=1}^{n} c_{i,j}$$

for every integer $1 \leq j \leq m$. To see this, take an integer j with $1 \leq j \leq m$, and let $\#(\{1 \leq i \leq n : a_i \geq j\}) = k$ for some $k \in \mathbf{N}$. If $k = 0$, then $a_i < j$, and thus $c_{i,j} = 0$ for all $1 \leq i \leq n$, and so

$$\sum_{i=1}^{n} c_{i,j} = \sum_{i=1}^{n} 0 = 0 = k,$$

as desired. If $k > 0$, then there are exactly k integers i, with $1 \leq i \leq n$, such that $a_i \geq j$. Denote these integers by $p_1, \ldots, p_k$. Then, $c_{p_1,j} = \cdots = c_{p_k,j} = 1$. If i is an integer such that $1 \leq i \leq n$ and $i \notin \{p_1, \ldots, p_k\}$, we must have $c_{i,j} = 0$, for otherwise $a_i \geq j$, a contradiction. Now, let f be a permutation of $\{i \in \mathbf{N} : 1 \leq i \leq n\}$ such that $f(i) = p_i$

for all $1 \leq i \leq k$ (the way f maps the other integers is irrelevant). By Proposition 7.1.11(c) and Definition 7.1.6, we have

$$\sum_{i=1}^{n} c_{i,j} = \sum_{i=1}^{n} c_{f(i),j}.$$

It follows that

$$\sum_{i=1}^{n} c_{i,j} = \sum_{i=1}^{k} c_{f(i),j} + \sum_{i=k+1}^{n} c_{f(i),j} = \sum_{i=1}^{k} 1 + \sum_{i=k+1}^{n} 0 = k,$$

as desired.

Now,

$$\sum_{j=1}^{m} \#(\{1 \leq i \leq n : a_i \geq j\}) = \sum_{j=1}^{m} \sum_{i=1}^{n} c_{i,j}.$$

By Corollary 7.1.14,

$$\sum_{j=1}^{m} \sum_{i=1}^{n} c_{i,j} = \sum_{i=1}^{n} \sum_{j=1}^{m} c_{i,j}.$$

For every integer $1 \leq i \leq n$, we have $c_{i,j} = 1$ if $1 \leq j \leq a_i$ and $c_{i,j} = 0$ if $a_i < j \leq m$, and so $\sum_{j=1}^{a_i} c_{i,j} = a_i$ and $\sum_{j=a_i+1}^{m} c_{i,j} = 0$, so that

$$\sum_{j=1}^{m} c_{i,j} = a_i.$$

Therefore,

$$\sum_{j=1}^{m} \#(\{1 \leq i \leq n : a_i \geq j\}) = \sum_{i=1}^{n} a_i.$$

7.2 Infinite Series

Exercise 7.2.1

Consider the series $\sum_{n=1}^{\infty}(-1)^n$. Note that the sequence $(a_n)_{n=1}^{\infty}$ defined by $a_n := 1$ for all $n \geq m$ is a non-negative ($1 \geq 0$) and decreasing sequence ($1 \leq 1$). But, $(a_n)_{n=1}^{\infty}$ does not converge to 0. Hence, by the alternating series test (Proposition 7.2.11), the series $\sum_{n=m}^{\infty}(-1)^n$ is divergent.[5]

Now, let us look at Example 1.2.2. In particular, let us focus on the series

$$S = 1-1+1-1+1-1+\ldots.$$

We see that

$$S = \sum_{n=1}^{\infty} -(-1)^n.$$

If S converges, then $-S$ converges by Proposition 7.2.13(b), which means that the series $\sum_{n=1}^{\infty}(-1)^n$ converges, a contradiction. Thus, S is divergent. Consequently, there is no real number assigned to this series, i.e., there is no real number r such that $S = r$. Hence, the expression $1 - S$, is nonsensical and we cannot write $S = 1 - S$ (subtraction has only been defined between real numbers so far).

The mistake in writing

$$S = (1-1)+(1-1)+(1-1)+\ldots$$

is that $(1-1)+(1-1)+(1-1)+\ldots$ is the series $\sum_{n=1}^{\infty}(1-1)$, and $\sum_{n=1}^{\infty}(1-1)$ converges (to 0) while S diverges. Before writing $S = (1-1)+(1-1)+(1-1)+\ldots$, we had to verify that $\sum_{n=1}^{\infty} -(-1)^n = \sum_{n=1}^{\infty}(1-1)$, which is not true.

Similarly,

$$1+(-1+1)+(-1+1)+\cdots = 1+\sum_{n=1}^{\infty}(-1+1) = 1+\sum_{n=1}^{\infty} 0 = 1+0 = 1 \neq S$$

since S diverges. To put it another way, note that $1+(-1+1)+(-1+1)+\ldots$ is the series $\sum_{n=1}^{\infty} b_n$ with $b_1 := 1$ and $b_n := (-1+1)$ for all $n > 1$; since $\sum_{n=1}^{\infty} b_n$ converges, it cannot be equal to the series $S = \sum_{n=1}^{\infty} -(-1)^n$ which diverges.

[5] For another argument, see Example 7.2.7.

Exercise 7.2.2

Let $\sum_{n=m}^{\infty} a_n$ be a formal series of real numbers. Let $S_N := \sum_{n=m}^{N} a_n$ denote the Nth partial sum of this series.

On the one hand, suppose that the series $\sum_{n=m}^{\infty} a_n$ converges. Then, the sequence $(S_N)_{N=m}^{\infty}$ also converges according to Definition 7.2.2. By Theorem 6.4.18, this means that $(S_N)_{N=m}^{\infty}$ is a Cauchy sequence. Take an arbitrary positive real ε. Since $(S_N)_{N=m}^{\infty}$ is a Cauchy sequence, there exists an integer $N \geq m$ such that $|S_p - S_q| \leq \varepsilon$ for all $p, q \geq N$. For all integers $p, q \geq N+1$, if $p \leq q$, then $|S_{p-1} - S_q| \leq \varepsilon$, and

$$\begin{aligned} |S_{p-1} - S_q| \leq \varepsilon &\implies \left| \sum_{n=m}^{p-1} a_n - \sum_{n=m}^{q} a_n \right| \leq \varepsilon \\ &\implies \left| \sum_{n=m}^{p-1} a_n - \left(\sum_{n=m}^{p-1} a_n + \sum_{n=p}^{q} a_n \right) \right| \leq \varepsilon \\ &\implies \left| \sum_{n=p}^{q} a_n \right| \leq \varepsilon; \end{aligned}$$

but if $p > q$, then

$$\left| \sum_{n=p}^{q} a_n \right| = 0.$$

So we have

$$\left| \sum_{n=p}^{q} a_n \right| \leq \varepsilon \quad \text{for all } \; p, q \geq N+1.$$

On the other hand, suppose that for every real $\varepsilon > 0$, there exists an integer $N \geq m$ such that $\left|\sum_{n=p}^{q} a_n\right| \leq \varepsilon$ for all $p, q \geq N$.

Let $\varepsilon > 0$. There exists an integer $N \geq m$ such that $\left|\sum_{n=p}^{q} a_n\right| \leq \varepsilon$ for all $p, q \geq N$. Take two integers $p, q \geq N$. If $p < q$, then $\sum_{n=p+1}^{q} a_n = S_q - S_p$, and so $|S_p - S_q| \leq \varepsilon$. If $p > q$, then $\sum_{n=q+1}^{p} a_n = S_p - S_q$, and so $|S_p - S_q| \leq \varepsilon$. If $p = q$, then $|S_p - S_q| = 0 \leq \varepsilon$. So we have

$$|S_p - S_q| \leq \varepsilon \quad \text{for all } \; p, q \geq N.$$

Given that ε is arbitrary, we deduce that $(S_N)_{N=m}^{\infty}$ is a Cauchy sequence. Thus, $(S_N)_{N=m}^{\infty}$ is convergent by Theorem 6.4.18. Finally, this means that the series $\sum_{n=m}^{\infty} a_n$ is convergent, by Definition 7.2.2.

Exercise 7.2.3

Let $\sum_{n=m}^{\infty} a_n$ be a convergent series of real numbers.

Let $\varepsilon > 0$. By Proposition 7.2.5, there is an integer $N \geq m$ such that $\left|\sum_{n=p}^{q} a_n\right| \leq \varepsilon$ for all $p, q \geq N$. So, for any integer $n \geq N$, we have $\left|\sum_{n=n}^{n} a_n\right| \leq \varepsilon$, that is, $|a_n| \leq \varepsilon$. As ε is arbitrary, we conclude that the sequence $(a_n)_{n=m}^{\infty}$ is convergent, and it converges to 0.

Exercise 7.2.4

Let $\sum_{n=m}^{\infty} a_n$ be a formal series of real numbers. Suppose that $\sum_{n=m}^{\infty} a_n$ is absolutely convergent. This means that the series $\sum_{n=m}^{\infty} |a_n|$ is convergent, according to Definition 7.2.8.

Let $\varepsilon > 0$ be a real number. Given that $\sum_{n=m}^{\infty} |a_n|$ converges, there exists an integer $N \geq m$ such that

$$\left| \sum_{n=p}^{q} |a_n| \right| \leq \varepsilon \quad \text{for all } \; p, q \geq N$$

by Proposition 7.2.5. For any $p, q \geq N$, we have $|a_n| \geq 0$ for all $p \leq n \leq q$; this implies that $\sum_{n=p}^{q} |a_n| \geq \sum_{n=p}^{q} 0 = 0$ by Lemma 7.1.4(f), and thus $\sum_{n=p}^{q} |a_n| = \left|\sum_{n=p}^{q} |a_n|\right|$. Therefore,

$$\sum_{n=p}^{q} |a_n| \leq \varepsilon \quad \text{for all } \; p, q \geq N.$$

According to Lemma 7.1.4(e), we have

$$\left| \sum_{n=p}^{q} a_n \right| \leq \sum_{n=p}^{q} |a_n| \quad \text{for all } \; p, q \geq N.$$

Therefore,

$$\left| \sum_{n=p}^{q} a_n \right| \leq \varepsilon \quad \text{for all } \; p, q \geq N.$$

From Proposition 7.2.5, we deduce that the series $\sum_{n=m}^{\infty} a_n$ is convergent.

Now, let

$$S_N := \sum_{n=m}^{N} a_n \quad \text{and} \quad T_N := \sum_{n=m}^{N} |a_n|$$

denote the Nth partial sums of the series $\sum_{n=m}^{\infty} a_n$ and $\sum_{n=m}^{\infty} |a_n|$ respectively, with $N \geq m$. Given that $-|a_n| \leq a_n \leq |a_n|$ for all integers $n \geq m$, we have

$$-T_N \leq S_N \leq T_N \text{ for all } N \geq m$$

by Lemma 7.1.4(f). In light of Definition 7.2.2, the sequences $(S_N)_{N=m}^{\infty}$ and $(T_N)_{N=m}^{\infty}$ converge since the series $\sum_{n=m}^{\infty} a_n$ and $\sum_{n=m}^{\infty} |a_n|$ converge. In addition, $(-T_N)_{N=m}^{\infty}$ converges by Theorem 6.1.19(c). Given that $-T_N \leq S_N \leq T_N$ for all $N \geq m$, we have $\lim_{N\to\infty} -T_N \leq \lim_{N\to\infty} S_N \leq \lim_{N\to\infty} T_N$ (this is an easily provable consequence of Lemma 6.4.13[6]). Thus,

$$-\lim_{N\to\infty} T_N \leq \lim_{N\to\infty} S_N \leq \lim_{N\to\infty} T_N,$$

that is,

$$-\sum_{n=m}^{\infty} |a_n| \leq \sum_{n=m}^{\infty} a_n \leq \sum_{n=m}^{\infty} |a_n|,$$

and so

$$\left| \sum_{n=m}^{\infty} a_n \right| \leq \sum_{n=m}^{\infty} |a_n|.$$

Exercise 7.2.5

(a) Let $\sum_{n=m}^{\infty} a_n$ be a series converging to a real x and $\sum_{n=m}^{\infty} b_n$ be a series converging to a real y.

For any integer $N \geq m$, let

$$S_N := \sum_{n=m}^{N} a_n \quad \text{and} \quad T_N := \sum_{n=m}^{N} b_n$$

denote the Nth partial sums of $\sum_{n=m}^{\infty} a_n$ and $\sum_{n=m}^{\infty} b_n$ respectively. We have:

$$S_N + T_N = \sum_{n=m}^{N} (a_n + b_n)$$

for all $N \geq m$. Since $(S_N)_{N=m}^{\infty}$ and $(T_N)_{N=m}^{\infty}$ converge to x and y respectively (Definition 7.2.2), $(S_N + T_N)_{N=m}^{\infty}$ converge to $x+y$. Thus,

$$\lim_{N\to\infty} (S_N + T_N) = \lim_{N\to\infty} \sum_{n=m}^{N} (a_n + b_n) = x + y.$$

But for any $N \geq m$, $\sum_{n=m}^{N} (a_n + b_n)$ is the Nth partial sum of $\sum_{n=m}^{\infty} (a_n + b_n)$. Therefore, $\sum_{n=m}^{\infty} (a_n + b_n)$ is a convergent series that converges to $x+y$, by Definition 7.2.2. So,

$$\sum_{n=m}^{\infty} (a_n + b_n) = \sum_{n=m}^{\infty} a_n + \sum_{n=m}^{\infty} b_n.$$

[6] Here is a sketch of a proof: use limsup or liminf first, then apply Proposition 6.4.12(f).

(b) Let $\sum_{n=m}^{\infty} a_n$ be a series converging to a real number x and let c be a real number.

For any integer $N \geq m$, let

$$S_N := \sum_{n=m}^{N} a_n$$

denote the Nth partial sum of $\sum_{n=m}^{\infty} a_n$. For any integer $N \geq m$, we have

$$cS_N = c \sum_{n=m}^{N} a_n = \sum_{n=m}^{N} (ca_n).$$

Thus, cS_N is the Nth partial sum of the series $\sum_{n=m}^{\infty}(ca_n)$. Since $\lim_{n\to\infty} S_N = x$ by Definition 7.2.2, we have

$$\lim_{n\to\infty} cS_N = c \lim_{n\to\infty} S_N = cx.$$

Therefore, $\sum_{n=m}^{\infty}(ca_n)$ is convergent, and converges to cx. In other words,

$$\sum_{n=m}^{\infty} (ca_n) = c \sum_{n=m}^{\infty} a_n.$$

(c) Let $\sum_{n=m}^{\infty} a_n$ be a series of real numbers and let $k \geq 0$ be an integer.

For any integer $N \geq m$, let $S_N := \sum_{n=m}^{N} a_n$ denote the Nth partial sum of $\sum_{n=m}^{\infty} a_n$. Consider the series $\sum_{n=m+k}^{\infty} a_n$; its Nth partial sum is $T_N := \sum_{n=m+k}^{N} a_n$ for $N \geq m+k$. Note that

$$S_N = \sum_{n=m}^{m+k-1} a_n + T_N \quad \text{for all } N \geq m+k. \tag{$\star$}$$

By Definition 7.2.2, the series $\sum_{n=m}^{\infty} a_n$ converges if and only if the sequence $(S_N)_{N=m}^{\infty}$ converges. Plus, $(S_N)_{N=m}^{\infty}$ converges if and only if $(S_N)_{N=m+k}^{\infty}$ converges (see Exercise 6.1.3). From $(\star)$ and Theorem 6.1.19(a), it is easy to see that $(S_N)_{N=m+k}^{\infty}$ converges if and only if $(T_N)_{N=m+k}^{\infty}$ converges. Lastly, Definition 7.2.2 asserts that $(T_N)_{N=m+k}^{\infty}$ converges if and only if $\sum_{n=m+k}^{\infty} a_n$ converges. In sum, if one of the two series $\sum_{n=m}^{\infty} a_n$ and $\sum_{n=m+k}^{\infty} a_n$ is convergent, then the other one is.

Now, suppose that one of the two series $\sum_{n=m}^{\infty} a_n$ and $\sum_{n=m+k}^{\infty} a_n$ is convergent, so that the other one is convergent as well. The sequences $(S_N)_{N=m}^{\infty}$ and $(S_N)_{N=m+k}^{\infty}$ both converge to the same limit (see Exercise 6.1.3), which is the sum of $\sum_{n=m}^{\infty} a_n$. In

addition, the sequence $(T_N)_{N=m+k}^{\infty}$ converges to the sum of $\sum_{n=m+k}^{\infty} a_n$. Therefore,

$$\begin{aligned}(\star) &\implies \lim_{N\to\infty} S_N = \lim_{N\to\infty}\left(\sum_{n=m}^{m+k-1} a_n + T_N\right)\\ &\implies \lim_{N\to\infty} S_N = \lim_{N\to\infty}\left(\sum_{n=m}^{m+k-1} a_n\right) + \lim_{N\to\infty} T_N\\ &\implies \sum_{n=m}^{\infty} a_n = \sum_{n=m}^{m+k-1} a_n + \sum_{n=m+k}^{\infty} a_n.\end{aligned}$$

(d) Let $\sum_{n=m}^{\infty} a_n$ be a series of real numbers converging to a real x and let k be an integer.

For any integer $N \geq m$, let $S_N := \sum_{n=m}^{N} a_n$ denote the Nth partial sum of $\sum_{n=m}^{\infty} a_n$. The sequence $(S_N)_{N=m}^{\infty}$ converges to x according to Definition 7.2.2. Consider the series $\sum_{n=m+k}^{\infty} a_{n-k}$; its Nth partial sum is $T_N := \sum_{n=m+k}^{N} a_{n-k}$ for $N \geq m+k$. Note that for all $N \geq m+k$, we have $N-k \geq m$ and

$$T_N = \sum_{n=m+k}^{N} a_{n-k} = \sum_{n=m}^{N-k} a_n = S_{N-k}$$

by Lemma 7.1.4(b).

First, suppose that $k \geq 0$. From Exercise 6.1.3, we know that $(S_N)_{N=m+k}^{\infty}$ also converges to x. Note that $(S_{(N-k)+k})_{N=m+k}^{\infty}$ is just the sequence $(S_N)_{N=m+k}^{\infty}$. Thus, by the results of Exercise 6.1.4, the sequence $(S_{N-k})_{N=m+k}^{\infty}$ also converges to x.[7]

Second, suppose that $k < 0$. Then $-k > 0$. From Exercise 6.1.4, we know that $(S_{N-k})_{N=m}^{\infty}$ also converges to x. Note that the sequence $(S_{N-k})_{N=(m+k)-k}^{\infty}$ is just the sequence $(S_{N-k})_{N=m}^{\infty}$. Since $(S_{N-k})_{N=m}^{\infty}$ converges to x, it follows from Exercise 6.1.3 that $(S_{N-k})_{N=m+k}^{\infty}$ also converges to x.

In sum, the sequence $(T_N)_{N=m+k}^{\infty}$ converges to x. That is to say, the series $\sum_{n=m+k}^{\infty} a_{n-k}$ converges to x by Definition 7.2.2.

Exercise 7.2.6

Let $(a_n)_{n=0}^{\infty}$ be a sequence of real numbers converging to 0. Consider the series $\sum_{n=0}^{\infty}(a_n - a_{n+1})$.

[7] If it is not clear, you can use the sequence $(U_N)_{N=m+k}^{\infty}$ defined by $U_N := S_{N-k}$ for all $N \geq m+k$ as a transition sequence. Observe that $(U_{N+k})_{N=m+k}^{\infty}$ is just the sequence $(S_{(N-k)+k})_{N=m+k}^{\infty}$, and since $(S_{(N-k)+k})_{N=m+k}^{\infty}$ converges to x, so does $(U_{N+k})_{N=m+k}^{\infty}$. It follows from Exercise 6.1.4 that $(U_N)_{N=m+k}^{\infty}$ converges to x as well. In other words, $(S_{N-k})_{N=m+k}^{\infty}$ converges to x.

For every integer $N \geq 0$, the Nth partial sum of this series is

$$S_N := \sum_{n=0}^{N} (a_n - a_{n+1}).$$

We will prove by induction on N that

$$S_N = a_0 - a_{N+1} \text{ for all } N \in \mathbf{N}.$$

The base case is easily validated as $S_0 = \sum_{n=0}^{0}(a_n - a_{n+1}) = a_0 - a_1$. Now, suppose inductively that $S_N = a_0 - a_{N+1}$ for some $n \in \mathbf{N}$. Then,

$$\begin{aligned} S_{N+1} &= \sum_{n=0}^{N+1} (a_n - a_{n+1}) \\ &= \sum_{n=0}^{N} (a_n - a_{n+1}) + a_{N+1} - a_{N+2} \\ &= (a_0 - a_{N+1}) + a_{N+1} - a_{N+2} && \text{by the induction hypothesis} \\ &= a_0 - a_{(N+1)+1}, \end{aligned}$$

and the induction is complete.

Given that $\lim_{n\to\infty} a_n = 0$, we have $\lim_{N\to\infty} a_{N+1} = 0$ (see Exercise 6.1.4). Thus,

$$\lim_{N\to\infty} (a_0 - a_{N+1}) = a_0 - 0 = a_0.$$

That is,

$$\lim_{N\to\infty} S_N = a_0.$$

By Definition 7.2.2, this means that the series $\sum_{n=0}^{\infty}(a_n - a_{n+1})$ converges to a_0.

If, instead, $(a_n)_{n=0}^{\infty}$ converged to an arbitrary real L, then we would have

$$\lim_{N\to\infty} a_{N+1} = L \quad \text{and} \quad S_N = a_0 - a_{N+1} \quad \text{for all } N \geq 0,$$

which would lead to

$$\lim_{N\to\infty} S_N = a_0 - L,$$

and thus the series $\sum_{n=0}^{\infty}(a_n - a_{n+1})$ would converge to $a_0 - L$ by Definition 7.2.2.

7.3 Sums of Non-negative Numbers

Exercise 7.3.1

Let $\sum_{n=m}^{\infty} a_n$ and $\sum_{n=m}^{\infty} b_n$ be two formal series of real numbers such that $|a_n| \leq b_n$ for all $n \geq m$. Note that $b_n \geq 0$ for all $n \geq m$.

Suppose that $\sum_{n=m}^{\infty} b_n$ is convergent. Then, by Proposition 7.3.1, there exists a real M such that

$$\sum_{n=m}^{N} b_n \leq M$$

for all $N \geq m$. For any integer $N \geq m$, we have

$$\sum_{n=m}^{N} |a_n| \leq \sum_{n=m}^{N} b_n$$

by Lemma 7.1.4(f), and so

$$\sum_{n=m}^{N} |a_n| \leq M.$$

According to Proposition 7.3.1, this implies that $\sum_{n=m}^{\infty} |a_n|$ converges. In other words, the series $\sum_{n=m}^{\infty} a_n$ is absolutely convergent.

By Proposition 7.2.9, the series $\sum_{n=m}^{\infty} a_n$ also converges, and we have

$$\left| \sum_{n=m}^{\infty} a_n \right| \leq \sum_{n=m}^{\infty} |a_n|.$$

Now, since $\sum_{n=m}^{\infty} |a_n|$ and $\sum_{n=m}^{\infty} b_n$ converge, the series $\sum_{n=m}^{\infty} (|a_n| - b_n)$ also converges by Proposition 7.2.13(a) and Proposition 7.2.13(b). This means that the sequence $\left(\sum_{n=m}^{N} (|a_n| - b_n)\right)_{N=m}^{\infty}$ converges, according to Definition 7.2.2. For all integers $N \geq m$, given that

$$\sum_{n=m}^{N} |a_n| \leq \sum_{n=m}^{N} b_n,$$

we have

$$\sum_{n=m}^{N} (|a_n| - b_n) \leq 0$$

by Lemma 7.1.4. Since $\left(\sum_{n=m}^{N} (|a_n| - b_n)\right)_{N=m}^{\infty}$ converges, this implies that

$$\lim_{N \to \infty} \sum_{n=m}^{N} (|a_n| - b_n) \leq 0,$$

that is,

$$\sum_{n=m}^{\infty} (|a_n| - b_n) \leq 0. \qquad (\star)$$

It follows from Proposition 7.2.13 that

$$\sum_{n=m}^{\infty} (|a_n| - b_n) = \sum_{n=m}^{\infty} |a_n| - \sum_{n=m}^{\infty} b_n.$$

So $(\star)$ implies that

$$\sum_{n=m}^{\infty} |a_n| \leq \sum_{n=m}^{\infty} b_n.$$

In conclusion, we have

$$\left| \sum_{n=m}^{\infty} a_n \right| \leq \sum_{n=m}^{\infty} |a_n| \leq \sum_{n=m}^{\infty} b_n.$$

Exercise 7.3.2

Let x be a real number.

Suppose that $|x| \geq 1$. Then by Lemma 6.5.2, $(x^n)_{n=0}^{\infty}$ diverges or converges to 1. In any case, the series $\sum_{n=0}^{\infty} x^n$ is divergent according to the zero test (Corollary 7.2.6).

Suppose that $|x| < 1$. As a preliminary step, we will show by induction on N that the formula

$$\sum_{n=0}^{N} x^n = \frac{1 - x^{N+1}}{1 - x}$$

holds for all natural numbers N. The base case is done easily:

$$\frac{1 - x^{0+1}}{1 - x} = \frac{1 - x}{1 - x} = 1 = x^0 = \sum_{n=0}^{0} x^n.$$

Next, suppose inductively that the formula holds for some $N \in \mathbf{N}$. Then,

$$\sum_{n=0}^{N+1} x^n = \sum_{n=0}^{N} x^n + x^{N+1} = \frac{1 - x^{N+1}}{1 - x} + x^{N+1} = \frac{1 - x^{N+2}}{1 - x}.$$

Thus, the formula also holds for $N + 1$, and the induction is closed.

Now, we know from Lemma 6.5.2 that $\lim_{n\to\infty} x^n = 0$, and so $\lim_{n\to\infty} x^{n+1} = 0$ (see Exercise 6.1.4). Using the limit laws, we observe that

$$\begin{aligned}\lim_{n\to\infty} x^{n+1} = 0 &\implies \lim_{n\to\infty} 1 - x^{n+1} = 1 \\ &\implies \lim_{n\to\infty} \frac{1-x^{n+1}}{1-x} = \frac{1}{1-x} \\ &\implies \lim_{N\to\infty} \frac{1-x^{N+1}}{1-x} = \frac{1}{1-x} \\ &\implies \lim_{N\to\infty} \sum_{n=0}^{N} x^n = \frac{1}{1-x}.\end{aligned}$$

Therefore, the series $\sum_{n=0}^{\infty} x^n$ is convergent and converges to $1/(1-x)$, that is,

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}.$$

Note that for any $n \in \mathbf{N}$, $|x^n| = |x|^n$. Since $||x|| < 1$, it follows from the above argument that $\sum_{n=0}^{\infty} |x^n|$ converges, and

$$\sum_{n=0}^{\infty} |x^n| = \frac{1}{1-|x|}.$$

Thus, $\sum_{n=0}^{\infty} x^n$ is absolutely convergent.

Exercise 7.3.3

Let $\sum_{n=0}^{\infty} a_n$ be an absolutely convergent series of real numbers such that $\sum_{n=0}^{\infty} |a_n| = 0$.

Suppose that there exists some $K \in \mathbf{N}$ such that $a_K = c$ for some real $c \neq 0$. Note that $|a_K| = |c| > 0$.

We claim that for all integers $N \geq K$, the Nth partial sum S_N of the series $\sum_{n=0}^{\infty} |a_n|$ is greater than or equal to $|c|$, and prove it as follows. For all $n \in \mathbf{N}$, $|a_n| \geq 0$. So, for any natural numbers $p \leq q$, $\sum_{n=p}^{q} |a_n| \geq \sum_{n=p}^{q} 0 = 0$ by Lemma 7.1.4(f). We also know from Definition 7.1.1 that $\sum_{n=p}^{q} |a_n| = 0$ for any natural numbers $p > q$. Hence, for any integer $N \geq K$,

$$S_N = \sum_{n=0}^{N} |a_n| = \sum_{n=0}^{K-1} |a_n| + |a_K| + \sum_{n=K+1}^{N} |a_n| \geq 0 + |c| + 0 = |c|.$$

Now, as $(S_N)_{N=K}^{\infty}$ converges, this implies that

$$\lim_{N\to\infty} S_N \geq |c| > 0,$$

so that

$$\sum_{n=0}^{\infty} |a_n| > 0,$$

a contradiction. So, we must have $|a_n| = 0$ for all $n \in \mathbf{N}$. Therefore, $a_n = 0$ for all $n \in \mathbf{N}$.

7.4 Rearrangement of Series

Exercise 7.4.1

Let $\sum_{n=0}^{\infty} a_n$ be an absolutely convergent series of real numbers. Let $f : \mathbf{N} \to \mathbf{N}$ be a strictly increasing function[8]. Note that f is injective because for any natural numbers $i < j$, we have $f(i) < f(j)$[9] and so $f(i) \neq f(j)$.

Let S_N and T_N denote the partial sums of $\sum_{n=0}^{\infty} |a_n|$ and $\sum_{n=0}^{\infty} |a_{f(n)}|$ respectively:

$$S_N = \sum_{n=0}^{N} |a_n| \quad \text{and} \quad T_N = \sum_{n=0}^{N} |a_{f(n)}| \quad \text{for all } N \in \mathbf{N}.$$

Since $\sum_{n=0}^{\infty} a_n$ is absolutely convergent, $(S_N)_{N=0}^{\infty}$ converges. Hence, $(S_N)_{N=0}^{\infty}$ is bounded by some positive real K. Fix a natural number N. Observe that

$$S_{f(N)} = \sum_{n=0}^{f(N)} |a_n| = \sum_{n \in \{i \in \mathbf{N} : 0 \leq i \leq f(N)\}} |a_n|.$$

Given that f is an increasing function, we have $0 \leq f(i) \leq f(N)$ for all integers $0 \leq i \leq N$, and so $f(\{i \in \mathbf{N} : 0 \leq i \leq N\}) \subseteq \{i \in \mathbf{N} : 0 \leq i \leq f(N)\}$. Thus,

$$\sum_{n \in \{i \in \mathbf{N} : 0 \leq i \leq f(N)\}} |a_n| \geq \sum_{n \in f(\{i \in \mathbf{N} : 0 \leq i \leq N\})} |a_n|$$

in light of Proposition 7.1.11[10]. Let $\tilde{f}$ be the function defined from $\{i \in \mathbf{N} : 0 \leq i \leq N\}$ to $f(\{i \in \mathbf{N} : 0 \leq i \leq N\})$ by $\tilde{f}(i) := f(i)$. Clearly, the function $\tilde{f}$ is bijective because f is injective and the codomain is the image of the domain. It follows from Proposition 7.1.11(c) that

$$\sum_{n \in f(\{i \in \mathbf{N} : 0 \leq i \leq N\})} |a_n| = \sum_{n \in \{i \in \mathbf{N} : 0 \leq i \leq N\}} |a_{\tilde{f}(n)}| = \sum_{n \in \{i \in \mathbf{N} : 0 \leq i \leq N\}} |a_{f(n)}|.$$

Observe that

$$\sum_{n \in \{i \in \mathbf{N} : 0 \leq i \leq N\}} |a_{f(n)}| = \sum_{n=0}^{N} |a_{f(n)}| = T_N.$$

[8] In the exercise prompt, f is said to be just an increasing function, but the precision in parentheses shows that f is meant to be a *strictly* increasing function. That is to say, for any integer $n \in \mathbf{N}$ we have not only $f(n+1) \geq f(n)$, but also $f(n+1) > f(n)$.

[9] This can easily be proven with a straightforward induction on j starting at $j = i+1$, fixing i.

[10] Propositions 7.1.11(e) and 7.1.11(h) can be used to show that if $X \subseteq Y$ are two finite subsets of $\mathbf{N}$ and $(b_n)_{n=0}^{\infty}$ is a sequence of non-negative terms, then $\sum_{i \in X} b_i \leq \sum_{i \in Y} b_i$.

Hence,

$$\sum_{n\in\{i\in\mathbf{N}:0\le i\le f(N)\}} |a_{f(n)}| \ge T_N,$$

that is

$$S_{f(N)} \ge T_N.$$

Therefore,

$$T_N \le K.$$

Since $(T_N)_{N=0}^{\infty}$ is a non-decreasing and bounded sequence, it converges, and so the series $\sum_{n=0}^{\infty} |a_{f(n)}|$ converges. This means that $\sum_{n=0}^{\infty} a_{f(n)}$ is an absolutely convergent series.

Even in the case where f is just a one-to-one function, this result holds. To see this, take an integer $N \in \mathbf{N}$ and let $M := \sup(f(\{i \in \mathbf{N} : 0 \le i \le N\}))$. Then, following the same line of reasoning as above, we can show that $T_N \le S_M$ since $f(\{i \in \mathbf{N} : 0 \le i \le N\}) \subseteq \{i \in \mathbf{N} : 0 \le i \le M\}$. It follows that $T_N \le K$, and hence, $(T_N)_{N=0}^{\infty}$ is bounded. Therefore, $(T_N)_{N=0}^{\infty}$ converges, and thus the series $\sum_{n=0}^{\infty} a_{f(n)}$ is absolutely convergent.

Exercise 7.4.2

Let $\sum_{n=0}^{\infty} a_n$ be an absolutely convergent series of real numbers. Let $f : \mathbf{N} \to \mathbf{N}$ be a bijection.

Since $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} |a_n|$ converge, the series $\sum_{n=0}^{\infty}(a_n + |a_n|)$ converges by Proposition 7.2.13(a) and the series $\sum_{n=0}^{\infty} -|a_n|$ converges by Proposition 7.2.13(b). Moreover, the terms of the series $\sum_{n=0}^{\infty}(a_n + |a_n|)$ are non-negative real numbers. Therefore, the series $\sum_{m=0}^{\infty}(a_{f(m)} + |a_{f(m)}|)$ is also convergent and has the same sum as $\sum_{n=0}^{\infty}(a_n + |a_n|)$, according to Proposition 7.4.1. From the same proposition, it follows that $\sum_{m=0}^{\infty} |a_{f(m)}|$ converges to the sum of $\sum_{n=0}^{\infty} |a_n|$. We have

$$\sum_{n=0}^{\infty} a_n = \sum_{n=0}^{\infty}(a_n + |a_n| - |a_n|).$$

Applying Proposition 7.2.13(a) and then Proposition 7.2.13(b) yields

$$\sum_{n=0}^{\infty} a_n = \sum_{n=0}^{\infty}(a_n + |a_n|) + \sum_{n=0}^{\infty} -|a_n| = \sum_{n=0}^{\infty}(a_n + |a_n|) - \sum_{n=0}^{\infty} |a_n|.$$

Hence,

$$\sum_{n=0}^{\infty} a_n = \sum_{m=0}^{\infty} (a_{f(m)} + |a_{f(m)}|) - \sum_{m=0}^{\infty} |a_{f(m)}|.$$

Finally, we have

$$\sum_{m=0}^{\infty} (a_{f(m)} + |a_{f(m)}|) - \sum_{m=0}^{\infty} |a_{f(m)}| = \sum_{m=0}^{\infty} (a_{f(m)} + |a_{f(m)}| - |a_{f(m)}|) = \sum_{m=0}^{\infty} a_{f(m)}$$

by Propositions 7.2.12(a) and 7.2.13(b), and so

$$\sum_{n=0}^{\infty} a_n = \sum_{m=0}^{\infty} a_{f(m)}.$$

We see that the series $\sum_{m=0}^{\infty} a_{f(m)}$ converges and has the same sum as $\sum_{n=0}^{\infty} a_n$.

7.5 The Root and Ratio Tests

Exercise 7.5.1

Let $(c_n)_{n=m}^{\infty}$ be a sequence of positive numbers. We are going to show that

$$\liminf_{n\to\infty} \frac{c_{n+1}}{c_n} \leq \liminf_{n\to\infty} {c_n}^{1/n}.$$

To begin with, note that $\frac{c_{n+1}}{c_n} > 0$ for all integers $n \geq m$. So, $\liminf_{n\to\infty} \frac{c_{n+1}}{c_n} \geq 0$ by Lemma 6.4.13. In particular, $\liminf_{n\to\infty} \frac{c_{n+1}}{c_n} \neq -\infty$. Thus, there are two cases to consider.

First, suppose that $\liminf_{n\to\infty} \frac{c_{n+1}}{c_n} = +\infty$. Take an arbitrary positive real number y. By Proposition 6.4.12(a), there exists an integer $N \geq m$ such that

$$\frac{c_{n+1}}{c_n} > y \qquad (\star)$$

for all integers $n \geq N$. Let $K := \max(N, 1)$. Then $(\star)$ also holds for all integers $n \geq K$ (we need positive integers n so that the nth root is well-defined). Using a simple induction, we see that

$$\frac{c_n}{c_K} \geq y^{n-K} \quad \text{for all } n \geq K.$$

So for all $n \geq K$, we have $c_n \geq c_K y^{n-K}$, so that

$${c_n}^{1/n} \geq (c_K y^{-K})^{1/n} y.$$

Thus,

$$\liminf_{n\to\infty} {c_n}^{1/n} \geq \liminf_{n\to\infty} (c_K y^{-K})^{1/n} y$$

by Lemma 6.4.13. But $\left((c_K y^{-K})^{1/n} y\right)_{n=K}^{\infty}$ is a convergent sequence that converges to y, since the sequence $\left((c_K y^{-K})^{1/n}\right)_{n=K}^{\infty}$ converges to 1 (see Lemma 6.5.3). Hence,

$$\liminf_{n\to\infty} (c_K y^{-K})^{1/n} y = \lim_{n\to\infty} (c_K y^{-K})^{1/n} y = y$$

by Proposition 6.4.12(f). So,

$$\liminf_{n\to\infty} {c_n}^{1/n} \geq y.$$

Given that y is an arbitrary positive real number, we must have $\liminf_{n\to\infty} {c_n}^{1/n} = +\infty$. It follows that

$$\liminf_{n\to\infty} \frac{c_{n+1}}{c_n} \leq \liminf_{n\to\infty} {c_n}^{1/n}.$$

Second, suppose that $\liminf_{n\to\infty} \frac{c_{n+1}}{c_n} = L$ where L is a positive real number. Let $0 < \varepsilon < L$ be an arbitrary real. Then $L - \varepsilon < L$, and so Proposition 6.4.12(a) guarantees the existence an integer $N \geq m$ for which

$$\frac{c_{n+1}}{c_n} > L - \varepsilon \qquad (\star\star)$$

for all $n \geq N$. Let $K := \max(N, 1)$. Then $(\star\star)$ also holds for all integers $n \geq K$. This implies by induction that

$$\frac{c_n}{c_K} \geq (L-\varepsilon)^{n-K}$$

for all $n \geq K$.

So, $c_n \geq c_K(L-\varepsilon)^{-K}(L-\varepsilon)^n$, so that

$$c_n{}^{1/n} \geq \left(c_K(L-\varepsilon)^{-K}\right)^{1/n}(L-\varepsilon)$$

for all $n \geq K$. Hence,

$$\liminf_{n\to\infty} c_n{}^{1/n} \geq \liminf_{n\to\infty}\left[\left(c_K(L-\varepsilon)^{-K}\right)^{1/n}(L-\varepsilon)\right]$$

by Lemma 6.4.13. Since

$$\lim_{n\to\infty}\left(c_K(L-\varepsilon)^{-K}\right)^{1/n} = 1,$$

we have

$$\liminf_{n\to\infty} c_n{}^{1/n} \geq L - \varepsilon \qquad (\star\star\star)$$

by Proposition 6.4.12(f). If $\liminf_{n\to\infty} c_n{}^{1/n} < L$, then we can find a small positive real number δ less than L such that $\liminf_{n\to\infty} c_n{}^{1/n} < L - \delta$ (for example, take δ as half the distance between these two numbers), which contradicts $(\star\star\star)$. Therefore, $\liminf_{n\to\infty} c_n{}^{1/n} \geq L$, that is

$$\liminf_{n\to\infty} \frac{c_{n+1}}{c_n} \leq \liminf_{n\to\infty} c_n{}^{1/n}.$$

Exercise 7.5.2

Let x be a real number with $|x| < 1$, and let q be a real number. For every integer $n \geq 1$, we have

$$|n^q x^n|^{1/n} = (n^q|x^n|)^{1/n} = (n^q|x|^n)^{1/n} = (n^q)^{1/n}|x| = (n^{1/n})^q|x|.$$

By Proposition 7.5.4, $\lim_{n\to\infty} n^{1/n} = 1$. We claim that $\lim_{n\to\infty}(n^{1/n})^q = 1$, and prove it as follows. There are three cases to consider.

In the first case, $q = 0$. Here, $(n^{1/n})^q = 1$ for all positive integers n, and so $\lim_{n\to\infty}(n^{1/n})^q = 1$.

In the second case, $q > 0$. Suppose that $\limsup_{n\to\infty}(n^{1/n})^q > 1$. Then, there exists a positive real number δ such that $\limsup_{n\to\infty}(n^{1/n})^q > 1+\delta$.[11] By Proposition 6.4.12(b), for any integer $N \geq 1$, there exists an integer $n \geq N$ such that $(n^{1/n})^q > 1+\delta$; thus, $n^{1/n} - 1 > (1+\delta)^{1/q} - 1$. Hence, the sequence $(n^{1/n})_{n=1}^{\infty}$ is not eventually $((1+\delta)^{1/q}-1)$-close to 1 (note that $(1+\delta)^{1/q} - 1 > 0$). This contradiction implies that $\limsup_{n\to\infty}(n^{1/n})^q \leq 1$.

Suppose that $\liminf_{n\to\infty}(n^{1/n})^q < 1$. Then, there exists a positive real number $\gamma < 1$ such that $\liminf_{n\to\infty}(n^{1/n})^q < 1-\gamma$. By Proposition 6.4.12(b), for any integer $N \geq 1$, there exists an integer $n \geq N$ such that $(n^{1/n})^q < 1-\gamma$; thus, $1 - n^{1/n} > 1 - (1-\gamma)^{1/q}$. Hence, the sequence $(n^{1/n})_{n=1}^{\infty}$ is not eventually $(1-(1-\gamma)^{1/q})$-close to 1 (note that $1-(1-\gamma)^{1/q} > 0$). This contradiction implies that $\liminf_{n\to\infty}(n^{1/n})^q \geq 1$.

Applying Proposition 6.4.12(c), we have:

$$1 \leq \liminf_{n\to\infty}(n^{1/n})^q \leq \limsup_{n\to\infty}(n^{1/n})^q \leq 1.$$

Thus, $\liminf_{n\to\infty}(n^{1/n})^q = \limsup_{n\to\infty}(n^{1/n})^q = 1$. By Proposition 6.4.12(f), the sequence $((n^{1/n})^q)_{n=1}^{\infty}$ converges to 1, as claimed.

In the third case, $q < 0$. Given that $-q > 0$, it follows from the second case that $\lim_{n\to\infty}(n^{1/n})^{-q} = 1$. For all positive integers n, $(n^{1/n})^q = 1/(n^{1/n})^{-q}$. Hence, we also have $\lim_{n\to\infty}(n^{1/n})^q = 1$. The claim is thus established across all cases.

Now, we have $\lim_{n\to\infty}|n^q x^n|^{1/n} = \lim_{n\to\infty}(n^{1/n})^q|x| = |x|$. Therefore,

$$\limsup_{n\to\infty}|n^q x^n|^{1/n} = \lim_{n\to\infty}|n^q x^n|^{1/n} = |x|,$$

and thus

$$\limsup_{n\to\infty}|n^q x^n|^{1/n} < 1.$$

According to the root test (Theorem 7.5.1), the series $\sum_{n=1}^{\infty} n^q x^n$ is absolutely convergent. Moreover, this implies that $\sum_{n=1}^{\infty} n^q x^n$ is convergent, and so $\lim_{n\to\infty} n^q x^n = 0$ by the zero test (Corollary 7.2.6).

[11] If $a > b$ are two real numbers, then there exists a positive real number c such that $a > b + c$, for otherwise, a and b are ε-close for any real $\varepsilon > 0$, and this implies that $a = b$ (see Proposition 4.3.7(a)), which is a contradiction. Of course, we may take c arbitrarily small by setting an upper bound $c < M$, where M is a positive real number. We can do this because if $a > b + c$, then $a > b + c'$ for any real $c' < c$.

Exercise 7.5.3

Let $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ be two series of real numbers defined by

$$a_n := \frac{1}{n} \quad \text{and} \quad b_n := \frac{1}{n^2}$$

for all integers $n \geq 1$.

We have:

$$\frac{|a_{n+1}|}{|a_n|} = \frac{a_{n+1}}{a_n} = 1 - \frac{1}{n+1} \quad \text{and} \quad |a_n|^{1/n} = {a_n}^{1/n} = \frac{1}{n^{1/n}}$$

while

$$\frac{|b_{n+1}|}{|b_n|} = \frac{b_{n+1}}{b_n} = \left(1 - \frac{1}{n+1}\right)^2 \quad \text{and} \quad |b_n|^{1/n} = {b_n}^{1/n} = \frac{1}{(n^{1/n})^2}$$

for all integers $n \geq 1$. Therefore,

$$\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \lim_{n\to\infty} {a_n}^{1/n} = 1 \quad \text{and} \quad \lim_{n\to\infty} \frac{b_{n+1}}{b_n} = \lim_{n\to\infty} {b_n}^{1/n} = 1,$$

in light of Proposition 7.5.4.

Nonetheless, we know from Corollary 7.3.7 that $\sum_{n=1}^{\infty} a_n$ diverges while $\sum_{n=1}^{\infty} b_n$ converges.